

FINAL TRIM AND STABILITY BOOKLET OF KB-28 (YARD NO. CHISTI/013)

AS AMENDED ON PAGE NO.4,52&53



SEE LETTER E-112402-150674

Rev.	Description	Date	Prepared	Review
<u>Issued for</u> <u>Approval</u>				
Client: K.B. SHIPPING & CO. (JAMNAGAR) Builder: CHISTI MARINE SERVICES		Project: PONTOON (YARD NO.CHISTI/013)		
Doc # CM20-V-110-0208-A		Document Title: FINAL TRIM AND STABILITY BOOKLET		Prepared OM
Scale NTS		Size A4		Checked MG
Sheet 1 of 205		Date 23-10-2020		Review 1 SR
				Review 2
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GENERAL PARTICULARS

Name of the Vessel	KB-28
Client	K.B. SHIPPING & CO. (JAMNAGAR)
Builder	CHISTI MARINE SERVICES
Yard No.	CHISTI/013
Type of Vessel	PONTOON
Reg. No.	JMR-0016
Port of Registry	JAMNAGAR
Class Notation	5SU, PONTOON, INDIAN COASTAL SERVICE DESCRIPTIVE NOTE "FOR CRANE OPERATION" "DECK STRENGTHENED FOR 10T/M²" Descriptive Notation : "Deck strengthened for 10 T/m ² ", "For Crane Operation"

Main Particulars

LOA	50.000 M
LBP	50.000 M
Breadth (Mld.)	18.000 M
Depth (Mld.)	03.000 M
Draft (Mld.)	2.200 M
Draft (Ext.)	2.212 M
Displacement(Ext.)	1893.06 T
Lightship	405.97 T
Deadweight	1487.09 T
Gross Tonnage	685
Net Tonnage	206

Stability Criteria

REGULATION

Intact Stability Criteria in accordance **IS CODE 2008**

- 1) The area under a righting lever curve up to the angle of maximum righting lever should not be less than 0.080 meter-radian.
- 2) The static angle of heel due to a uniformly distributed wind load of 0.54 kPa (wind speed 30 m/s) should not exceed an angle corresponding to half freeboard for the relevant loading condition, where the lever of wind heeling moment is measured from the centroid of the windage area to half the draught.
- 3) The minimum range of stability should be:
For $L \leq 100\text{m}$: 20°
For $L \geq 150\text{m}$: 15°
For intermediate length by interpolation

Intact stability criteria during cargo lifting

The following intact stability criteria are to be complied with:

- $\theta_C \leq 15^\circ$
- $GZC \leq 0,6 GZ_{MAX}$
- $A1 \geq 0.4 ATOT$

where:

θ_C : Heeling angle of equilibrium, corresponding to the first intersection between heeling and righting arms (see Fig 1)

GZC , GZ_{MAX} : Defined in Fig 1

$A1$: Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θ_C to the heeling angle equal to the lesser of:

- heeling angle θ_R of loss of stability, corresponding to the second intersection between heeling and righting arms (see Fig 1)
- heeling angle θ_F , corresponding to flooding of unprotected openings

$ATOT$: Total area, in m.rad, below the righting lever curve.

In the above formula, the heeling arm, corresponding to the cargo lifting, is to be obtained, in m, from the following formula:

$$b = (Pd - Zz) / \Delta$$

where:

P : Cargo lifting mass, in t

d : Transversal distance, in m, of lifting cargo to the longitudinal plane (see Fig 1)

Z : Mass, in t, of ballast used for righting the pontoon, if applicable (see Fig 1)

z : Transversal distance, in m, of the centre of gravity of Z to the longitudinal plane (see Fig 1)

Δ : Displacement, in t, at the loading condition

The above check is to be carried out considering the most unfavourable situations of cargo lifting combined with the lesser initial metacentric height GM . The vertical position of the centre of gravity of cargo lifting is to be assumed in correspondence of the suspension point.

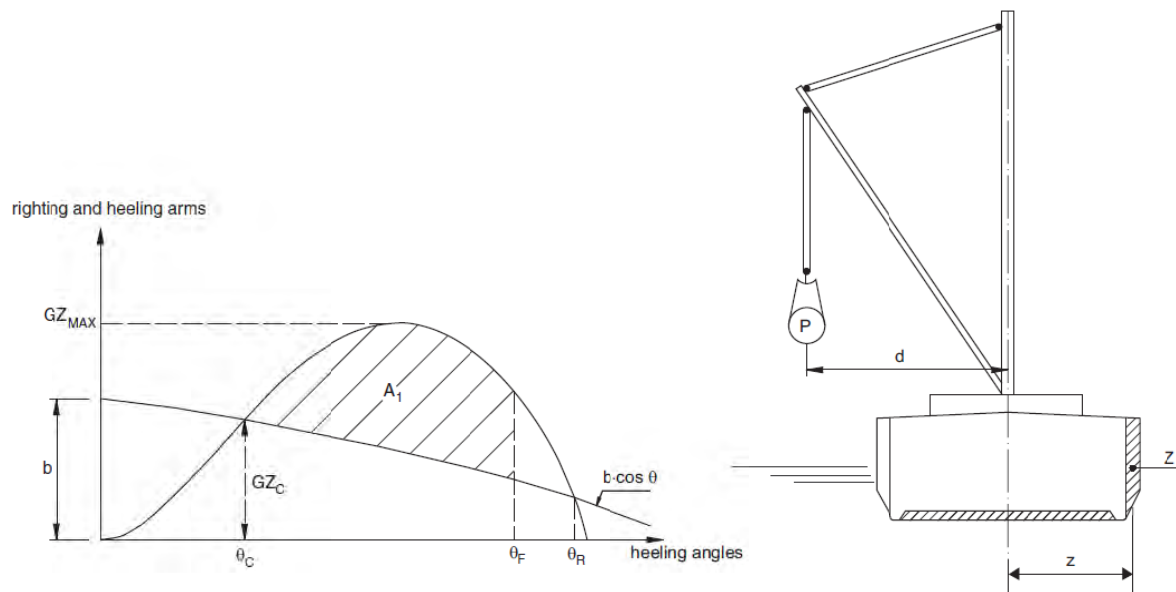


FIGURE 1: CARGO LIFTING

Intact stability criteria in the event of sudden loss of cargo during lifting

This additional requirement is compulsory when counterweights or ballasting of the ship are necessary or when deemed necessary by the Society taking into account the ship dimensions and the weights lifted.

The case of a hypothetical loss of cargo during lifting due to a break of the lifting cable is to be considered.

In this case, the following intact stability criteria are to be compiled with

- $A_2 / A_1 \geq 1$
- $\theta_2 - \theta_3 \geq 20^\circ$

where:

A_1 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θ_1 to the heeling angle θ_C (see Fig 2)

A_2 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θ_C to the heeling angle θ_2 (see Fig 2)

A_3 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θ_C to the heeling angle θ_3 (see Fig 2)

θ_1 : Heeling angle of equilibrium during lifting (see Fig 2)

θ_2 : Heeling angle corresponding to the lesser of θ_R and θ_F

θ_C : Heeling angle of equilibrium, corresponding to the first intersection between heeling and righting arms (see Fig 2)

θ_3 : Maximum heeling angle due to roll, at which $A_3 = A_1$, to be taken not greater than 30° (angle in correspondence of which the loaded cargo on deck is assumed to shift (see Fig 2)

θ_R : Heeling angle of loss of stability, corresponding to the second intersection between heeling and righting arms (see Fig 2).

θ_F : Heeling angle at which progressive flooding may occur (see Fig 2)

In the above formulae, the heeling arm, induced on the ship by the cargo loss, is to be obtained, in m, from the following formula

$$b = Zz \cos\theta/\Delta$$

HEELING AND RIGHTING ARMS

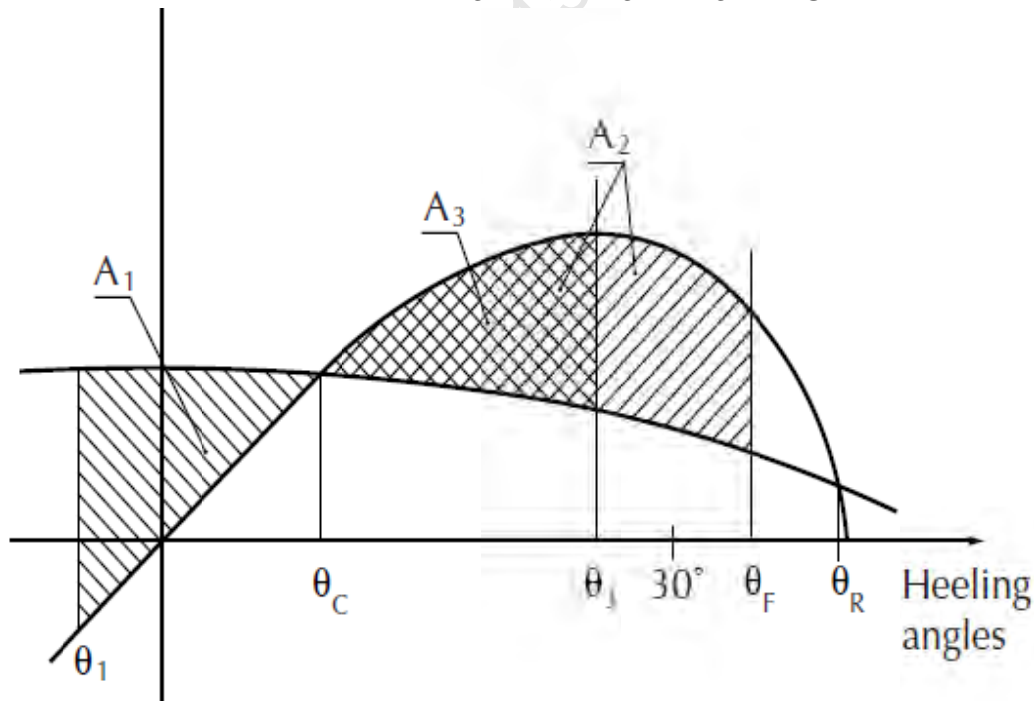


FIGURE 2: CARGO LOSS

Notes to the Master

General precautions against capsizing

1. Compliance with the required minimum stability criteria does not insure immunity against capsizing, regardless of the circumstances, or absolve the Master from his responsibilities. Masters should therefore exercise prudence and good seamanship, having regard to the season of the year, weather forecasts, the navigational zone, and should take appropriate action as to speed and course warranted by the prevailing circumstances.
2. Loading conditions shown in this booklet are typical of those which may be encountered in service and are worked out to be used for guidance only. Further the booklet contains sufficient data to enable the ship's staff to compute the ship's stability in various loading conditions.
3. Excessive trim by forward is to be avoided in all cases as this may lead to difficulties of maneuverability and course keeping as also infringe the minimum Bow-height requirements.
4. All longitudinal levers, namely LCB, LCG, and LCF are computed with reference to the A.P of Ship.
5. In this booklet hydrostatic particulars and cross curves are given at following trims.

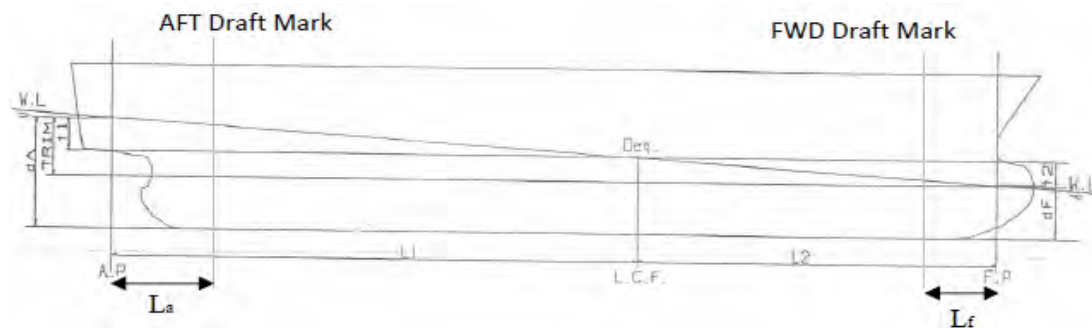
-1.00m, -0.50m, 0.00m, 0.50m, 1.00m

These trims are the expected operational trims.

Notes on Use of Cross Curves of Stability

The purpose of the Cross-Curves of Stability is to enable Statical Stability curves to be drawn for the ship in any loading condition. Intact stability is then determined on the basis of this Curve. WORKED OUT EXAMPLE demonstrates the use of Cross Curves.

Draft and Trim Calculation



The following data are required for ascertaining trim and draft:

- a) Longitudinal Centre of gravity of the ship calculated as follows:

$$LCG = \frac{\text{Summation of longitudinal weight moments}}{\text{Displacement}}$$

- b) Longitudinal centre of buoyancy, LCB
- c) Displacement, Δ
- d) Moment to change trim, MCT1cm
- e) Longitudinal centre of floatation, LCF

LCB, LCF and MCT1cm are taken from the hydrostatic tables (trim = 0 m) corresponding to the actual displacement. The trim is calculated according to the following formula:

$$\text{Trim} = \frac{\Delta * (LCB - LCG)}{MCT1cm * 100}$$

The Extreme drafts at AP (TAP) and at FP (TFP) are calculated according to the following formulae:

$$T_{AP} = \frac{LCF * \text{Trim}}{LBP} + T (\text{extreme})$$

$$T_{FP} = T_{AP} - \text{Trim}$$

Where T is the Extreme draft taken from the hydrostatic tables (at trim = 0) corresponding to the actual displacement.

The draft at the Aft and Forward reading marks are as follows:

$$T_a = T_{AP} - \frac{\text{Trim} * L_a}{LBP}$$

$$T_f = T_{FP} + \frac{\text{Trim} * L_f}{LBP}$$

Where,

T_a = Draft at Aft Draft Mark

T_f = Draft at Fwd Draft Mark

L_a = Distance of Aft Draft Mark from AP and it is consider positive when Aft Draft Mark is Fwd of AP

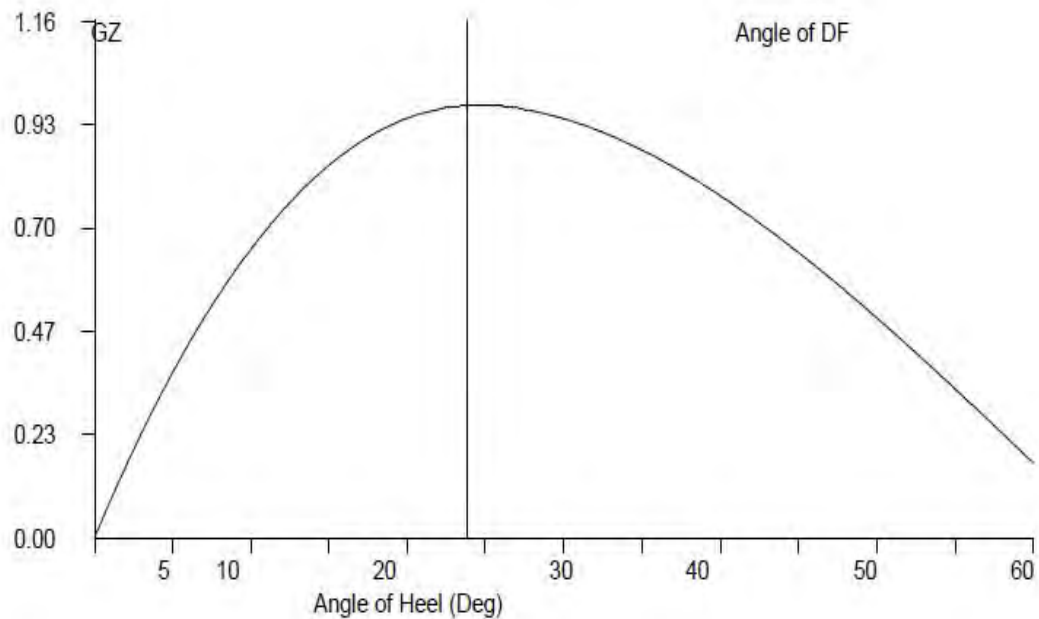
L_f = Distance of Fwd Draft Mark from FP and it is consider positive when Fwd Draft Mark is Aft of FP

GZ Calculation:

- 1) $GZ = KN - KGo \sin (\theta)$
- 2) Area under 0-10 = $10 * 0.01745 * [GZ0 + GZ10] / 2$
- 3) Area under 10-20 = $10 * 0.01745 * [GZ10 + GZ20] / 2$
- 4) Area under 20-30 = $10 * 0.01745 * [GZ20 + GZ30] / 2$
- 5) Area under 30-40 = $10 * 0.01745 * [GZ30 + GZ40] / 2$
- 6) Area under 40-50 = $10 * 0.01745 * [GZ40 + GZ50] / 2$
- 7) Area under 50-60 = $10 * 0.01745 * [GZ50 + GZ60] / 2$
- 8) Area upto 40 deg / Angle of DF = sum of 2+3+4+5
- 9) Max Gz = Read from Gz Curve

Plot GZ values against the angles of heel as above. At 57.3 degree erect an ordinate equal to GM fluid on the same scale as GZ ordinates. Join the point to the origin. This is the Tangent to the GZ Curve at the Origin. Find the

Maxima of GZ curve, ordinates under GZ curve upto 30 degrees, 40 degrees/angle of flooding & between 30 degrees to 40 degrees/angle of flooding. If maximum GZ occurs at an angle less than 30 degrees then area upto this angle must be calculated.



Worked Out Examples**1 Loading the Vessel**

List out the weights, LCG, VCG, TCG and work out longitudinal moment and vertical moment for each load and finally sum up the entire loads to get Displacement, LCG and VCG of the vessel.

For this worked out Example,

Condition 1: Lightship + Crane + Ballast

Displacement	905.29	Tonnes
LCG	24.214	M Fwd of AP
VCG (KG)	4.100	M above Base Line
TCG	0.000	M Off centerline

2 Free Surface Correction

A tank which is partially filled, adversely affects the stability of the vessel. This is known as Free Surface Effect and is expressed as 'Loss of GM' or 'Virtual Rise in KG' and is calculated as follows:

$$\text{Free Surface Correction (M)} = \frac{\text{Sum of Free Surface Moments of all Tanks}}{\text{Displacement of Vessel}}$$

$$\text{Free Surface moment for a Tank} = \text{Moment of Inertia (M}^4\text{) for a Tank} \times \text{Density of Liquid in Tank}$$

$$\text{KGo (Fluid)} = \text{KG (Solid)} + \text{Free Surface Correction.}$$

Hydrostatic Particulars

The trim of a particular loading condition is obtained by

$$\text{Trim} = (\text{LCB} - \text{LCG}) \times \text{displacement} / (\text{MCT} \times 100) + \text{Trim}^*$$

Where,

Trim* = Trim at which hydrostatic particulars have been computed.

Condition 1 : Lightship + Crane + Ballast

TANK NAME	VOL(Cu.M.)	SpGr	Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT	FSM
				METER	T-M	METER	T-M	METER	T-M	T-M
07_WBTK.P	48.39	1.025	49.60	47.660	2363.94	-5.971	-296.16	1.880	93.25	72.32
07_WBTK.C	48.89	1.025	50.11	47.663	2388.39	0.000	0.00	1.872	93.81	72.33
07_WBTK.S	48.39	1.025	49.60	47.660	2363.94	5.971	296.16	1.880	93.25	72.32
TOTAL TANK			149.31	47.661	7116.26	0.000	0.00	1.877	280.30	216.97

FIXED WEIGHT			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
LIGHT SHIP			405.97	25.000	10,149.25	0.000	0.00	3.000	1,217.91
CRANE			350.00	13.300	4,655.00	0.000	0.00	6.325	2,213.75
TOTAL FIXED WEIGHT			755.97	19.583	14,804.25	0.000	0.00	4.539	3,431.66

ITEM			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
TOTAL TANK			149.310	47.661	7116.265	0.000	0.000	1.877	280.302
TOTAL FIXED WEIGHT			755.970	19.583	14804.250	0.000	0.000	4.539	3431.660
TOTAL WEIGHT			905.280	24.214	21920.515	0.000	0.000	4.100	3711.962

Condition 1 : Lightship + Crane + Ballast

Free Surface Correction for VCG = FSM/Displacement

	0.240	m
VCGc	4.340	m

From the MaxVCG table,

Displacement	905.28	T
Allowable Max VCG	15.969	m

VCGc < Maxvcg

PASS

Note : The Master should always check the obtained VCG value against Maximum limiting VCG curve for intact
Refer to the Hydrostatic Table for even trim & Even Heel Condition of Stability Booklet.

Displacement =	905.280	T
Draft ,Tm @ 25.0f =	1.113	m
LCB =	25.000	m
LCF =	25.000	m
MCT =	31.477	T-m/cm
KMT =	26.280	m

The vessels trim is to be determined as follow,

$$\text{Trim} = (\text{LCB} - \text{LCG}) \times \text{Displacement} / (\text{MCT} \times 100)$$

0.226 m, (+) is aft trim, (-) is fwd trim

Determining the Forward perpendicular, Midship and aft Perpendicular Draft

LBP = 50m	
Draft @ 25.00 (MS), Tmid	1.113 m
Draft @ 50.0f (FP), Tfwd	=Tm - Trim/2 1.000 m
Draft@0(AP), Taft	=Tm + Trim/2 1.226 m

Condition 1 : Lightship + Crane + Ballast**From HYDROSTATICS**

Hydrostatic Particulars			Drafts & Trims	
MCT	31.477	T-m	Draft (Mean)	1.113 m
LCB	25.000	m	Draft AP (USK)	1.226 m
			Draft FP (USK)	1.000 m
Metacentric Height				
KMt	26.280	m		
KG	4.100	m		
KG0	4.340	m		
GMt	21.940	m	Trim	0.226 m

The condition shown above is first worked using Even Keel hydrostatic particulars Program then interpolates the hydrostatic particulars from the values available for the displacement

From HYDROSTATICS

Hydrostatic Particulars			Drafts & Trims	
MCT	30.750	T-m	Draft (Mean)	1.113 m
LCB	24.198	m	Draft AP (USK)	1.228 m
			Draft FP (USK)	0.997 m
Metacentric Height				
KMt	26.292	m		
KG	4.100	m		
KG0	4.340	m		
GMt	21.952	m	Trim	0.231 m

Trim is calculated as follows:

$$\text{Trim} = (\text{LCB} - \text{LCG}) \times \text{displacement} / (\text{MCT} \times 100) + \text{Trim}^*$$

Where Trim* is trim at which hydrostatics were calculated.

Condition 1 : Lightship + Crane + Ballast**Intact Stability (Primary Criterion)**

Displacement	905.28	T	
KG0	4.340	M	
GMt	21.952	M	0.150 M

Residual Arm Curve

Angle (Deg)	0	15	25	35	45	55
KN (M)	0.000	5.140	5.613	5.495	5.115	4.545
GZ (M)	0.000	4.017	3.779	3.006	2.046	0.990

	Computed values	Minimum Required
Residual Area from abs 0 deg to MAXRA	0.8402 m-rad	0.080 m-rad
Angle from Equilibrium to RAZero or flood	LARGE m-rad	20.00 Deg.
Angle from Equilibrium to 50% DK Imm Angle	6.02 m-rad	0.00 Deg.

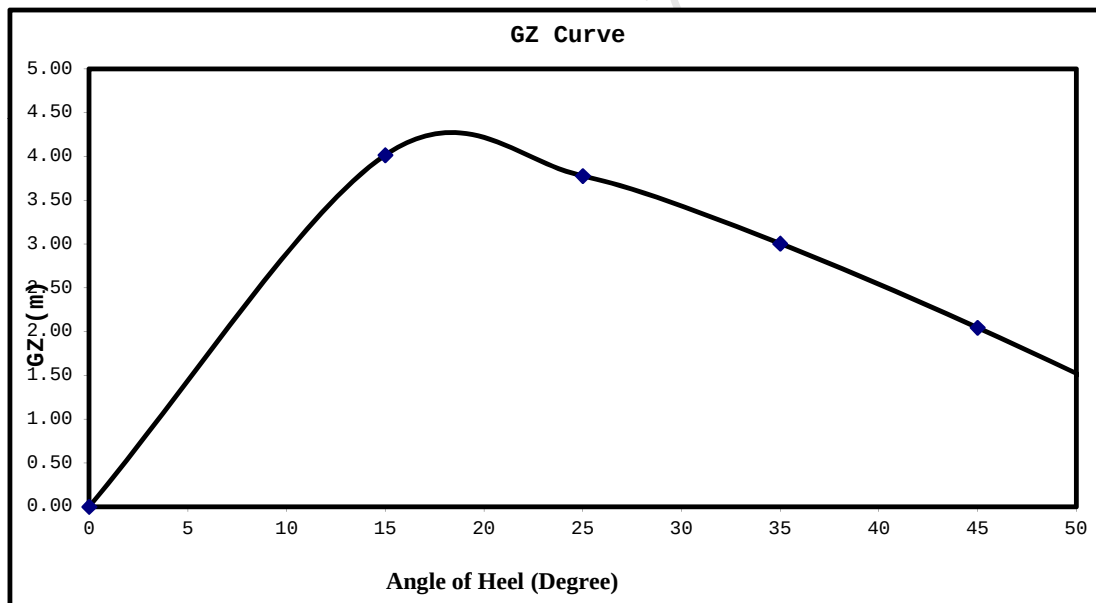
Plot GZ values against the angles of heel as above.

At 57.3 degree erect an ordinate equal to GM fluid on the same scale as GZ ordinates. Join the point to the origin.

This is the Tangent to the GZ Curve at the Origin. Find the Maxima of GZ

curve, ordinates under GZ curve upto 30 degrees, 40 degrees/angle of flooding

If maximum GZ occurs at an angle less than 30 degrees then area upto this angle must be calculated.



Worked Out Example Contd.

Mean Draft	1.113	M
LCF from AP	24.799	M
LCB from AP	24.198	M
MCT	30.750	T-M/Cm
Trim	0.231	M
Draft (AP)	1.228	M
Draft (FP)	0.997	M
KG	4.100	M
KGo	4.340	M
GoMt	21.952	M

LOADLINE DRAFT: Check that Mean Draft (Midship) does not exceed the Summer Draft (Ext) for the corresponding season, as per the assigned loadline.

2020

[illegible]

Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
	METER	T-M	METER	T-M	METER	T-M
405.970	25.000	10,149.250	0.000	0.000	3.000	1,217.910

[illegible]

	m
VCGc	m

Displacement	T
Allowable Max VCG	m

Note : The Master should always check the obtained VCG value against Maximum limiting VCG curve for intact
Refer to the Hydrostatic Table for even trim & Even Heel Condition of Stability Booklet.

Displacement =	T
Draft ,Tm @ 25.0f =	m
LCB =	m
LCF =	m
MCT =	T-m/cm
KMT =	m

$$\text{Trim} = (\text{LCB} - \text{LCG}) \times \text{Displacement} / (\text{MCTX} \times 100)$$

m, (+) is aft trim, (-) is fwd trim

LBP = 50m

Draft @ 25.00 (MS), Tmid m

Draft @ 50.0f (FP), T_{fwd} = T_m - Trim/2 m

Draft @ 0 (AP), T_{aft} = T_m + Trim/2 m

Definitions, Symbols and Sign Conventions:

AP	Fr.0
FP	Fr.100
Midship	Fr.50
Frame Spacing	500mm throughout

List of Symbols

Symbol	Unit	Description
LOA	M	Length Overall
LBP	M	Length between perpendiculars
B (MLD)	M	Moulded Breadth
D (MLD)	M	Moulded Depth
TPCm	T/Cm	Tonnes per 1 Cm immersion
MCTCm	T-M/Cm	Moment to change Trim by 1 Cm
Draft AP (USK)	M	Draft at AP wrt Keel line
Draft FP (USK)	M	Draft at FP wrt Keel line
Draft (Mean)	M	Draft wrt Keel line at Midship
Draft (Mean)	M	[Draft AP(USK) + Draft FP(USK)] x 0.5
VCG, KG	M	Vertical Center of Gravity above Base Line before correcting for free surface
KGo	M	Vertical Center of Gravity above Base Line after correcting for free surface
KB	M	Vertical Center of Buoyancy from Base Line
KMt	M	Transverse Metacenter above Base Line
GoMt	M	Transverse Metacenter above KGo
GZ	M	Righting Arm
FSM	T-M	Free Surface Moment
sp.gr.		Specific Gravity
LCG	M	Longitudinal Center of Gravity
LCB	M	Longitudinal Center of Buoyancy
LCF	M	Longitudinal Center of Floatation

Sign Convention

LCB, LCG, LCF	Fwd of AP is denoted by f and aft of AP is denoted by a
TCG	port is negative and stbd is positive
Trim by Aft	Positive
Trim by Ford	Negative

Metric Conversion Table

<u>MULTIPLY BY</u>	<u>TO CONVERT FROM</u>	<u>TO OBTAIN</u>	
0.03937	millimeters	Inches	25.400
0.39370	Centimeters	Inches	2.5400
3.28080	Meters	Feet	0.3050
2.20460	kilograms	Pounds	0.4540
0.00098	kilogram's	tons(2240 lb.)	1016.1
0.98420	Metric tonnes (i.e. tonnes of 1000 kg.)	tons(2240 lb.)	1.0160
2.49980	Metric tonnes per centimeter of immersion	tons per inch (immersion)	0.4000
8.20140	moment to change trim one centimeter one (tonne meter units)	moment to change trim inch(foot ton units)	0.1220
187.976	Meter Radians	Feet degrees	0.0053
	<u>TO OBTAIN</u>	<u>TO CONVERT FROM</u>	<u>MULTIPLY</u>

Relation between Weight and volume

1000mm cubed	1 Cubic Centimeters
1 Cubic Centimeter of Fresh Water (S.G.1.0)	1 Gram
1000 cubic centimeter of fresh Water (S.G.1.0)	1 Kilogram (1000 Gm.)
1 Cubic Meter of fresh water (S.G.1.0)	1 Tonne (1000 Kg.)
1 Cubic Meter of fresh water (S.G.1.0)	1 tonne
1 Cubic Meter of Salt water (S.G. 1.025)	1.025 Tonnes
1 Tonne of Salt Water (S.G.1.025)	0.975 Cubic Meters

Angle of Downflooding

1. The angle of flooding is the angle of immersion of the sill or coaming of the opening on or above the freeboard deck through which progressive downflooding can occur as the vessel heels.
2. If this opening has a weathertight closing device which cannot be kept closed and secured at all times while the vessel is underway (e.g. engine room air supply trunk), or has no weathertight closing device, progressive downflooding will occur when the sill of the opening immerses. This angle is to be marked on the GZ curve if it comes within the apparent positive range of the GZ curve, and the GZ curve terminates at this angle: i.e., the actual angle of flooding becomes the range of the GZ curve.
3. If an opening is fitted with a weathertight closing device which can be kept closed and secured at all times while the vessel is underway, progressive downflooding can only occur through the opening if its closing device is not closed and secured. The angle of immersion of the lower sill of this opening is a 'potential' angle of flooding. If the closing appliance is not closed and secured, downflooding will occur when the lower sill of this opening reaches the water level.
If the lower sill of the opening through which downflooding can occur immerses at 40 degree or smaller angles of heel, its closing appliances must be kept closed and secured at all times when the vessel is underway or at sea where this opening is used for access in the normal working of the vessel, an alternate access should be used in heavy weather. In moderate weather, the opening may be used for access, but the closing appliances must be closed and secured again immediately after use.

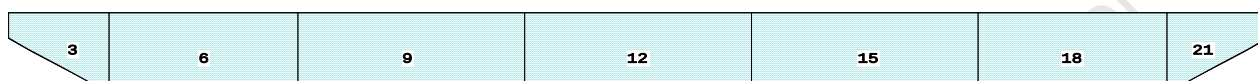
As there is no any unprotected opening exists above freeboard deck, hence the downflooding angle is considered at 50 deg.

01-08-20 21:16:44
GHS 17.22

Cybermarine Technologies PTE. Ltd.
PONTOON

CAPACITY PLAN AND TABLE

Profile View



Plan View

1	4	7	10	13	16	19
2	5	8	11	14	17	20
3	6	9	12	15	18	21

Tanks

1 01_WBTK.P	4 02_WBTK.P	8 03_WBTK.C	12 04_WBTK.S	16 06_WBTK.P	20 07_WBTK.C
2 01_WBTK.C	5 02_WBTK.C	9 03_WBTK.S	13 05_WBTK.P	17 06_WBTK.C	21 07_WBTK.S
3 01_WBTK.S	6 02_WBTK.S	10 04_WBTK.P	14 05_WBTK.C	18 06_WBTK.S	
	7 03_WBTK.P	11 04_WBTK.C	15 05_WBTK.S	19 07_WBTK.P	

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TANK STATUS							
Trim: zero, Heel: zero							
Part-----	Cu.M.----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
01_WBTK.P	48.4	1.025	49.60	2.340f	5.971p	1.880	72.32*
01_WBTK.C	48.9	1.025	50.11	2.336f	0.000	1.872	72.33*
01_WBTK.S	48.4	1.025	49.60	2.340f	5.971s	1.880	72.32*
02_WBTK.P	131.4	1.025	134.72	7.749f	5.980p	1.509	135.61*
02_WBTK.C	132.4	1.025	135.67	7.749f	0.000	1.499	135.61*
02_WBTK.S	131.4	1.025	134.72	7.749f	5.980s	1.509	135.61*
03_WBTK.P	157.7	1.025	161.60	16.000f	5.980p	1.509	162.73*
03_WBTK.C	158.8	1.025	162.73	16.000f	0.000	1.500	162.73*
03_WBTK.S	157.7	1.025	161.60	16.000f	5.980s	1.509	162.73*
04_WBTK.P	157.7	1.025	161.60	25.000f	5.980p	1.509	162.73*
04_WBTK.C	158.8	1.025	162.73	25.000f	0.000	1.500	162.73*
04_WBTK.S	157.7	1.025	161.60	25.000f	5.980s	1.509	162.73*
05_WBTK.P	157.7	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.C	158.8	1.025	162.73	34.000f	0.000	1.500	162.73*
05_WBTK.S	157.7	1.025	161.60	34.000f	5.980s	1.509	162.73*
06_WBTK.P	131.4	1.025	134.72	42.251f	5.980p	1.509	135.61*
06_WBTK.C	132.4	1.025	135.67	42.251f	0.000	1.499	135.61*
06_WBTK.S	131.4	1.025	134.72	42.251f	5.980s	1.509	135.61*
07_WBTK.P	48.4	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	48.9	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	48.4	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			2,566.63	25.000f	0.000	1.549	2712.14
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

HYDROSTATIC PARTICULARS

31-07-20 12:18:46 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

HYDROSTATIC PROPERTIES

Trim: Fwd 1.000/50.000, No Heel, Fixed VCG = 0.000

Draft@ 25.000f	Displacement ---Weight(MT)	Buoyancy-Ctr. ----LCB-----VCB	Weight/ -----cm	Moment/ -----LCF-----cm trim	KML	KMT
0.112	133.29	37.689f 0.175	5.02 33.490f	6.52 244.36	96.881	
0.212	188.30	36.107f 0.211	5.98 31.124f	10.90 289.46	82.456	
0.312	252.92	34.523f 0.246	6.94 28.751f	16.92 334.36	71.772	
0.412	327.16	32.938f 0.281	7.91 26.375f	24.82 379.30	63.564	
0.512	407.76	31.583f 0.319	8.13 26.034f	26.65 326.70	53.012	
0.612	489.54	30.651f 0.361	8.23 25.975f	27.51 280.91	45.226	
0.712	572.35	29.971f 0.406	8.33 25.923f	28.38 247.92	39.491	
0.812	656.10	29.452f 0.452	8.42 25.886f	29.26 222.97	35.068	
0.912	740.70	29.044f 0.499	8.50 25.870f	30.12 203.29	31.535	
1.012	826.02	28.717f 0.548	8.57 25.880f	30.81 186.46	28.540	
1.112	912.01	28.450f 0.597	8.63 25.878f	31.57 173.02	26.147	
1.212	998.69	28.226f 0.646	8.70 25.875f	32.35 161.94	24.171	
1.312	1,086.07	28.037f 0.696	8.78 25.872f	33.17 152.69	22.519	
1.412	1,174.15	27.875f 0.746	8.85 25.869f	34.03 144.90	21.124	
1.512	1,262.91	27.735f 0.797	8.91 25.870f	34.71 137.41	19.834	
1.612	1,352.12	27.609f 0.848	8.94 25.783f	35.14 129.91	18.700	
1.712	1,441.67	27.493f 0.899	8.97 25.694f	35.58 123.36	17.712	
1.812	1,531.58	27.384f 0.950	9.01 25.603f	36.03 117.61	16.848	
1.912	1,621.83	27.283f 1.000	9.05 25.508f	36.51 112.52	16.088	
2.012	1,712.42	27.187f 1.051	9.08 25.436f	36.88 107.68	15.374	
2.112	1,803.32	27.097f 1.102	9.11 25.351f	37.30 103.41	14.750	
2.212	1,894.54	27.011f 1.153	9.14 25.263f	37.74 99.58	14.190	
2.312	1,986.08	26.928f 1.203	9.17 25.174f	38.19 96.11	13.689	
2.412	2,077.95	26.849f 1.254	9.21 25.081f	38.65 92.98	13.238	
2.512	2,170.15	26.772f 1.305	9.23 25.013f	39.01 89.86	12.803	
2.612	2,258.72	26.664f 1.353	8.40 22.760f	29.59 65.48	11.408	
2.712	2,338.10	26.490f 1.395	7.48 20.257f	21.09 45.09	10.043	
2.812	2,408.27	26.272f 1.431	6.56 17.755f	14.46 30.01	8.792	
2.912	2,469.20	26.032f 1.462	5.63 15.252f	9.46 19.15	7.632	
3.012	2,520.92	25.786f 1.488	4.71 12.750f	5.86 11.62	6.543	

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.

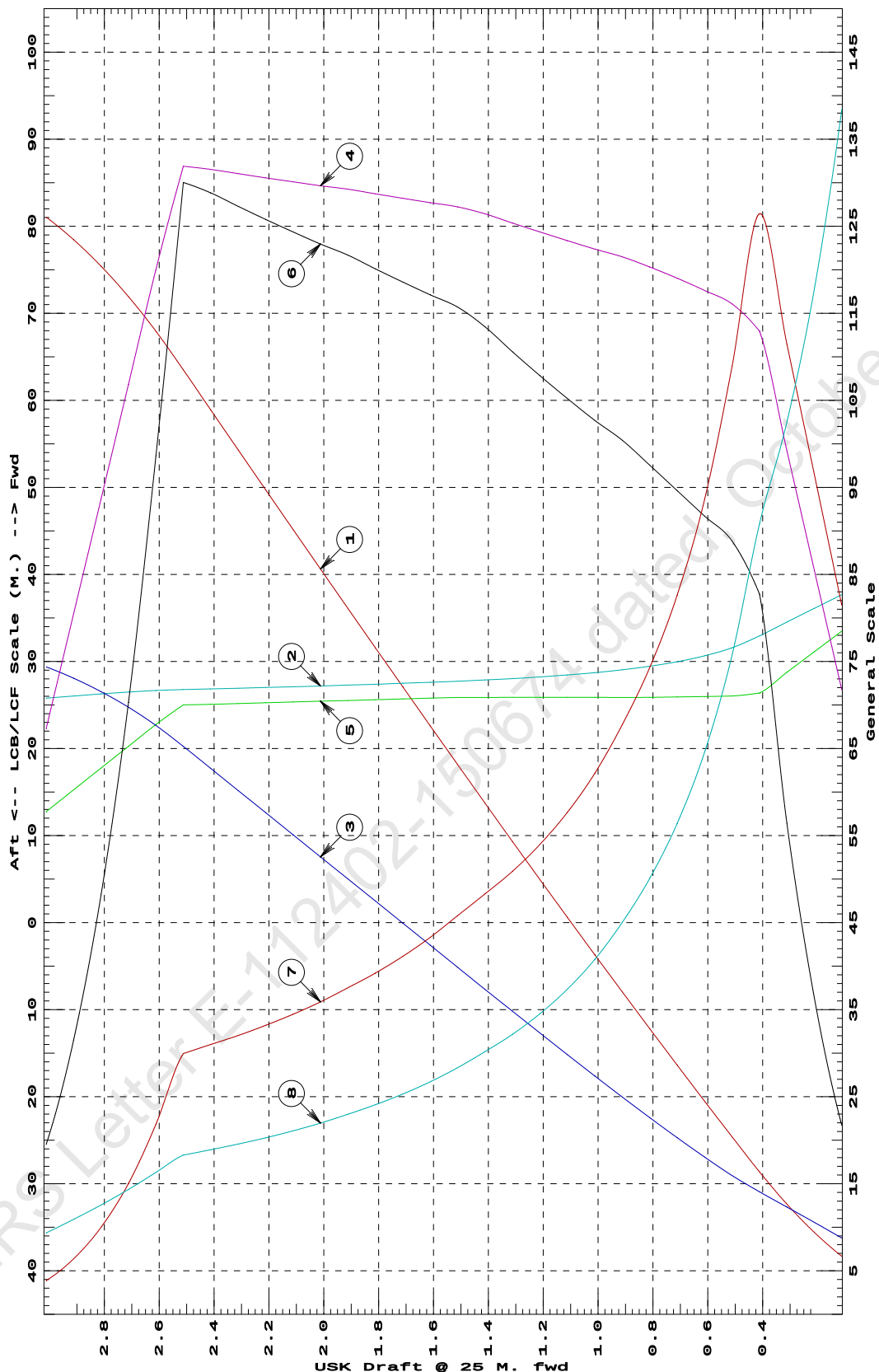
Trim is per 50.00m.

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GHS 17.22

Cybermarine Technologies PTE. Ltd.
PONTON

HYDROSTATIC PROPERTIES at 1 M. FWD TRIM



- ① Displacement 1=20 MT
- ② LCB (use top scale)
- ③ VCB (KB) 1=.02 M.
- ④ Immersion 1=.07 MT/cm
- ④ WPA 1=6.83 Sq.M.
- ⑤ LCF (use top scale)
- ⑥ Moment/Trim 1=.3 M.-MT/cm
- ⑦ KML 1=3 M.
- ⑧ KMT 1=.7 M.

Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

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GHS 17.22 PONT00N

HYDROSTATIC PROPERTIES

Trim: Fwd 0.500/50.000, No Heel, Fixed VCG = 0.000

Draft@	Displacement	Buoyancy-Ctr.	Weight/	Moment/					
25.000f	---Weight(MT)---	LCB----	VCB-----	cm-----	LCF----	cm trim----	KML-----	KMT-----	
0.112	94.61	35.657f	0.098	5.88	30.439f	10.72	566.32	155.843	
0.212	162.65	32.417f	0.132	7.74	25.617f	24.15	742.47	120.816	
0.312	240.79	30.208f	0.173	7.89	25.602f	25.06	520.34	85.097	
0.412	320.44	29.059f	0.218	8.04	25.566f	25.94	404.70	66.389	
0.512	401.45	28.349f	0.266	8.16	25.509f	26.81	333.87	54.610	
0.612	483.61	27.862f	0.315	8.27	25.457f	27.70	286.38	46.395	
0.712	566.72	27.508f	0.365	8.35	25.449f	28.52	251.60	40.187	
0.812	650.57	27.242f	0.415	8.42	25.440f	29.21	224.49	35.343	
0.912	735.11	27.034f	0.466	8.49	25.439f	29.96	203.76	31.632	
1.012	820.35	26.868f	0.517	8.56	25.438f	30.74	187.34	28.689	
1.112	906.31	26.733f	0.567	8.63	25.436f	31.56	174.08	26.309	
1.212	992.99	26.620f	0.619	8.71	25.451f	32.35	162.89	24.309	
1.312	1,080.33	26.524f	0.670	8.77	25.439f	33.07	153.05	22.576	
1.412	1,168.34	26.443f	0.722	8.83	25.438f	33.86	144.89	21.132	
1.512	1,257.02	26.372f	0.773	8.90	25.437f	34.68	137.92	19.898	
1.612	1,346.39	26.310f	0.825	8.97	25.435f	35.53	131.95	18.836	
1.712	1,436.46	26.255f	0.877	9.04	25.448f	36.37	126.58	17.889	
1.812	1,527.11	26.206f	0.929	9.08	25.389f	36.90	120.81	16.991	
1.912	1,618.11	26.157f	0.981	9.12	25.304f	37.34	115.37	16.198	
2.012	1,709.44	26.109f	1.033	9.15	25.217f	37.79	110.53	15.496	
2.112	1,801.11	26.062f	1.085	9.19	25.127f	38.26	106.21	14.872	
2.212	1,893.13	26.014f	1.137	9.22	25.034f	38.75	102.33	14.317	
2.312	1,985.42	25.968f	1.189	9.23	25.012f	38.91	97.97	13.759	
2.412	2,077.73	25.925f	1.240	9.23	25.009f	38.96	93.75	13.255	
2.512	2,170.06	25.886f	1.292	9.23	25.006f	39.02	89.90	12.799	
2.612	2,262.39	25.850f	1.343	9.23	25.004f	39.08	86.36	12.384	
2.712	2,354.73	25.817f	1.394	9.24	25.001f	39.14	83.10	12.006	
2.812	2,445.61	25.772f	1.445	8.50	23.000f	30.67	62.71	10.846	
2.912	2,521.33	25.617f	1.485	6.65	18.000f	15.11	29.97	8.622	
3.012	2,578.59	25.395f	1.516	4.80	13.000f	6.19	12.01	6.555	

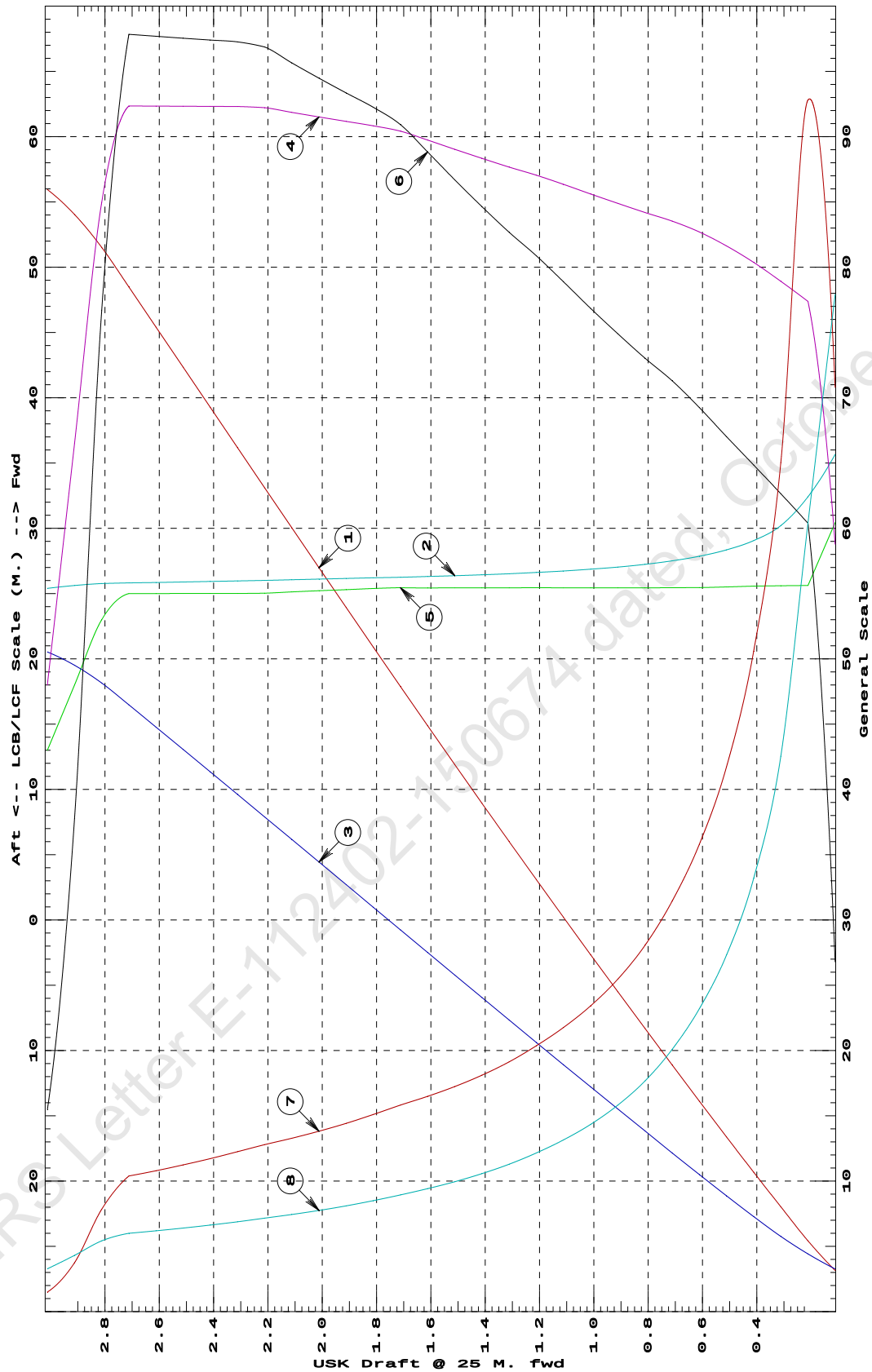
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

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GHS 17.22

Cybermarine Technologies PTE. Ltd.
PONTON

HYDROSTATIC PROPERTIES at 0.5 M. FWD TRIM



- ① Displacement 1=30 MT
② LCB (use top scale)
③ VCB (KB) 1=.03 M.
④ Immersion 1=.1 MT/cm
⑤ WPA 1=9.76 Sq.M.
⑥ LCF (use top scale)
⑦ Moment/Trim 1=.4 M.-MT/cm
⑧ KML 1=8 M.
⑨ KMT 1=2 M.

Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

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GHS 17.22 PONT00N

HYDROSTATIC PROPERTIES
No Trim, No Heel, Fixed VCG = 0.000

Draft@	Displacement	Buoyancy-Ctr.		Weight/		Moment/			
25.000f	---Weight(MT)---	LCB	VCB	cm	LCF	cm	trim	KML	KMT
0.112	83.79	25.000f	0.044	7.57	25.000f	23.17	1382.58	223.643	
0.212	160.29	25.000f	0.095	7.73	25.000f	24.11	752.15	122.246	
0.312	238.38	25.000f	0.145	7.89	25.000f	25.08	526.13	85.934	
0.412	318.09	25.000f	0.196	8.06	25.000f	26.09	410.04	67.317	
0.512	399.40	25.000f	0.248	8.20	25.000f	26.98	337.80	55.685	
0.612	481.78	25.000f	0.300	8.27	25.000f	27.70	287.43	46.641	
0.712	564.87	25.000f	0.351	8.35	25.000f	28.44	251.70	40.223	
0.812	648.69	25.000f	0.403	8.42	25.000f	29.21	225.14	35.449	
0.912	733.24	25.000f	0.454	8.50	25.000f	30.02	204.72	31.774	
1.012	818.51	25.000f	0.506	8.56	25.000f	30.72	187.65	28.736	
1.112	904.44	25.000f	0.558	8.63	25.000f	31.47	173.99	26.304	
1.212	991.06	25.000f	0.610	8.70	25.000f	32.26	162.74	24.297	
1.312	1,078.37	25.000f	0.661	8.77	25.000f	33.07	153.35	22.621	
1.412	1,166.40	25.000f	0.713	8.84	25.000f	33.93	145.46	21.208	
1.512	1,255.11	25.000f	0.765	8.90	25.000f	34.66	138.07	19.915	
1.612	1,344.46	25.000f	0.818	8.97	25.000f	35.45	131.83	18.822	
1.712	1,434.46	25.000f	0.870	9.03	25.000f	36.27	126.41	17.872	
1.812	1,525.13	25.000f	0.922	9.10	25.000f	37.12	121.69	17.042	
1.912	1,616.48	25.000f	0.974	9.17	25.000f	38.01	117.57	16.315	
2.012	1,708.51	25.000f	1.027	9.23	25.000f	38.71	113.30	15.610	
2.112	1,800.77	25.000f	1.079	9.23	25.000f	38.78	107.66	14.924	
2.212	1,893.06	25.000f	1.132	9.23	25.000f	38.84	102.58	14.309	
2.312	1,985.36	25.000f	1.184	9.23	25.000f	38.90	97.98	13.756	
2.412	2,077.68	25.000f	1.236	9.23	25.000f	38.97	93.78	13.257	
2.512	2,170.03	25.000f	1.287	9.24	25.000f	39.04	89.95	12.804	
2.612	2,262.38	25.000f	1.339	9.24	25.000f	39.09	86.38	12.385	
2.712	2,354.73	25.000f	1.390	9.24	25.000f	39.14	83.10	12.003	
2.812	2,447.09	25.000f	1.442	9.24	25.000f	39.19	80.07	11.654	
2.912	2,539.44	25.000f	1.493	9.24	25.000f	39.24	77.26	11.334	
3.012	2,631.79	25.000f	1.544	9.24	25.000f	39.29	74.65	11.040	

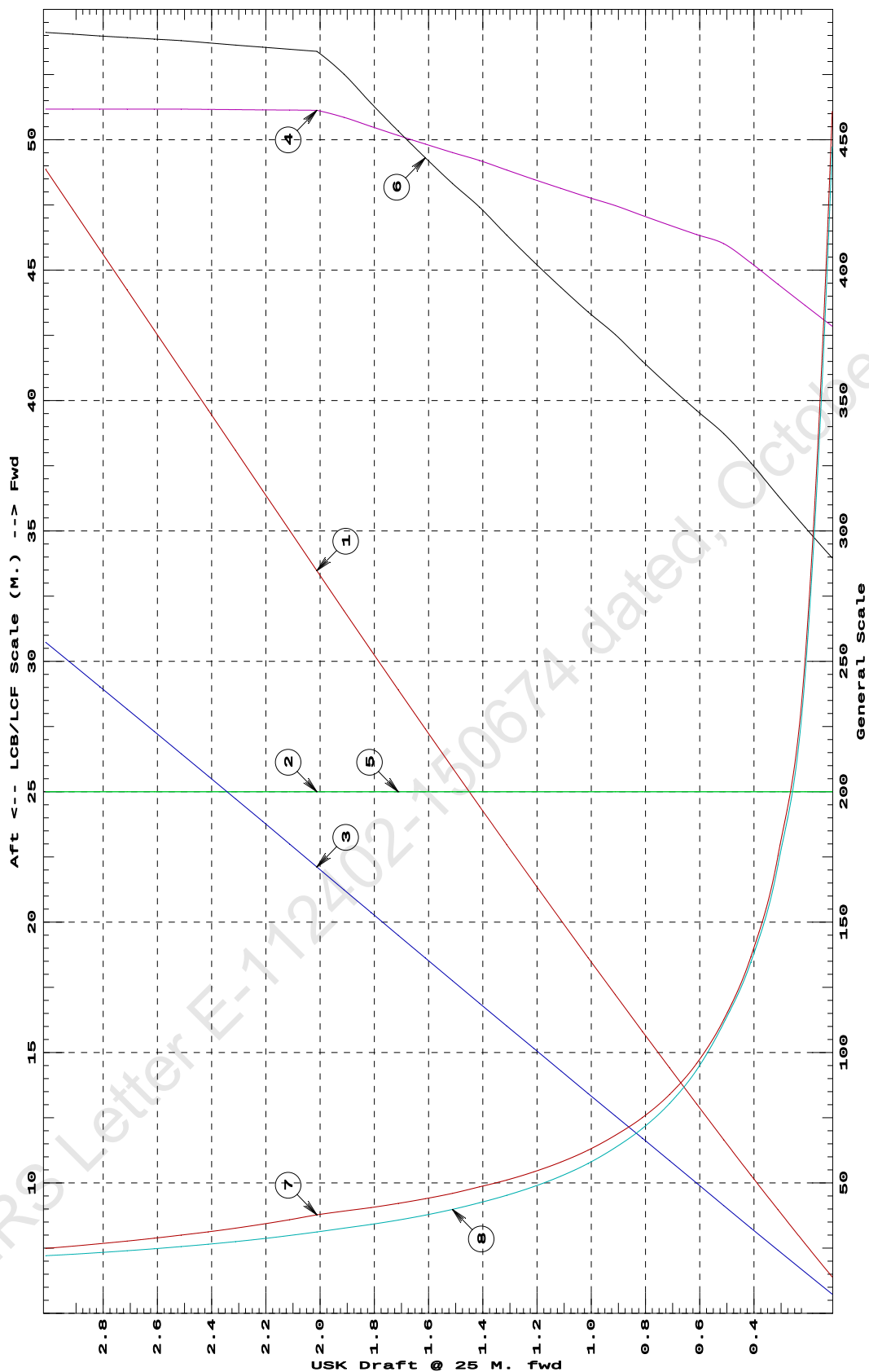
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

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GHS 17.22

Cybermarine Technologies PTE. Ltd.
PONTON

HYDROSTATIC PROPERTIES at LEVEL TRIM



- ① Displacement 1=6 MT
- ② LCB (use top scale)
- ③ VCB (KB) 1=.006 M.
- ④ Immersion 1=.02 MT/cm
- ④ WPA 1=1.95 Sq.M.
- ⑤ LCF (use top scale)
- ⑥ Moment/Trim 1=.08 M.-MT/cm
- ⑦ KML 1=3 M.
- ⑧ KMT 1=.5 M.

Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

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GHS 17.22 PONT00N

HYDROSTATIC PROPERTIES

Trim: Aft 0.500/50.000, No Heel, Fixed VCG = 0.000

Draft@	Displacement	Buoyancy-Ctr.		Weight/		Moment/			
25.000f	---Weight(MT)---	LCB----	VCB-----	cm-----	LCF----	cm trim----	KML-----	KMT	
0.112	94.61	14.343f	0.098	5.88	19.561f	10.72	566.32	155.836	
0.212	162.65	17.583f	0.132	7.74	24.383f	24.15	742.47	120.811	
0.312	240.79	19.792f	0.173	7.89	24.398f	25.06	520.34	85.098	
0.412	320.44	20.941f	0.218	8.04	24.434f	25.94	404.70	66.390	
0.512	401.45	21.651f	0.266	8.16	24.491f	26.81	333.87	54.611	
0.612	483.61	22.138f	0.315	8.27	24.543f	27.70	286.38	46.395	
0.712	566.72	22.492f	0.365	8.35	24.551f	28.52	251.60	40.186	
0.812	650.57	22.758f	0.415	8.42	24.560f	29.21	224.49	35.343	
0.912	735.11	22.966f	0.466	8.49	24.561f	29.96	203.76	31.632	
1.012	820.35	23.132f	0.517	8.56	24.562f	30.74	187.34	28.689	
1.112	906.31	23.267f	0.567	8.63	24.564f	31.56	174.08	26.309	
1.212	992.99	23.380f	0.619	8.71	24.549f	32.35	162.89	24.308	
1.312	1,080.33	23.476f	0.670	8.77	24.561f	33.07	153.05	22.576	
1.412	1,168.34	23.557f	0.722	8.83	24.562f	33.86	144.89	21.132	
1.512	1,257.02	23.628f	0.773	8.90	24.563f	34.68	137.92	19.898	
1.612	1,346.39	23.690f	0.825	8.97	24.565f	35.53	131.95	18.836	
1.712	1,436.46	23.745f	0.877	9.04	24.552f	36.37	126.58	17.889	
1.812	1,527.11	23.794f	0.929	9.08	24.611f	36.90	120.81	16.991	
1.912	1,618.10	23.843f	0.981	9.12	24.696f	37.34	115.37	16.198	
2.012	1,709.44	23.891f	1.033	9.15	24.783f	37.79	110.53	15.496	
2.112	1,801.11	23.938f	1.085	9.19	24.873f	38.26	106.21	14.872	
2.212	1,893.13	23.986f	1.137	9.22	24.966f	38.75	102.33	14.317	
2.312	1,985.42	24.032f	1.189	9.23	24.988f	38.91	97.97	13.758	
2.412	2,077.73	24.075f	1.240	9.23	24.991f	38.96	93.75	13.255	
2.512	2,170.06	24.114f	1.292	9.23	24.994f	39.02	89.90	12.799	
2.612	2,262.39	24.150f	1.343	9.23	24.996f	39.08	86.36	12.384	
2.712	2,354.73	24.183f	1.394	9.24	24.999f	39.14	83.10	12.006	
2.812	2,445.61	24.228f	1.445	8.50	27.000f	30.67	62.71	10.846	
2.912	2,521.33	24.383f	1.485	6.65	32.000f	15.11	29.97	8.622	
3.012	2,578.59	24.605f	1.516	4.80	37.000f	6.19	12.01	6.555	

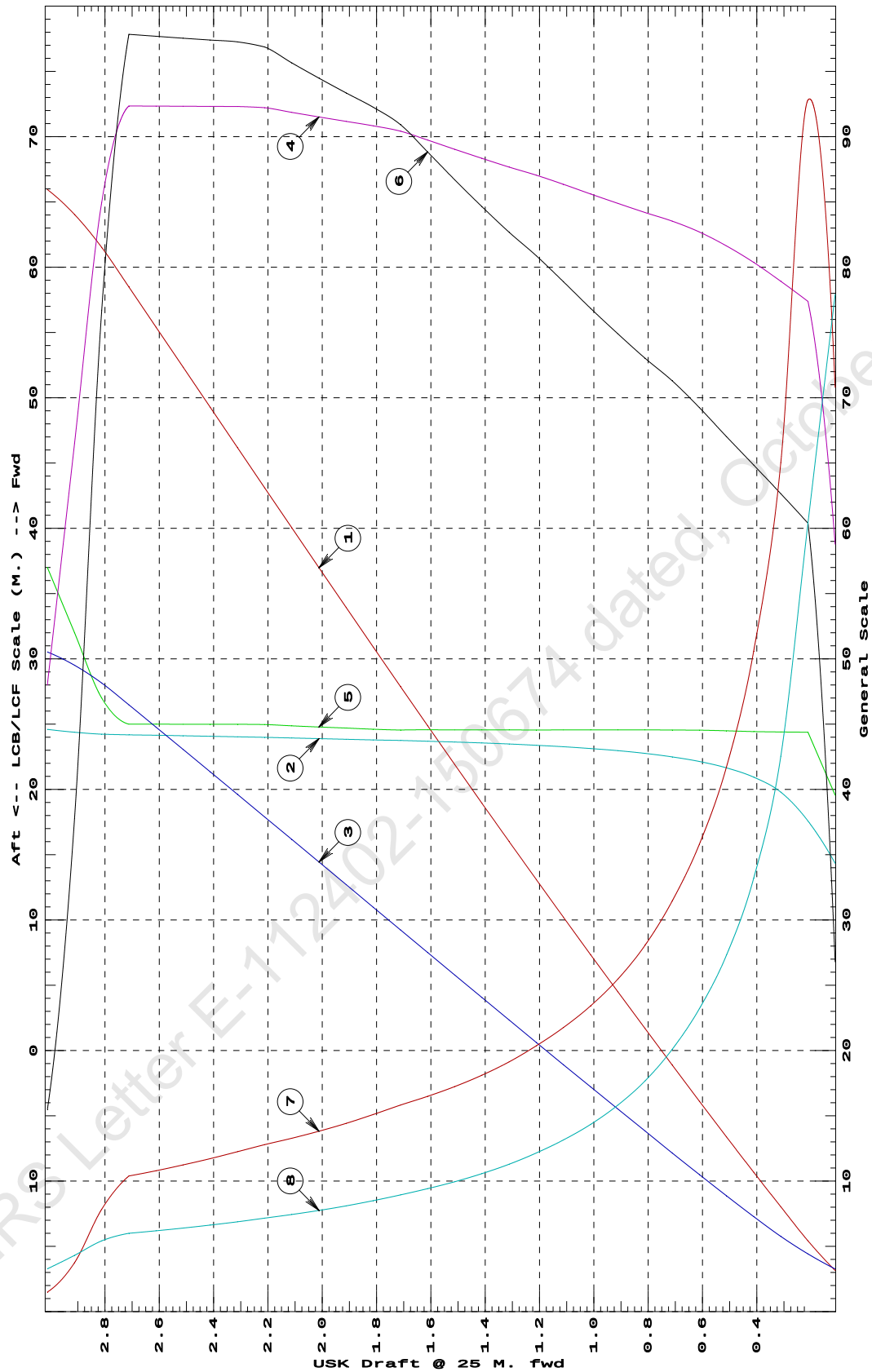
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

31-07-20 12:18:46
GHS 17.22

Cybermarine Technologies PTE. Ltd.
PONTON

HYDROSTATIC PROPERTIES at 0.5 M. AFT TRIM



- ① Displacement 1=30 MT
② LCB (use top scale)
③ VCB (KB) 1=.03 M.
- ④ Immersion 1=.1 MT/cm
⑤ WPA 1=9.76 Sq.M.
⑥ LCF (use top scale)
- ⑦ Moment/Trim 1=.4 M.-MT/cm
⑧ KML 1=8 M.
⑨ KMT 1=2 M.

Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

31-07-20 12:18:46 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

HYDROSTATIC PROPERTIES

Trim: Aft 1.000/50.000, No Heel, Fixed VCG = 0.000

Draft@ 25.000f	Displacement ---Weight(MT)	Buoyancy-Ctr. ----LCB-----VCB	Weight/ -----cm	Moment/ -----LCF-----cm trim	KML	KMT
0.112	133.29	12.311f 0.175	5.02 16.510f	6.52 244.36	96.880	
0.212	188.30	13.893f 0.211	5.98 18.876f	10.90 289.46	82.454	
0.312	252.92	15.477f 0.246	6.94 21.249f	16.92 334.36	71.771	
0.412	327.16	17.062f 0.281	7.91 23.625f	24.82 379.31	63.562	
0.512	407.76	18.417f 0.319	8.13 23.966f	26.65 326.70	53.012	
0.612	489.54	19.349f 0.361	8.23 24.026f	27.51 280.91	45.226	
0.712	572.35	20.029f 0.406	8.33 24.077f	28.38 247.92	39.491	
0.812	656.10	20.548f 0.452	8.42 24.114f	29.26 222.97	35.069	
0.912	740.70	20.956f 0.499	8.50 24.130f	30.12 203.29	31.535	
1.012	826.02	21.283f 0.548	8.57 24.120f	30.81 186.46	28.540	
1.112	912.01	21.550f 0.597	8.63 24.122f	31.57 173.02	26.147	
1.212	998.69	21.774f 0.646	8.70 24.125f	32.35 161.94	24.171	
1.312	1,086.07	21.963f 0.696	8.78 24.128f	33.17 152.69	22.519	
1.412	1,174.15	22.125f 0.746	8.85 24.131f	34.03 144.90	21.125	
1.512	1,262.91	22.265f 0.797	8.91 24.130f	34.71 137.41	19.834	
1.612	1,352.12	22.391f 0.848	8.94 24.217f	35.14 129.91	18.700	
1.712	1,441.67	22.507f 0.899	8.97 24.306f	35.58 123.36	17.712	
1.812	1,531.58	22.616f 0.950	9.01 24.397f	36.03 117.61	16.848	
1.912	1,621.83	22.717f 1.000	9.05 24.492f	36.51 112.52	16.088	
2.012	1,712.42	22.813f 1.051	9.08 24.564f	36.88 107.68	15.374	
2.112	1,803.32	22.903f 1.102	9.11 24.649f	37.30 103.41	14.749	
2.212	1,894.54	22.989f 1.153	9.14 24.737f	37.74 99.58	14.190	
2.312	1,986.08	23.072f 1.203	9.17 24.826f	38.19 96.11	13.689	
2.412	2,077.95	23.151f 1.254	9.21 24.919f	38.65 92.98	13.238	
2.512	2,170.15	23.228f 1.305	9.23 24.987f	39.01 89.86	12.803	
2.612	2,258.72	23.336f 1.353	8.40 27.240f	29.59 65.48	11.408	
2.712	2,338.10	23.510f 1.395	7.48 29.743f	21.09 45.09	10.042	
2.812	2,408.27	23.728f 1.431	6.56 32.245f	14.46 30.01	8.792	
2.912	2,469.21	23.968f 1.462	5.63 34.748f	9.46 19.15	7.632	
3.012	2,520.92	24.214f 1.488	4.71 37.250f	5.86 11.62	6.543	

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.

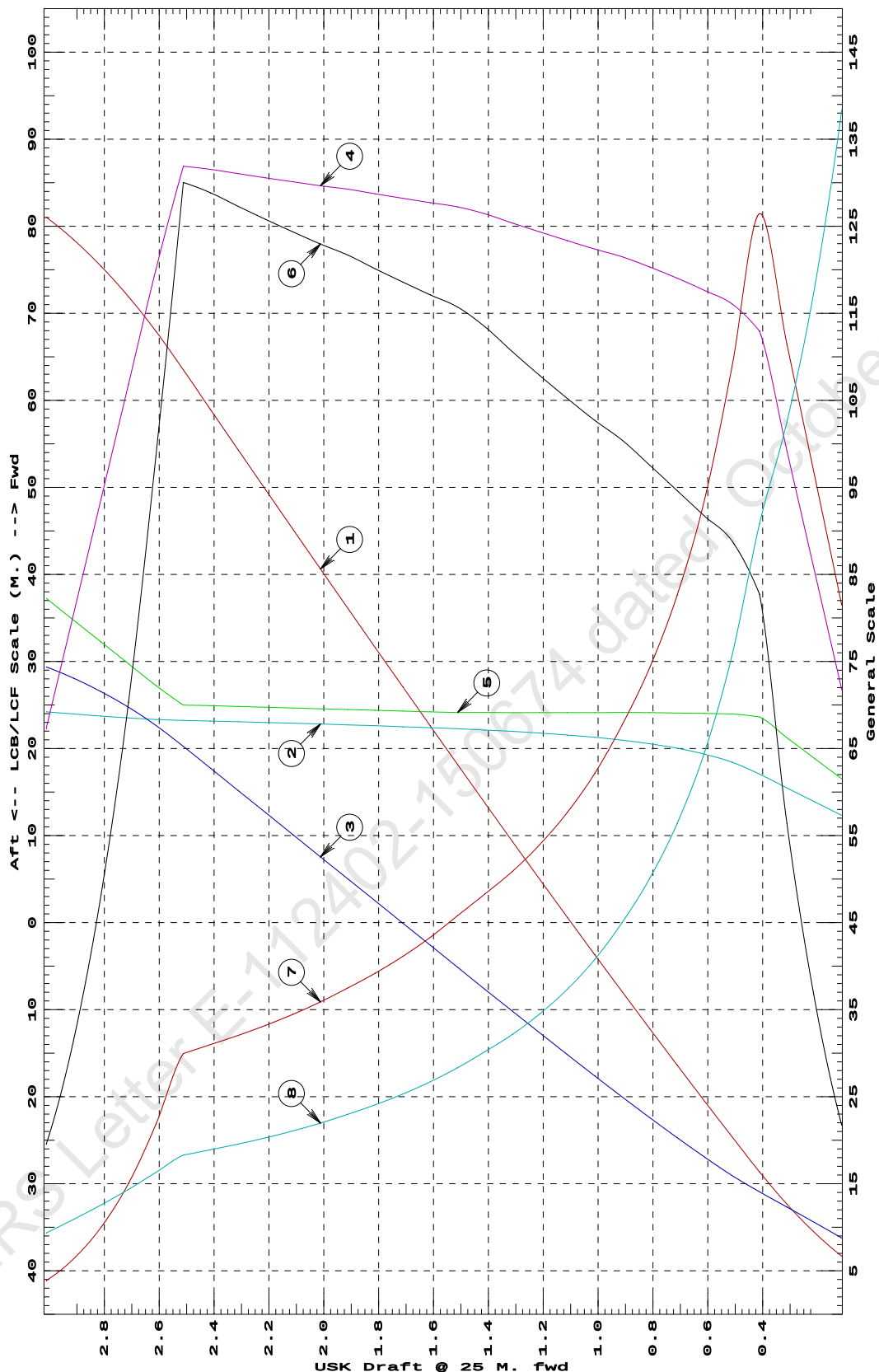
Trim is per 50.00m.

Draft is from USK.

31-07-20 12:18:46
GHS 17.22

Cybermarine Technologies PTE. Ltd.
PONTON

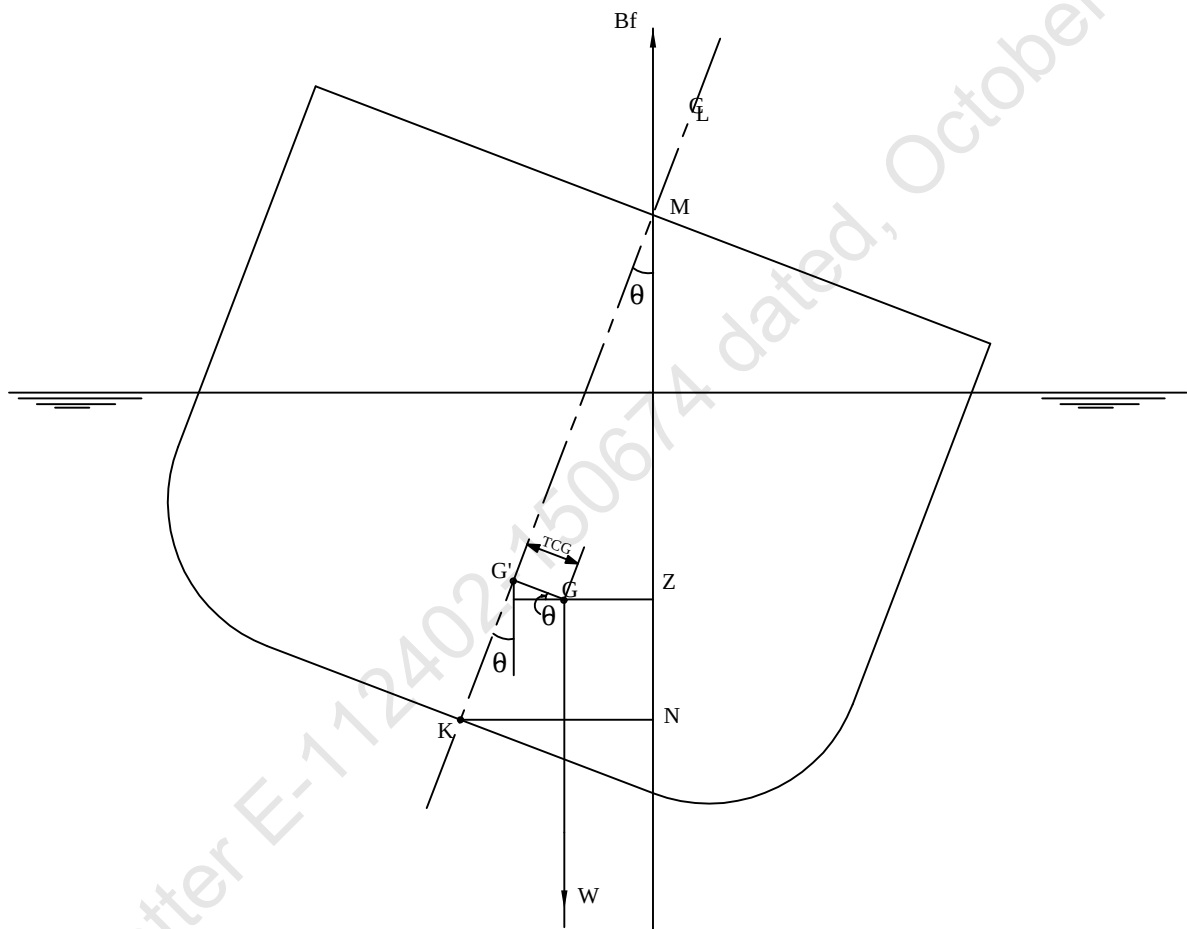
HYDROSTATIC PROPERTIES at 1 M. AFT TRIM



- ① Displacement 1=20 MT
- ② LCB (use top scale)
- ③ VCB (KB) 1=.02 M.
- ④ Immersion 1=.07 MT/cm
- ④ WPA 1=6.83 Sq.M.
- ⑤ LCF (use top scale)
- ⑥ Moment/Trim 1=.3 M.-MT/cm
- ⑦ KML 1=3 M.
- ⑧ KMT 1=.7 M.

Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

KN CONCEPT



$$\underline{GZ = KN - KG' \sin \theta - TCG \cos \theta}$$

01-08-20 21:16:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

VOLUME USED FOR KN COMPUTATION

Condition Graphic

Profile View



Plan View



CROSS CURVE TABLES

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

CROSS CURVES OF STABILITY

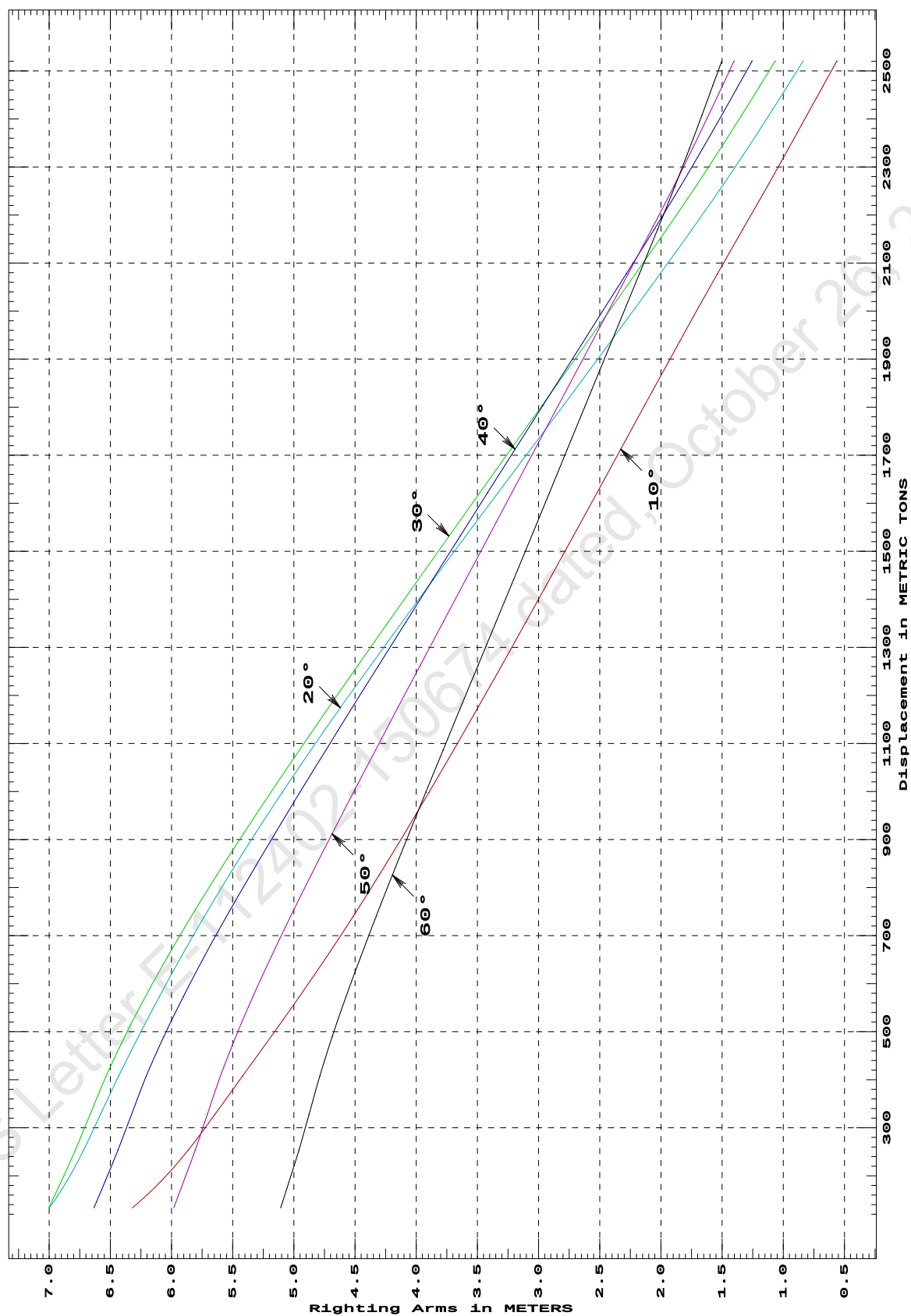
Showing righting arms in heel at VCG = 0.00

Trim: Fwd 1.000/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
133.29	6.320s	6.999s	7.003s	6.634s	5.981s	5.107s
188.30	6.087s	6.857s	6.900s	6.542s	5.901s	5.038s
252.92	5.865s	6.719s	6.786s	6.438s	5.806s	4.959s
327.16	5.646s	6.581s	6.666s	6.327s	5.707s	4.876s
407.76	5.419s	6.431s	6.534s	6.205s	5.600s	4.788s
489.54	5.186s	6.270s	6.382s	6.063s	5.474s	4.686s
572.35	4.956s	6.100s	6.214s	5.904s	5.335s	4.574s
656.10	4.731s	5.920s	6.031s	5.732s	5.184s	4.453s
740.70	4.513s	5.729s	5.836s	5.548s	5.025s	4.326s
826.02	4.303s	5.528s	5.629s	5.355s	4.859s	4.194s
912.01	4.097s	5.316s	5.412s	5.155s	4.687s	4.059s
998.69	3.895s	5.094s	5.186s	4.949s	4.512s	3.921s
1,086.07	3.695s	4.861s	4.955s	4.739s	4.333s	3.781s
1,174.15	3.497s	4.620s	4.717s	4.524s	4.151s	3.638s
1,262.91	3.300s	4.371s	4.475s	4.306s	3.966s	3.494s
1,352.12	3.105s	4.116s	4.229s	4.086s	3.781s	3.349s
1,441.67	2.910s	3.858s	3.982s	3.864s	3.594s	3.203s
1,531.58	2.716s	3.596s	3.732s	3.641s	3.406s	3.057s
1,621.83	2.523s	3.331s	3.480s	3.417s	3.218s	2.911s
1,712.42	2.329s	3.065s	3.227s	3.192s	3.029s	2.765s
1,803.32	2.134s	2.798s	2.972s	2.965s	2.840s	2.618s
1,894.54	1.938s	2.532s	2.717s	2.738s	2.649s	2.471s
1,986.08	1.740s	2.268s	2.462s	2.511s	2.459s	2.323s
2,077.95	1.538s	2.006s	2.207s	2.283s	2.268s	2.175s
2,170.15	1.332s	1.746s	1.954s	2.056s	2.077s	2.027s
2,258.72	1.132s	1.503s	1.718s	1.843s	1.897s	1.888s
2,338.10	0.958s	1.294s	1.514s	1.660s	1.743s	1.767s
2,408.27	0.806s	1.115s	1.338s	1.502s	1.610s	1.664s
2,469.20	0.674s	0.963s	1.190s	1.368s	1.497s	1.576s
2,520.92	0.561s	0.836s	1.066s	1.255s	1.402s	1.502s
Distances in METERS.---Specific Gravity = 1.025.-----						

31-07-20 12:20:24
GHS 17.22Cybermarine Technologies PTE. Ltd.
PONTON

CROSS CURVES OF STABILITY - Stbd Heel
at 1 M. FWD TRIM (initial)



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

CROSS CURVES OF STABILITY

Showing righting arms in heel at VCG = 0.00

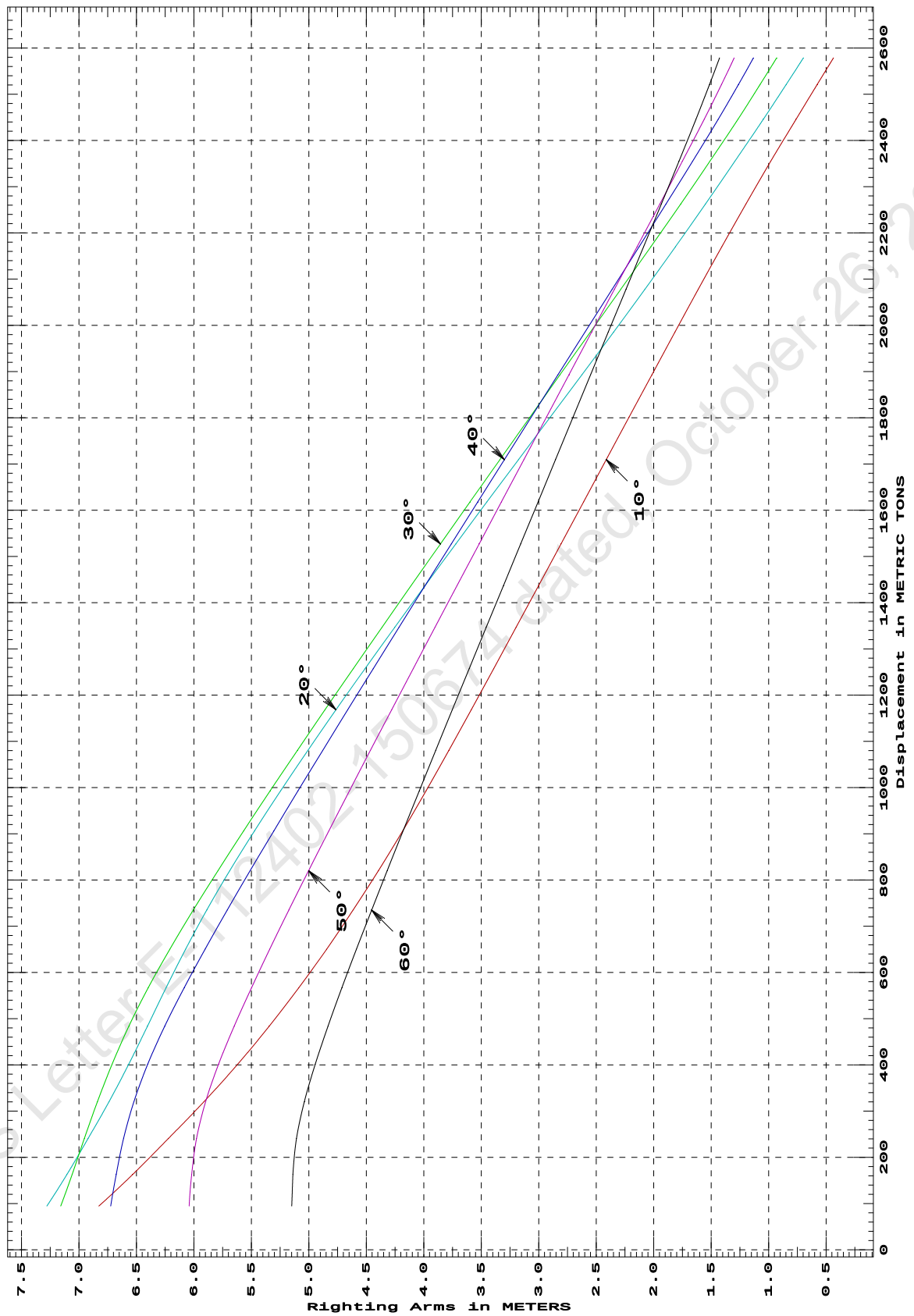
Trim: Fwd 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
94.61	6.827s	7.278s	7.158s	6.723s	6.041s	5.149s
162.65	6.533s	7.105s	7.059s	6.676s	6.020s	5.139s
240.79	6.221s	6.920s	6.953s	6.612s	5.977s	5.107s
320.44	5.912s	6.738s	6.839s	6.522s	5.897s	5.039s
401.45	5.619s	6.567s	6.712s	6.401s	5.785s	4.944s
483.61	5.346s	6.404s	6.566s	6.254s	5.649s	4.833s
566.72	5.088s	6.242s	6.397s	6.085s	5.499s	4.712s
650.57	4.845s	6.071s	6.209s	5.902s	5.340s	4.585s
735.11	4.615s	5.887s	6.005s	5.710s	5.173s	4.454s
820.35	4.395s	5.688s	5.790s	5.510s	5.002s	4.318s
906.31	4.184s	5.473s	5.567s	5.305s	4.826s	4.180s
992.99	3.980s	5.246s	5.336s	5.094s	4.647s	4.039s
1,080.33	3.782s	5.009s	5.100s	4.880s	4.465s	3.896s
1,168.34	3.587s	4.764s	4.859s	4.662s	4.280s	3.752s
1,257.02	3.393s	4.512s	4.614s	4.441s	4.093s	3.605s
1,346.39	3.198s	4.253s	4.364s	4.217s	3.903s	3.457s
1,436.46	3.003s	3.990s	4.111s	3.990s	3.712s	3.308s
1,527.11	2.807s	3.722s	3.856s	3.761s	3.519s	3.157s
1,618.11	2.610s	3.451s	3.598s	3.530s	3.325s	3.006s
1,709.44	2.413s	3.177s	3.338s	3.299s	3.130s	2.854s
1,801.11	2.215s	2.901s	3.077s	3.066s	2.934s	2.702s
1,893.13	2.016s	2.624s	2.815s	2.833s	2.738s	2.549s
1,985.42	1.816s	2.349s	2.551s	2.598s	2.540s	2.396s
2,077.73	1.613s	2.077s	2.287s	2.363s	2.343s	2.242s
2,170.06	1.408s	1.811s	2.025s	2.128s	2.145s	2.088s
2,262.39	1.199s	1.551s	1.766s	1.894s	1.947s	1.934s
2,354.73	0.983s	1.295s	1.512s	1.662s	1.751s	1.781s
2,445.61	0.763s	1.048s	1.268s	1.440s	1.562s	1.633s
2,521.33	0.578s	0.848s	1.074s	1.263s	1.411s	1.515s
2,578.59	0.436s	0.699s	0.931s	1.133s	1.301s	1.428s
Distances in METERS.---Specific Gravity = 1.025.-----						

31-07-20 12:20:24
GHS 17.22

Cybermarine Technologies PTE. Ltd.
PONTON

CROSS CURVES OF STABILITY - Stbd Heel
at 0.5 M. FWD TRIM (initial)



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

CROSS CURVES OF STABILITY

Showing righting arms in heel at VCG = 0.00

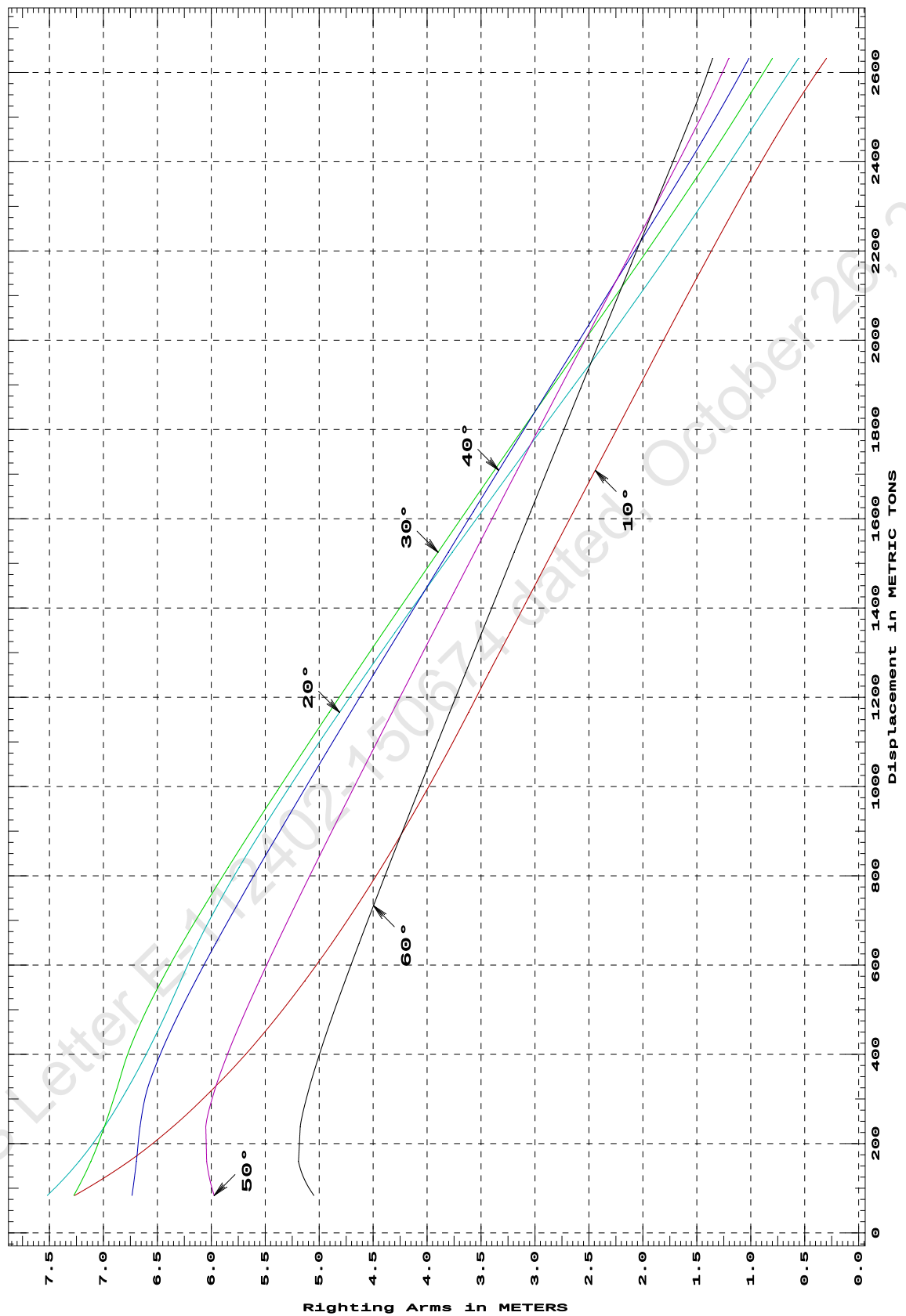
Trim: zero at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
83.79	7.268s	7.519s	7.274s	6.735s	5.973s	5.049s
160.29	6.763s	7.226s	7.116s	6.694s	6.043s	5.192s
238.38	6.354s	6.990s	6.990s	6.660s	6.051s	5.176s
318.09	6.003s	6.788s	6.881s	6.597s	5.971s	5.100s
399.40	5.685s	6.606s	6.776s	6.474s	5.851s	4.999s
481.78	5.397s	6.438s	6.634s	6.319s	5.709s	4.883s
564.87	5.132s	6.283s	6.462s	6.146s	5.555s	4.760s
648.69	4.884s	6.127s	6.269s	5.960s	5.392s	4.630s
733.24	4.648s	5.945s	6.061s	5.764s	5.224s	4.497s
818.51	4.425s	5.743s	5.844s	5.562s	5.050s	4.360s
904.44	4.211s	5.525s	5.619s	5.355s	4.873s	4.221s
991.06	4.006s	5.297s	5.387s	5.143s	4.693s	4.080s
1,078.37	3.807s	5.058s	5.149s	4.927s	4.509s	3.936s
1,166.40	3.615s	4.812s	4.907s	4.708s	4.324s	3.791s
1,255.11	3.427s	4.559s	4.660s	4.486s	4.135s	3.643s
1,344.46	3.231s	4.300s	4.410s	4.261s	3.945s	3.495s
1,434.46	3.036s	4.035s	4.156s	4.033s	3.753s	3.345s
1,525.13	2.840s	3.767s	3.899s	3.803s	3.559s	3.193s
1,616.48	2.642s	3.493s	3.640s	3.571s	3.363s	3.040s
1,708.51	2.442s	3.216s	3.377s	3.336s	3.165s	2.886s
1,800.77	2.241s	2.937s	3.113s	3.100s	2.966s	2.731s
1,893.06	2.041s	2.656s	2.848s	2.864s	2.767s	2.576s
1,985.36	1.840s	2.374s	2.582s	2.627s	2.568s	2.421s
2,077.68	1.638s	2.100s	2.316s	2.390s	2.369s	2.265s
2,170.03	1.433s	1.832s	2.049s	2.153s	2.169s	2.110s
2,262.38	1.225s	1.570s	1.786s	1.916s	1.969s	1.954s
2,354.73	1.010s	1.314s	1.529s	1.679s	1.769s	1.798s
2,447.09	0.789s	1.063s	1.280s	1.449s	1.571s	1.642s
2,539.44	0.554s	0.813s	1.038s	1.228s	1.380s	1.491s
2,631.79	0.298s	0.558s	0.799s	1.016s	1.204s	1.354s
Distances in METERS.---Specific Gravity = 1.025.-----						

31-07-20 12:20:24
GHS 17.22

Cybermarine Technologies PTE. Ltd.
PONTON

CROSS CURVES OF STABILITY - Stbd Heel
at LEVEL TRIM (initial)



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

CROSS CURVES OF STABILITY

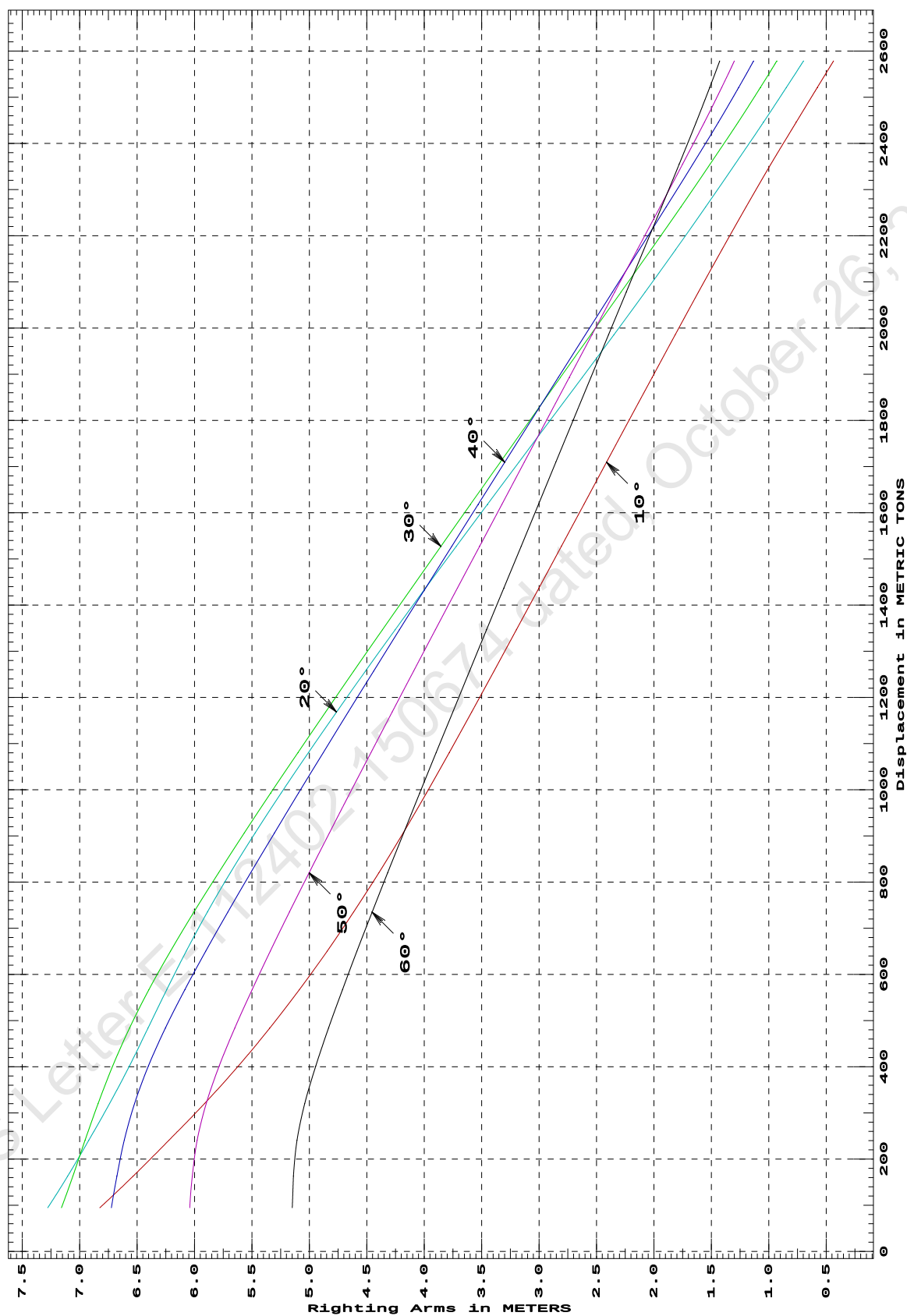
Showing righting arms in heel at VCG = 0.00

Trim: Aft 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
94.61	6.824s	7.277s	7.158s	6.723s	6.041s	5.149s
162.65	6.535s	7.105s	7.059s	6.676s	6.020s	5.139s
240.79	6.222s	6.920s	6.953s	6.612s	5.977s	5.107s
320.44	5.912s	6.738s	6.839s	6.522s	5.897s	5.040s
401.45	5.619s	6.567s	6.712s	6.401s	5.785s	4.944s
483.61	5.346s	6.405s	6.566s	6.254s	5.649s	4.833s
566.72	5.088s	6.242s	6.397s	6.085s	5.499s	4.712s
650.57	4.845s	6.071s	6.209s	5.902s	5.340s	4.585s
735.11	4.615s	5.887s	6.005s	5.710s	5.173s	4.454s
820.35	4.395s	5.688s	5.790s	5.510s	5.002s	4.318s
906.31	4.184s	5.473s	5.567s	5.305s	4.826s	4.180s
992.99	3.980s	5.246s	5.336s	5.094s	4.647s	4.039s
1,080.33	3.782s	5.009s	5.100s	4.880s	4.465s	3.896s
1,168.34	3.587s	4.764s	4.859s	4.662s	4.280s	3.752s
1,257.02	3.393s	4.511s	4.614s	4.441s	4.093s	3.605s
1,346.39	3.198s	4.253s	4.364s	4.216s	3.903s	3.457s
1,436.46	3.003s	3.990s	4.111s	3.990s	3.712s	3.308s
1,527.11	2.807s	3.722s	3.855s	3.761s	3.519s	3.157s
1,618.10	2.610s	3.451s	3.598s	3.530s	3.325s	3.006s
1,709.44	2.413s	3.177s	3.338s	3.299s	3.130s	2.854s
1,801.11	2.215s	2.901s	3.077s	3.066s	2.934s	2.702s
1,893.13	2.016s	2.624s	2.815s	2.833s	2.738s	2.549s
1,985.42	1.816s	2.349s	2.551s	2.598s	2.540s	2.396s
2,077.73	1.613s	2.077s	2.287s	2.363s	2.343s	2.242s
2,170.06	1.408s	1.811s	2.025s	2.128s	2.145s	2.088s
2,262.39	1.198s	1.551s	1.766s	1.894s	1.947s	1.934s
2,354.73	0.983s	1.295s	1.512s	1.662s	1.751s	1.781s
2,445.61	0.763s	1.048s	1.268s	1.440s	1.562s	1.633s
2,521.33	0.578s	0.848s	1.074s	1.263s	1.411s	1.515s
2,578.59	0.436s	0.698s	0.931s	1.133s	1.301s	1.428s
Distances in METERS.---Specific Gravity = 1.025.-----						

31-07-20 12:20:24
GHS 17.22Cybermarine Technologies PTE. Ltd.
PONTON

CROSS CURVES OF STABILITY - Stbd Heel
at 0.5 M. AFT TRIM (initial)



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

CROSS CURVES OF STABILITY

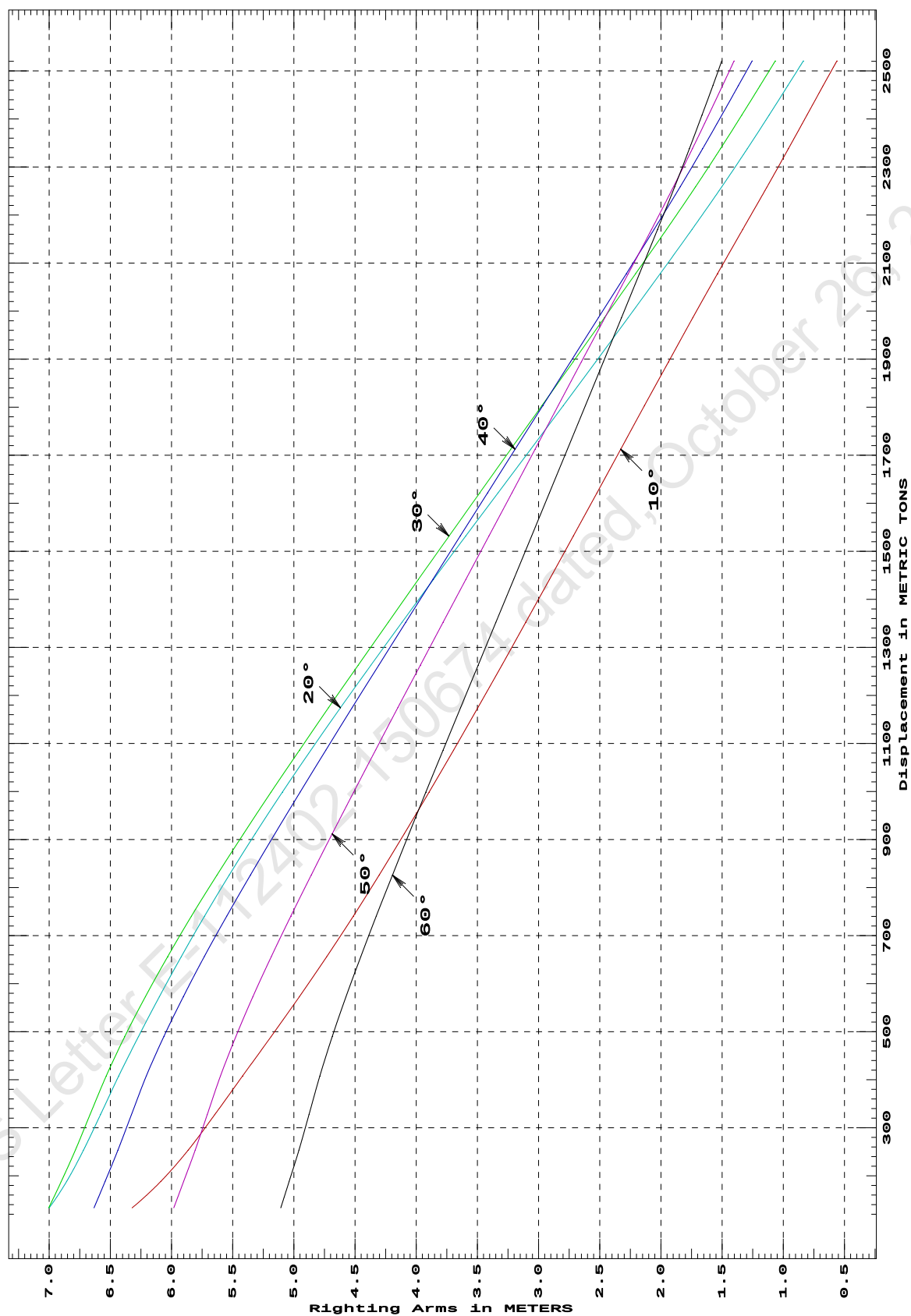
Showing righting arms in heel at VCG = 0.00

Trim: Aft 1.000/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
133.29	6.322s	6.999s	7.004s	6.633s	5.981s	5.107s
188.30	6.087s	6.857s	6.900s	6.542s	5.901s	5.038s
252.92	5.864s	6.719s	6.786s	6.438s	5.807s	4.959s
327.16	5.646s	6.580s	6.666s	6.327s	5.708s	4.876s
407.76	5.419s	6.431s	6.533s	6.205s	5.600s	4.787s
489.54	5.186s	6.270s	6.382s	6.063s	5.474s	4.686s
572.35	4.956s	6.100s	6.214s	5.904s	5.335s	4.574s
656.10	4.731s	5.920s	6.031s	5.732s	5.184s	4.453s
740.70	4.513s	5.729s	5.836s	5.548s	5.025s	4.326s
826.02	4.303s	5.528s	5.629s	5.355s	4.858s	4.194s
912.01	4.097s	5.316s	5.411s	5.155s	4.687s	4.059s
998.69	3.895s	5.093s	5.186s	4.949s	4.511s	3.921s
1,086.07	3.695s	4.861s	4.954s	4.738s	4.333s	3.780s
1,174.15	3.497s	4.619s	4.717s	4.524s	4.151s	3.638s
1,262.91	3.300s	4.370s	4.474s	4.306s	3.966s	3.494s
1,352.12	3.105s	4.116s	4.229s	4.085s	3.780s	3.348s
1,441.67	2.910s	3.858s	3.981s	3.863s	3.593s	3.203s
1,531.58	2.716s	3.596s	3.732s	3.640s	3.406s	3.057s
1,621.83	2.523s	3.330s	3.480s	3.416s	3.218s	2.911s
1,712.42	2.329s	3.065s	3.227s	3.192s	3.029s	2.764s
1,803.32	2.134s	2.798s	2.973s	2.966s	2.840s	2.618s
1,894.54	1.938s	2.532s	2.717s	2.738s	2.650s	2.471s
1,986.08	1.740s	2.268s	2.462s	2.511s	2.458s	2.323s
2,077.95	1.538s	2.006s	2.207s	2.283s	2.268s	2.175s
2,170.15	1.332s	1.746s	1.954s	2.056s	2.077s	2.027s
2,258.72	1.132s	1.503s	1.718s	1.843s	1.898s	1.888s
2,338.10	0.958s	1.294s	1.513s	1.660s	1.743s	1.768s
2,408.27	0.806s	1.115s	1.339s	1.502s	1.610s	1.664s
2,469.21	0.674s	0.963s	1.189s	1.367s	1.497s	1.576s
2,520.92	0.561s	0.836s	1.065s	1.254s	1.402s	1.503s
Distances in METERS.---Specific Gravity = 1.025.-----						

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PONT00N

CROSS CURVES OF STABILITY - Stbd Heel
at 1 M. AFT TRIM (initial)



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

DEADWEIGHT TABLE

Draft	Disp	Dwt	
M	T	T	
0.612	481.78	75.81	
0.712	564.87	158.90	
0.812	648.69	242.72	
0.912	733.24	327.27	
1.012	818.51	412.54	
1.112	904.44	498.47	
1.212	991.06	585.09	
1.312	1078.37	672.40	
1.412	1166.40	760.43	
1.512	1255.11	849.14	
1.612	1344.46	938.49	
1.712	1434.46	1028.49	
1.812	1525.13	1119.16	
1.912	1616.48	1210.51	
2.012	1708.51	1302.54	
2.112	1800.77	1394.80	
2.212	1893.06	1487.09	Full Load Displacement

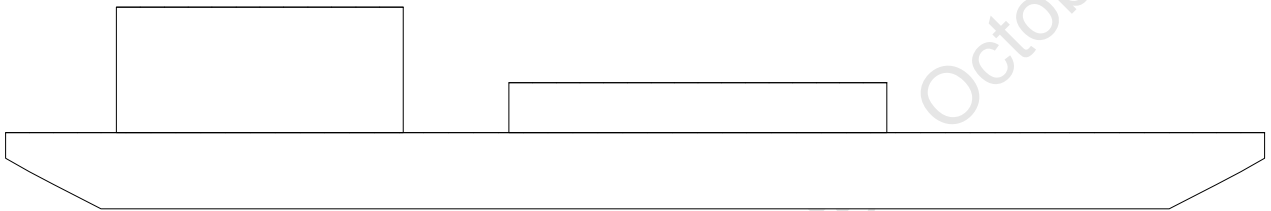
01-08-20 22:26:06 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

USK draft refers to the line:

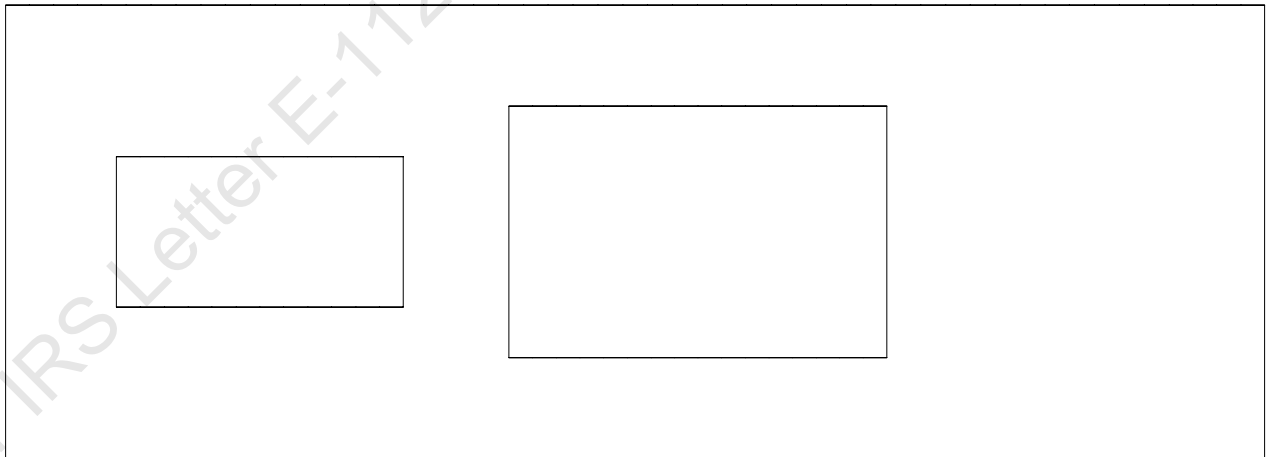
0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

Condition Graphic

Profile View



Plan View



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MAXIMUM VCG vs. DISPLACEMENT

Heeling moment is present from: wind

Trim = Fwd 1.000/50.000 at zero heel (trim righting arm held at zero)

Displacement --- Margins ---

METRIC TONS Max VCG LIM1 LIM2 LIM3

400.00	18.489	122%	0d	7d
600.00	17.349	115%	0d	6d
800.00	16.087	112%	0d	5d
1,000.00	14.631	73%	0d	4d
1,200.00	13.079	51%	0d	3d
1,400.00	11.429	30%	0d	3d
9.697 1,600.00	9.770	12%	0d	2d
7.619 1,800.00	7.778	0%	1d	1d
2,000.00	5.433	0%	5d	1d

Heeling moment is present from: wind

Trim = Fwd 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement --- Margins ---

METRIC TONS Max VCG LIM1 LIM2 LIM3

400.00	18.792	138%	0d	10d
600.00	17.834	103%	0d	7d
800.00	16.573	83%	0d	6d
1,000.00	15.128	70%	0d	5d
1,200.00	13.510	57%	0d	4d
1,400.00	11.824	35%	0d	3d
1,600.00	10.137	7%	0d	3d
1,800.00	7.959	0%	2d	2d
2,000.00	5.618	3%	6d	1d

Heeling moment is present from: wind

Trim = zero at zero heel (trim righting arm held at zero)

Displacement --- Margins ---

METRIC TONS Max VCG LIM1 LIM2 LIM3

400.00	18.899	137%	0d	14d
600.00	17.990	117%	0d	10d
800.00	16.745	80%	0d	7d
1,000.00	15.270	69%	0d	6d
1,200.00	13.683	59%	0d	5d
1,400.00	11.943	37%	0d	4d
1,600.00	10.176	8%	0d	3d
1,800.00	8.007	0%	2d	3d
2,000.00	5.806	0%	5d	2d

Heeling moment is present from: wind

Trim = Aft 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement --- Margins ---

METRIC TONS Max VCG LIM1 LIM2 LIM3

400.00	18.793	138%	0d	10d
600.00	17.835	103%	0d	7d
800.00	16.573	83%	0d	6d
1,000.00	15.129	70%	0d	5d
1,200.00	13.510	57%	0d	4d

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GHS 17.22 PONT00N

1,400.00	11.825	35%	0d	3d
1,600.00	10.136	7%	0d	3d
1,800.00	7.958	0%	2d	2d
2,000.00	5.599	4%	6d	1d

Heeling moment is present from: wind
Trim = Aft 1.000/50.000 at zero heel (trim righting arm held at zero)

Displacement --- Margins ---
METRIC TONS Max VCG LIM1 LIM2 LIM3

400.00	18.491	119%	0d	7d
600.00	17.392	114%	0d	6d
800.00	16.095	112%	0d	5d
1,000.00	14.632	73%	0d	4d
1,200.00	13.079	51%	0d	3d
1,400.00	11.430	31%	0d	3d
9.699 1,600.00	9.769	12%	0d	2d
7.107 1,800.00	7.781	0%	1d	1d
2,000.00	5.434	0%	5d	1d

Distances in METERS.---Specific Gravity = 1.025.---d = degrees.

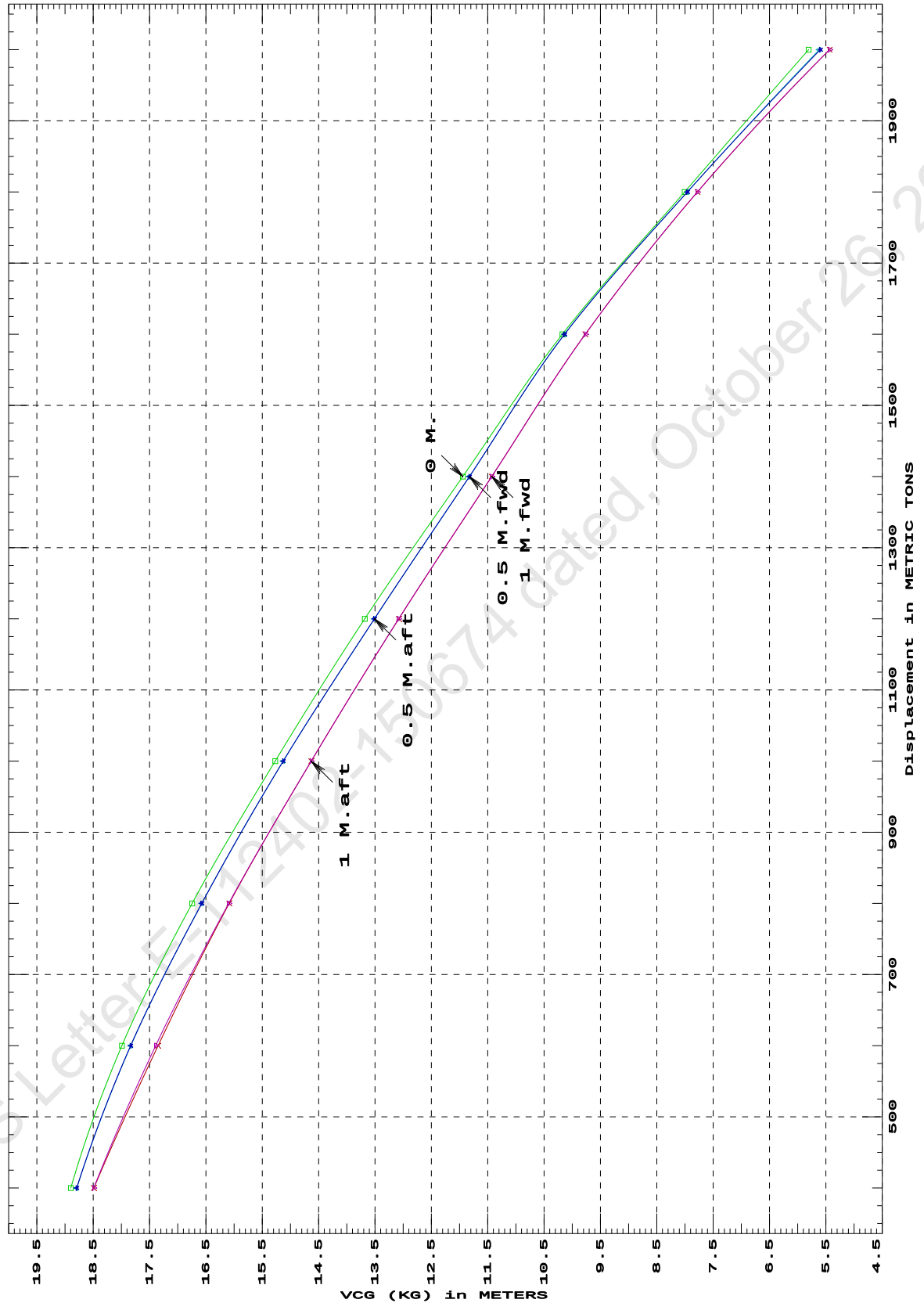
LIM-----IS CODE 2008 CRITERION-----Min/Max
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad
(2) Angle from Equilibrium to RAzero or Flood > 20.00 deg
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg

MAX VCG DOES NOT COVER CRANE OPERATION/LIFTING OF WEIGHTS AND IS VALID ONLY
FOR UNMANNED TOW, WITH CRANE IN STOWED POSITION
MAX DECK CARGO HEIGHT = 4M ABOVE DECK LEVEL

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GHS 17.22

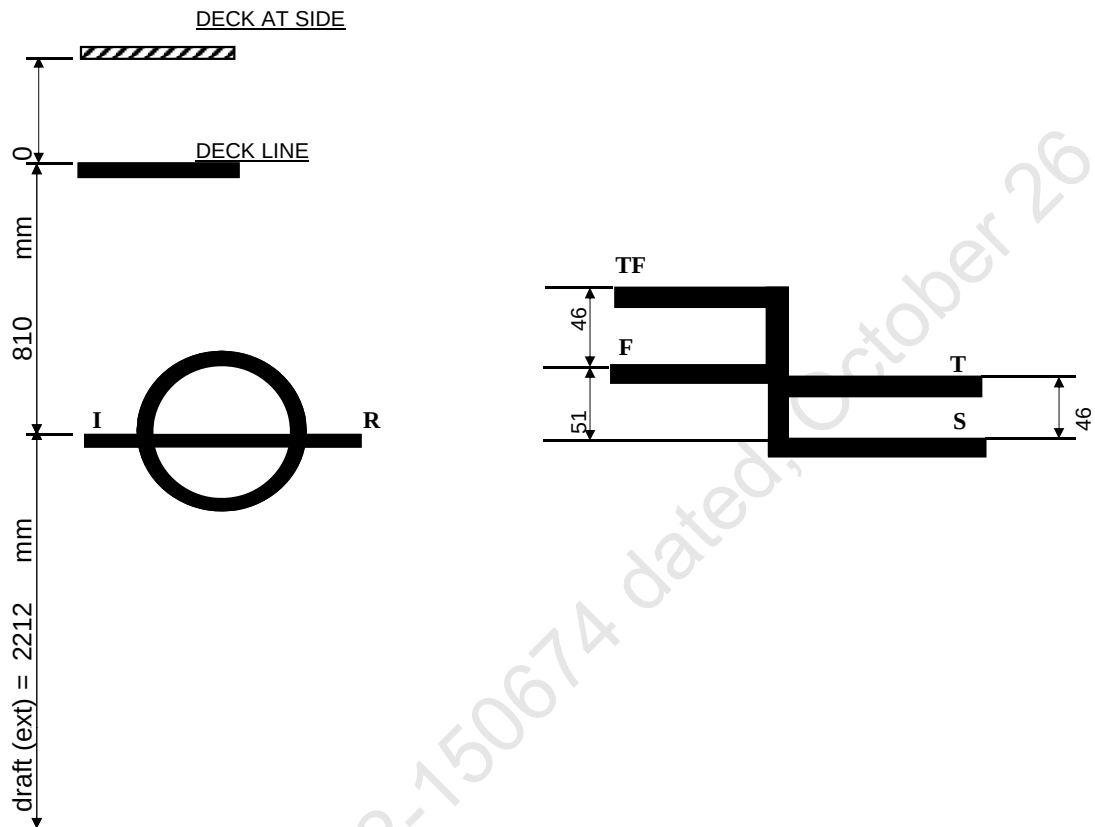
Cybermarine Technologies PTE. Ltd.
PONTOON

IS CODE 2008 criterion MAXIMUM VCG (KG)
at various trims (initial)



Specific Gravity = 1.025
Trim is per 50 M.
"K" = BPL

TABLE OF FREEBOARDS



LOADLINE	FREEBOARD (mm)	DRAFT (ext) KL (m)	DISPLACEMENT (ext) T	DEADWEIGHT T
SUMMER	810	2.212	1893.06	1487.09
FRESH WATER	759	2.263	1892.81	1486.84
TROPICAL	764	2.258	1935.52	1529.55
TROPICAL FRESH WATER	713	2.309	1934.23	1528.26

Deck Plate Thickness = 10 mm

Keel Plate Thickness = 12 mm

LOADING CONDITIONS

SUMMARY OF CONDITIONS					
CONDITION NO.	1	2	3	4	5
CONDITION	Condition 00 : Lightship Condition	Condition 1 : Lightship + Crane + Ballast	Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo Crane Lifting 51.30T @ 12m Radius Aft	Condition 3 : Lightship + Load at Stbd + Ballast + Deck Cargo Crane Lifting 51.30T @ 12m Radius Stbd	Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo Crane Lifting 51.30T @ 12m Radius Fwd
TOTAL TANK LOAD (T)	0.00	149.32	472.52	472.52	472.52
DWT CONST.(T)	0.00	0.00	0.00	0.00	0.00
DEADWEIGHT (T)	0.00	0.00	0.00	0.00	0.00
LIGHT WEIGHT (T)	405.97	405.97	405.97	405.97	405.97
DISPLACEMENT (T)	405.76	905.29	1893.08	1893.08	1893.21
FWD DRAFT (M)	0.52	1.00	2.11	2.26	2.42
AFT DRAFT (M)	0.52	1.23	2.32	2.16	2.01
MEAN DRAFT (M)	0.52	1.11	2.21	2.21	2.21
TRIM (M)	0.00	0.231a	0.208a	0.102f	0.413f
L.C.G. (FROM AP) (M)	25.000f	24.214f	24.589f	25.202f	25.815f
L.C.B. (FROM AP) (M)	25.000f	24.198f	24.577f	25.208f	25.838f
M.C.T. (T-CM)	26.79	30.75	37.33	37.33	37.33
KG (M)	3.00	4.10	3.99	3.99	3.99
GM _T (M)	51.85	21.95	10.03	10.03	10.03
MAX. GZ (M)	5.54	4.06	1.32	1.32	1.31
Note:For details of deadweight constant refer respective loading condition					

SUMMARY OF CONDITIONS					
CONDITION NO.	6	7	8	9	10
CONDITION	Condition 5 : Lightship + Crane + Ballast + Deck Cargo Towing Condition with Max. draft 2.212m	Condition 6 : Lightship + Max. Deck Cargo Max. deck cargo loading condition	Condition 7 : Lightship + Load at Aft + Ballast + Deck Cargo Crane Lifting 51.30T @ 12m Radius Aft	Condition 8 : Lightship + Load at Stbd + Ballast + Deck Cargo Crane Lifting 51.30T @ 12m Radius Stbd	Condition 9 : Lightship + Load at Fwd + Ballast + Deck Cargo Crane Lifting 51.30T @ 12m Radius Fwd
TOTAL TANK LOAD (T)	422.40	0.00	373.31	373.31	373.31
DWT CONST.(T)	0.00	0.00	0.00	0.00	0.00
DEADWEIGHT (T)	0.00	0.00	0.00	0.00	0.00
LIGHT WEIGHT (T)	405.97	405.97	405.97	405.97	405.97
DISPLACEMENT (T)	1893.05	1893.06	1893.06	1893.06	1893.06
FWD DRAFT (M)	2.20	2.21	2.12	2.20	2.28
AFT DRAFT (M)	2.22	2.21	2.31	2.23	2.14
MEAN DRAFT (M)	2.21	2.21	2.21	2.21	2.21
TRIM (M)	0.016a	0.00	0.194a	0.028a	0.137f
L.C.G. (FROM AP) (M)	24.968f	25.000f	24.619f	24.944f	25.269f
L.C.B. (FROM AP) (M)	24.967f	25.000f	24.607f	24.942f	25.278f
M.C.T. (T-CM)	37.45	37.11	37.25	37.25	37.25
KG (M)	3.68	4.57	4.19	4.19	4.19
GM _T (M)	10.38	9.74	9.91	9.91	9.91
MAX. GZ (M)	1.41	1.26	1.29	1.30	1.30
Note: For details of deadweight constant refer respective loading condition					

SUMMARY OF CONDITIONS			
CONDITION NO.	11	12	13
CONDITION	Condition 10 : Lightship + Load at Aft + Ballast Crane Lifting 36.6T @ 18m Radius Aft	Condition 11 : Lightship + Load at Stbd + Ballast Crane Lifting 36.6T @ 18m Radius Stbd	Condition 12 : Lightship + Load at Fwd + Ballast Crane Lifting 36.60T @ 18m Radius Fwd
TOTAL TANK LOAD (T)	472.52	472.52	472.52
DWT CONST.(T)	0.00	0.00	0.00
DEADWEIGHT (T)	0.00	0.00	0.00
LIGHT WEIGHT (T)	405.97	405.97	405.97
DISPLACEMENT (T)	1893.06	1893.05	1893.06
FWD DRAFT (M)	2.15	2.24	2.33
AFT DRAFT (M)	2.28	2.19	2.10
MEAN DRAFT (M)	2.21	2.21	2.21
TRIM (M)	0.127a	0.051f	0.229f
L.C.G. (FROM AP) (M)	24.751f	25.099f	25.447f
L.C.B. (FROM AP) (M)	24.742f	25.103f	25.464f
M.C.T. (T-CM)	37.04	37.04	37.04
KG (M)	4.76	4.76	4.76
GM _r (M)	9.26	9.26	9.26
MAX. GZ (M)	1.16	1.16	1.16
Note:For details of deadweight constant refer respective loading condition			

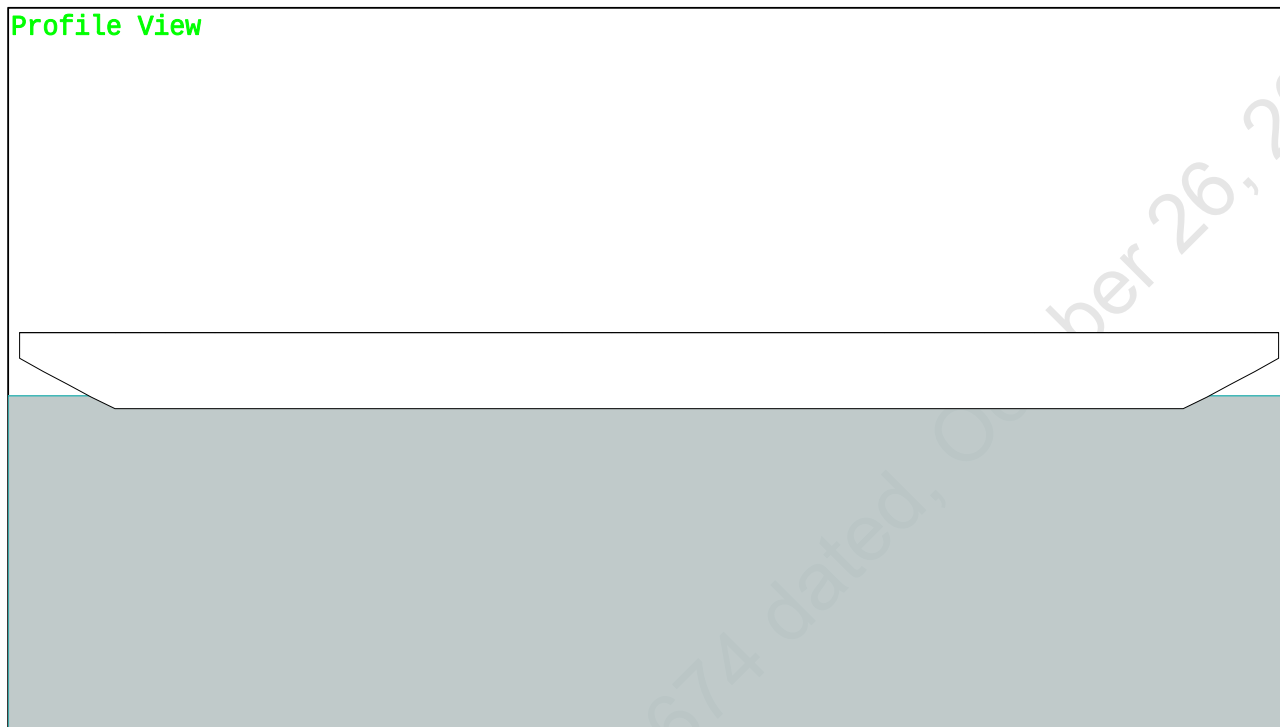
22-10-20 23:28:00 Cybermarine Technologies PTE. Ltd.
GHS 17.22

PONTOON

Condition 00 : Lightship
Lightship condition

Condition Graphic - Draft: 0.520 Trim: zero Heel: zero

Profile View



Plan View



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PONTOON

Condition 00 : Lightship
Lightship condition

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 0.520 @ 50.00f, 0.520 @ 25.00f, 0.520 @ 0.00

Trim: zero, Heel: zero

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
WEIGHT	405.97	25.000f	0.000	3.000	
Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----
Total Tanks----->		0.00			0.00
		Displ(MT)----	LCB-----	TCB-----	VCB-----
HULL	1.025	405.76	25.000f	0.000	0.252

Righting Arms:		0.000	0.000		
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----BMT
Total Waterplane---->	1.025	800.8	25.000f	0.000	332.92 54.603
		MT/cm-----	m.-MT/cm-----	GML-----	GMT
		8.21	26.79	330.18	51.854
Distances in METERS.-----			Moments in m.-MT.		

SUMMARY OF LOADING

0.0 Cu.M. (0%) SALT WATER

DRAFT AT FP (M) = 0.520
DRAFT AT MS (M) = 0.520
DRAFT AT AP (M) = 0.520
DRAFT AT FWD MARK (M) = 0.520
DRAFT AT AFT MARK (M) = 0.520

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PONTOON

Condition 00 : Lightship
Lightship condition

HYDROSTATIC PROPERTIES

No Trim, No Heel, VCG = 3.000

LCF Displacement Buoyancy-Ctr. Weight/ Moment/
Draft---Weight(MT)---LCB---VCB---cm---LCF---cm trim---GML---GMT
0.520 405.76 25.000f 0.252 8.21 25.000f 26.79 330.17 51.854
Distances in METERS.---Specific Gravity = 1.025.---Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 0.520 @ 50.00f, 0.520 @ 25.00f, 0.520 @ 0.00

Trim: zero, Heel: zero

Part-----LPA*SF-----HCP-----Arm----Pressure-----Moment
HULL 121.1 1.277 1.534 0.05500 10.21
Distances in METERS.---Pressure in MT/Sqm.---Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

LCG = 25.000f TCG = 0.000 VCG = 3.000

Origin Depth---	Degrees of Trim---	Heel---	Displacement Weight(MT)---	Residual Arms in Trim--in Heel---	Res. 50% DeckImm Area---Angle
0.460	0.00	4.97p	405.85	0.000 -3.954	0.0000
0.508	0.00	0.00	406.14	0.000 -0.025	-0.1727 13.835
0.508	0.00	0.03s	405.97	0.000 0.000	-0.1727 13.808
0.459	0.00	5.03s	406.16	0.000 3.926	-0.0012 8.808
-0.134	0.00	13.83s	405.91	0.000 5.427	0.7551 0.000
-0.239	0.00	15.03s	405.97	0.000 5.477	0.8686 -1.192
-0.668	0.00	19.53s	405.75	0.000 5.543	1.3045 -5.699
-1.234	0.00	25.03s	405.99	0.000 5.445	1.8344 -11.192
-2.302	0.00	35.03s	405.80	0.000 4.917	2.7453 -21.192
-3.303	0.00	45.03s	406.01	0.000 4.035	3.5316 -31.192
-4.204	0.00	55.03s	406.09	0.000 2.954	4.1444 -41.192
-4.609	0.00	60.03s	405.86	0.000 2.364	4.3766 -46.192

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: No tank loads are present.

Note: The Residual Righting Arms shown above are in excess of the
wind heeling arms derived from these moments (in m.-MT):

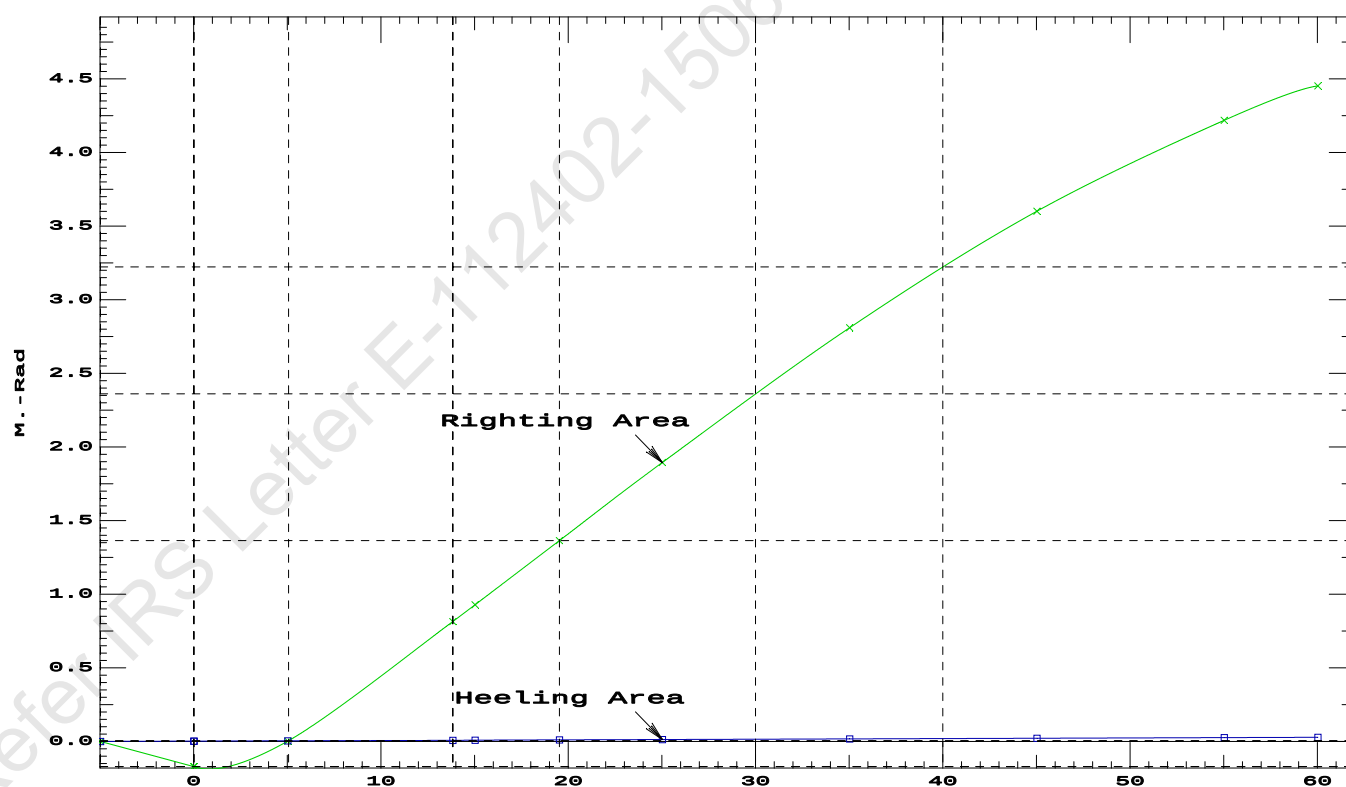
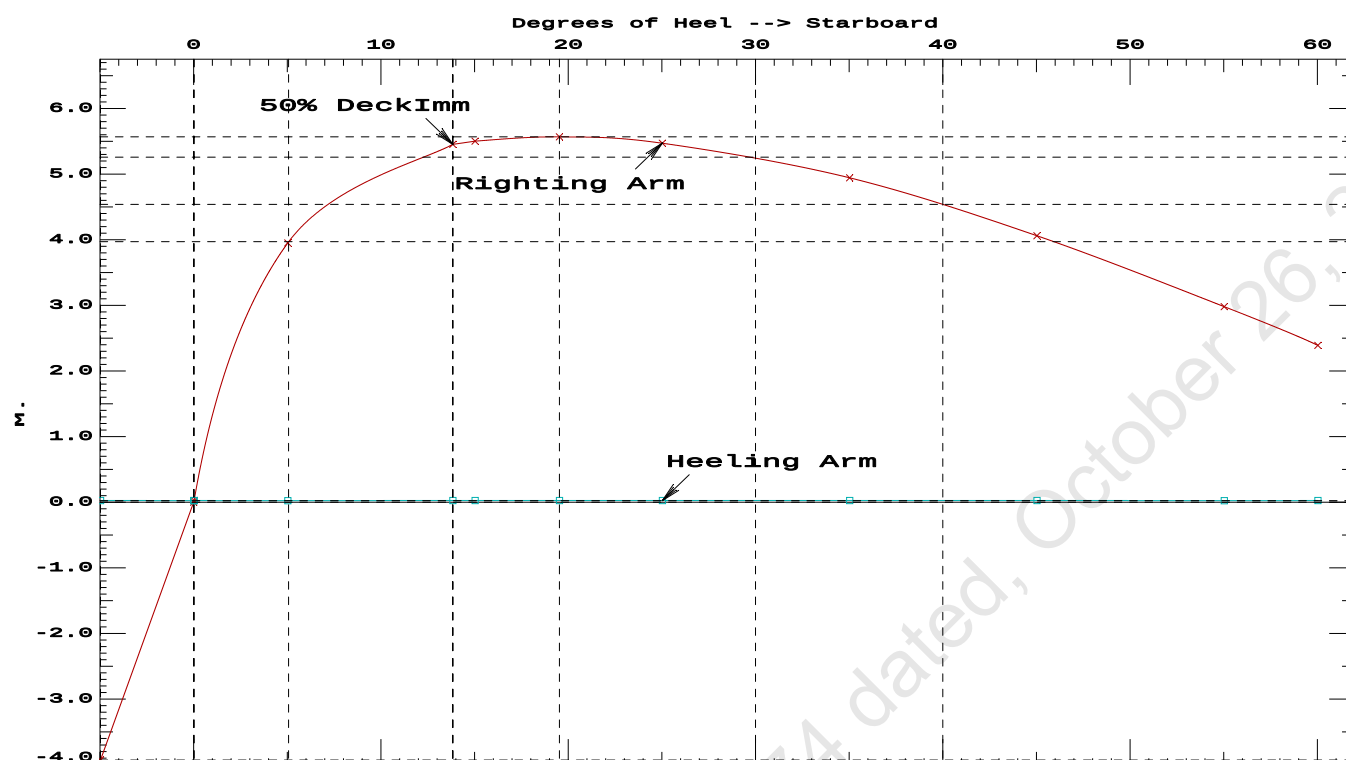
Stbd heeling moment = 10.21 (constant)

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max-----Attained
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 1.4772 P
(2) Angle from Equilibrium to RAZero or Flood > 20.00 deg LARGE
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 13.81 P

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Cybermarine Technologies PTE. Ltd.

PONT00N

Condition 00 : LightshipLightship condition

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GHS 17.22

PONTOON

Condition 00 : Lightship
Lightship condition

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

LCG = 25.000f TCG = 0.000 VCG = 3.000

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. 50% DeckImm Area	Res. 50% DeckImm Angle
0.459	0.00	5.03s	406.09	0.000	-3.926	0.0000	
0.508	0.00	0.03s	405.94	0.000	0.000	-0.1713	
0.508	0.00	0.00	406.03	0.000	0.025	-0.1713	13.836
0.461	0.00	4.97p	406.15	0.000	3.952	0.0012	8.863
-0.134	0.00	13.84p	405.97	0.000	5.478	0.7690	0.000
-0.234	0.00	14.97p	405.97	0.000	5.526	0.8782	-1.137
-0.668	0.00	19.53p	405.94	0.000	5.593	1.3233	-5.697
-1.228	0.00	24.97p	405.99	0.000	5.497	1.8528	-11.137
-2.296	0.00	34.97p	406.02	0.000	4.971	2.7730	-21.137
-3.297	0.00	44.97p	406.11	0.000	4.090	3.5688	-31.137
-4.199	0.00	54.97p	406.09	0.000	3.010	4.1914	-41.137
-4.975	0.00	64.97p	405.86	0.000	1.806	4.6135	-51.137
-5.305	0.00	69.97p	406.08	0.000	1.175	4.7437	-56.137

Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.

Note: No tank loads are present.

Note: The Residual Righting Arms shown above are in excess of the
wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 10.21 (constant)

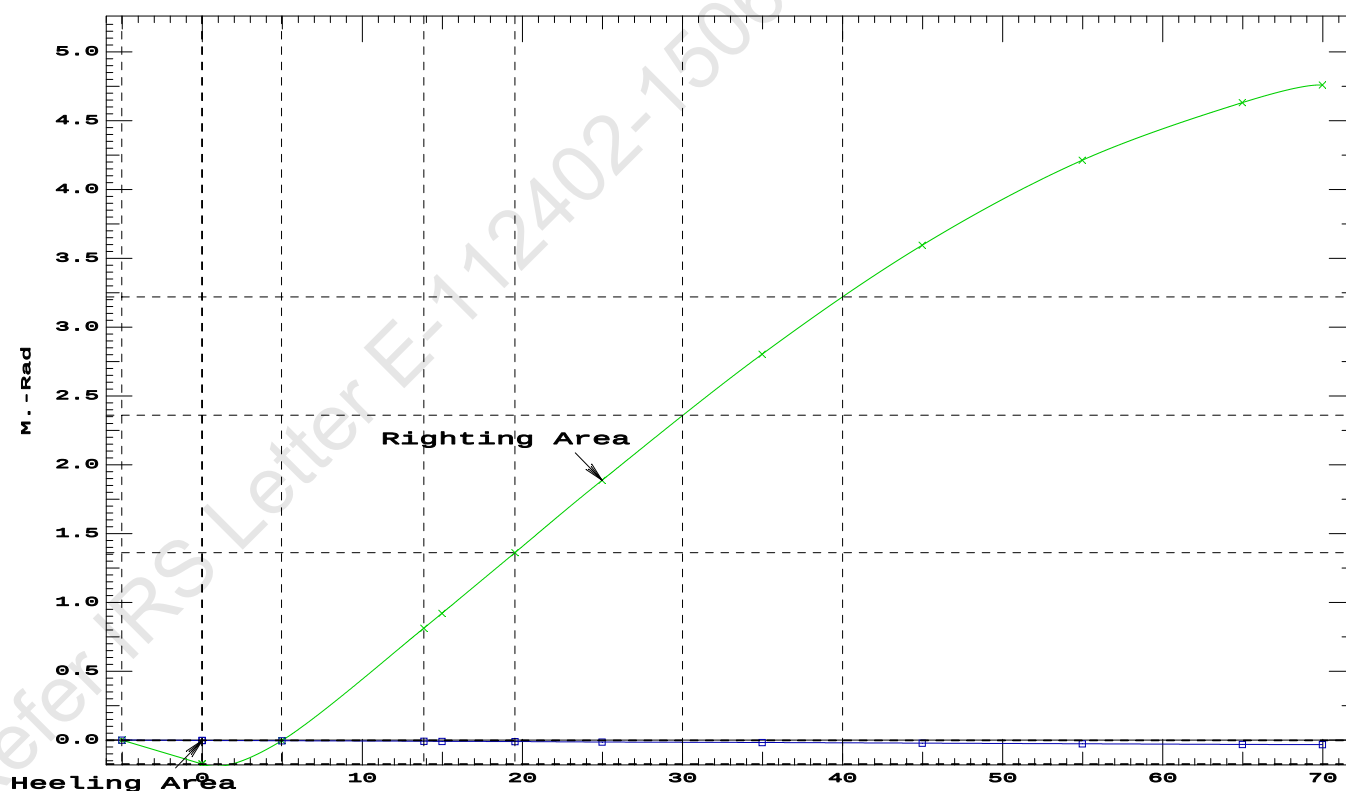
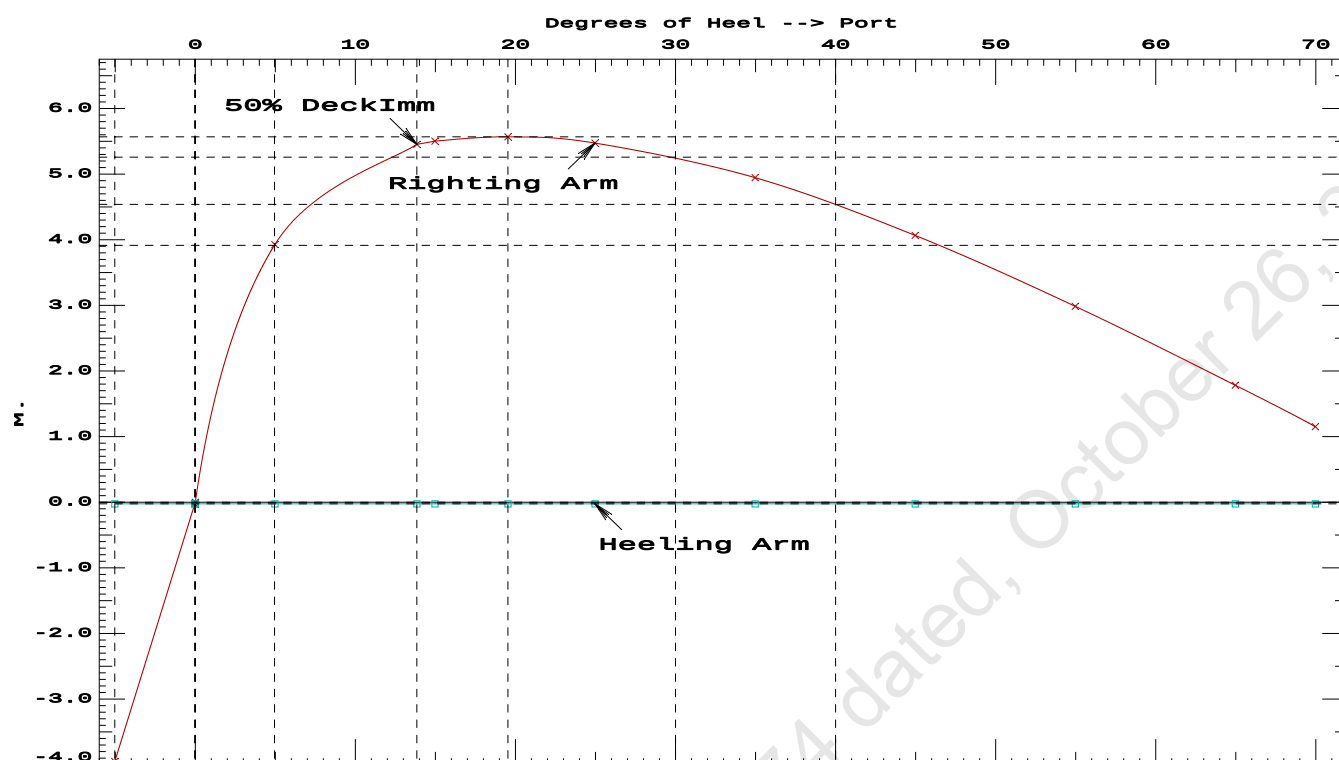
LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max-----Attained

(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	1.4946 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	LARGE
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	13.86 P

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GHS 17.22

PONTOON

Condition 00 : Lightship
Lightship condition



22-10-20 23:28:06 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONT00N

Condition 1 : Lightship + Crane + Ballast

Condition Graphic - Draft: 0.997 @ 50.000f, 1.228 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

19 07_WBTK.P 20 07_WBTK.C 21 07_WBTK.S

Condition 1 : Lightship + Crane + Ballast

USK draft: 0.997 @ 50.00f, 1.113 @ 25.00f, 1.228 @ 0.00
Trim: Aft 0.231/50.000, Heel: zero

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

DRAFT AT FP (M) = 0.997
DRAFT AT MS (M) = 1.113
DRAFT AT AP (M) = 1.228
DRAFT AT FWD MARK (M) = 1.015
DRAFT AT AFT MARK (M) = 1.210

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GHS 17.22

PONTOON

Condition 1 : Lightship + Crane + Ballast

HYDROSTATIC PROPERTIES

Trim: Aft 0.231/50.000, No Heel, VCG = 4.100

LCF Displacement Buoyancy-Ctr. Weight/ Moment/
Draft----Weight(MT)----LCB-----VCB-----cm-----LCF---cm trim----GML-----GMT
1.113 905.29 24.198f 0.560 8.63 24.799f 30.75 169.85 21.952
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 217.0 m.-MT

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 0.997 @ 50.00f, 1.113 @ 25.00f, 1.228 @ 0.00

Trim: Aft 0.231/50.000, Heel: zero

Part-----LPA*SF-----HCP-----Arm----Pressure-----Moment
HULL 94.0 0.967 1.514 0.05500 7.83
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.214f TCG = 0.000 VCG = 4.100

Free Surface Adjustment: 0.240

Adjusted CG: LCG = 24.215f TCG = 0.000 VCG = 4.340

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. 50% DeckImm Area	Angle
1.202	0.27a	4.98p	905.28	0.000	-1.922	0.0000
1.216	0.27a	0.00	905.29	0.000	-0.009	6.039
1.216	0.27a	0.02s	905.29	0.000	0.000	6.019
1.201	0.27a	5.02s	905.28	0.000	1.920	1.019
1.194	0.27a	6.04s	905.30	0.000	2.294	0.000
0.929	0.39a	15.02s	905.29	0.000	4.017	-8.981
0.817	0.46a	17.63s	905.29	0.000	4.063	-11.594
0.494	0.66a	25.02s	905.29	0.000	3.779	-18.981
0.053	0.93a	35.02s	905.29	0.000	3.006	-28.981
-0.381	1.20a	45.02s	905.29	0.000	2.046	-38.981
-0.798	1.44a	55.02s	905.29	0.000	0.990	-48.981
-0.998	1.54a	60.02s	905.29	0.000	0.443	-53.981

Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 217.0 m.-MT was applied to artificially modify the CG.

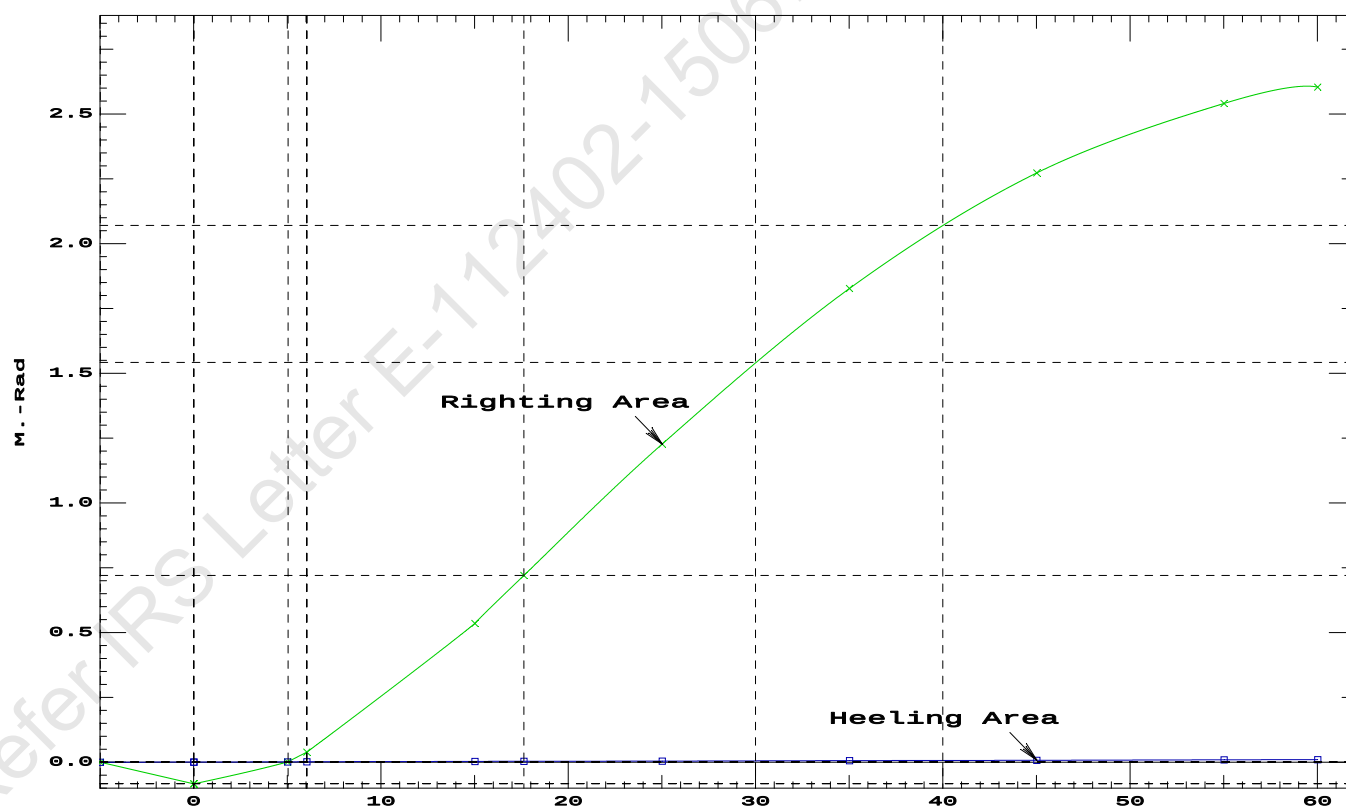
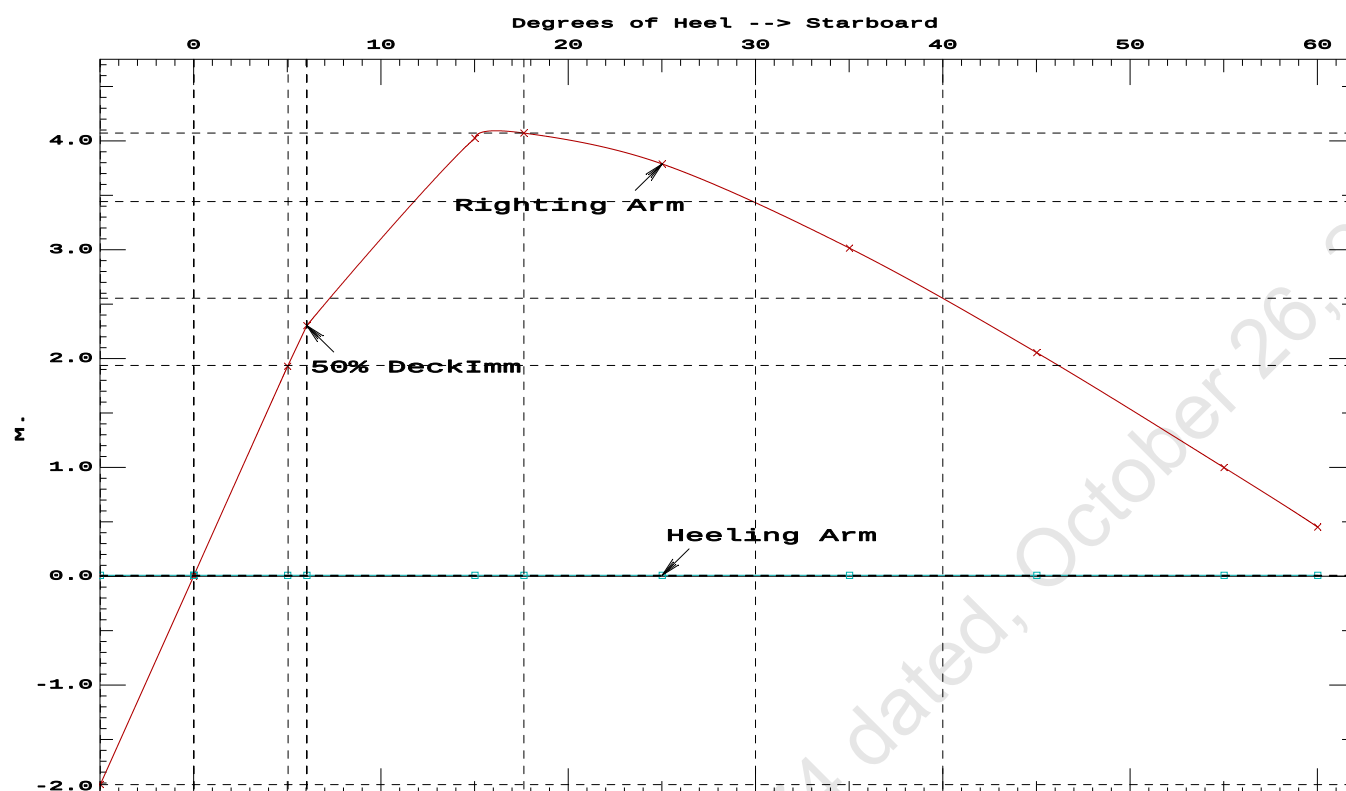
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 7.83 (constant)

LIM-----INTACT CRITERIA-IS CODE 2008PART B ----Min/Max-----Attained
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 0.8402 P
(2) Angle from Equilibrium to RAZero or Flood > 20.00 deg LARGE
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 6.02 P

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GHS 17.22

PONT00N

Condition 1 : Lightship + Crane + Ballast

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GHS 17.22

PONTOON

Condition 1 : Lightship + Crane + Ballast

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.214f TCG = 0.000 VCG = 4.100

Free Surface Adjustment: 0.240

Adjusted CG: LCG = 24.215f TCG = 0.000 VCG = 4.340

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. 50% DeckImm Area	Angle
1.201	0.27a	5.02s	905.28	0.000	-1.920	0.0000	
1.216	0.27a	0.02s	905.29	0.000	0.000	-0.0838	
1.216	0.27a	0.00	905.28	0.000	0.009	-0.0838	6.039
1.202	0.27a	4.98p	905.28	0.000	1.922	0.0001	1.059
1.194	0.27a	6.04p	905.30	0.000	2.311	0.0392	0.000
0.931	0.39a	14.98p	905.29	0.000	4.032	0.5738	-8.941
0.816	0.46a	17.64p	905.29	0.000	4.080	0.7623	-11.605
0.495	0.66a	24.98p	905.29	0.000	3.799	1.2710	-18.941
0.054	0.93a	34.98p	905.29	0.000	3.027	1.8732	-28.941
-0.379	1.20a	44.98p	905.29	0.000	2.067	2.3205	-38.941
-0.797	1.44a	54.98p	905.29	0.000	1.012	2.5906	-48.941
-1.160	1.62a	64.18p	905.28	0.000	0.000	2.6724	-58.140
-1.191	1.63a	64.98p	905.29	0.000	-0.089	2.6717	-58.941
-1.377	1.70a	69.98p	905.29	0.000	-0.643	2.6398	-63.941

Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 217.0 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 7.83 (constant)

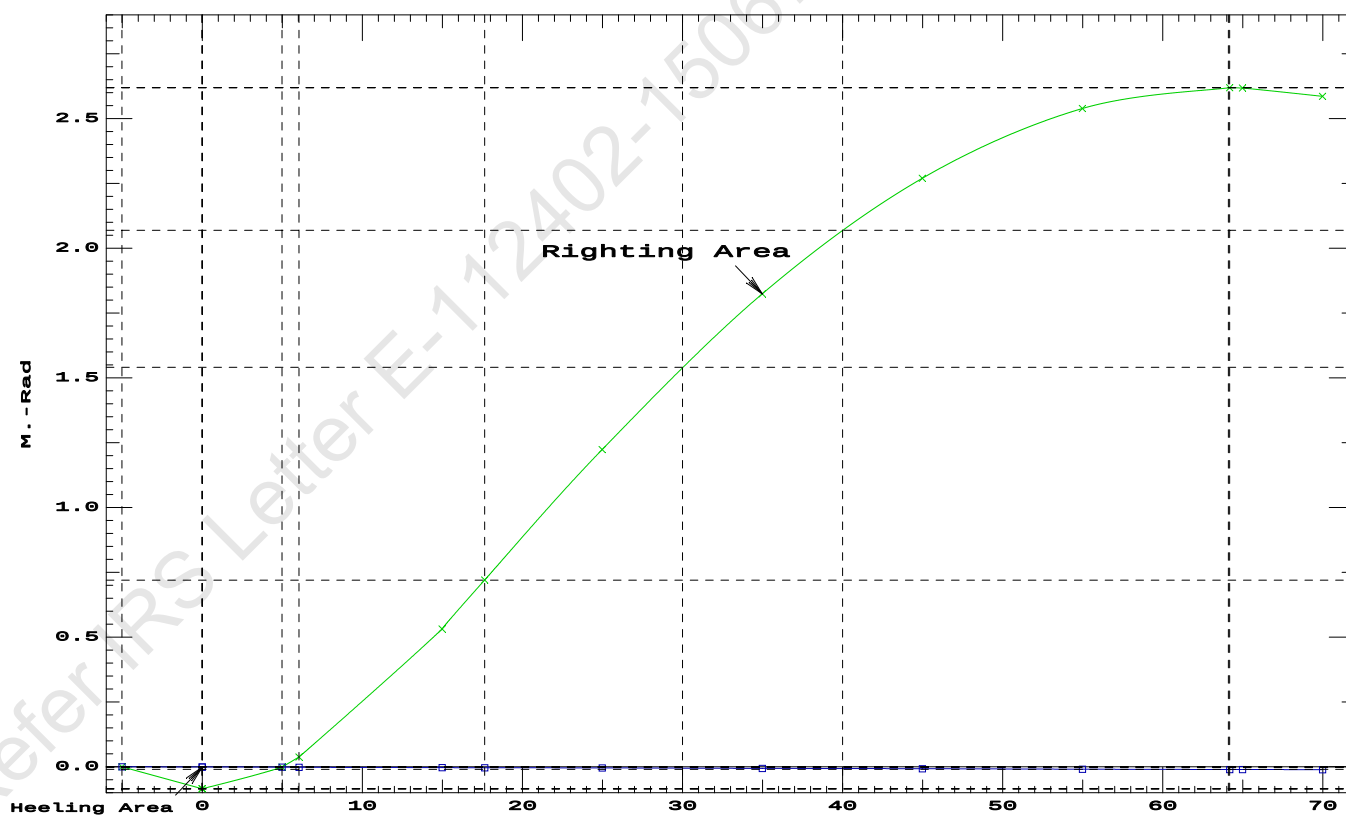
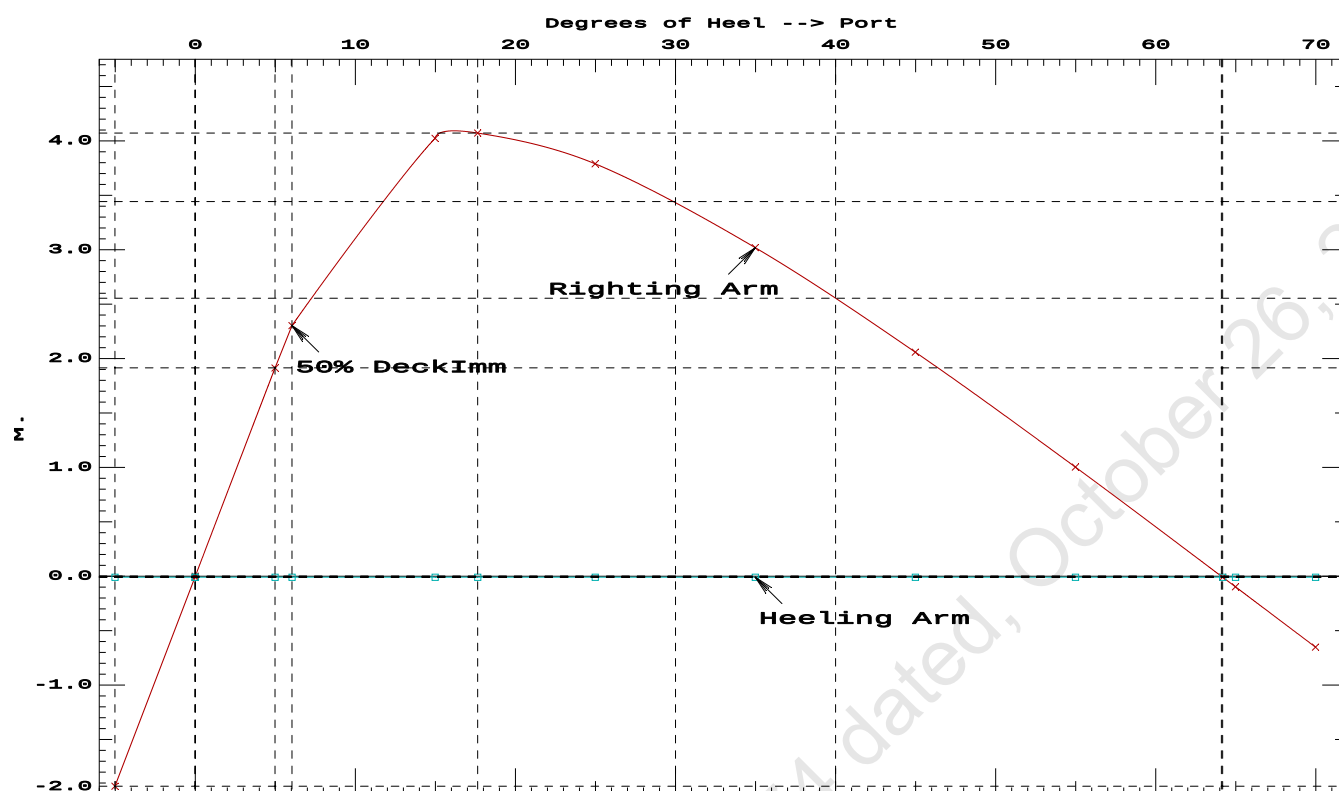
LIM-----INTACT CRITERIA-IS CODE 2008PART B ----Min/Max-----Attained

(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.8461 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	64.20 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	6.06 P

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GHS 17.22

PONTOON

Condition 1 : Lightship + Crane + Ballast

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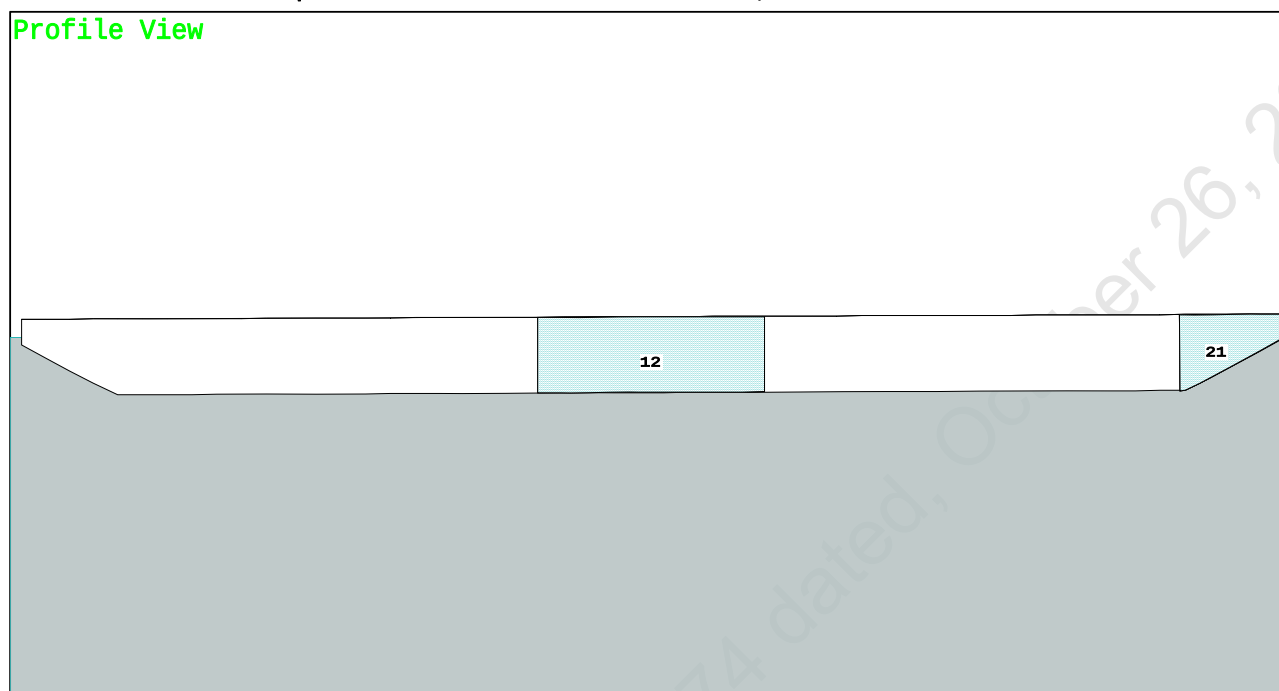
GHS 17.22

PONT00N

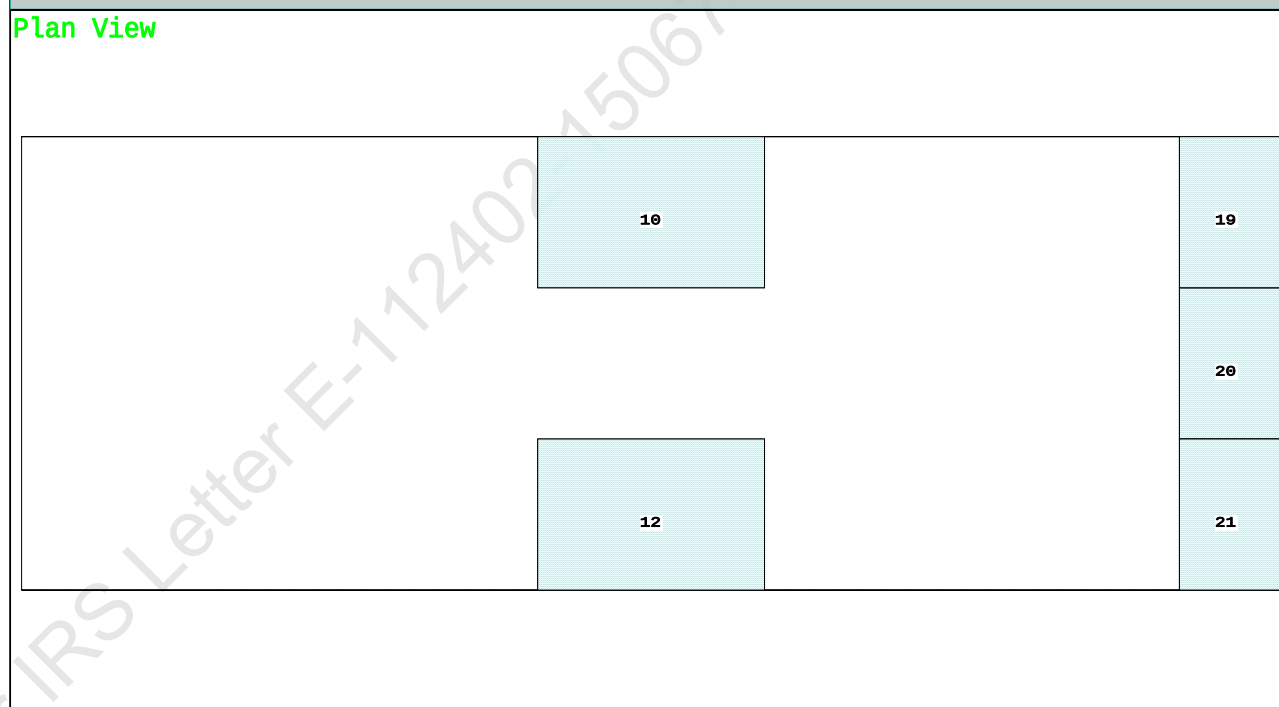
Condition 2 : Lightship + Load at Aft + Ballast + Deck CargoCrane Lifting 40.0T @ 29m Radius Aft

Condition Graphic - Draft: 2.108 @ 50.000f, 2.316 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

10 04_WBTK.P 12 04_WBTK.S 19 07_WBTK.P 20 07_WBTK.C 21 07_WBTK.S

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PONTOON

Condition 2 : Lightship + Load at Aft + Ballast + Deck CargoCrane Lifting 40.0T @ 29m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.108 @ 50.00f, 2.212 @ 25.00f, 2.316 @ 0.00

Trim: Aft 0.208/50.000, Heel: zero

Part-----	Weight(MT)-----	LCG-----	TCG-----	VCG-----	FSM-----
LIGHT SHIP	405.97	25.000f	0.000	3.000	
Crane	350.00	13.300f	0.000	6.325	
Crane Load	40.00	15.700a	0.000	21.478	
Deck Cargo	624.59	27.500f	0.000	4.000	
Total Fixed----->	1,420.56	22.071f	0.000	4.779	
Load-----	SpGr-----	Weight(MT)-----	LCG-----	TCG-----	VCG-----
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s
Total Tanks----->		472.52	32.161f	0.000	1.626
Total Weight----->		1,893.08	24.589f	0.000	3.992
		Displ(MT)-----	LCB-----	TCB-----	VCB-----
HULL	1.025	1,893.08	24.577f	0.000	1.133

Righting Arms:		0.000	0.000		
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----
Total Waterplane----->	1.025	900.4	24.995f	0.000	101.45
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----
		9.23	37.33	98.59	10.031
Distances in METERS.-----			Moments in m.-MT.		

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

461.0 Cu.M. (18%) SALT WATER

350.00 MT Crane

40.00 MT Crane Load

624.59 MT Deck Cargo

DRAFT AT FP (M) = 2.108
 DRAFT AT MS (M) = 2.212
 DRAFT AT AP (M) = 2.316
 DRAFT AT FWD MARK (M) = 2.124
 DRAFT AT AFT MARK (M) = 2.300

HYDROSTATIC PROPERTIES

Trim: Aft 0.208/50.000, No Heel, VCG = 3.992

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/					
Draft----	Weight(MT)-----	LCB-----	VCB-----	cm-----	LCF-----	cm trim-----	GML-----	GMT-----	
2.212	1,893.08	24.577f	1.133	9.23	24.995f	37.33	98.59	10.031	
Distances in METERS.-----					Specific Gravity = 1.025.-----				
					Moment in m.-MT.				
					Trim is per 50.00m.				

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 542.4 m.-MT

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PONTOON

Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo
Crane Lifting 40.0T @ 29m Radius Aft

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 2.108 @ 50.00f, 2.212 @ 25.00f, 2.316 @ 0.00

Trim: Aft 0.208/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	40.5	0.407	1.481	0.05500	3.30
CRANE	57.0	3.238	4.312	0.05500	13.52
DECKCARGO.C	29.4	1.810	2.884	0.05500	4.66
Total wind heeling moment to starboard-->					21.48
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.589f TCG = 0.000 VCG = 3.992

Free Surface Adjustment: 0.287

Adjusted CG: LCG = 24.590f TCG = 0.000 VCG = 4.279

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. 50% DeckImm Area	Angle
2.304	0.24a	0.00	1,893.1	0.000	-0.011	0.0000	2.238
2.304	0.24a	0.06s	1,893.1	0.000	0.000	-0.0000	2.176
2.303	0.24a	2.24s	1,893.1	0.000	0.385	0.0073	-0.000
2.297	0.25a	5.06s	1,893.1	0.000	0.875	0.0384	-2.824
2.444	0.36a	10.06s	1,893.1	0.000	1.284	0.1376	-7.824
2.597	0.43a	13.02s	1,893.1	0.000	1.320	0.2054	-10.779
2.718	0.47a	15.06s	1,893.1	0.000	1.303	0.2522	-12.824
3.041	0.59a	20.06s	1,893.1	0.000	1.172	0.3619	-17.824
3.371	0.75a	25.06s	1,893.1	0.000	0.953	0.4553	-22.824
3.678	0.91a	30.06s	1,893.1	0.000	0.687	0.5272	-27.824
3.960	1.07a	35.06s	1,893.1	0.000	0.397	0.5747	-32.824
4.214	1.22a	40.06s	1,893.1	0.000	0.093	0.5961	-37.824
4.285	1.27a	41.56s	1,893.1	0.000	0.000	0.5974	-39.323
4.438	1.37a	45.06s	1,893.1	0.000	-0.218	0.5907	-42.824
4.627	1.51a	50.06s	1,893.1	0.000	-0.531	0.5581	-47.824
4.781	1.63a	55.06s	1,893.1	0.000	-0.842	0.4982	-52.824
4.896	1.74a	60.06s	1,893.1	0.000	-1.148	0.4113	-57.824
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.							

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 542.4 m.-MT was applied to artificially modify the CG.

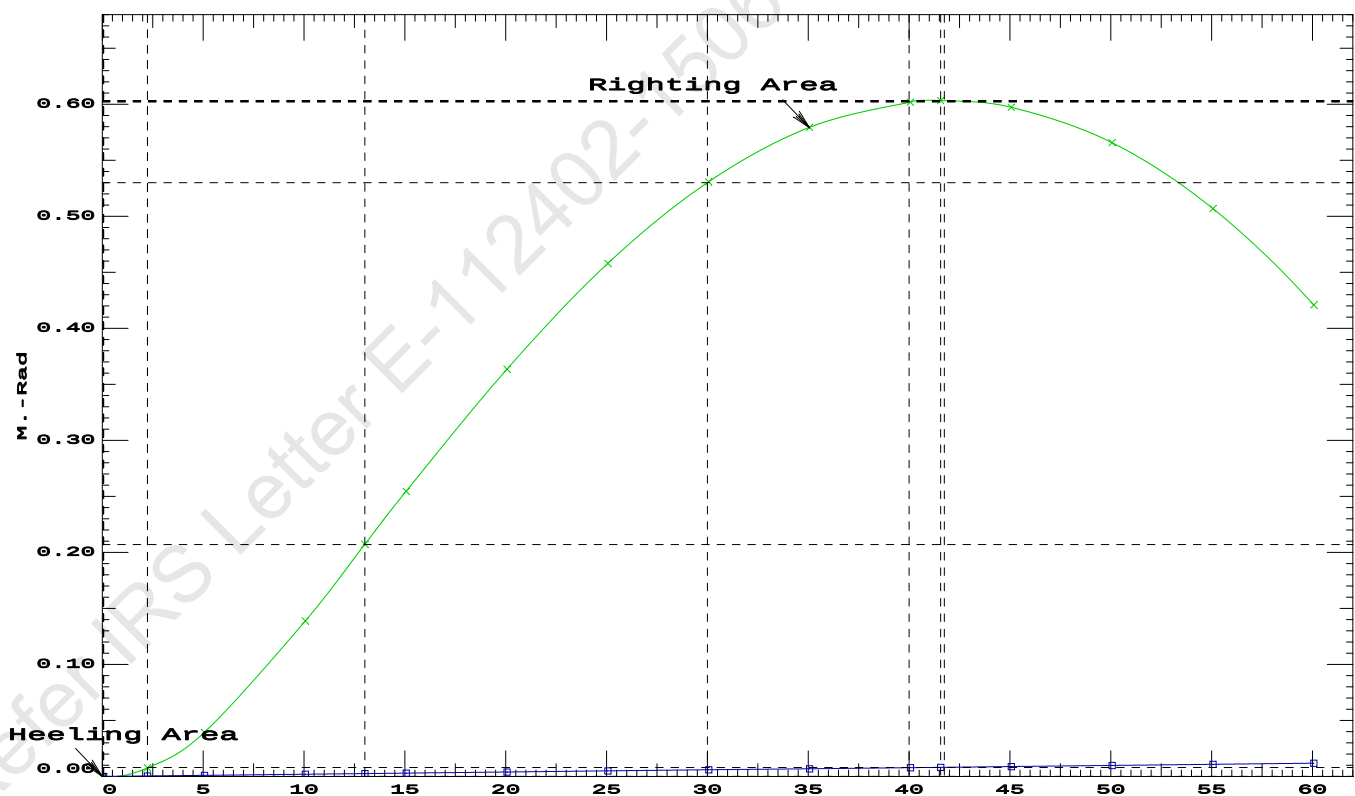
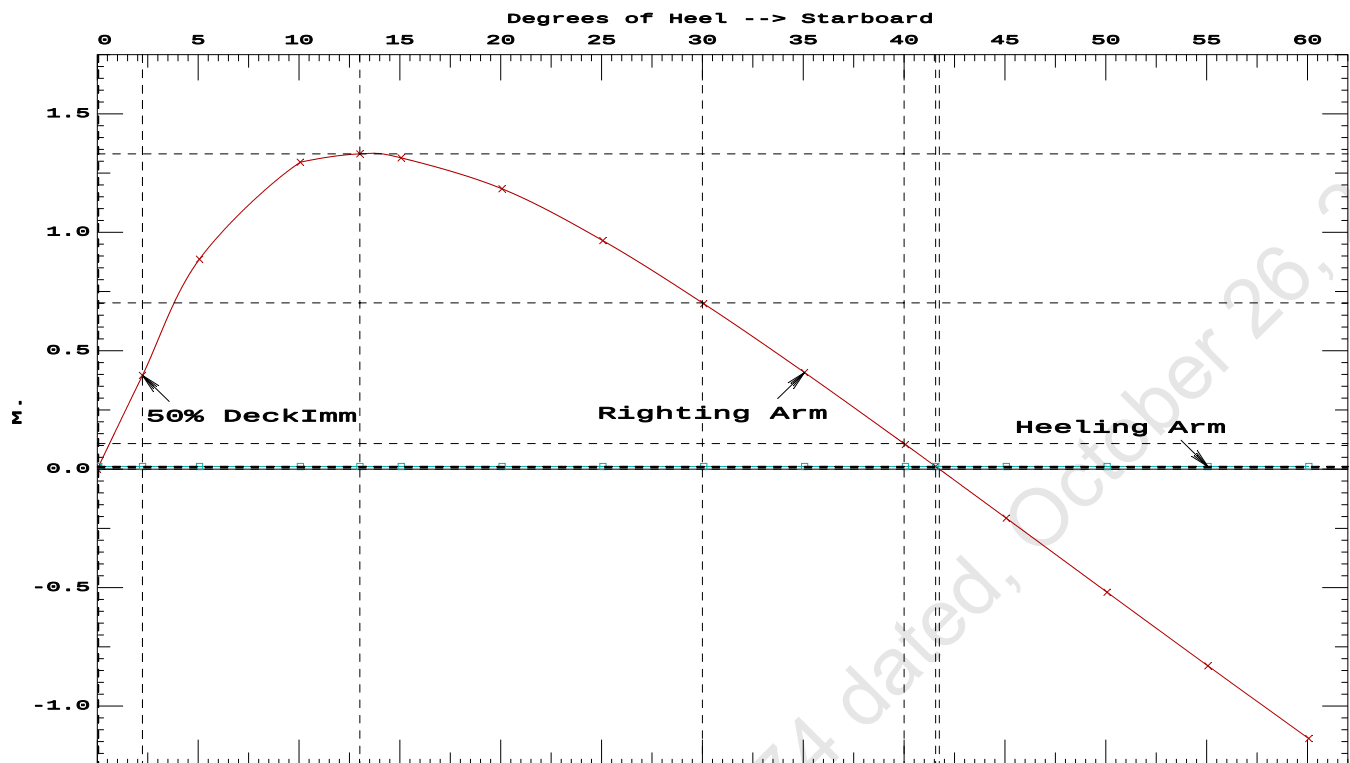
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.48 (constant)

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2-----Min/Max-----Attained

(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.2054 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	41.50 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	2.18 P

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GHS 17.22 PONT00N

Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo
Crane Lifting 40.0T @ 29m Radius Aft



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GHS 17.22

PONTOON

Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo
Crane Lifting 40.0T @ 29m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.589f TCG = 0.000 VCG = 3.992

Free Surface Adjustment: 0.287

Adjusted CG: LCG = 24.590f TCG = 0.000 VCG = 4.279

Origin Depth---	Degrees of Trim----	Heel----	Displacement Weight(MT)---	Residual Arms in Trim--	in Heel---	Res. Area
2.304	0.24a	0.00	1,893.1	0.000	0.000	0.0000
2.297	0.25a	5.00s	1,893.1	0.000	0.876	0.0382
2.441	0.36a	10.00s	1,893.1	0.000	1.293	0.1379
2.597	0.43a	13.02s	1,893.1	0.000	1.330	0.2076
2.714	0.47a	15.00s	1,893.1	0.000	1.314	0.2535
3.037	0.59a	20.00s	1,893.1	0.000	1.185	0.3642
3.367	0.75a	25.00s	1,893.1	0.000	0.967	0.4588
3.675	0.91a	30.00s	1,893.1	0.000	0.701	0.5319
3.957	1.07a	35.00s	1,893.1	0.000	0.411	0.5806
4.212	1.22a	40.00s	1,893.1	0.000	0.107	0.6033
4.293	1.27a	41.73s	1,893.1	0.000	0.000	0.6049
4.435	1.37a	45.00s	1,893.1	0.000	-0.204	0.5991
4.625	1.51a	50.00s	1,893.1	0.000	-0.516	0.5677
4.779	1.63a	55.00s	1,893.1	0.000	-0.828	0.5090
4.895	1.74a	60.00s	1,893.1	0.000	-1.134	0.4234

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 542.4 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

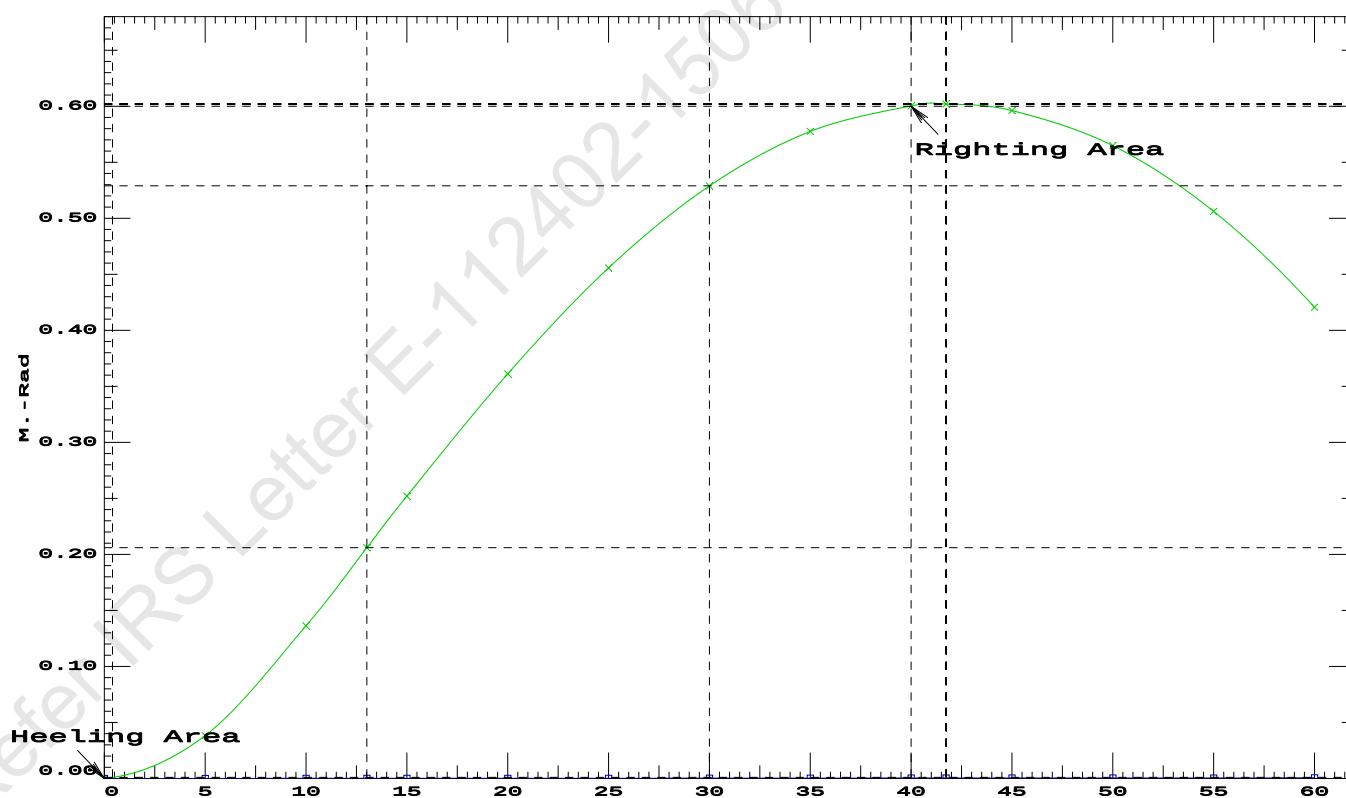
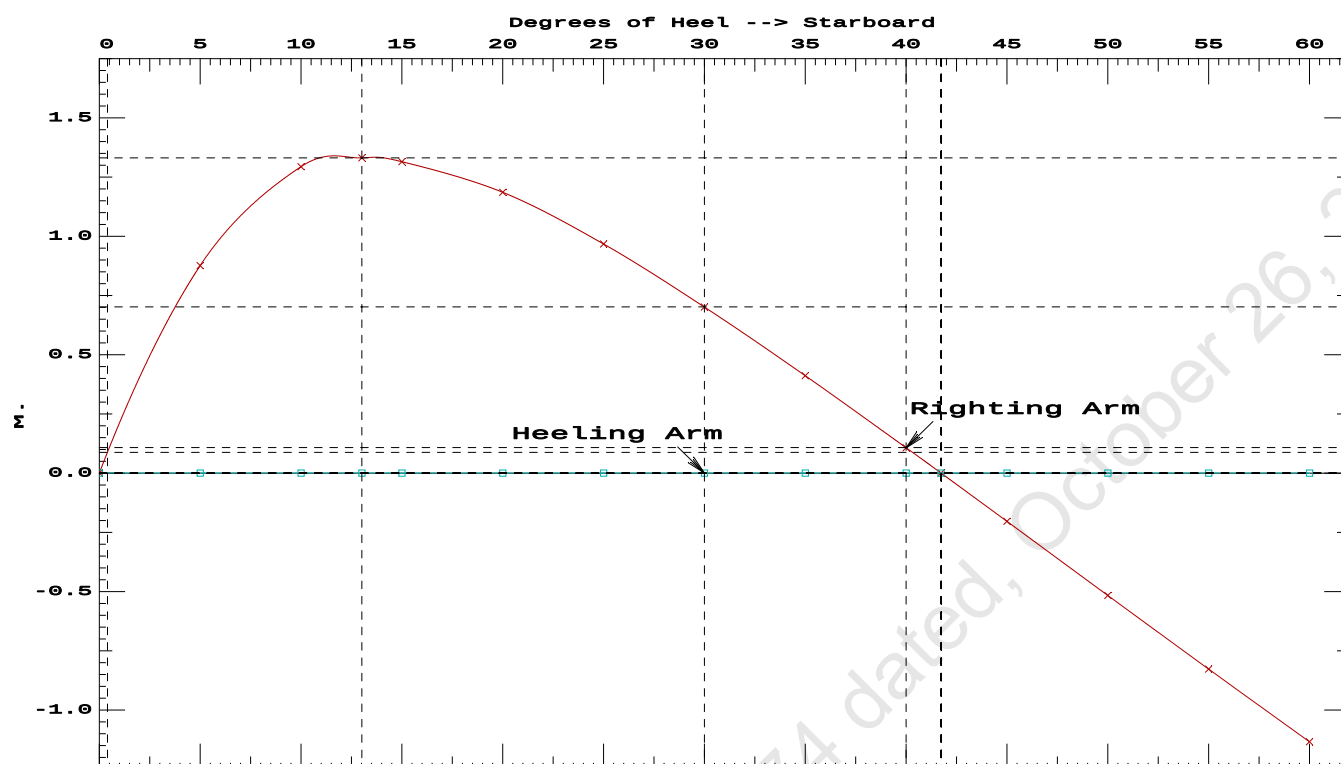
LIM-----STABILITY CRITERIA FOR CRANE OPERAT-----Min/Max-----Attained

(1) Absolute Angle at Equilibrium	<	15.00 deg	0.00 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	251854 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	1573.29 P

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PONTOON

Condition 2 : Lightship + Load at Aft + Ballast + Deck CargoCrane Lifting 40.0T @ 29m Radius Aft

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GHS 17.22 PONTON

Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo**Crane Lifting 40.0T @ 29m Radius Aft**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.108 @ 50.00f, 2.212 @ 25.00f, 2.316 @ 0.00

Trim: Aft 0.208/50.000, Heel: 0.00 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			350.00	13.300f	0.000	6.325	
Crane Load			40.00	15.700a	0.000	21.478	
Deck Cargo			624.59	27.500f	0.000	4.000	
Total Fixed----->			1,420.56	22.071f	0.000	4.779	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p	1.509	162.73*
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			472.52	32.161f	0.000	1.626	542.42
Total Weight----->			1,893.08	24.589f	0.000	3.992	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,893.08	24.577f	0.001	1.133	

	Righting Arms:			0.000	0.001s		
	External Arms:			0.000	0.001		
	Residual Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->		1.025	900.4	24.995f	0.000	101.45	12.890
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			9.23		37.33	98.59	10.031
Distances in METERS.-----							
-----Moments in m.-MT.							

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

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After hook load drop							
WEIGHT and DISPLACEMENT and WATERPLANE STATUS							
USK draft: 2.282 @ 50.00f, 2.169 @ 25.00f, 2.055 @ 0.00							
Trim: Fwd 0.227/50.000, Heel: 0.00 deg.							
Part-----	Weight(MT)----		LCG-----	TCG-----	VCG-----		
LIGHT SHIP	405.97	25.000f	0.000	3.000			
Crane	350.00	13.300f	0.000	6.325			
Deck Cargo	624.59	27.500f	0.000	4.000			
Total Fixed----->	1,380.56	23.165f	0.000	4.295			
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p	1.509	162.73*
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			472.52	32.161f	0.000	1.626	542.42
Total Weight----->			1,853.08	25.459f	0.000	3.615	
	Displ(MT)----		LCB-----	TCB-----	VCB-----		
HULL	1.025	1,853.08	25.470f	0.001	1.110		

Righting Arms:			0.000	0.001s			
External Arms:			0.000	0.001			
Residual Righting Arms:			0.000	0.000			
Part-----	SpGr-----		WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		900.4	25.006f	0.000	103.61	13.165
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			9.23	37.47	101.11	10.661	
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

Theta3 is 0.07 degrees, as demonstrated below.

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 23.165f TCG = 0.000 VCG = 4.295

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
2.043	0.26f	0.00	1,853.1	0.000	0.000	0.0000
2.043	0.26f	0.07s	1,853.1	0.000	0.013	0.0000
2.029	0.27f	5.00s	1,853.1	0.000	0.957	0.0418
2.057	0.38f	10.00s	1,853.1	0.000	1.495	0.1541
2.216	0.50f	15.00s	1,853.1	0.000	1.599	0.2922
2.407	0.64f	20.00s	1,853.1	0.000	1.533	0.4301
2.574	0.81f	25.00s	1,853.1	0.000	1.366	0.5573
2.718	0.97f	30.00s	1,853.1	0.000	1.147	0.6673

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1380.56 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 0.1 > 1.000 LARGE

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GHS 17.22 PONT00N

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 23.165f TCG = 0.000 VCG = 4.295

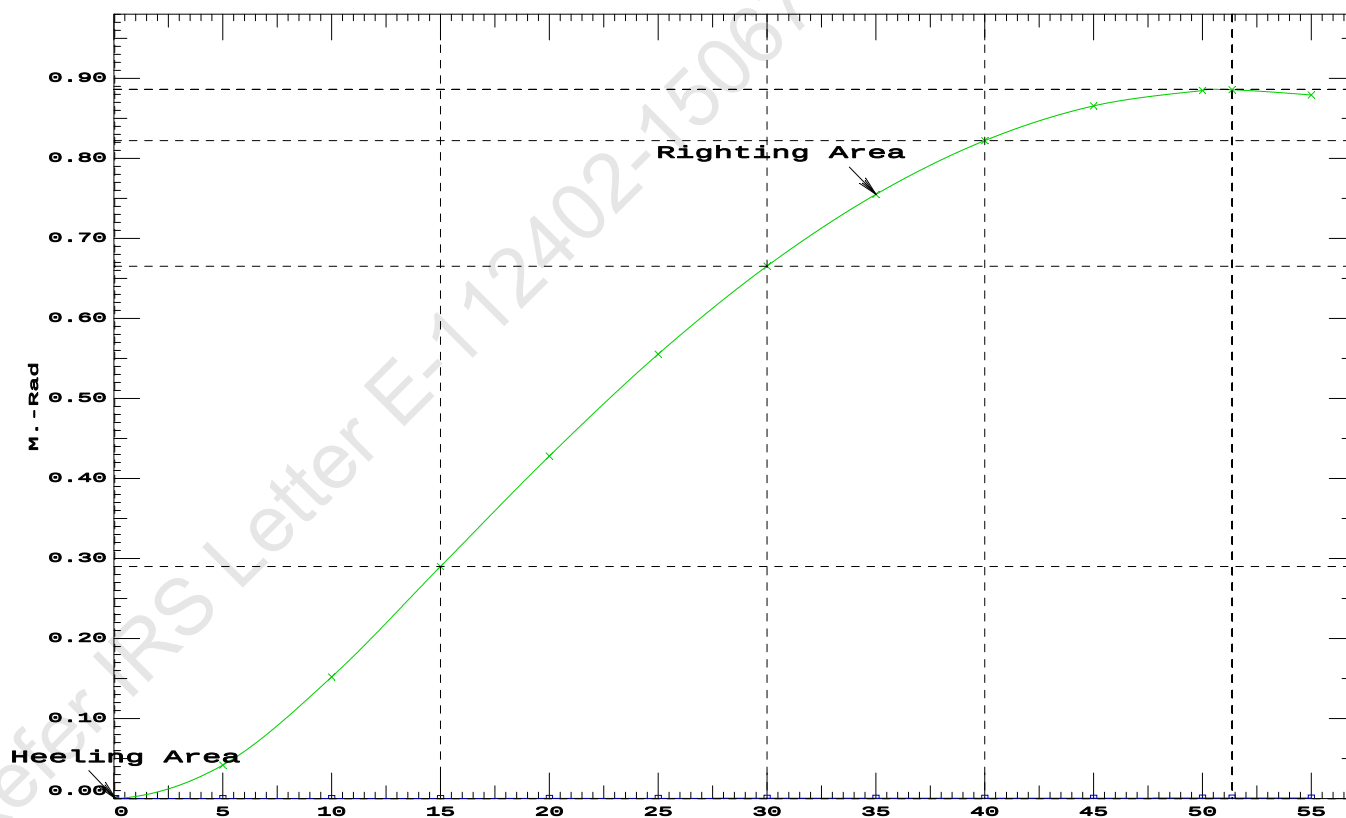
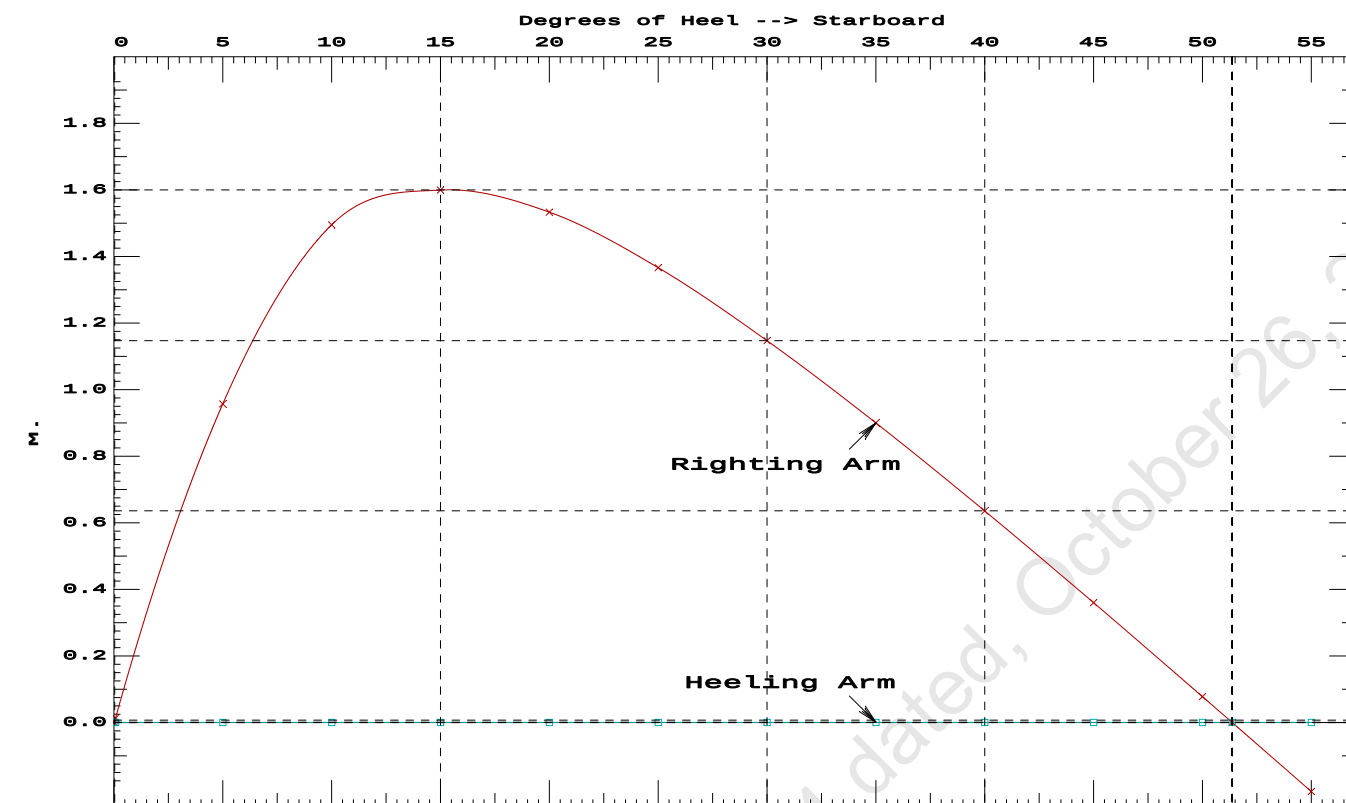
Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
2.043	0.26f	0.00	1,853.1	0.000	0.000	0.0000
2.043	0.26f	0.07s	1,853.1	0.000	0.013	0.0000
2.029	0.27f	5.00s	1,853.1	0.000	0.957	0.0418
2.057	0.38f	10.00s	1,853.1	0.000	1.495	0.1541
2.216	0.50f	15.00s	1,853.1	0.000	1.599	0.2922
2.407	0.64f	20.00s	1,853.1	0.000	1.533	0.4301
2.574	0.81f	25.00s	1,853.1	0.000	1.366	0.5573
2.718	0.97f	30.00s	1,853.1	0.000	1.147	0.6673
2.839	1.14f	35.00s	1,853.1	0.000	0.900	0.7568
2.935	1.30f	40.00s	1,853.1	0.000	0.635	0.8239
3.008	1.46f	45.00s	1,853.1	0.000	0.359	0.8674
3.057	1.61f	50.00s	1,853.1	0.000	0.077	0.8865
3.066	1.65f	51.36s	1,853.1	0.000	0.000	0.8874
3.082	1.75f	55.00s	1,853.1	0.000	-0.208	0.8808

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1380.56 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero > 1.000 LARGE
(2) Angle from abs 0.1 deg to RAZero > 20.00 deg 51.29 P

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PONTOON

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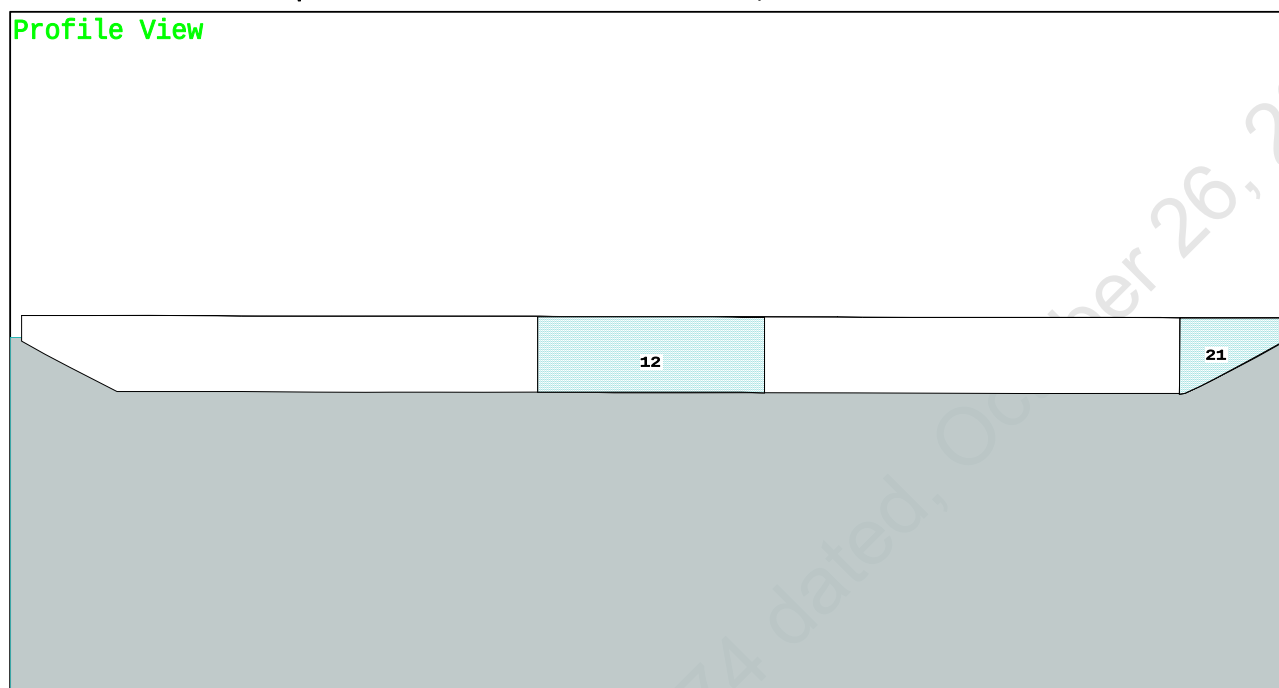
GHS 17.22

PONT00N

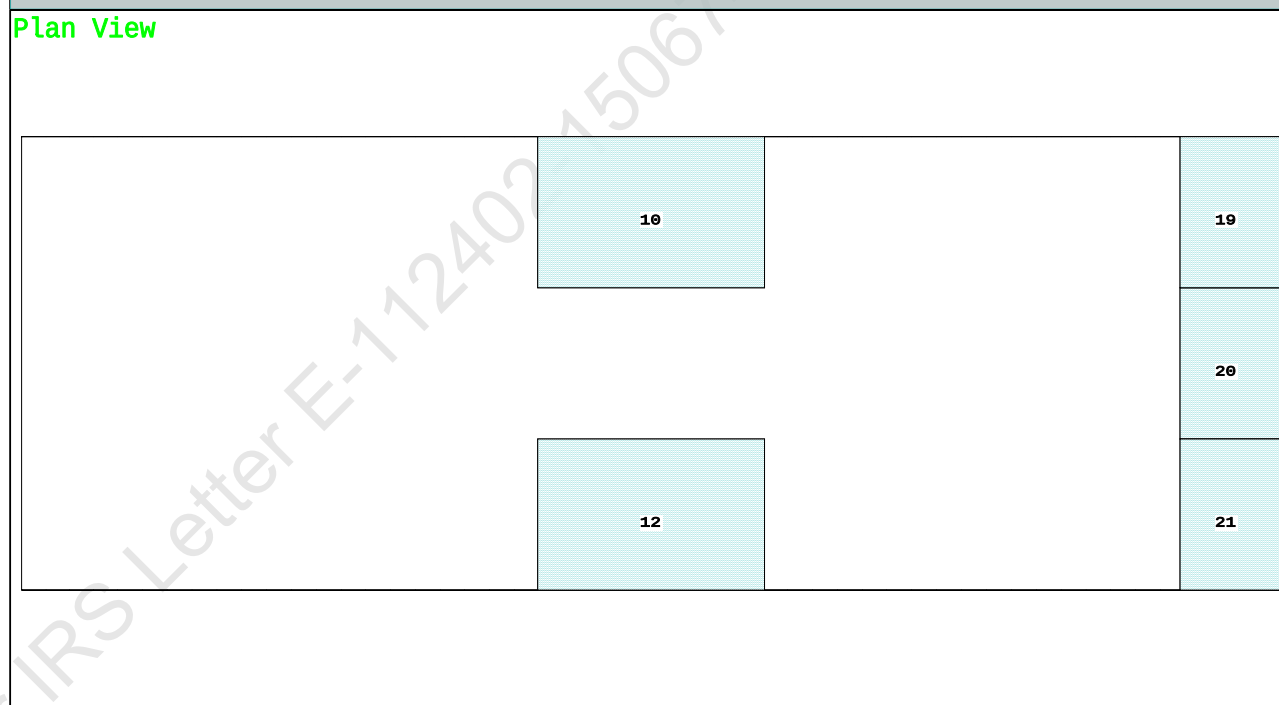
Condition 3 : Lightship + Load at Stbd + Ballast + Deck CargoCrane Lifting 40.0T @ 29m Radius Stbd

Condition Graphic - Draft: 2.263 @ 50.000f, 2.161 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

10 04_WBTK.P

12 04_WBTK.S

19 07_WBTK.P

20 07_WBTK.C

21 07_WBTK.S

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PONTOON

Condition 3 : Lightship + Load at Stbd + Ballast + Deck CargoCrane Lifting 40.0T @ 29m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.263 @ 50.00f, 2.212 @ 25.00f, 2.161 @ 0.00

Trim: Fwd 0.102/50.000, Heel: zero

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			350.00	13.300f	0.000	6.325	
Crane Load			40.00	13.300f	0.000	21.478	
Deck Cargo			624.59	27.500f	0.000	4.000	
Total Fixed----->			1,420.56	22.887f	0.000	4.779	
	Load----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p	1.509	162.73*
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			472.52	32.161f	0.000	1.626	542.42
Total Weight----->			1,893.08	25.202f	0.000	3.992	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,893.08	25.208f	0.000	1.132	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->		1.025	900.4	25.003f	0.000	101.45	12.891
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			9.23		37.33	98.59	10.030
Distances in METERS.-----				Moments in m.-MT			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

461.0 Cu.M. (18%) SALT WATER

350.00 MT Crane

40.00 MT Crane Load

624.59 MT Deck Cargo

DRAFT AT FP (M) = 2.263
 DRAFT AT MS (M) = 2.212
 DRAFT AT AP (M) = 2.161
 DRAFT AT FWD MARK (M) = 2.255
 DRAFT AT AFT MARK (M) = 2.169

HYDROSTATIC PROPERTIES

Trim: Fwd 0.102/50.000, No Heel, VCG = 3.992

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/				
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----	LCF-----	cm trim-----	GML-----	GMT
2.212	1,893.08	25.208f	1.132	9.23	25.003f	37.33	98.59	10.030
Distances in METERS.-----				Specific Gravity = 1.025.-----				
				Moment in m.-MT.				
				Trim is per 50.00m.				

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 542.4 m.-MT

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PONTOON

Condition 3 : Lightship + Load at Stbd + Ballast + Deck Cargo
Crane Lifting 40.0T @ 29m Radius Stbd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 2.263 @ 50.00f, 2.212 @ 25.00f, 2.161 @ 0.00

Trim: Fwd 0.102/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	40.5	0.406	1.479	0.05500	3.29
CRANE	57.0	3.330	4.404	0.05500	13.81
DECKCARGO.C	29.4	1.795	2.868	0.05500	4.64
Total wind heeling moment to starboard-->					21.74
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.202f TCG = 0.000 VCG = 3.992

Free Surface Adjustment: 0.287

Adjusted CG: LCG = 25.201f TCG = 0.000 VCG = 4.279

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. 50% DeckImm Area	Angle
2.149	0.12f	0.00	1,893.1	0.000	-0.011	2.410
2.149	0.12f	0.06s	1,893.1	0.000	0.000	2.347
2.146	0.12f	2.41s	1,893.1	0.000	0.415	0.000
2.136	0.12f	5.06s	1,893.1	0.000	0.877	-2.653
2.207	0.18f	10.06s	1,893.0	0.000	1.288	-7.653
2.319	0.21f	13.01s	1,893.1	0.000	1.323	-10.600
2.410	0.23f	15.06s	1,893.1	0.000	1.306	-12.653
2.655	0.29f	20.06s	1,893.1	0.000	1.177	-17.653
2.882	0.37f	25.06s	1,893.1	0.000	0.958	-22.653
3.087	0.45f	30.06s	1,893.1	0.000	0.692	-27.653
3.266	0.52f	35.06s	1,893.1	0.000	0.401	-32.653
3.420	0.60f	40.06s	1,893.1	0.000	0.097	-37.653
3.463	0.62f	41.63s	1,893.1	0.000	0.000	-39.224
3.547	0.67f	45.06s	1,893.1	0.000	-0.214	-42.653
3.647	0.74f	50.06s	1,893.1	0.000	-0.527	-47.653
3.719	0.80f	55.06s	1,893.1	0.000	-0.839	-52.653
3.764	0.86f	60.06s	1,893.1	0.000	-1.146	-57.653
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.						

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 542.4 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.74 (constant)

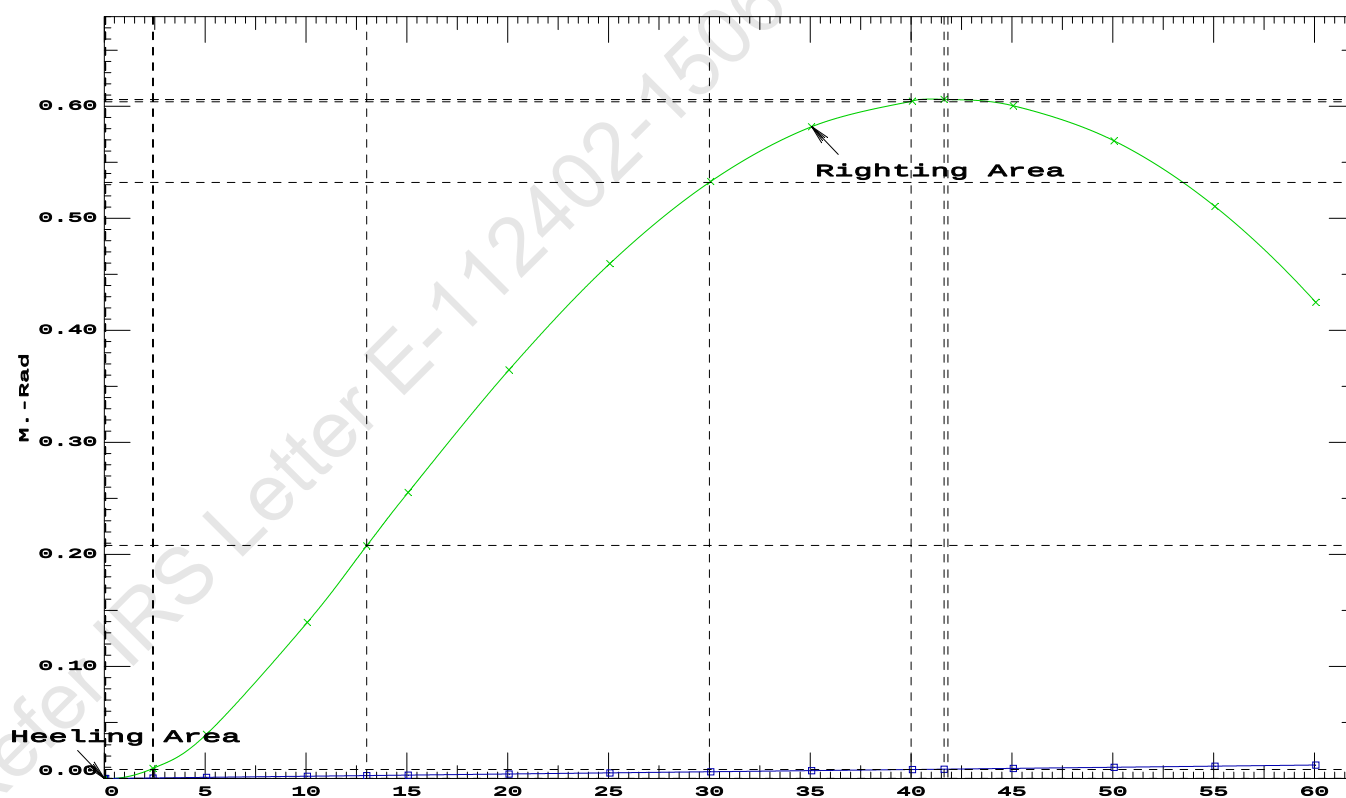
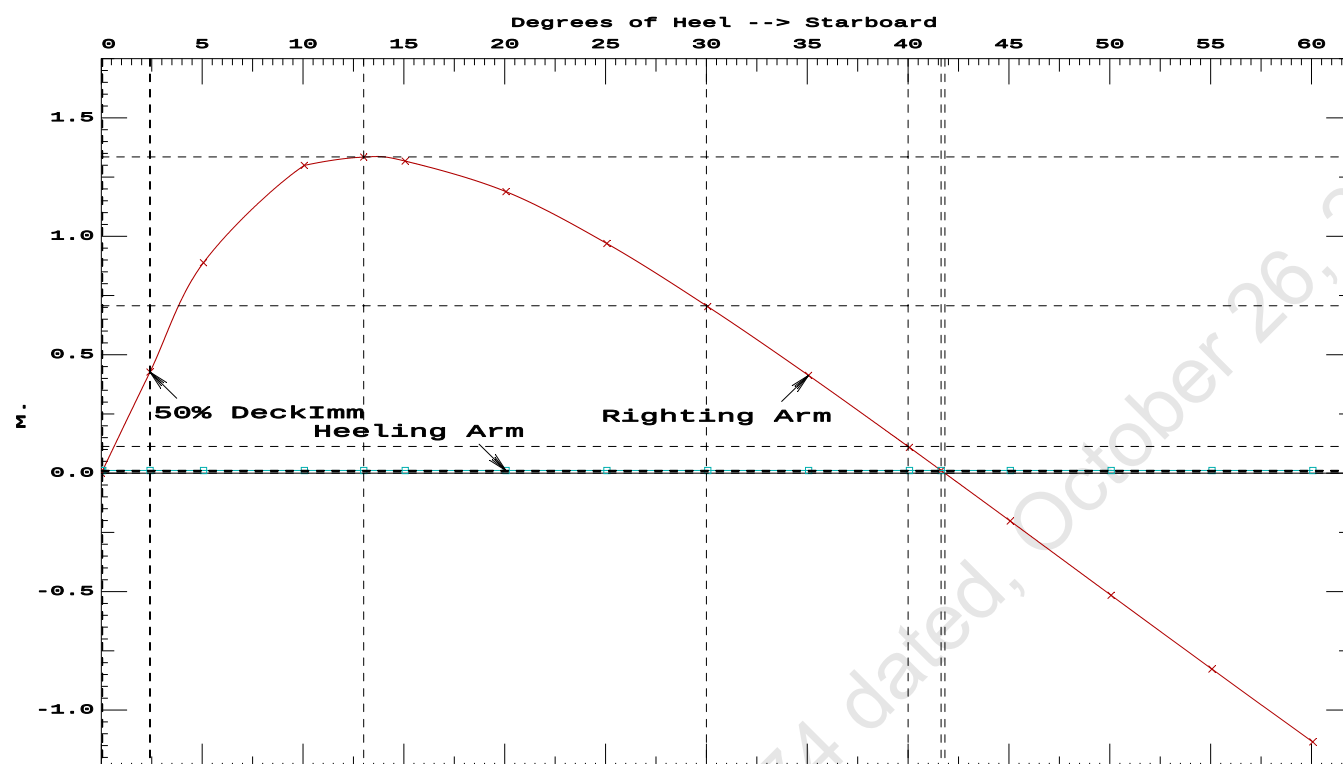
LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2-----Min/Max-----Attained

(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.2058 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	41.57 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	2.35 P

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PONTOON

Condition 3 : Lightship + Load at Stbd + Ballast + Deck CargoCrane Lifting 40.0T @ 29m Radius Stbd

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PONTOON

Condition 3 : Lightship + Load at Stbd + Ballast + Deck Cargo
Crane Lifting 40.0T @ 29m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.202f TCG = 0.000 VCG = 3.992

Free Surface Adjustment: 0.287

Adjusted CG: LCG = 25.201f TCG = 0.017p VCG = 4.278

Origin Depth---	Degrees of Trim----	Heel----	Displacement Weight(MT)---	Residual Arms in Trim--	in Heel---	Res. Area
2.143	0.12f	3.38s	1,893.1	0.000	0.000	0.0000
2.162	0.16f	8.38s	1,893.1	0.000	0.636	0.0278
2.313	0.21f	12.87s	1,893.1	0.000	0.738	0.0831
2.335	0.21f	13.38s	1,893.1	0.000	0.737	0.0897
2.573	0.27f	18.38s	1,893.1	0.000	0.647	0.1510
2.808	0.34f	23.38s	1,893.1	0.000	0.453	0.1997
3.021	0.42f	28.38s	1,893.1	0.000	0.198	0.2286
3.156	0.48f	31.90s	1,893.1	0.000	0.000	0.2347
3.209	0.50f	33.38s	1,893.1	0.000	-0.086	0.2336
3.371	0.58f	38.38s	1,893.1	0.000	-0.388	0.2131
3.507	0.65f	43.38s	1,893.1	0.000	-0.698	0.1658
3.616	0.72f	48.38s	1,893.1	0.000	-1.011	0.0913
3.697	0.78f	53.38s	1,893.1	0.000	-1.325	-0.0107
3.752	0.84f	58.38s	1,893.1	0.000	-1.635	-0.1399
3.779	0.89f	63.38s	1,893.1	0.000	-1.939	-0.2959

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 542.4 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1160.00

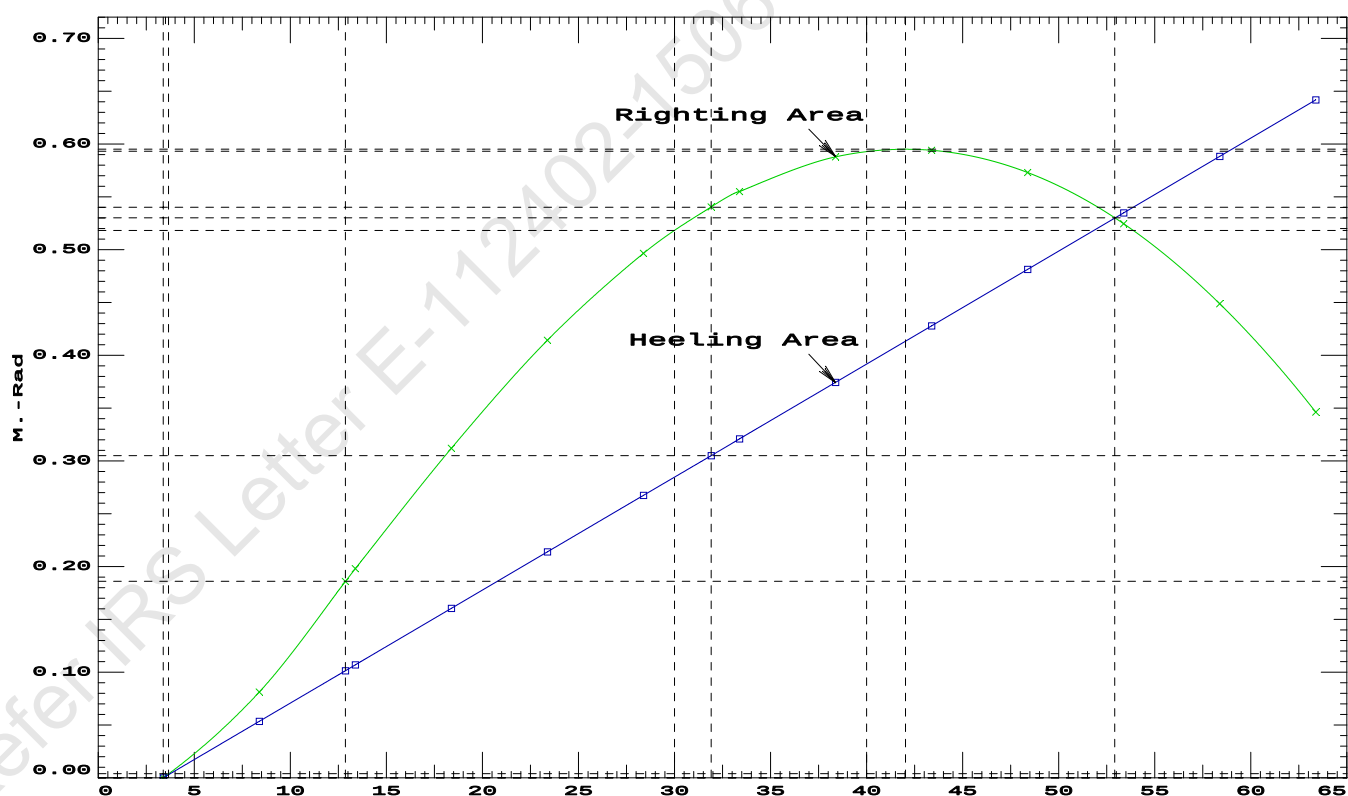
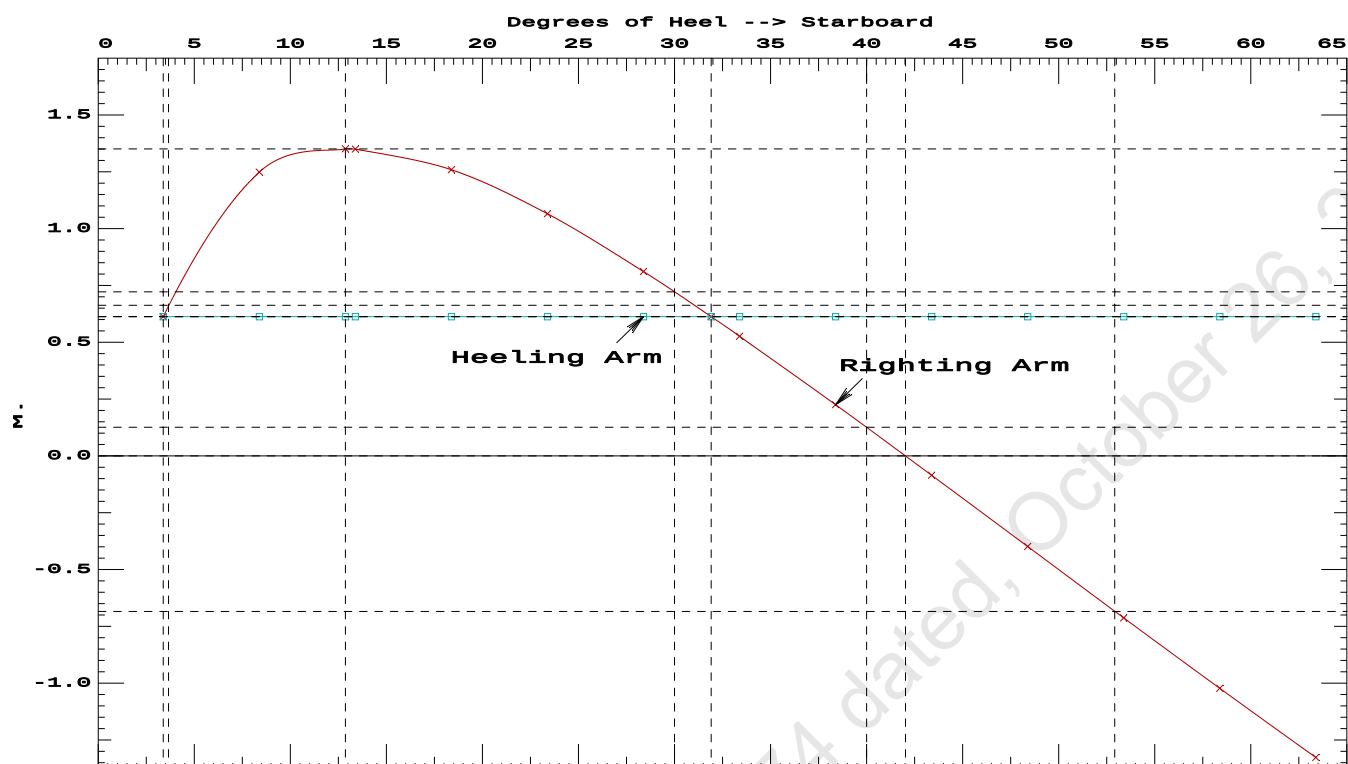
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	3.38 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	120.5 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	1.770 P

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PONTOON

Condition 3 : Lightship + Load at Stbd + Ballast + Deck CargoCrane Lifting 40.0T @ 29m Radius Stbd

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Condition 3 : Lightship + Load at Stbd + Ballast + Deck Cargo**Crane Lifting 40.0T @ 29m Radius Stbd**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.263 @ 50.00f, 2.211 @ 25.00f, 2.159 @ 0.00

Trim: Fwd 0.104/50.000, Heel: Stbd 3.38 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			350.00	13.300f	0.000	6.325	
Crane Load			40.00	13.300f	0.000	21.478	
Deck Cargo			624.59	27.500f	0.000	4.000	
Total Fixed----->			1,420.56	22.887f	0.000	4.779	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p	1.509	162.73*
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			472.52	32.161f	0.000	1.626	542.42
Total Weight----->			1,893.08	25.202f	0.000	3.992	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,893.08	25.208f	0.782s	1.155	

	Righting Arms:			0.000	0.613s		
	External Arms:			0.000	0.613s		
	Residual Righting Arms:			0.000	0.000s		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->		1.025	896.8	25.026f	0.049s	99.93	12.809
			MT/cm				
			9.19				

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 3.38

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After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 2.282 @ 50.00f, 2.169 @ 25.00f, 2.055 @ 0.00
Trim: Fwd 0.227/50.000, Heel: 0.00 deg.

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----			
LIGHT SHIP	405.97	25.000f	0.000	3.000			
Crane	350.00	13.300f	0.000	6.325			
Deck Cargo	624.59	27.500f	0.000	4.000			
Total Fixed----->	1,380.56	23.165f	0.000	4.295			
Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----	
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p	1.509	162.73*
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			472.52	32.161f	0.000	1.626	542.42
Total Weight----->			1,853.08	25.459f	0.000	3.615	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL	1.025		1,853.08	25.470f	0.001	1.110	

Righting Arms:			0.000	0.001s			
External Arms:			0.000	0.001			
Residual Righting Arms:			0.000	0.000			
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----	
Total Waterplane---->	1.025	900.4	25.006f	0.000	103.61	13.165	
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----		
		9.23	37.47	101.11	10.661		
Distances in METERS.-----Moments in m.-MT.							

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: 3389.28; t3: 30.0; t: 15.0
limmarg: 1427.65; t3: 15.0; t: 7.5
limmarg: 379.40; t3: 7.5; t: 3.8
limmarg: 19.99; t3: 3.7; t: 1.9
limmarg: -71.39; t3: 1.8; t: 0.9
limmarg: -35.91; t3: 2.7; t: 0.5
limmarg: -10.13; t3: 3.2; t: 0.2
limmarg: 1.40; t3: 3.4; t: 0.1
limmarg: -4.45; t3: 3.3; t: 0.1
limmarg: 1.40; t3: 3.4; t: 0.0
limmarg: 1.40; t3: 3.4; t: 0.0

Theta3 is 3.40 degrees, as demonstrated below.

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 23.165f TCG = 0.000 VCG = 4.295

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
2.036	0.26f	3.38s	1,853.1	0.000	-0.649	0.0000
2.043	0.26f	0.00	1,853.1	0.000	0.000	-0.0191
2.036	0.26f	3.40p	1,853.1	0.000	0.654	0.0003
2.029	0.27f	5.00p	1,853.1	0.000	0.957	0.0227
2.057	0.38f	10.00p	1,853.1	0.000	1.495	0.1320
2.216	0.50f	15.00p	1,853.1	0.000	1.600	0.2702
2.407	0.64f	20.00p	1,853.1	0.000	1.534	0.4082
2.574	0.80f	25.00p	1,853.1	0.000	1.367	0.5355
2.718	0.97f	30.00p	1,853.1	0.000	1.148	0.6456

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1380.56 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Res. Area Ratio from abs -3.4 deg to abs 3.4 > 1.000 1.014 P

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 23.165f TCG = 0.000 VCG = 4.295

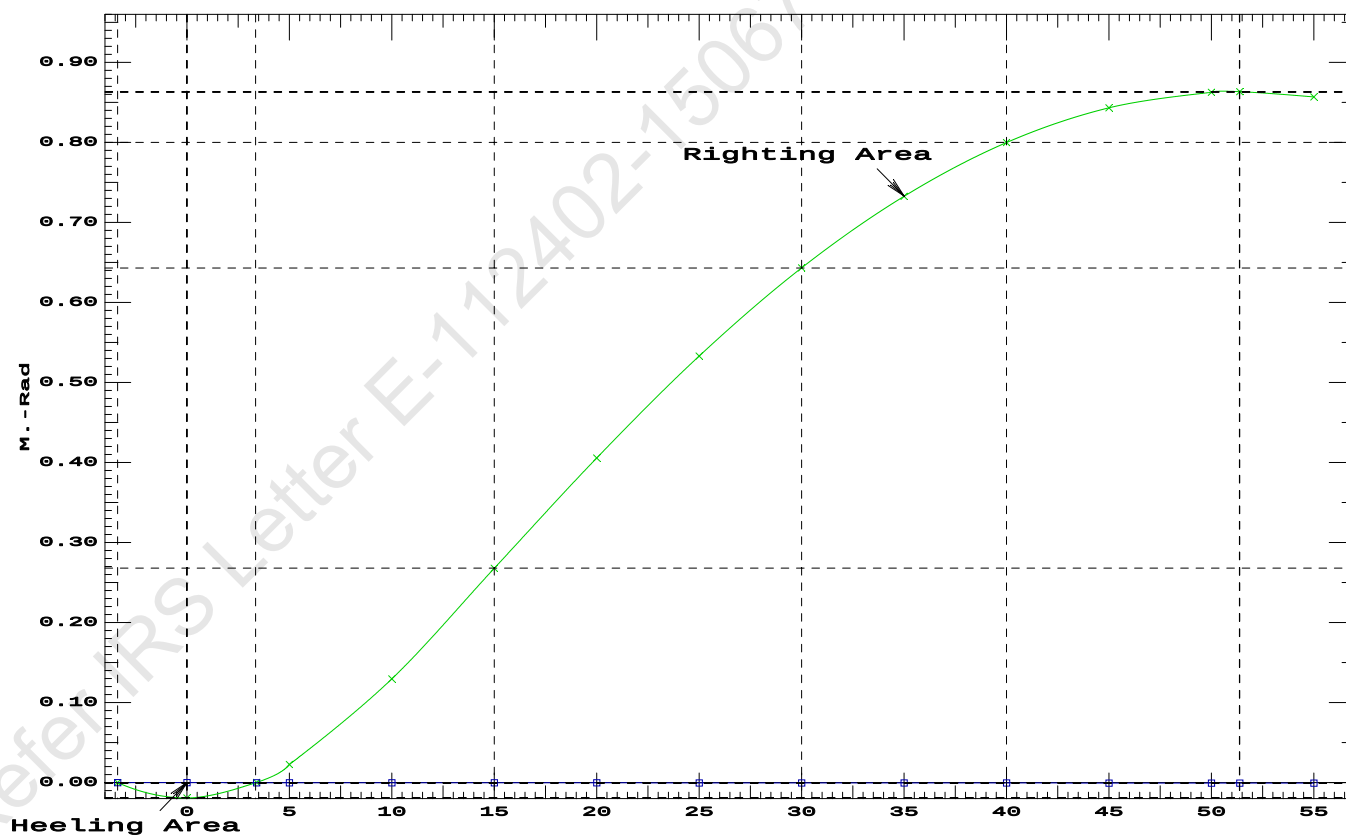
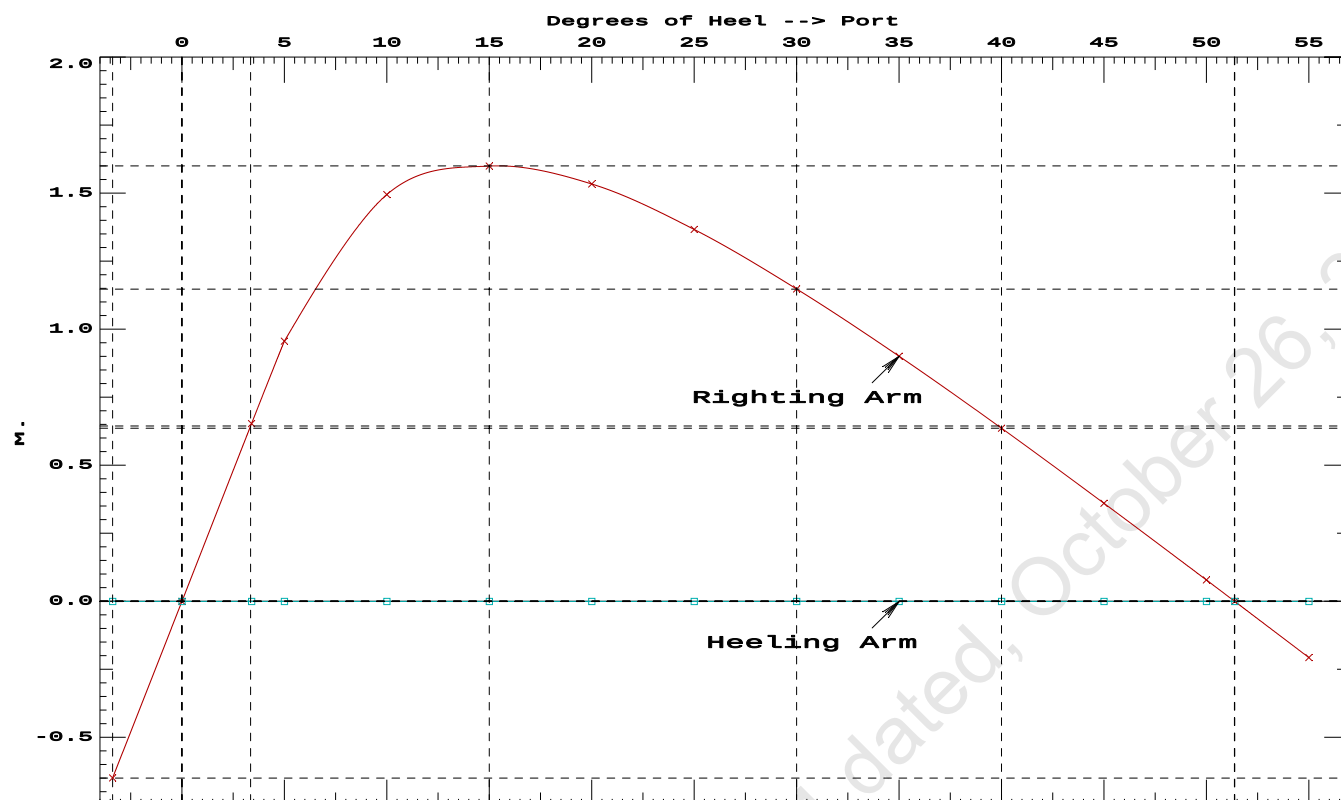
Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
2.036	0.26f	3.38s	1,853.1	0.000	-0.649	0.0000
2.043	0.26f	0.00	1,853.1	0.000	0.000	-0.0191
2.036	0.26f	3.40p	1,853.1	0.000	0.654	0.0003
2.029	0.27f	5.00p	1,853.1	0.000	0.957	0.0227
2.057	0.38f	10.00p	1,853.1	0.000	1.495	0.1320
2.216	0.50f	15.00p	1,853.1	0.000	1.600	0.2702
2.407	0.64f	20.00p	1,853.1	0.000	1.534	0.4082
2.574	0.80f	25.00p	1,853.1	0.000	1.367	0.5355
2.718	0.97f	30.00p	1,853.1	0.000	1.148	0.6456
2.839	1.14f	35.00p	1,853.1	0.000	0.901	0.7352
2.935	1.30f	40.00p	1,853.1	0.000	0.636	0.8025
3.008	1.46f	45.00p	1,853.1	0.000	0.361	0.8460
3.057	1.61f	50.00p	1,853.1	0.000	0.079	0.8653
3.066	1.65f	51.38p	1,853.1	0.000	0.000	0.8662
3.082	1.75f	55.00p	1,853.1	0.000	-0.206	0.8597

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1380.56 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Res. Area Ratio from abs -3.4 deg to RAzero > 1.000 46.259 P
(2) Angle from abs 3.4 deg to RAzero > 20.00 deg 47.98 P

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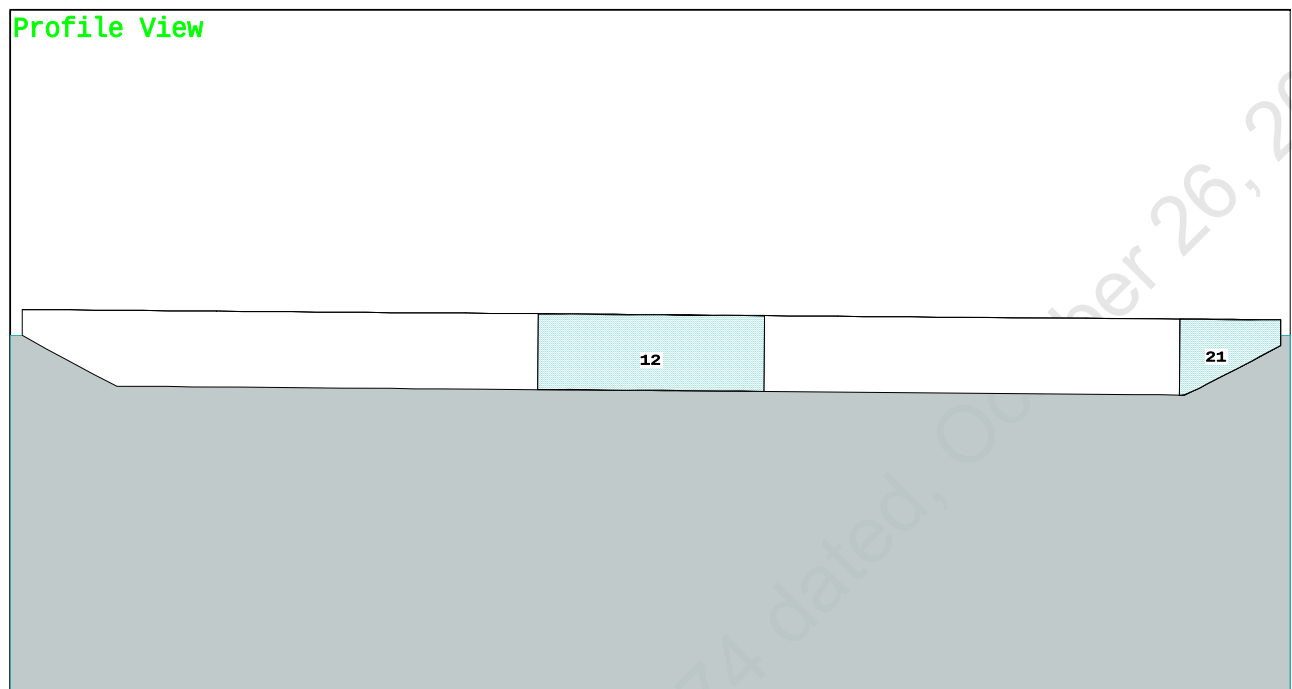
GHS 17.22

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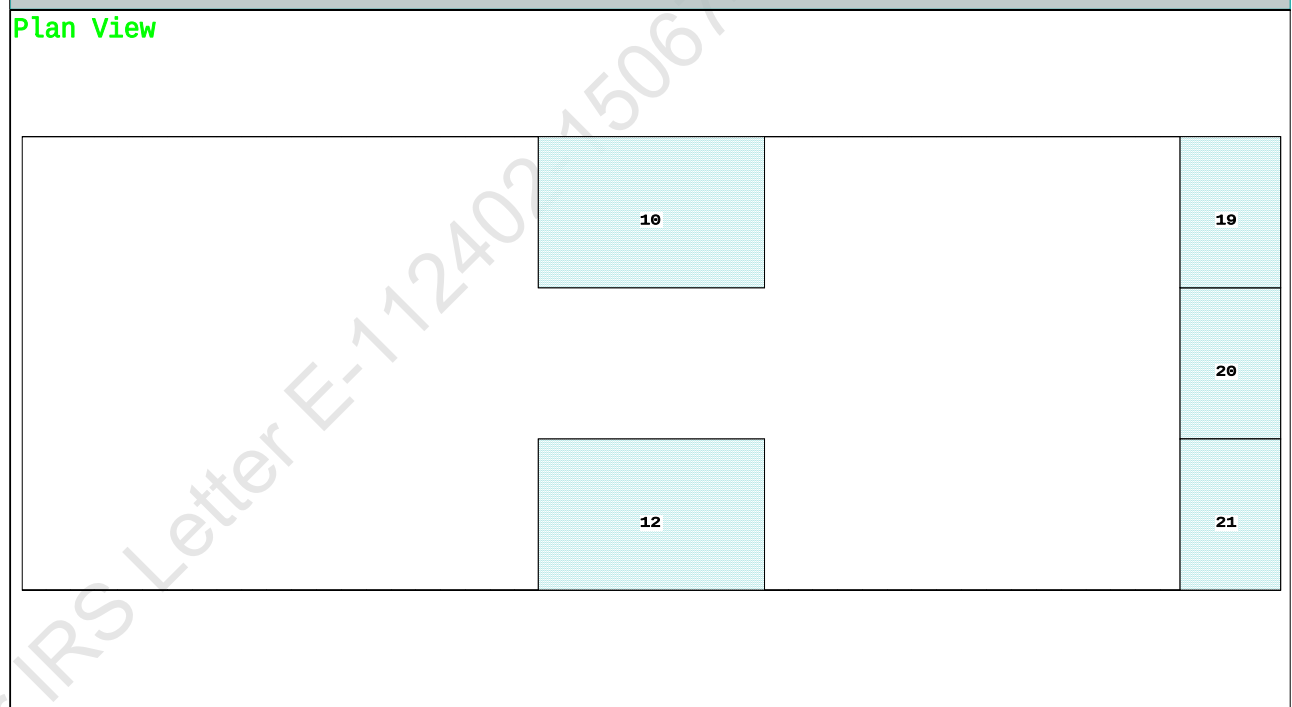
Condition 4 : Lightship + Load at Fwd + Ballast + Deck CargoCrane Lifting 40.0T @ 29m Radius Fwd

Condition Graphic - Draft: 2.419 @ 50.000f, 2.006 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

10 04_WBTK.P 12 04_WBTK.S 19 07_WBTK.P 20 07_WBTK.C 21 07_WBTK.S

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PONTOON

Condition 4 : Lightship + Load at Fwd + Ballast + Deck CargoCrane Lifting 40.0T @ 29m Radius Fwd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.419 @ 50.00f, 2.212 @ 25.00f, 2.006 @ 0.00

Trim: Fwd 0.413/50.000, Heel: zero

Part-----			Weight(MT)-----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			350.00	13.300f	0.000	6.325	
Crane Load			40.00	42.300f	0.000	21.478	
Deck Cargo			624.59	27.500f	0.000	4.000	
Total Fixed----->			1,420.56	23.704f	0.000	4.779	
	Load-----	SpGr-----	Weight(MT)-----	LCG-----	TCG-----	VCG-----	FSM-----
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p	1.509	162.73*
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			472.52	32.161f	0.000	1.626	542.42
Total Weight----->			1,893.08	25.815f	0.000	3.992	
			Displ(MT)-----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,893.21	25.838f	0.000	1.135	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->		1.025	900.5	25.011f	0.000	101.45	12.890
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			9.23	37.33	98.60	10.034	
Distances in METERS.-----				Moments in m.-MT			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

461.0 Cu.M. (18%) SALT WATER

350.00 MT Crane

40.00 MT Crane Load

624.59 MT Deck Cargo

DRAFT AT FP (M) = 2.419
 DRAFT AT MS (M) = 2.212
 DRAFT AT AP (M) = 2.006
 DRAFT AT FWD MARK (M) = 2.386
 DRAFT AT AFT MARK (M) = 2.039

HYDROSTATIC PROPERTIES

Trim: Fwd 0.413/50.000, No Heel, VCG = 3.992

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/				
Draft----	Weight(MT)-----	LCB-----	VCB-----	cm-----	LCF-----	cm trim-----	GML-----	GMT
2.212	1,893.21	25.838f	1.135	9.23	25.011f	37.33	98.60	10.034
Distances in METERS.-----				Specific Gravity = 1.025.-----				
				Moment in m.-MT.				
				Trim is per 50.00m.				

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 542.4 m.-MT

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PONTOON

Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo
Crane Lifting 40.0T @ 29m Radius Fwd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 2.419 @ 50.00f, 2.212 @ 25.00f, 2.006 @ 0.00

Trim: Fwd 0.413/50.000, Heel: zero

Part-----	LPA*SF-----	HCP-----	Arm----	Pressure-----	Moment
HULL	40.5	0.414	1.490	0.05500	3.32
CRANE	57.0	3.423	4.499	0.05500	14.11
DECKCARGO.C	29.4	1.779	2.856	0.05500	4.62
Total wind heeling moment to starboard----->					22.04
Distances in METERS.-----Pressure in MT/Sqm.-----					Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.815f TCG = 0.000 VCG = 3.992

Free Surface Adjustment: 0.287

Adjusted CG: LCG = 25.812f TCG = 0.000 VCG = 4.279

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---	Trim----	Heel----	Weight(MT)---	Area----
			in Trim--in Heel-->	Angle
1.994	0.47f	0.00	1,893.1	0.0000
1.993	0.47f	0.06s	1,893.1	-0.0000
1.989	0.48f	1.91s	1,893.1	0.0052
1.974	0.50f	5.06s	1,893.1	0.0381
1.976	0.72f	10.06s	1,893.1	0.1361
2.046	0.85f	13.02s	1,893.1	0.2033
2.108	0.94f	15.06s	1,893.2	0.2495
2.266	1.18f	20.06s	1,893.1	0.3578
2.393	1.49f	25.06s	1,893.1	0.4496
2.494	1.80f	30.06s	1,893.1	0.5199
2.570	2.11f	35.06s	1,893.1	0.5657
2.623	2.42f	40.06s	1,893.1	0.5855
2.632	2.49f	41.27s	1,893.1	0.5863
2.653	2.71f	45.06s	1,893.1	-0.235
2.663	2.99f	50.06s	1,893.1	-0.547
2.654	3.23f	55.06s	1,893.1	-0.856
2.630	3.44f	60.06s	1,893.1	-1.160
Distances in METERS.-----Specific Gravity = 1.025.-----				Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 542.4 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 22.04 (constant)

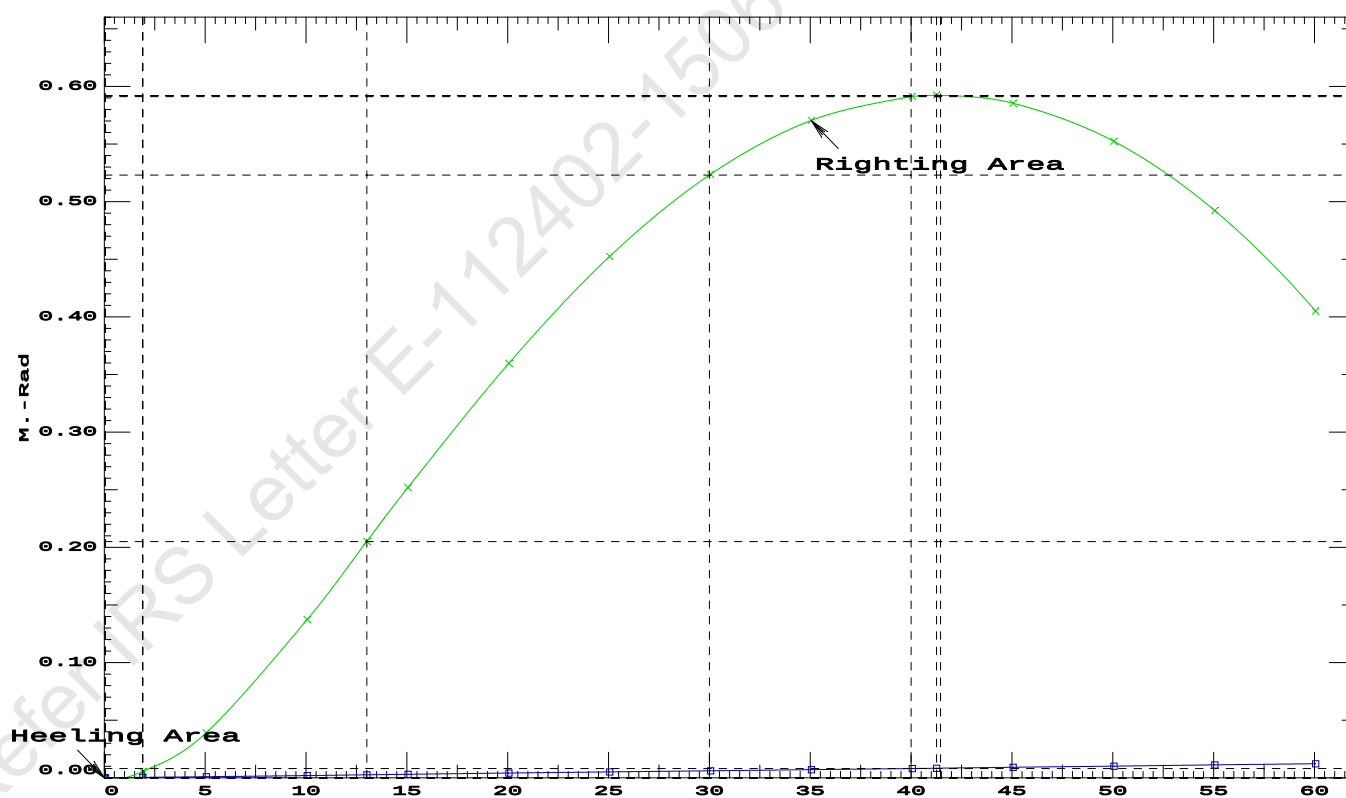
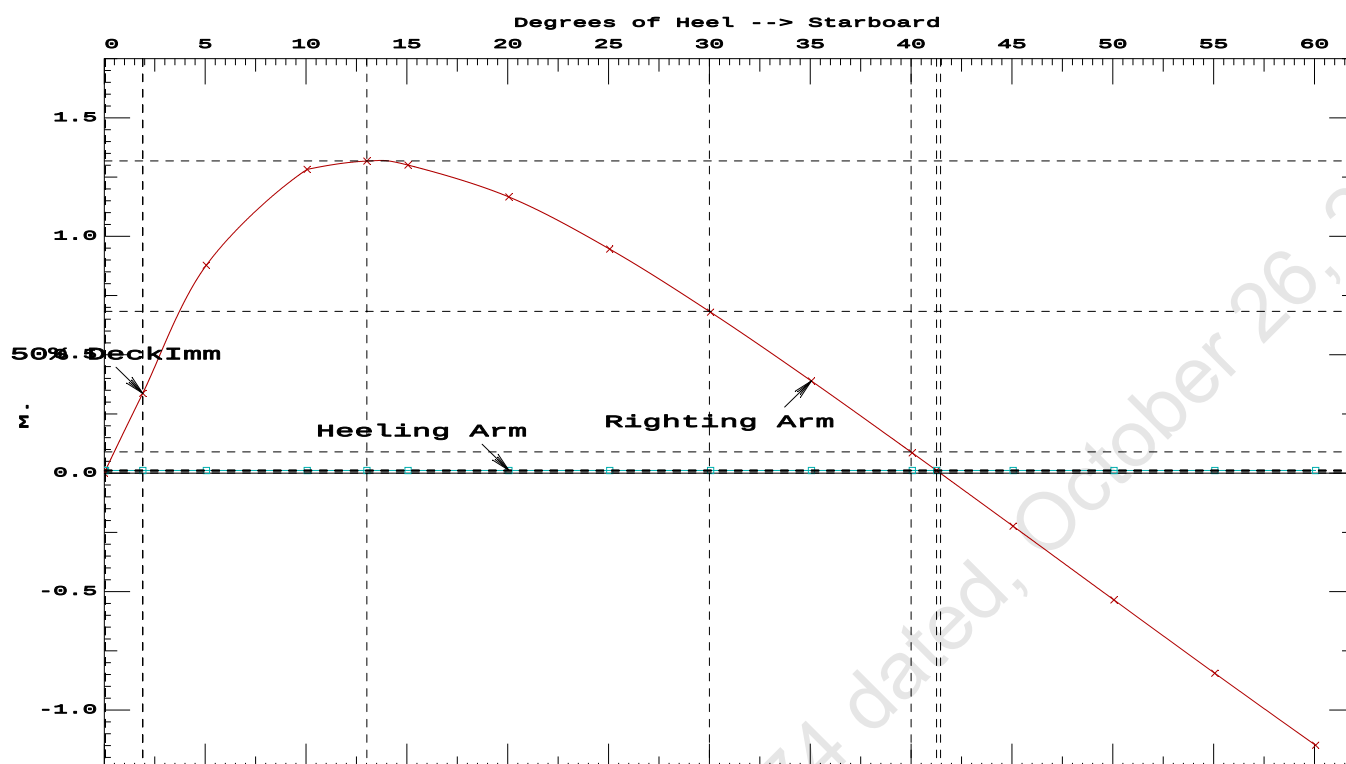
LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2-----Min/Max-----Attained

(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.2033 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	41.21 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	1.85 P

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PONTOON

Condition 4 : Lightship + Load at Fwd + Ballast + Deck CargoCrane Lifting 40.0T @ 29m Radius Fwd

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PONTOON**Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo**
Crane Lifting 40.0T @ 29m Radius Fwd**RESIDUAL RIGHTING ARMS vs HEEL ANGLE**

Total CG: LCG = 25.815f TCG = 0.000 VCG = 3.992

Free Surface Adjustment: 0.287

Adjusted CG: LCG = 25.812f TCG = 0.000 VCG = 4.279

Origin Depth---	Degrees of Trim----	Heel----	Displacement Weight(MT)---	Residual Arms in Trim--in Heel---	Res. Area
1.993	0.47f	0.00	1,893.1	0.000 0.000	0.0000
1.975	0.49f	5.00s	1,893.1	0.000 0.869	0.0379
1.976	0.72f	10.00s	1,893.1	0.000 1.280	0.1365
2.046	0.85f	13.02s	1,893.1	0.000 1.317	0.2055
2.106	0.93f	15.00s	1,893.2	0.000 1.300	0.2509
2.264	1.18f	20.00s	1,893.1	0.000 1.168	0.3603
2.392	1.48f	25.00s	1,893.1	0.000 0.948	0.4532
2.493	1.80f	30.00s	1,893.1	0.000 0.682	0.5247
2.570	2.11f	35.00s	1,893.1	0.000 0.392	0.5717
2.623	2.42f	40.00s	1,893.1	0.000 0.089	0.5928
2.634	2.50f	41.44s	1,893.1	0.000 0.000	0.5939
2.653	2.71f	45.00s	1,893.1	0.000 -0.221	0.5870
2.663	2.98f	50.00s	1,893.1	0.000 -0.532	0.5542
2.655	3.23f	55.00s	1,893.1	0.000 -0.842	0.4942
2.630	3.44f	60.00s	1,893.1	0.000 -1.146	0.4075

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 542.4 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

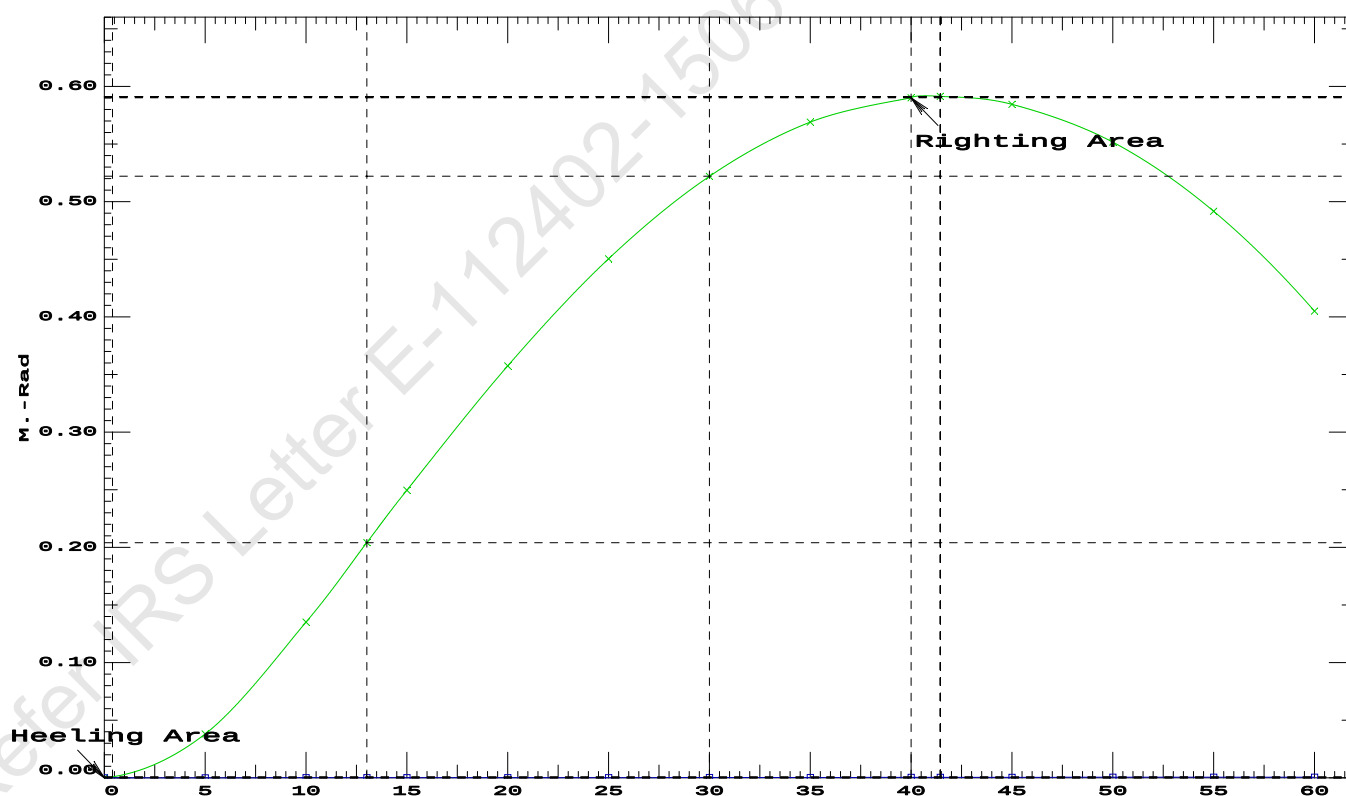
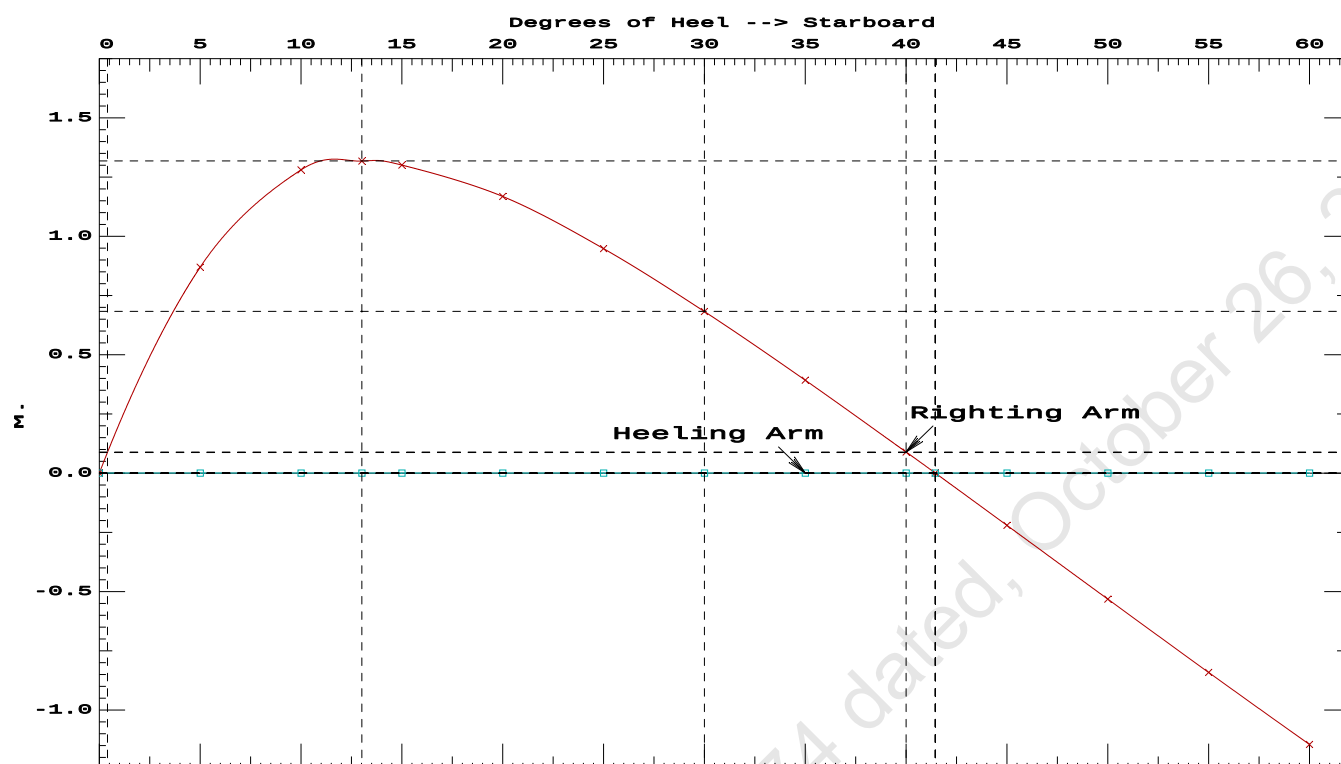
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	0.00 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	249321 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	1555.46 P

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PONTOON

Condition 4 : Lightship + Load at Fwd + Ballast + Deck CargoCrane Lifting 40.0T @ 29m Radius Fwd

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Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo**Crane Lifting 40.0T @ 29m Radius Fwd**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.419 @ 50.00f, 2.212 @ 25.00f, 2.005 @ 0.00

Trim: Fwd 0.413/50.000, Heel: 0.00 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			350.00	13.300f	0.000	6.325	
Crane Load			40.00	42.300f	0.000	21.478	
Deck Cargo			624.59	27.500f	0.000	4.000	
Total Fixed----->			1,420.56	23.704f	0.000	4.779	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p	1.509	162.73*
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			472.52	32.161f	0.000	1.626	542.42
Total Weight----->			1,893.08	25.815f	0.000	3.992	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,893.08	25.838f	0.001	1.135	

	Righting Arms:			0.000	0.001s		
	External Arms:			0.000	0.001		
	Residual Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->		1.025	900.5	25.011f	0.000	101.46	12.891
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			9.23		37.33	98.60	10.034
Distances in METERS.-----							
-----Moments in m.-MT							

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

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After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 2.282 @ 50.00f, 2.169 @ 25.00f, 2.055 @ 0.00
Trim: Fwd 0.227/50.000, Heel: 0.00 deg.

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----
LIGHT SHIP	405.97	25.000f	0.000	3.000
Crane	350.00	13.300f	0.000	6.325
Deck Cargo	624.59	27.500f	0.000	4.000
Total Fixed----->	1,380.56	23.165f	0.000	4.295

Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p	1.509 162.73*
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s	1.509 162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880 72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872 72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880 72.32*
Total Tanks----->			472.52	32.161f	0.000	1.626 542.42
Total Weight----->			1,853.08	25.459f	0.000	3.615

Displ(MT)----	LCB-----	TCB-----	VCB-----
HULL	1.025	1,853.07	25.470f 0.001 1.110

Righting Arms:	0.000	0.001s
External Arms:	0.000	0.001
Residual Righting Arms:	0.000	0.000

Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025	900.4	25.006f	0.000	103.61	13.165
		MT/cm-----	m.-MT/cm-----		GML-----	GMT-----
		9.23	37.47		101.11	10.661

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: INFINITY; t3: 30; t: 15.00
limmarg: INFINITY; t3: 15; t: 7.50
limmarg: INFINITY; t3: 8; t: 3.75
limmarg: INFINITY; t3: 4; t: 1.88
limmarg: INFINITY; t3: 2; t: 0.94
limmarg: INFINITY; t3: 1; t: 0.47
limmarg: INFINITY; t3: 1; t: 0.23
limmarg: INFINITY; t3: 1; t: 0.12
limmarg: INFINITY; t3: 1; t: 0.06
limmarg: INFINITY; t3: 1; t: 0.03
limmarg: INFINITY; t3: 1; t: 0.01

Theta3 is 0.97 degrees, as demonstrated below.

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 23.165f TCG = 0.000 VCG = 4.295

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
2.043	0.26f	0.00	1,853.1	0.000	0.000	0.0000
2.042	0.26f	0.97s	1,853.1	0.000	0.188	0.0016
2.029	0.27f	5.00s	1,853.1	0.000	0.957	0.0419
2.057	0.38f	10.00s	1,853.1	0.000	1.495	0.1542
2.216	0.50f	15.00s	1,853.1	0.000	1.599	0.2923
2.407	0.64f	20.00s	1,853.1	0.000	1.533	0.4302
2.574	0.81f	25.00s	1,853.1	0.000	1.366	0.5574
2.718	0.97f	30.00s	1,853.1	0.000	1.147	0.6674

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1380.56 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 1 > 1.000 LARGE

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 23.165f TCG = 0.000 VCG = 4.295

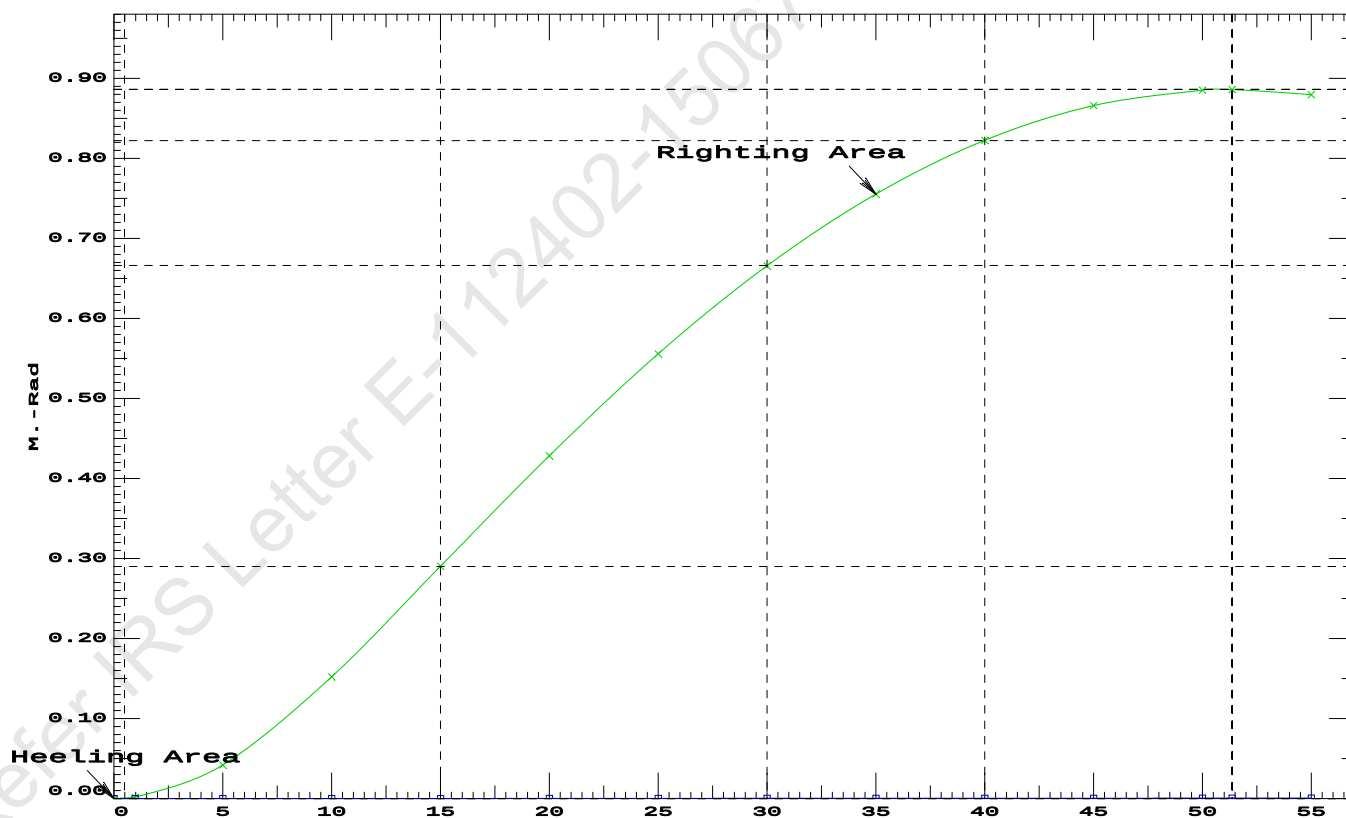
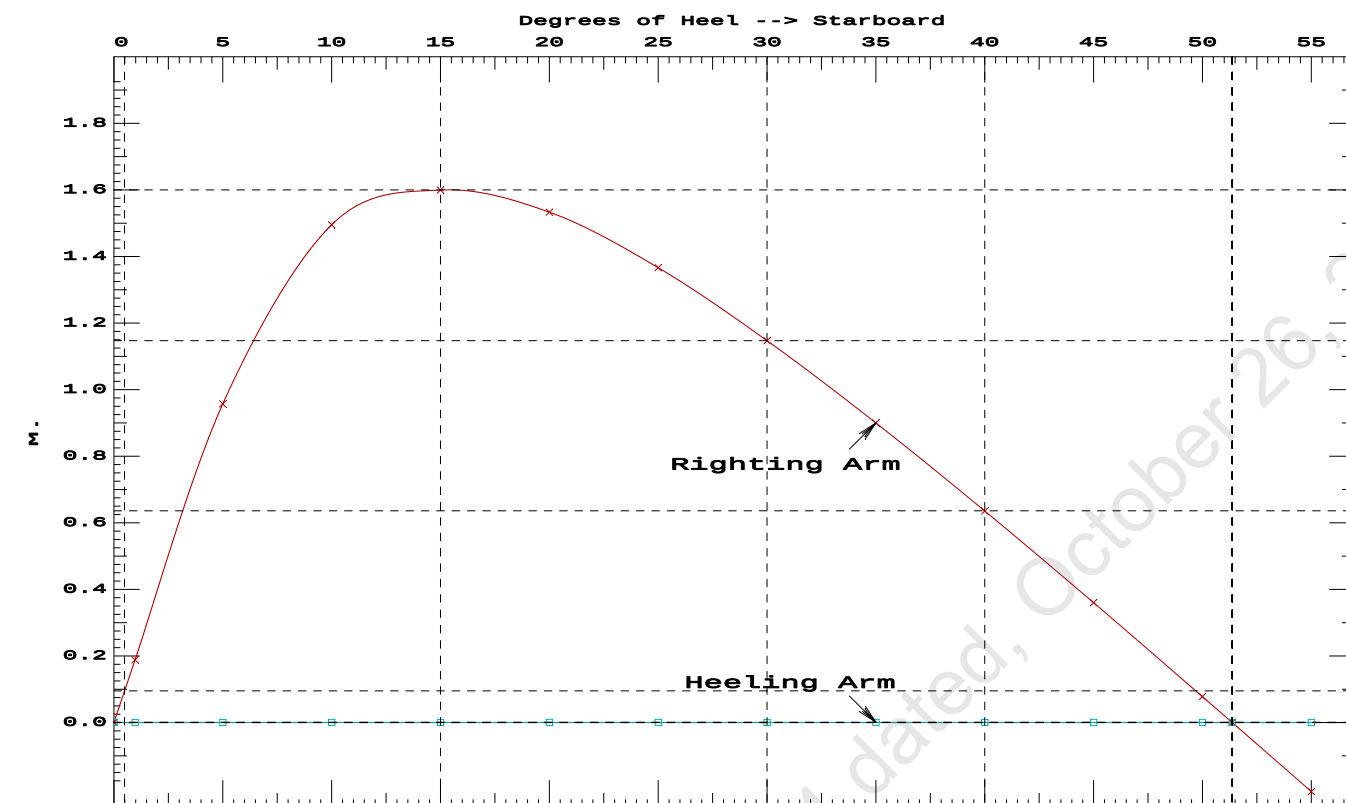
Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
2.043	0.26f	0.00	1,853.1	0.000	0.000	0.0000
2.042	0.26f	0.97s	1,853.1	0.000	0.188	0.0016
2.029	0.27f	5.00s	1,853.1	0.000	0.957	0.0419
2.057	0.38f	10.00s	1,853.1	0.000	1.495	0.1542
2.216	0.50f	15.00s	1,853.1	0.000	1.599	0.2923
2.407	0.64f	20.00s	1,853.1	0.000	1.533	0.4302
2.574	0.81f	25.00s	1,853.1	0.000	1.366	0.5574
2.718	0.97f	30.00s	1,853.1	0.000	1.147	0.6674
2.839	1.14f	35.00s	1,853.1	0.000	0.900	0.7569
2.935	1.30f	40.00s	1,853.1	0.000	0.635	0.8240
3.008	1.46f	45.00s	1,853.1	0.000	0.359	0.8675
3.057	1.61f	50.00s	1,853.1	0.000	0.077	0.8866
3.066	1.65f	51.36s	1,853.1	0.000	0.000	0.8875
3.082	1.75f	55.00s	1,853.1	0.000	-0.208	0.8809

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1380.56 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero > 1.000 LARGE
(2) Angle from abs 1 deg to RAZero > 20.00 deg 50.39 P

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PONTOON

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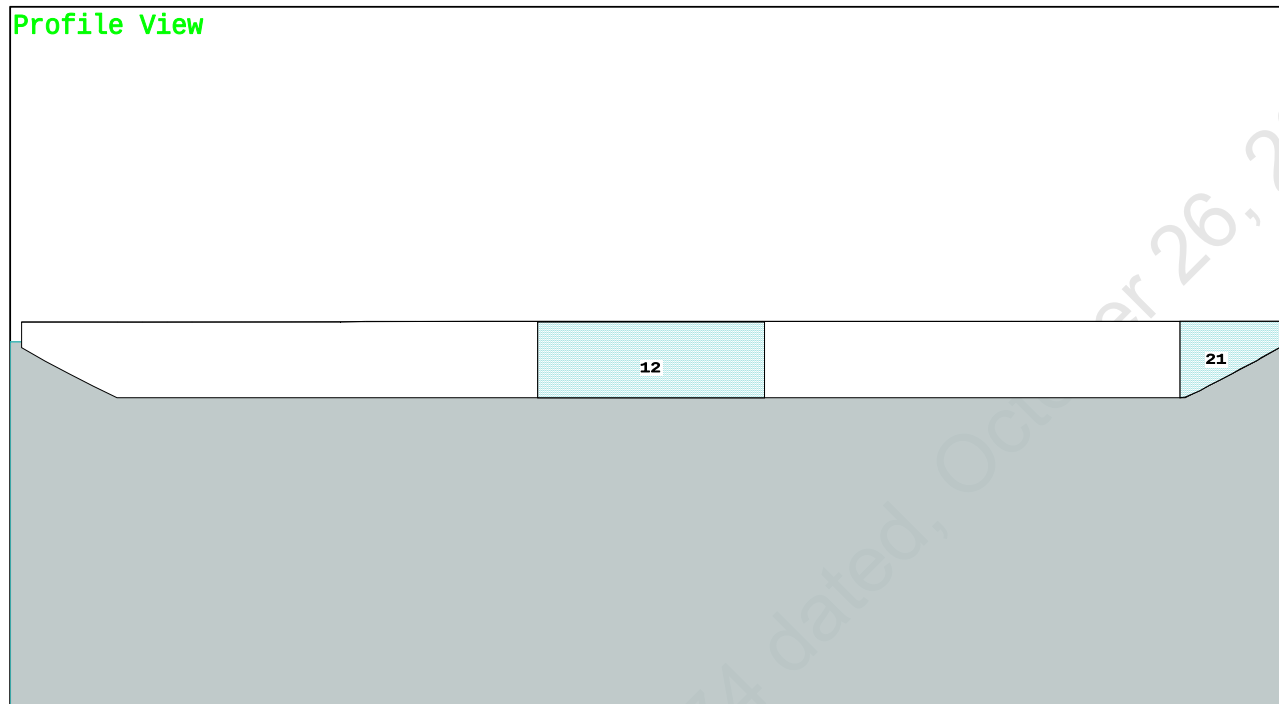
GHS 17.22

PONT00N

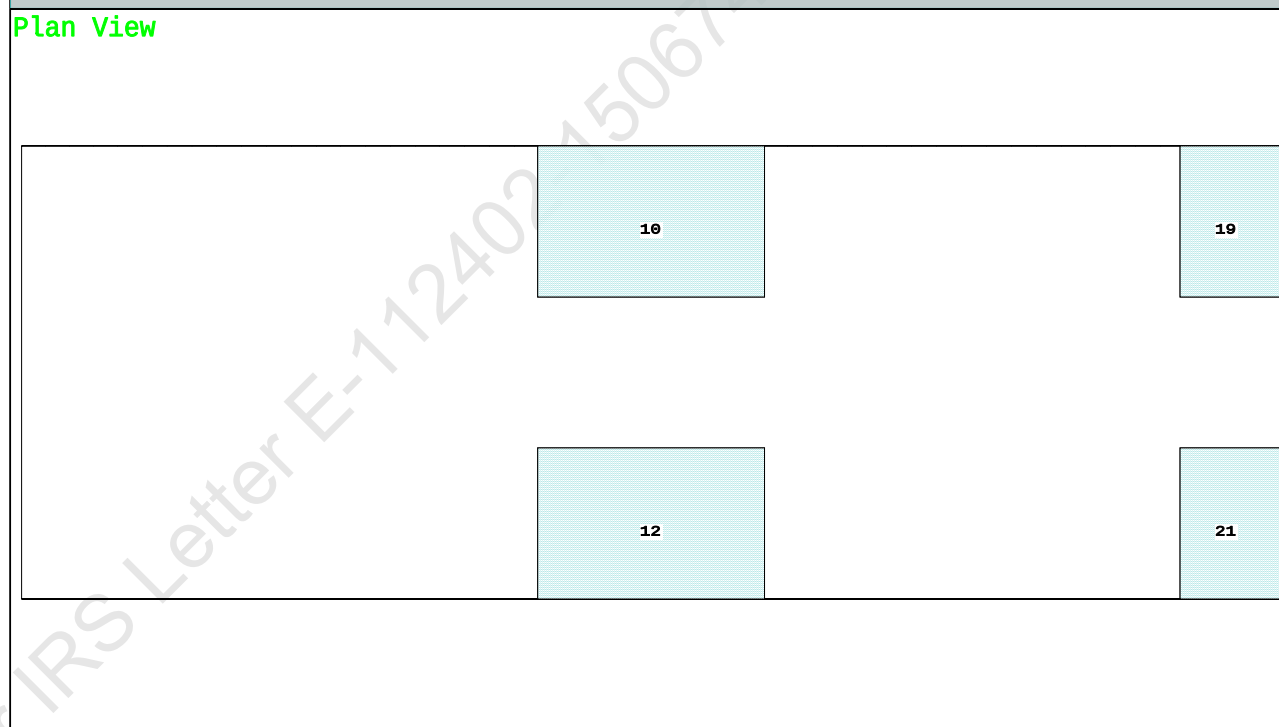
Condition 5 : Lightship + Crane + Ballast + Deck Cargo

Condition Graphic - Draft: 2.204 @ 50.000f, 2.220 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

10 04_WBTK.P 12 04_WBTK.S 19 07_WBTK.P 21 07_WBTK.S

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PONTOON

Condition 5 : Lightship + Crane + Ballast + Deck Cargo

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.204 @ 50.00f, 2.212 @ 25.00f, 2.220 @ 0.00

Trim: Aft 0.016/50.000, Heel: zero

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			350.00	13.300f	0.000	6.325	
Deck Cargo			714.68	27.500f	0.000	4.000	
Total Fixed----->			1,470.65	23.430f	0.000	4.277	
	Load----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p	1.509	162.73*
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	30.322f	0.000	1.596	470.09
Total Weight----->			1,893.05	24.968f	0.000	3.679	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,893.05	24.967f	0.000	1.132	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->		1.025	900.4	25.000f	0.000	101.45	12.929
			MT/cm-----	m.-	MT/cm-----	GML-----	GMT-----
			9.23		37.45	98.90	10.381
Distances in METERS.-----				Moments in m.-MT			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

412.1 Cu.M. (16%) SALT WATER

350.00 MT Crane

714.68 MT Deck Cargo

DRAFT AT FP (M) = 2.204
 DRAFT AT MS (M) = 2.212
 DRAFT AT AP (M) = 2.220
 DRAFT AT FWD MARK (M) = 2.205
 DRAFT AT AFT MARK (M) = 2.219

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GHS 17.22

PONTOON

Condition 5 : Lightship + Crane + Ballast + Deck Cargo

HYDROSTATIC PROPERTIES

Trim: Aft 0.016/50.000, No Heel, VCG = 3.679

LCF Displacement Buoyancy-Ctr. Weight/ Moment/
Draft----Weight(MT)----LCB----VCB-----cm----LCF---cm trim----GML-----GMT
2.212 1,893.05 24.967f 1.132 9.23 25.000f 37.45 98.90 10.381
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 470.1 m.-MT

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 2.204 @ 50.00f, 2.212 @ 25.00f, 2.220 @ 0.00

Trim: Aft 0.016/50.000, Heel: zero

Part-----	LPA*SF-----	HCP-----	Arm----	Pressure-----	Moment
HULL	40.5	0.405	1.478	0.05500	3.29
CRANE	57.0	3.295	4.368	0.05500	13.69
DECKCARGO.C	29.4	1.801	2.874	0.05500	4.65
Total wind heeling moment to starboard----->					21.63
Distances in METERS.-----Pressure in MT/Sqm.-----					Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.968f TCG = 0.000 VCG = 3.679

Free Surface Adjustment: 0.248

Adjusted CG: LCG = 24.968f TCG = 0.000 VCG = 3.927

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---Heel---	Weight(MT)---	in Trim--in Heel---	Area----	Angle
2.198 0.02a 4.94p	1,893.1	0.000 -0.909	0.0000	
2.208 0.02a 0.00	1,893.1	0.000 -0.011	-0.0396	2.550
2.208 0.02a 0.06s	1,893.1	0.000 0.000	-0.0396	2.489
2.205 0.02a 2.55s	1,893.0	0.000 0.455	-0.0298	0.000
2.197 0.02a 5.06s	1,893.1	0.000 0.908	0.0001	-2.511
2.297 0.03a 10.06s	1,893.1	0.000 1.350	0.1009	-7.511
2.462 0.03a 13.80s	1,893.0	0.000 1.405	0.1920	-11.247
2.527 0.04a 15.06s	1,893.1	0.000 1.399	0.2229	-12.511
2.802 0.05a 20.06s	1,893.1	0.000 1.299	0.3423	-17.511
3.068 0.06a 25.06s	1,893.1	0.000 1.109	0.4481	-22.511
3.312 0.07a 30.06s	1,893.1	0.000 0.870	0.5347	-27.511
3.531 0.08a 35.06s	1,893.1	0.000 0.605	0.5992	-32.511
3.723 0.09a 40.06s	1,893.1	0.000 0.325	0.6399	-37.511
3.886 0.10a 45.06s	1,893.1	0.000 0.036	0.6557	-42.511
3.905 0.11a 45.69s	1,893.1	0.000 0.000	0.6559	-43.137
4.021 0.12a 50.06s	1,893.0	0.000 -0.256	0.6462	-47.511
4.124 0.13a 55.06s	1,893.1	0.000 -0.549	0.6110	-52.511
4.196 0.13a 60.06s	1,893.0	0.000 -0.840	0.5504	-57.511
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

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PONTOON

Condition 5 : Lightship + Crane + Ballast + Deck Cargo

Note: The Residual Righting Arms shown above are in excess of the
wind heeling arms derived from these moments (in m.-MT):

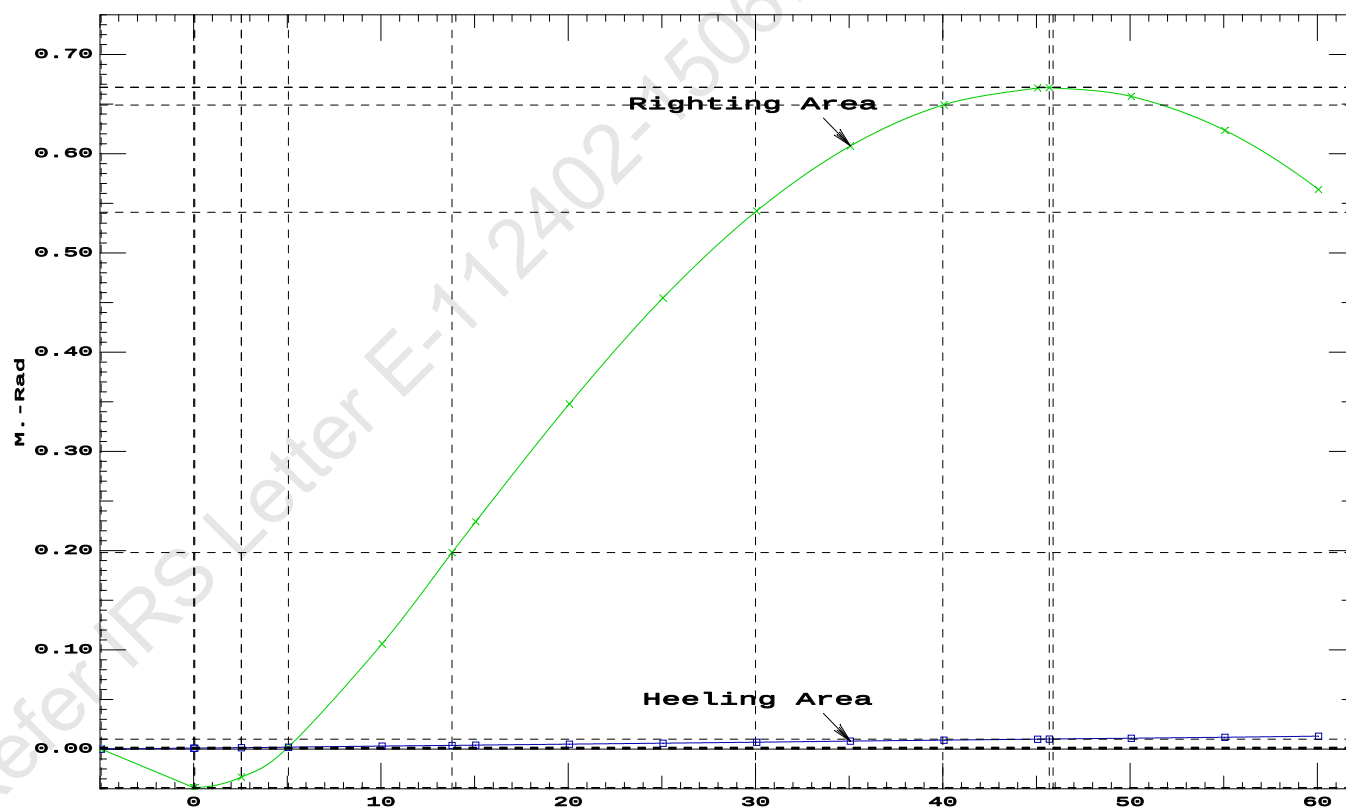
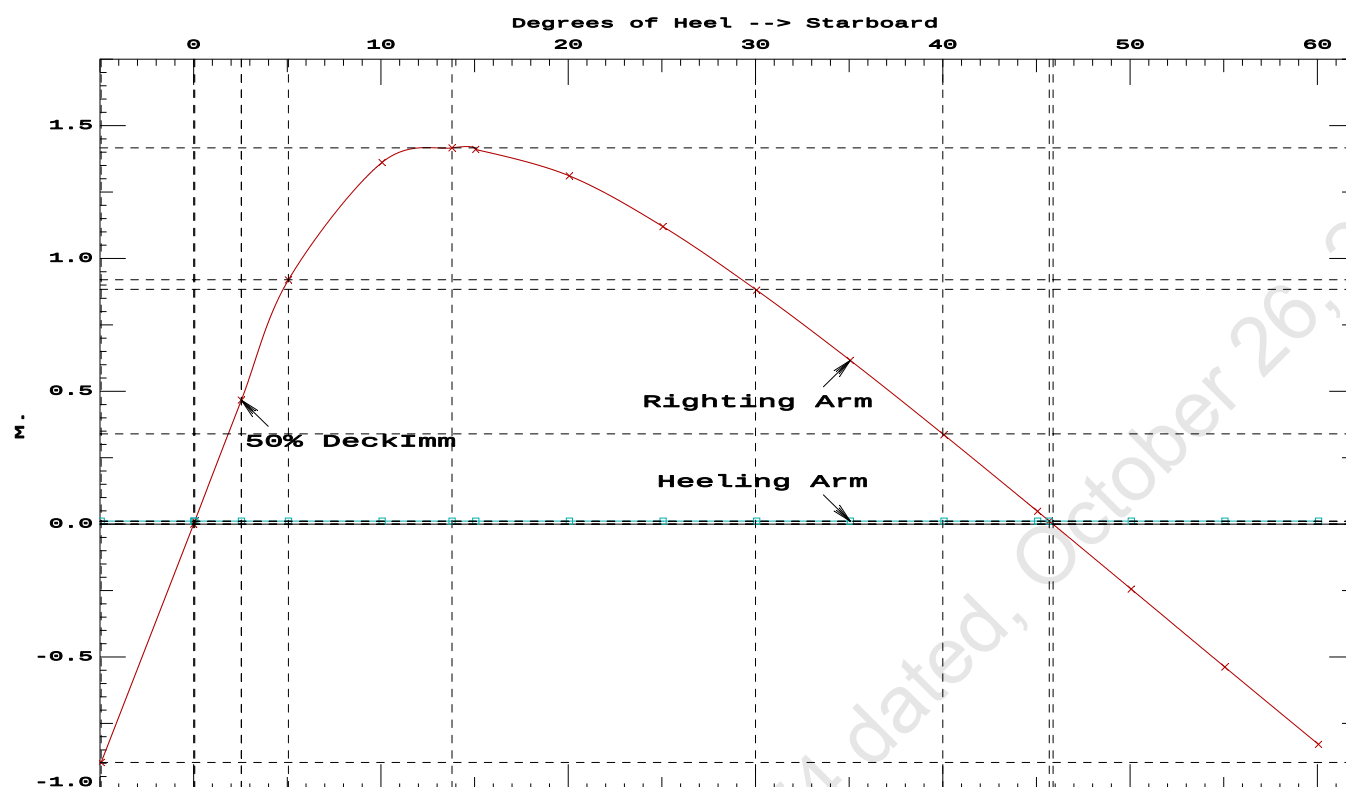
Stbd heeling moment = 21.63 (constant)

LIM-----	INTACT CRITERIA-IS CODE 2008PART B	----Min/Max-----	Attained
(1)	Residual Area from abs 0 deg to MaxRA	> 0.0800 m.-Rad	0.2317 P
(2)	Angle from Equilibrium to RAzero or Flood	> 20.00 deg	45.63 P
(3)	Angle from Equilibrium to 50% Dk Imm Angle	> 0.00 deg	2.49 P

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PONTOON

Condition 5 : Lightship + Crane + Ballast + Deck Cargo

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PONTOON**Condition 5 : Lightship + Crane + Ballast + Deck Cargo****RESIDUAL RIGHTING ARMS vs HEEL ANGLE**

Total CG: LCG = 24.968f TCG = 0.000 VCG = 3.679

Free Surface Adjustment: 0.248

Adjusted CG: LCG = 24.968f TCG = 0.000 VCG = 3.927

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. 50% DeckImm Area	Angle
2.197	0.02a	5.06s	1,893.1	0.000	-0.908	0.0000	
2.208	0.02a	0.06s	1,893.1	0.000	0.000	-0.0396	
2.208	0.02a	0.00	1,893.1	0.000	0.011	-0.0396	2.550
2.205	0.02a	2.55p	1,893.0	0.000	0.478	-0.0287	0.000
2.198	0.02a	4.94p	1,893.1	0.000	0.909	0.0002	-2.390
2.292	0.03a	9.94p	1,893.1	0.000	1.368	0.1017	-7.390
2.480	0.03a	14.14p	1,893.0	0.000	1.427	0.2059	-11.589
2.795	0.05a	19.94p	1,893.1	0.000	1.325	0.3480	-17.390
3.306	0.07a	29.94p	1,893.1	0.000	0.898	0.5449	-27.390
3.718	0.09a	39.94p	1,893.1	0.000	0.354	0.6559	-37.390
3.916	0.11a	46.07p	1,893.1	0.000	0.000	0.6750	-43.522
4.018	0.12a	49.94p	1,893.1	0.000	-0.226	0.6674	-47.390
4.195	0.13a	59.94p	1,893.1	0.000	-0.811	0.5769	-57.390
4.246	0.15a	69.94p	1,893.1	0.000	-1.375	0.3859	-67.390

Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

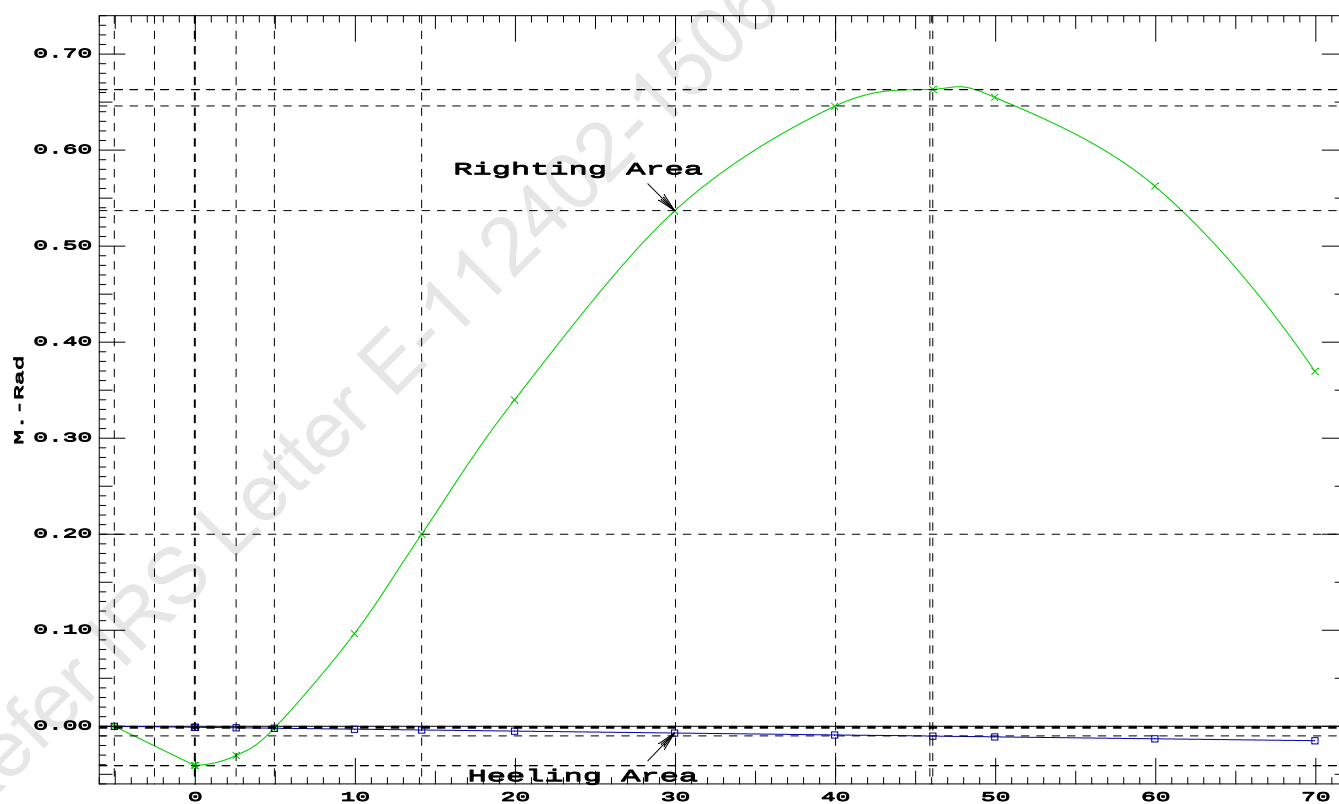
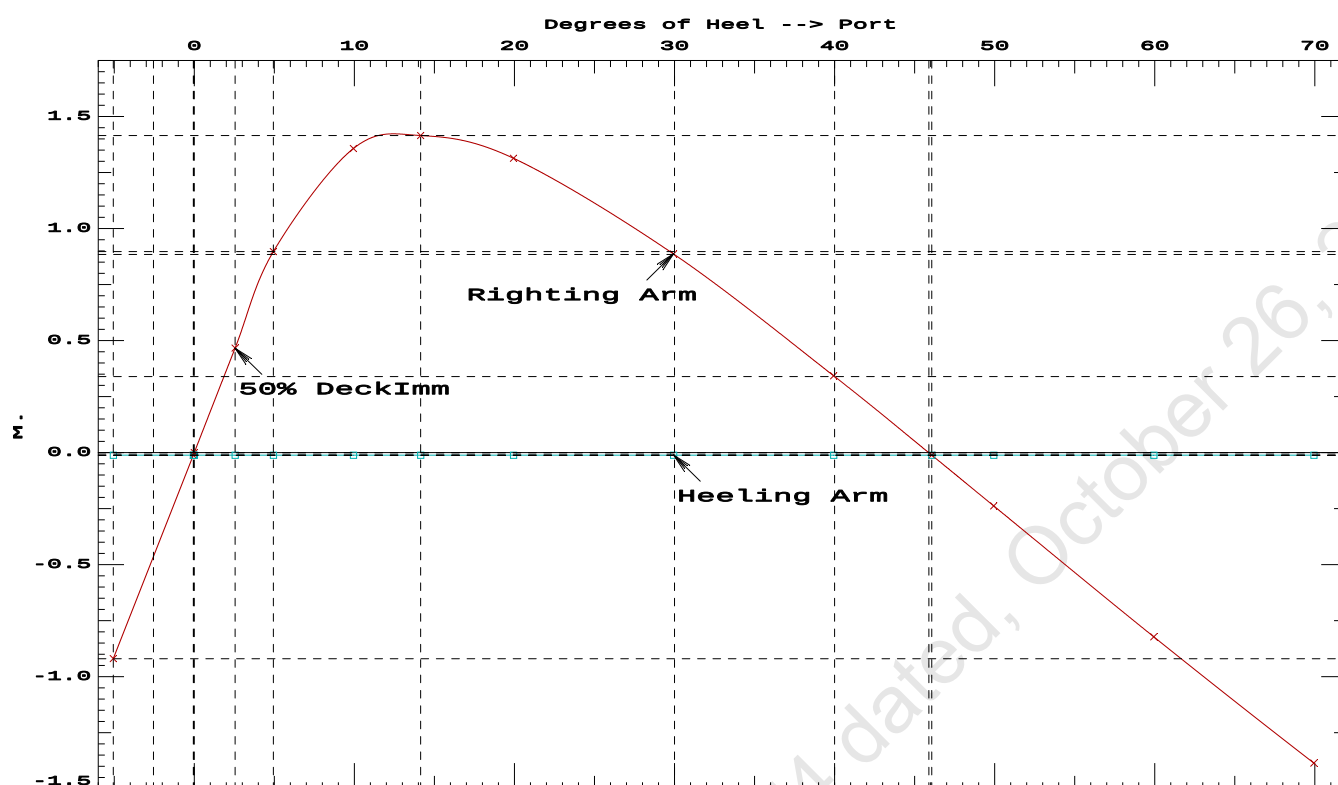
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.63 (constant)

LIM-----	INTACT CRITERIA-IS CODE 2008PART B	Min/Max-----	Attained
(1)	Residual Area from abs 0 deg to MaxRA	> 0.0800 m.-Rad	0.2456 P
(2)	Angle from Equilibrium to RAZero or Flood	> 20.00 deg	46.13 P
(3)	Angle from Equilibrium to 50% Dk Imm Angle	> 0.00 deg	2.61 P

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PONT00N

Condition 5 : Lightship + Crane + Ballast + Deck Cargo

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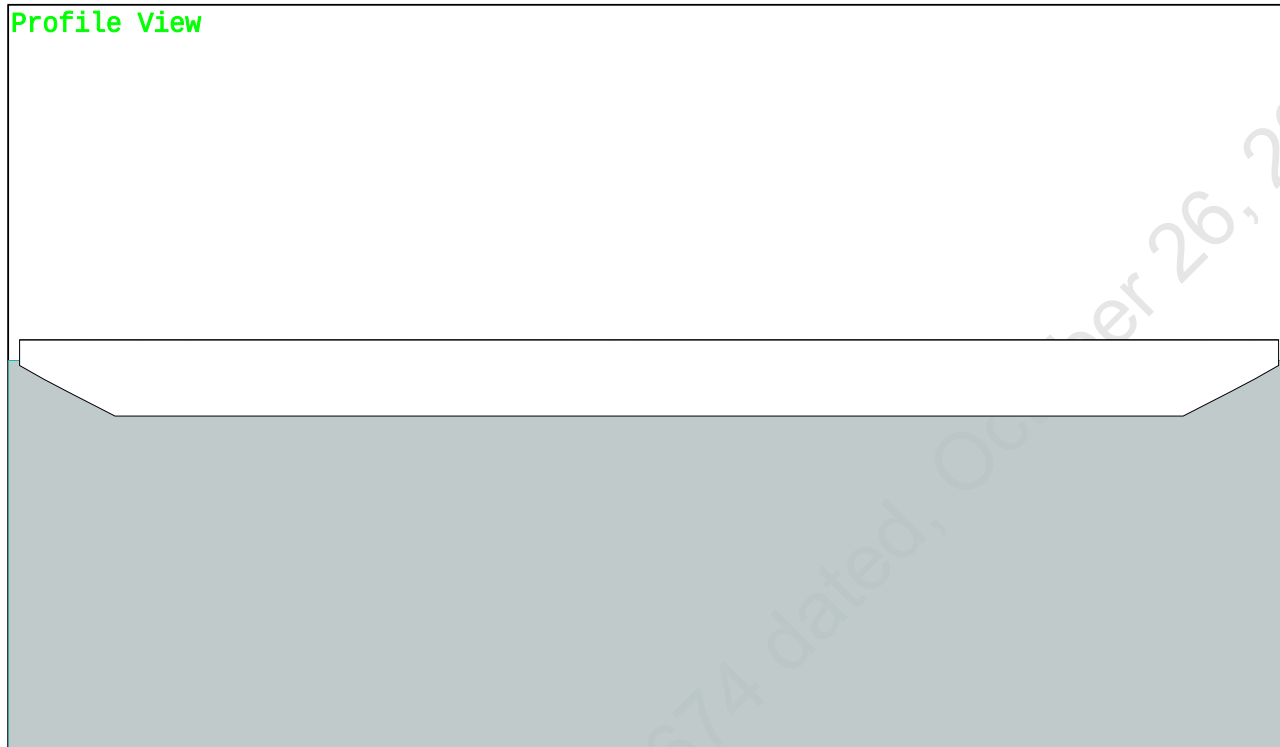
GHS 17.22

PONT00N

Condition 6 : Lightship + Max. Deck Cargo

Condition Graphic - Draft: 2.212 Trim: zero Heel: zero

Profile View



Plan View



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GHS 17.22

PONTOON

Condition 6 : Lightship + Max. Deck Cargo

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.212 @ 50.00f, 2.212 @ 25.00f, 2.212 @ 0.00

Trim: zero, Heel: zero

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
LIGHT SHIP	405.97	25.000f	0.000	3.000	
Deck Cargo	1,487.09	25.000f	0.000	5.000	
Total Weight----->	1,893.06	25.000f	0.000	4.571	
Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----
Total Tanks----->		0.00			0.00
		Displ(MT)----	LCB-----	TCB-----	VCB-----
HULL	1.025	1,893.06	25.000f	0.000	1.132

Righting Arms:		0.000	0.000		
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----BMT
Total Waterplane---->	1.025	900.4	25.000f	0.000	101.45 13.177
		MT/cm-----	m.-MT/cm-----	GML-----	GMT
		9.23	37.11	98.01	9.738
Distances in METERS.-----			Moments in m.-MT.		

SUMMARY OF LOADING

0.0 Cu.M. (0%) SALT WATER

1,487.09 MT Deck Cargo

DRAFT AT FP (M) = 2.212
DRAFT AT MS (M) = 2.212
DRAFT AT AP (M) = 2.212
DRAFT AT FWD MARK (M) = 2.212
DRAFT AT AFT MARK (M) = 2.212

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PONTOON

Condition 6 : Lightship + Max. Deck Cargo

HYDROSTATIC PROPERTIES

No Trim, No Heel, VCG = 4.571

LCF Displacement Buoyancy-Ctr. Weight/ Moment/
Draft----Weight(MT)----LCB-----VCB-----cm-----LCF---cm trim----GML-----GMT
2.212 1,893.06 25.000f 1.132 9.23 25.000f 37.11 98.01 9.738
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 2.212 @ 50.00f, 2.212 @ 25.00f, 2.212 @ 0.00

Trim: zero, Heel: zero

Part-----LPA*SF-----HCP-----Arm-----Pressure-----Moment
HULL 40.5 0.405 1.478 0.05500 3.29
DECKCARGO.C 117.6 2.800 3.873 0.05500 25.05
Total wind heeling moment to starboard-----> 28.34
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

LCG = 25.000f TCG = 0.000 VCG = 4.571

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. 50% DeckImm Area	Res. 50% DeckImm Angle
2.190	0.00	4.91p	1,893.1	0.000	-0.853	0.0000	
2.200	0.00	0.00	1,893.1	0.000	-0.015	-0.0372	2.576
2.200	0.00	0.09s	1,893.1	0.000	0.000	-0.0372	2.489
2.197	0.00	2.58s	1,893.1	0.000	0.427	-0.0279	-0.000
2.189	0.00	5.09s	1,893.1	0.000	0.852	0.0001	-2.511
2.390	0.00	12.59s	1,893.1	0.000	1.257	0.1505	-10.011
2.512	0.00	15.09s	1,893.1	0.000	1.227	0.2050	-12.511
3.045	0.00	25.09s	1,893.1	0.000	0.831	0.3897	-22.511
3.496	0.00	35.09s	1,893.1	0.000	0.230	0.4836	-32.511
3.632	0.00	38.64s	1,893.1	0.000	0.000	0.4908	-36.061
3.842	0.00	45.09s	1,893.1	0.000	-0.425	0.4670	-42.511
4.068	0.01f	55.09s	1,893.1	0.000	-1.082	0.3355	-52.511
4.138	0.00	60.09s	1,893.1	0.000	-1.403	0.2270	-57.511

Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.

Note: No tank loads are present.

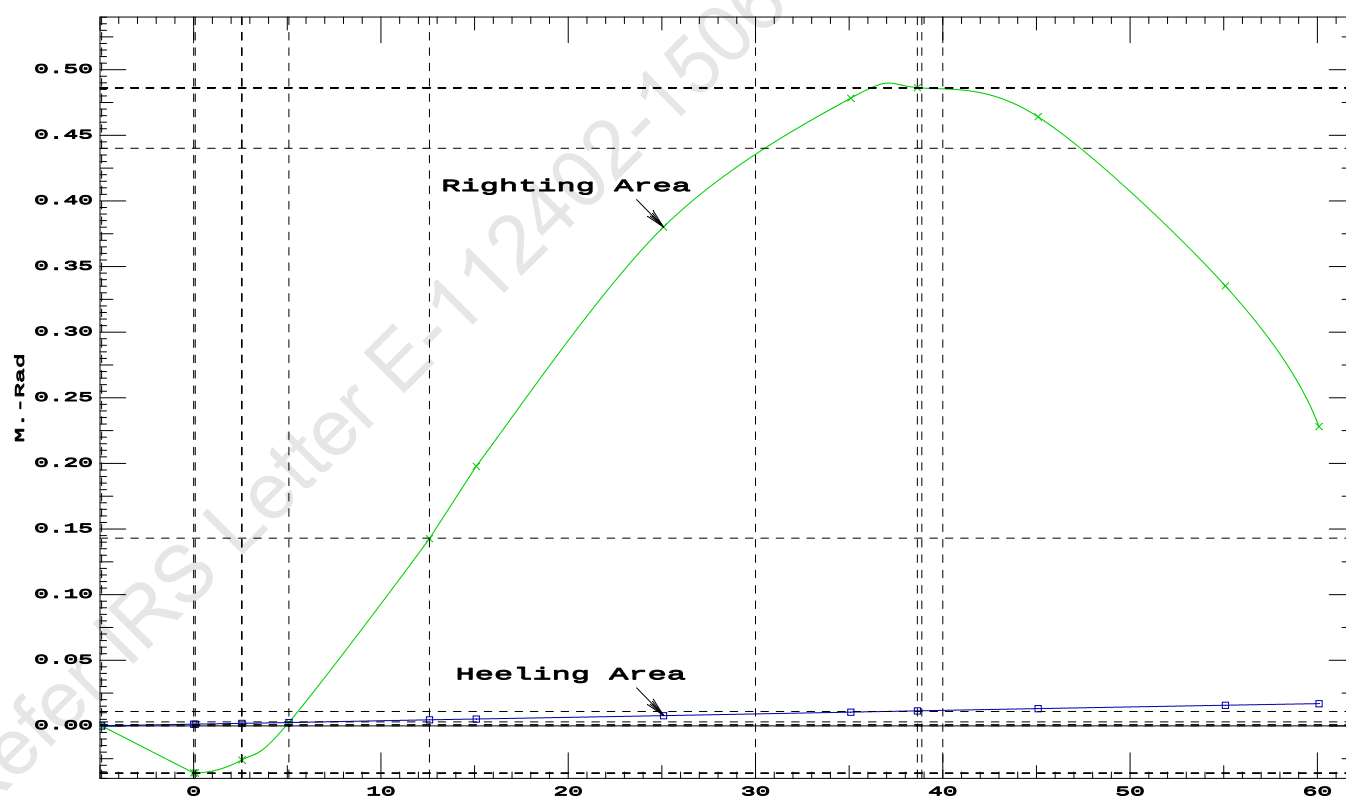
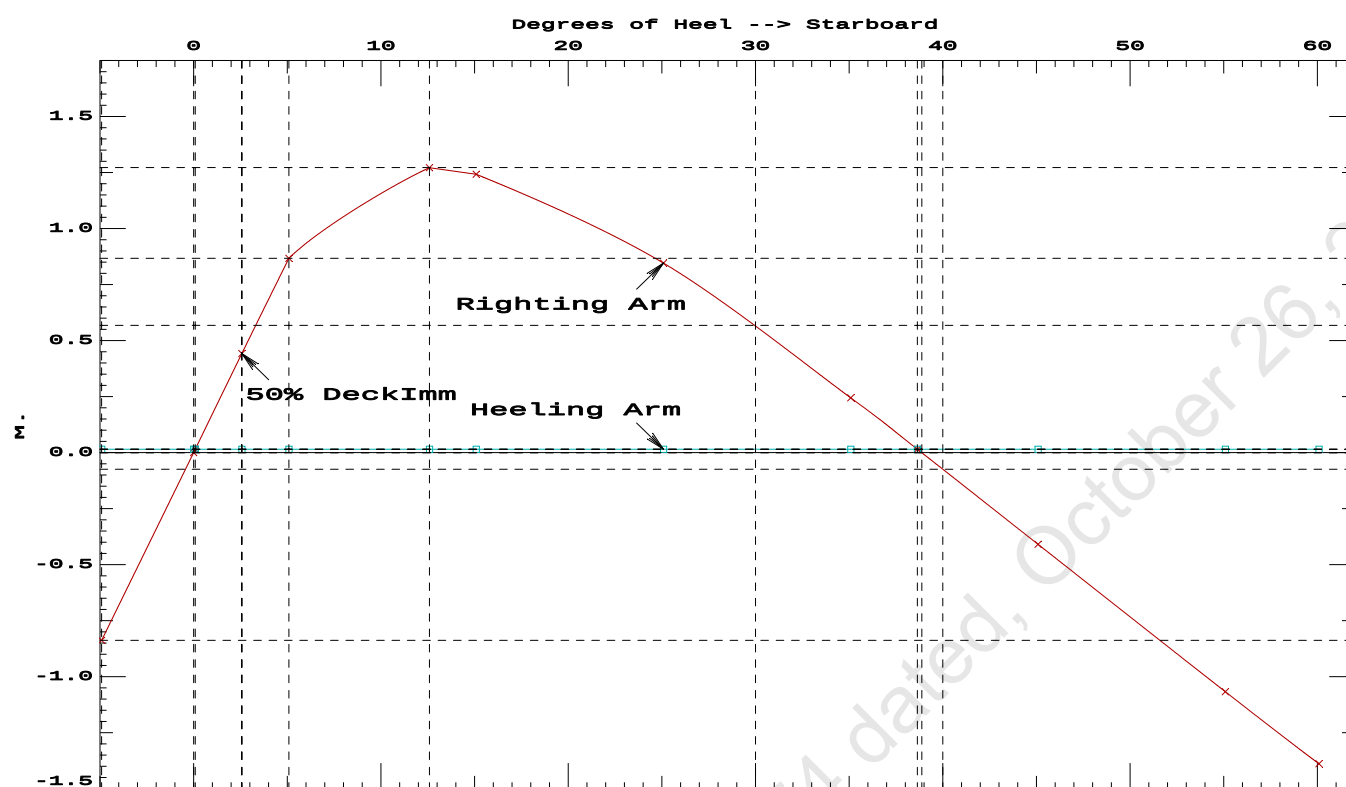
Note: The Residual Righting Arms shown above are in excess of the
wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 28.34 (constant)

LIM-----INTACT CRITERIA-IS CODE 2008PART B ----Min/Max-----Attained
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 0.1877 P
(2) Angle from Equilibrium to RAZero or Flood > 20.00 deg 38.55 P
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 2.49 P

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PONTOON

Condition 6 : Lightship + Max. Deck Cargo

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PONTOON

Condition 6 : Lightship + Max. Deck Cargo

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

LCG = 25.000f TCG = 0.000 VCG = 4.571

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. 50% DeckImm Area	Res. 50% DeckImm Angle
2.189	0.00	5.09s	1,893.3	0.000	-0.852	0.0000	
2.200	0.00	0.09s	1,893.1	0.000	0.000	-0.0372	
2.200	0.00	0.00	1,893.1	0.000	0.015	-0.0372	2.576
2.197	0.00	2.58p	1,892.9	0.000	0.457	-0.0265	0.000
2.190	0.00	4.91p	1,893.2	0.000	0.852	0.0002	-2.337
2.382	0.00	12.41p	1,893.1	0.000	1.287	0.1532	-9.837
2.503	0.00	14.91p	1,893.1	0.000	1.260	0.2091	-12.337
3.036	0.00	24.91p	1,893.1	0.000	0.870	0.4003	-22.337
3.489	0.00	34.91p	1,893.1	0.000	0.271	0.5012	-32.337
3.648	0.00	39.10p	1,893.1	0.000	0.000	0.5111	-36.520
3.837	0.00	44.91p	1,893.1	0.000	-0.383	0.4918	-42.337
4.069	0.01a	54.91p	1,893.1	0.000	-1.041	0.3675	-52.337
4.172	0.01f	64.91p	1,893.1	0.000	-1.674	0.1302	-62.337
4.180	0.00	69.91p	1,893.1	0.000	-1.974	-0.0291	-67.337

Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.

Note: No tank loads are present.

Note: The Residual Righting Arms shown above are in excess of the
wind heeling arms derived from these moments (in m.-MT):

Stbd heeling moment = 28.34 (constant)

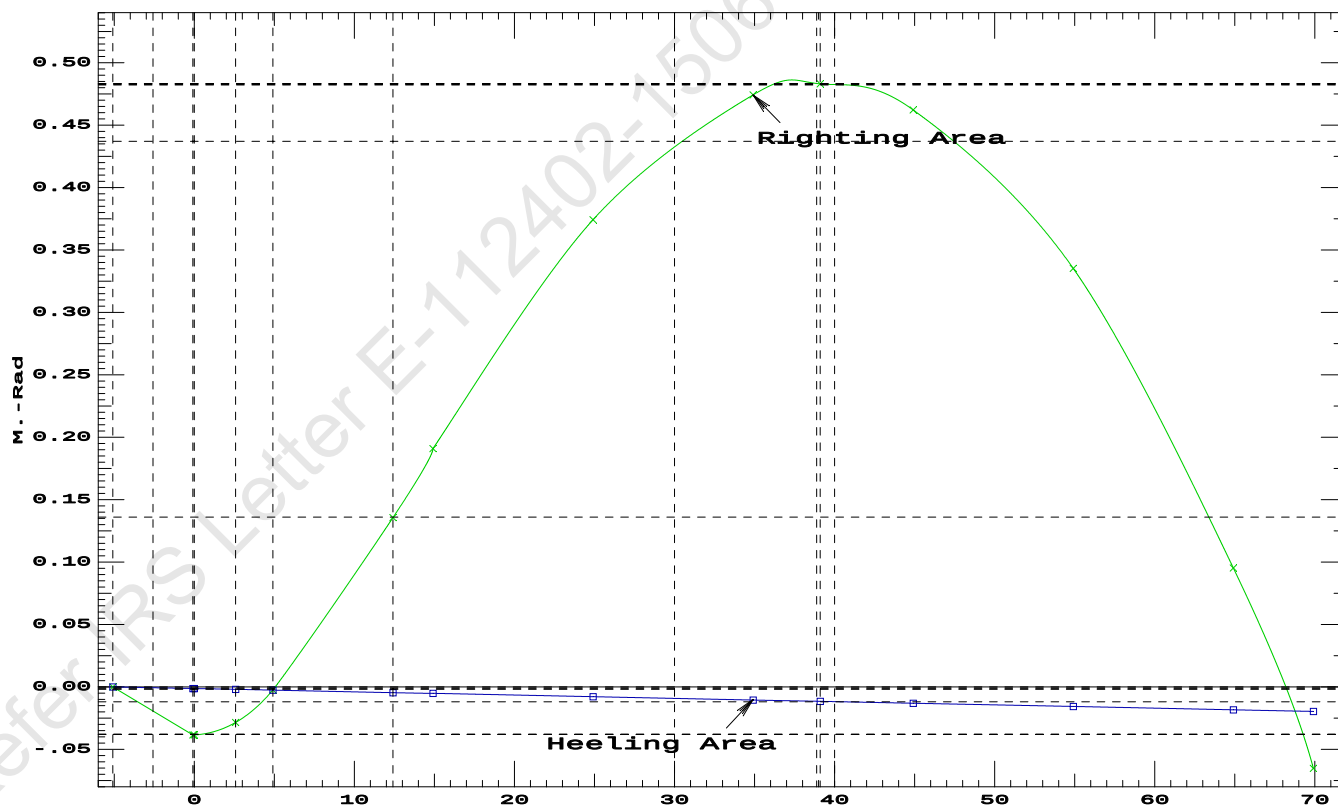
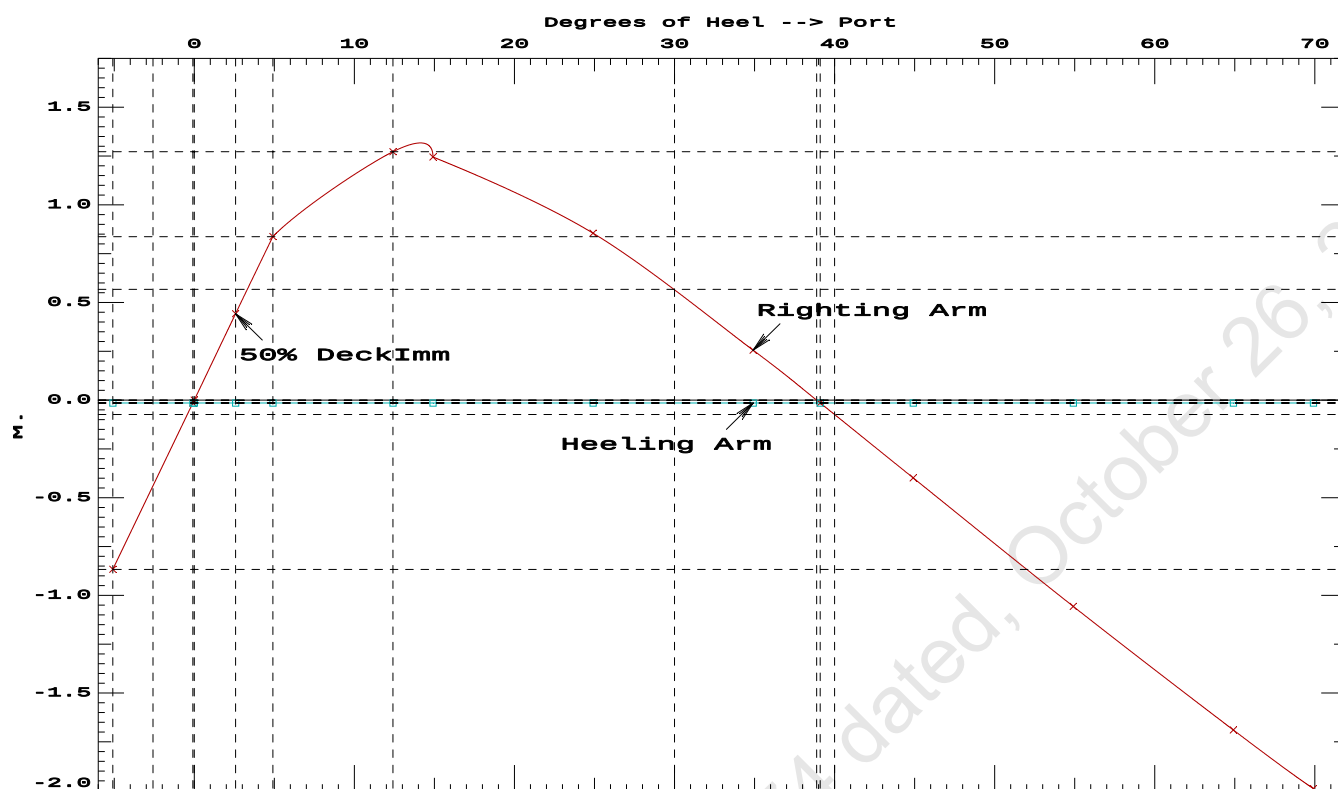
LIM-----INTACT CRITERIA-IS CODE 2008PART B ----Min/Max-----Attained

(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.1903 P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00 deg	39.18 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	2.66 P

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PONTOON

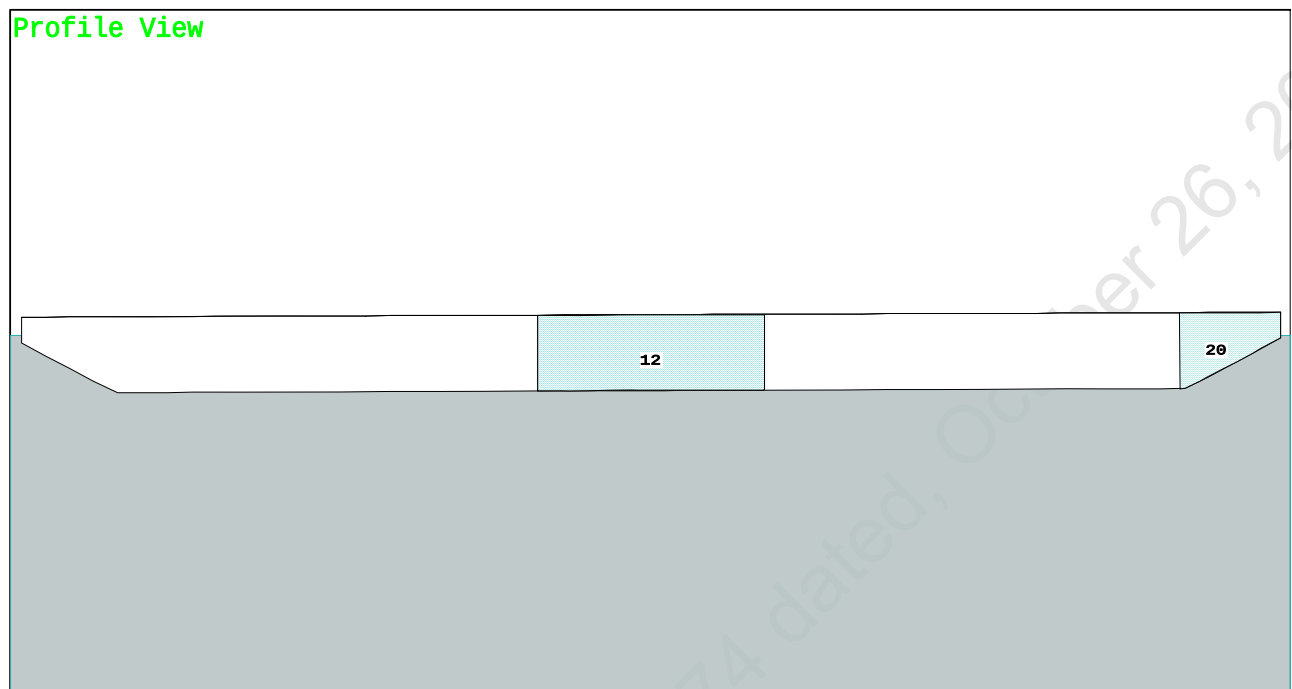
Condition 6 : Lightship + Max. Deck Cargo

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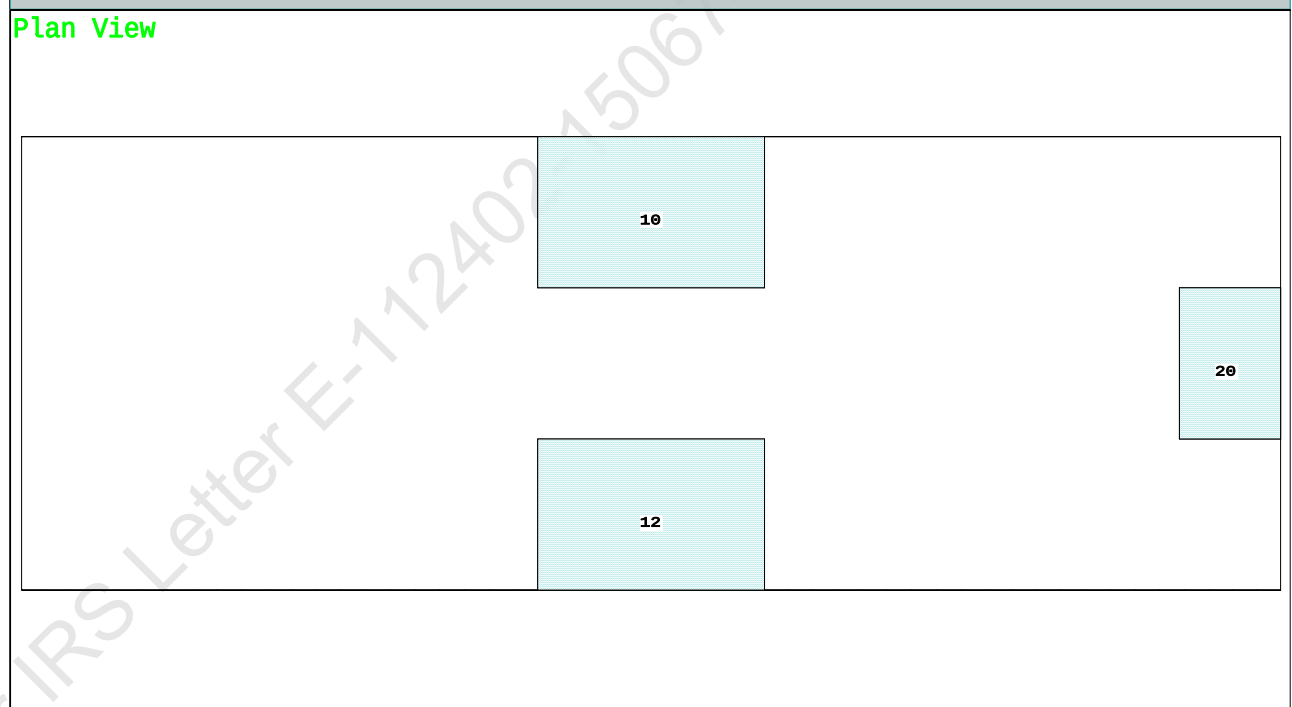
Condition 7 : Lightship + Load at Aft + Ballast + Deck Cargo
Crane Lifting 51.3T @ 12m Radius Aft

Condition Graphic - Draft: 2.115 @ 50.000f, 2.309 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

10 04_WBTK.P 12 04_WBTK.S 20 07_WBTK.C

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PONTOON

Condition 7 : Lightship + Load at Aft + Ballast + Deck CargoCrane Lifting 51.3T @ 12m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.115 @ 50.00f, 2.212 @ 25.00f, 2.309 @ 0.00

Trim: Aft 0.194/50.000, Heel: zero

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	4.500a	0.000	33.218	
Deck Cargo			912.48	27.500f	0.000	4.000	
Total Fixed----->			1,519.75	23.778f	0.000	4.838	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p	1.509	162.73*
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s	1.509	162.73*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
Total Tanks----->			373.31	28.042f	0.000	1.558	397.78
Total Weight----->			1,893.06	24.619f	0.000	4.191	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,893.06	24.607f	0.000	1.133	

Righting Arms: 0.000 0.000

Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->	1.025	900.4	24.995f	0.000	101.45	12.967
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
		9.23	37.25	98.39	9.909	

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

364.2 Cu.M. (15%) SALT WATER

150.00 MT Crane

51.30 MT Crane Load

912.48 MT Deck Cargo

DRAFT AT FP (M) = 2.115
 DRAFT AT MS (M) = 2.212
 DRAFT AT AP (M) = 2.309
 DRAFT AT FWD MARK (M) = 2.131
 DRAFT AT AFT MARK (M) = 2.293

HYDROSTATIC PROPERTIES

Trim: Aft 0.194/50.000, No Heel, VCG = 4.191

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/				
Draft----	Weight(MT)-----	LCB-----	VCB-----	cm-----	LCF-----	cm trim-----	GML-----	GMT
2.212	1,893.06	24.607f	1.133	9.23	24.995f	37.25	98.39	9.909
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.								
Trim is per 50.00m.								

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 397.8 m.-MT

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Condition 7 : Lightship + Load at Aft + Ballast + Deck Cargo
Crane Lifting 51.3T @ 12m Radius Aft

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 2.115 @ 50.00f, 2.212 @ 25.00f, 2.309 @ 0.00

Trim: Aft 0.194/50.000, Heel: zero

Part-----	LPA*SF-----	HCP-----	Arm----	Pressure-----	Moment
HULL	40.5	0.407	1.481	0.05500	3.30
CRANE	57.0	3.242	4.316	0.05500	13.53
DECKCARGO.C	29.4	1.810	2.884	0.05500	4.66
Total wind heeling moment to starboard----->					21.49
Distances in METERS.-----Pressure in MT/Sqm.-----					Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.619f TCG = 0.000 VCG = 4.191

Free Surface Adjustment: 0.210

Adjusted CG: LCG = 24.620f TCG = 0.000 VCG = 4.401

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm	
Depth---	Trim----	Heel----	Weight(MT)---	Area----	
			in Trim--in Heel-->	Angle	
2.297	0.22a	0.00	1,893.1	0.0000	2.262
2.297	0.22a	0.06s	1,893.1	0.0000	2.198
2.296	0.22a	2.26s	1,893.1	0.0074	0.000
2.289	0.23a	5.06s	1,893.1	0.0379	-2.802
2.432	0.34a	10.06s	1,893.1	0.1358	-7.802
2.569	0.39a	12.76s	1,893.1	0.1964	-10.496
2.704	0.44a	15.06s	1,893.1	0.2481	-12.802
3.023	0.55a	20.06s	1,893.1	0.3546	-17.802
3.348	0.70a	25.06s	1,893.1	0.4439	-22.802
3.651	0.85a	30.06s	1,893.1	0.5110	-27.802
3.929	1.00a	35.06s	1,893.1	0.5528	-32.802
4.179	1.14a	40.06s	1,893.1	0.5678	-37.802
4.190	1.15a	40.29s	1,893.1	0.5678	-38.032
4.398	1.28a	45.06s	1,893.1	0.5552	-42.802
4.583	1.41a	50.06s	1,893.1	0.5147	-47.802
4.733	1.52a	55.06s	1,893.1	0.4464	-52.802
4.845	1.62a	60.06s	1,893.1	0.3505	-57.802
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.					

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 397.8 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.49 (constant)

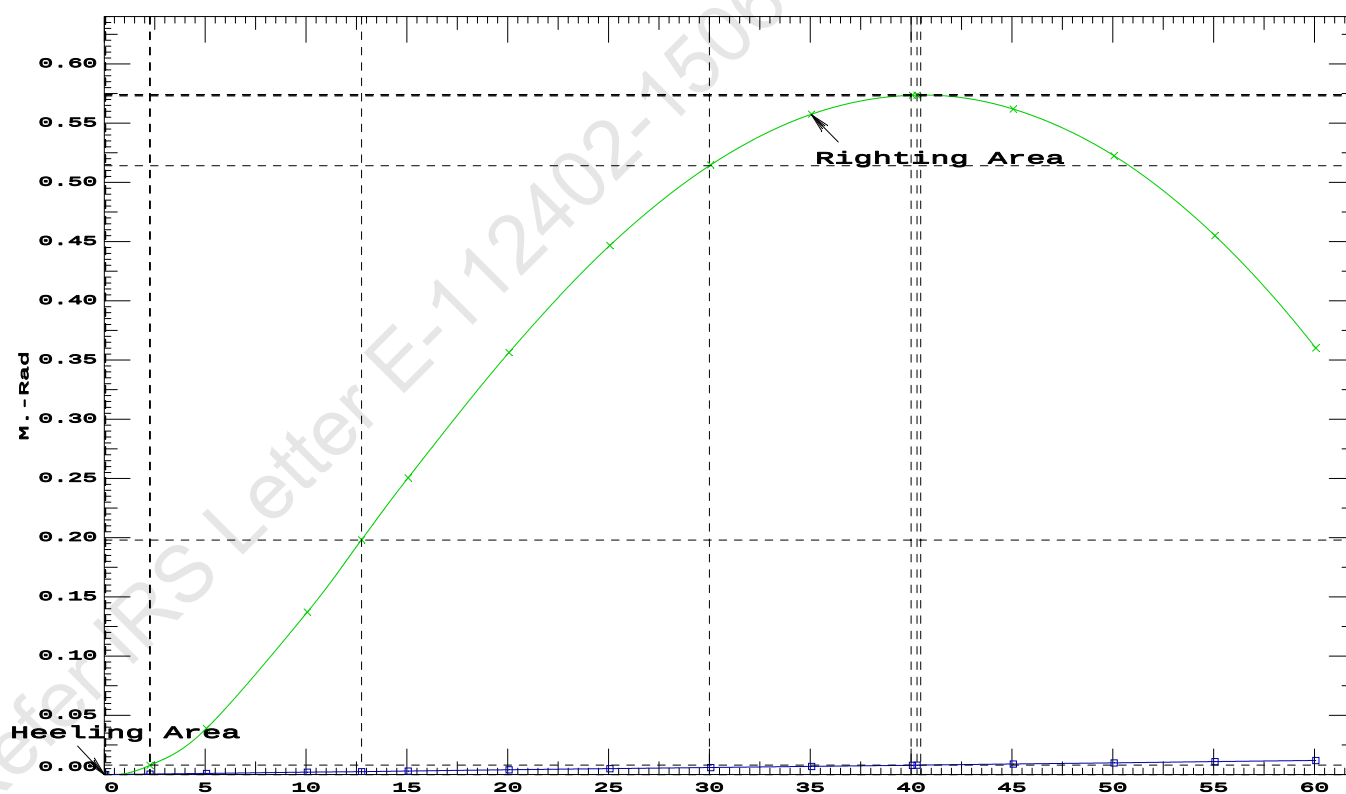
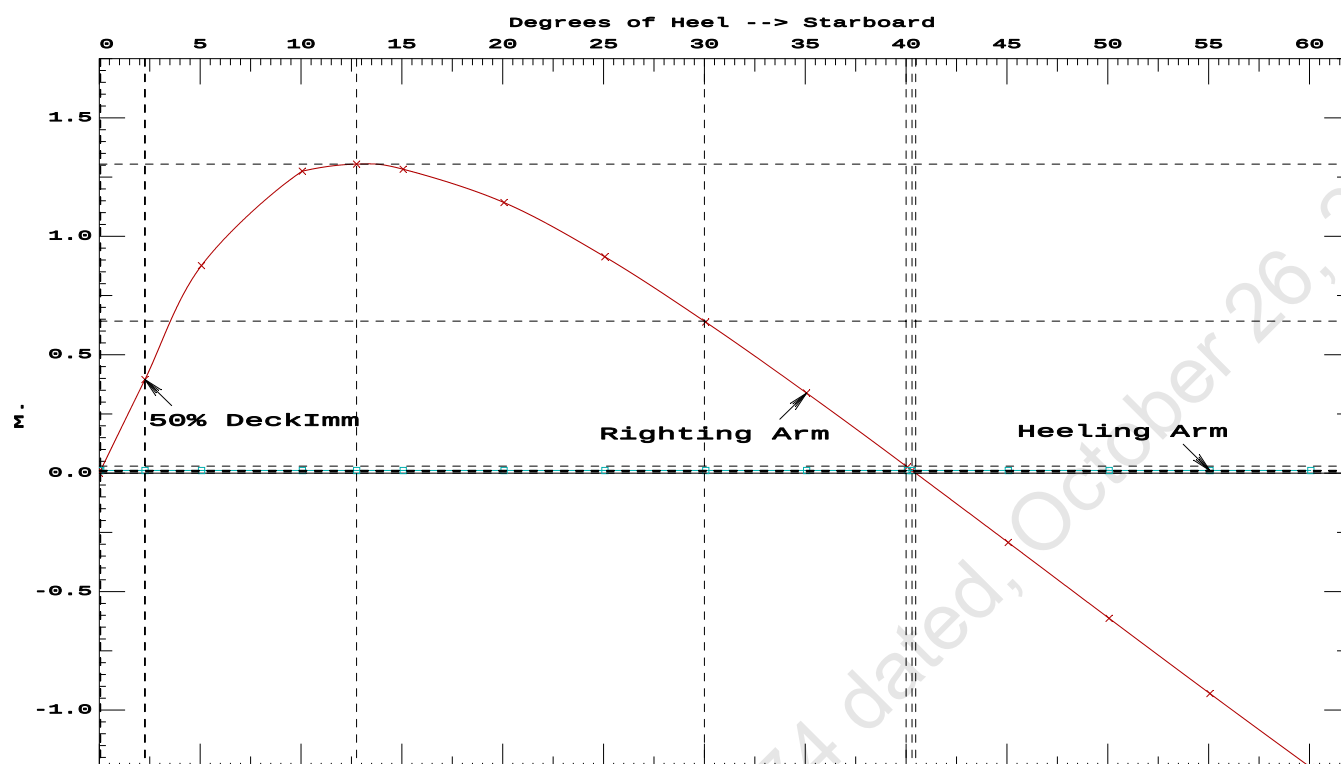
LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2-----Min/Max-----Attained

(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.1964 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	40.23 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	2.20 P

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PONT00N

Condition 7 : Lightship + Load at Aft + Ballast + Deck CargoCrane Lifting 51.3T @ 12m Radius Aft

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GHS 17.22

PONTOON**Condition 7 : Lightship + Load at Aft + Ballast + Deck Cargo**
Crane Lifting 51.3T @ 12m Radius Aft**RESIDUAL RIGHTING ARMS vs HEEL ANGLE**

Total CG: LCG = 24.619f TCG = 0.000 VCG = 4.191

Free Surface Adjustment: 0.210

Adjusted CG: LCG = 24.620f TCG = 0.000 VCG = 4.401

Origin Depth---	Degrees of Trim----	Heel----	Displacement Weight(MT)---	Residual Arms in Trim--in Heel---	Res. Area
2.297	0.22a	0.00	1,893.1	0.000 0.000	0.0000
2.289	0.23a	5.00s	1,893.1	0.000 0.866	0.0378
2.430	0.33a	10.00s	1,893.1	0.000 1.273	0.1361
2.569	0.39a	12.76s	1,893.1	0.000 1.304	0.1986
2.700	0.44a	15.00s	1,893.1	0.000 1.283	0.2493
3.019	0.55a	20.00s	1,893.1	0.000 1.144	0.3569
3.344	0.70a	25.00s	1,893.1	0.000 0.916	0.4475
3.648	0.85a	30.00s	1,893.1	0.000 0.641	0.5158
3.926	0.99a	35.00s	1,893.1	0.000 0.341	0.5588
4.176	1.14a	40.00s	1,893.1	0.000 0.029	0.5750
4.197	1.15a	40.46s	1,893.1	0.000 0.000	0.5752
4.395	1.28a	45.00s	1,893.1	0.000 -0.289	0.5637
4.581	1.41a	50.00s	1,893.1	0.000 -0.609	0.5245
4.731	1.52a	55.00s	1,893.1	0.000 -0.927	0.4574
4.843	1.62a	60.00s	1,893.1	0.000 -1.240	0.3629

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 397.8 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

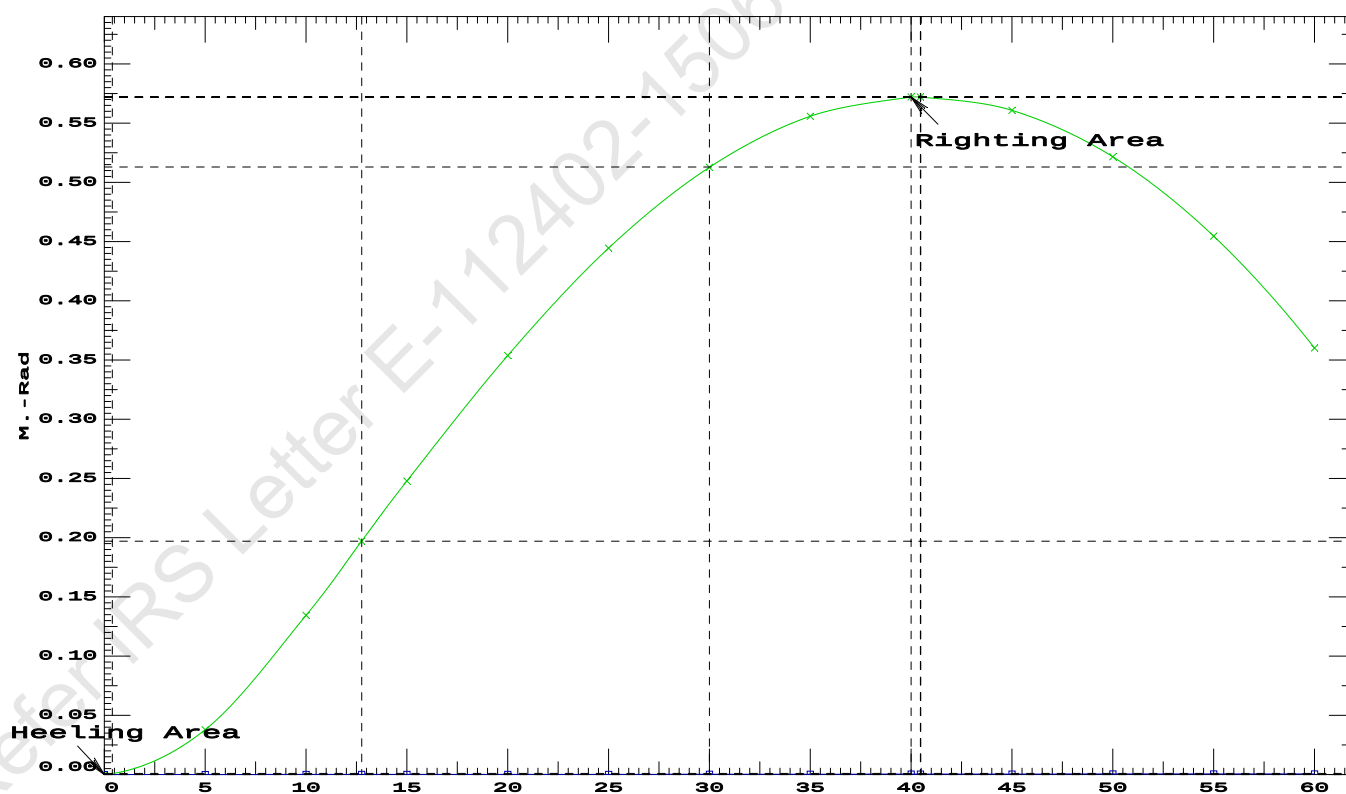
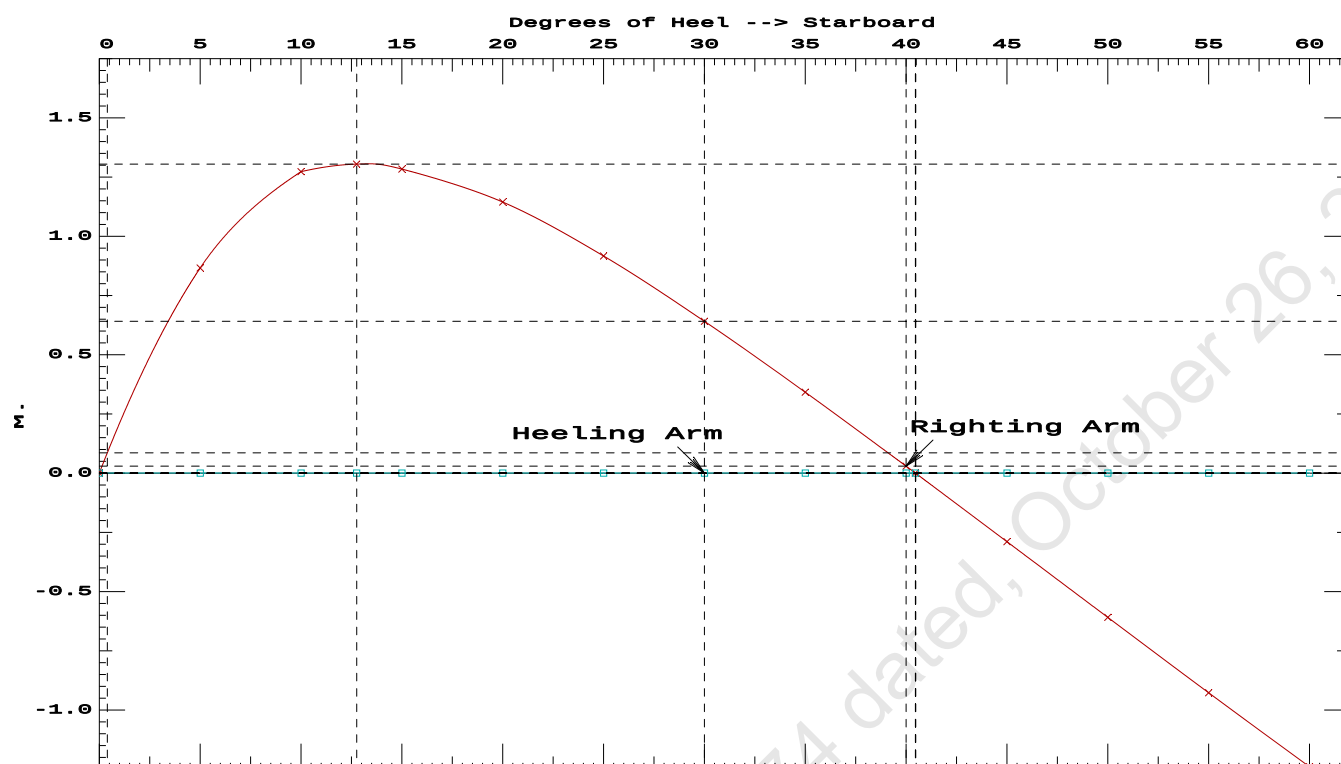
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	0.00 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	246871 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	1542.89 P

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PONTOON

Condition 7 : Lightship + Load at Aft + Ballast + Deck CargoCrane Lifting 51.3T @ 12m Radius Aft

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GHS 17.22 PONTON

Condition 7 : Lightship + Load at Aft + Ballast + Deck CargoCrane Lifting 51.3T @ 12m Radius Aft

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.115 @ 50.00f, 2.212 @ 25.00f, 2.309 @ 0.00

Trim: Aft 0.194/50.000, Heel: 0.00 deg.

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP	405.97	25.000f	0.000	3.000	
Crane	150.00	7.500f	0.000	5.200	
Crane Load	51.30	4.500a	0.000	33.218	
Deck Cargo	912.48	27.500f	0.000	4.000	
Total Fixed----->	1,519.75	23.778f	0.000	4.838	
Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000
Total Tanks----->			373.31	28.042f	0.000
Total Weight----->			1,893.06	24.619f	0.000
			Displ(MT)----	LCB-----	TCB-----
HULL	1.025		1,893.06	24.607f	0.001
			Righting Arms:	0.000	0.001s
			External Arms:	0.000	0.001
			Residual Righting Arms:	0.000	0.000
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----
Total Waterplane---->	1.025	900.4	24.995f	0.000	101.45
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----
		9.23	37.25	98.39	9.909
Distances in METERS.-----					Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

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After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 2.262 @ 50.00f, 2.156 @ 25.00f, 2.051 @ 0.00
Trim: Fwd 0.211/50.000, Heel: 0.00 deg.

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----
LIGHT SHIP	405.97	25.000f	0.000	3.000
Crane	150.00	7.500f	0.000	5.200
Deck Cargo	912.48	27.500f	0.000	4.000
Total Fixed----->	1,468.45	24.766f	0.000	3.846

Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p	1.509
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s	1.509
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872
Total Tanks----->			373.31	28.042f	0.000	1.558
Total Weight----->			1,841.76	25.430f	0.000	3.382

Displ(MT)----	LCB-----	TCB-----	VCB-----
HULL	1.025	1,841.76	25.440f
		0.001	1.104

Righting Arms:	0.000	0.001s
External Arms:	0.000	0.001
Residual Righting Arms:	0.000	0.000

Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025	900.3	25.006f	0.000	104.24	13.324
		MT/cm-----	m.-MT/cm-----		GML-----	GMT-----
		9.23	37.56		101.96	11.045

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

Theta3 is 0.07 degrees, as demonstrated below.

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 24.766f TCG = 0.000 VCG = 3.846

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
2.039	0.24f	0.00	1,841.8	0.000	0.000	0.0000
2.039	0.24f	0.07s	1,841.8	0.000	0.014	0.0000
2.025	0.25f	5.00s	1,841.8	0.000	0.983	0.0429
2.053	0.35f	10.00s	1,841.8	0.000	1.560	0.1594
2.211	0.46f	15.00s	1,841.8	0.000	1.692	0.3045
2.246	0.48f	15.88s	1,841.8	0.000	1.695	0.3304
2.401	0.59f	20.00s	1,841.8	0.000	1.648	0.4514
2.568	0.75f	25.00s	1,841.8	0.000	1.499	0.5897
2.712	0.90f	30.00s	1,841.8	0.000	1.296	0.7120

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1468.45 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 0.1 > 1.000 LARGE

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GHS 17.22 PONTON

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 24.766f TCG = 0.000 VCG = 3.846

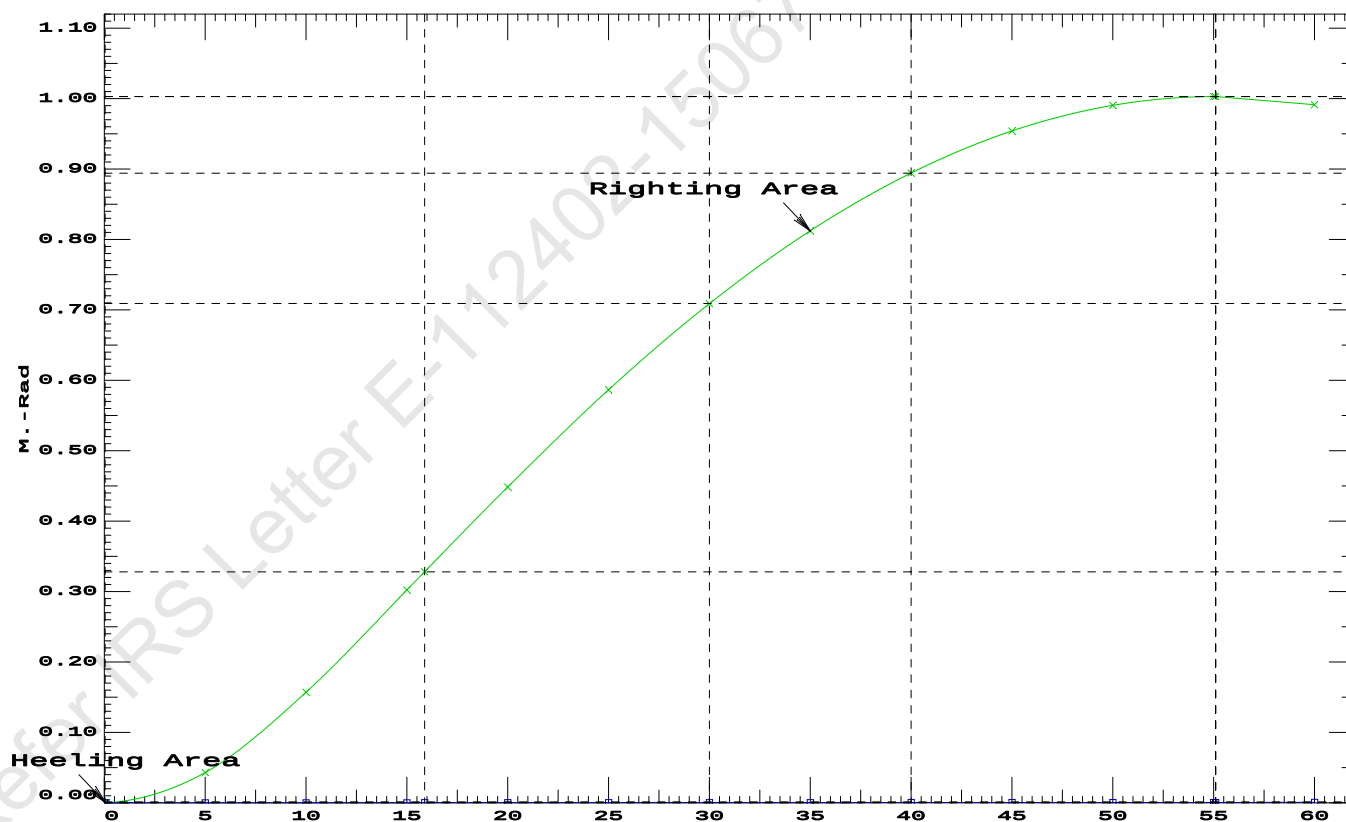
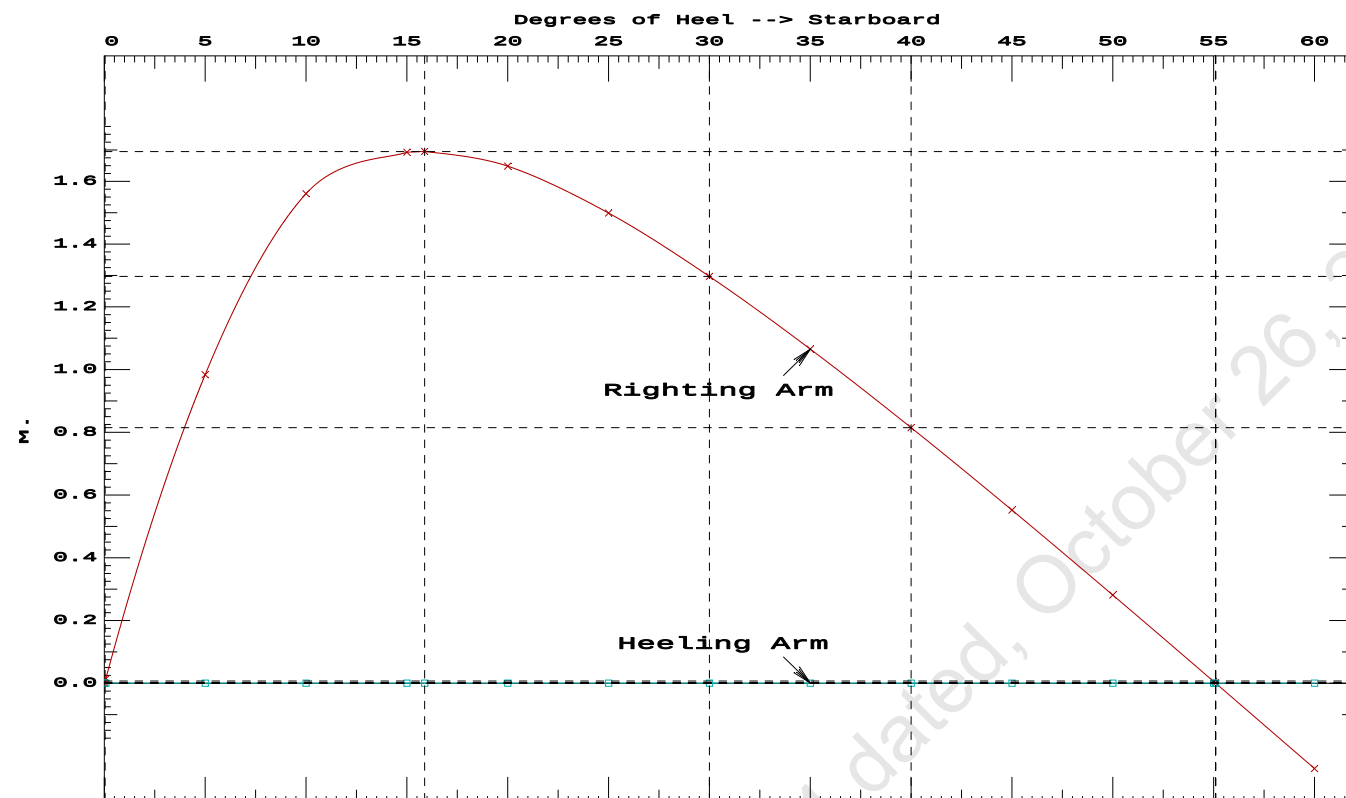
Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
2.039	0.24f	0.00	1,841.8	0.000	0.000	0.0000
2.039	0.24f	0.07s	1,841.8	0.000	0.014	0.0000
2.025	0.25f	5.00s	1,841.8	0.000	0.983	0.0429
2.053	0.35f	10.00s	1,841.8	0.000	1.560	0.1594
2.211	0.46f	15.00s	1,841.8	0.000	1.692	0.3045
2.246	0.48f	15.88s	1,841.8	0.000	1.695	0.3304
2.401	0.59f	20.00s	1,841.8	0.000	1.648	0.4514
2.568	0.75f	25.00s	1,841.8	0.000	1.499	0.5897
2.712	0.90f	30.00s	1,841.8	0.000	1.296	0.7120
2.833	1.05f	35.00s	1,841.8	0.000	1.065	0.8152
2.930	1.20f	40.00s	1,841.8	0.000	0.814	0.8974
3.003	1.35f	45.00s	1,841.8	0.000	0.551	0.9570
3.052	1.49f	50.00s	1,841.8	0.000	0.280	0.9934
3.077	1.61f	55.00s	1,841.8	0.000	0.005	1.0059
3.078	1.62f	55.09s	1,841.8	0.000	0.000	1.0059
3.081	1.73f	60.00s	1,841.8	0.000	-0.272	0.9942

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1468.45 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero > 1.000 LARGE
(2) Angle from abs 0.1 deg to RAZero > 20.00 deg 55.02 P

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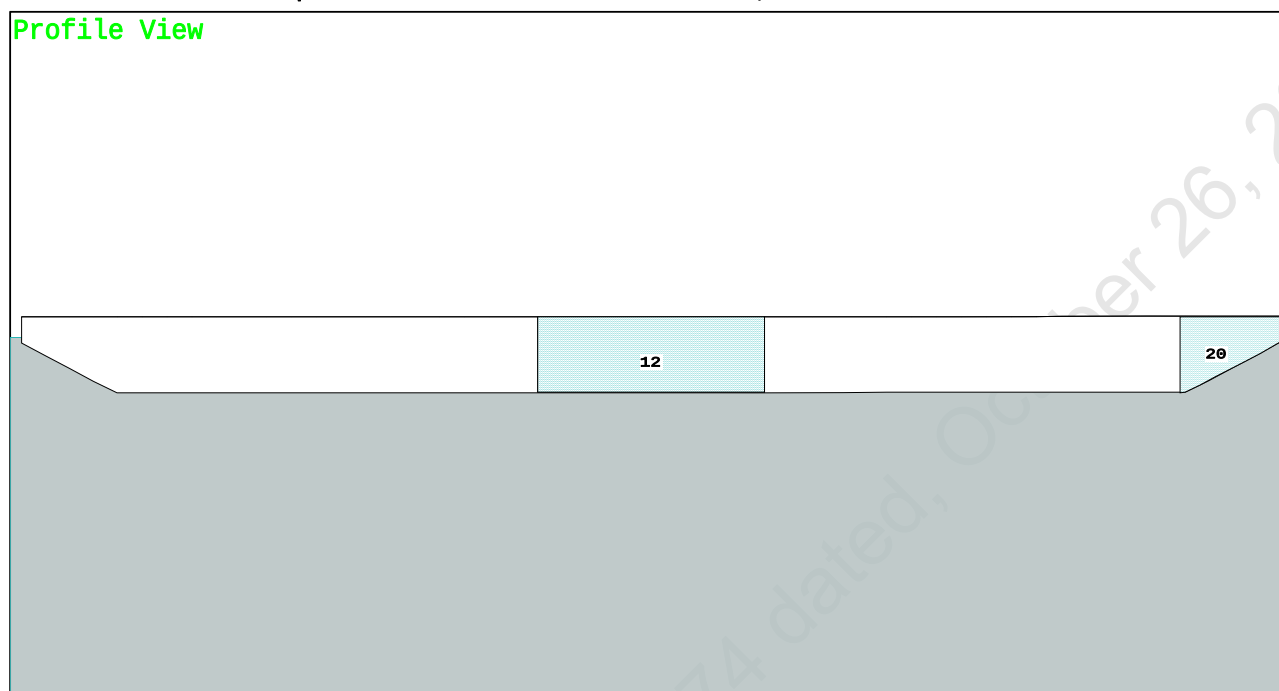
GHS 17.22

PONT00N

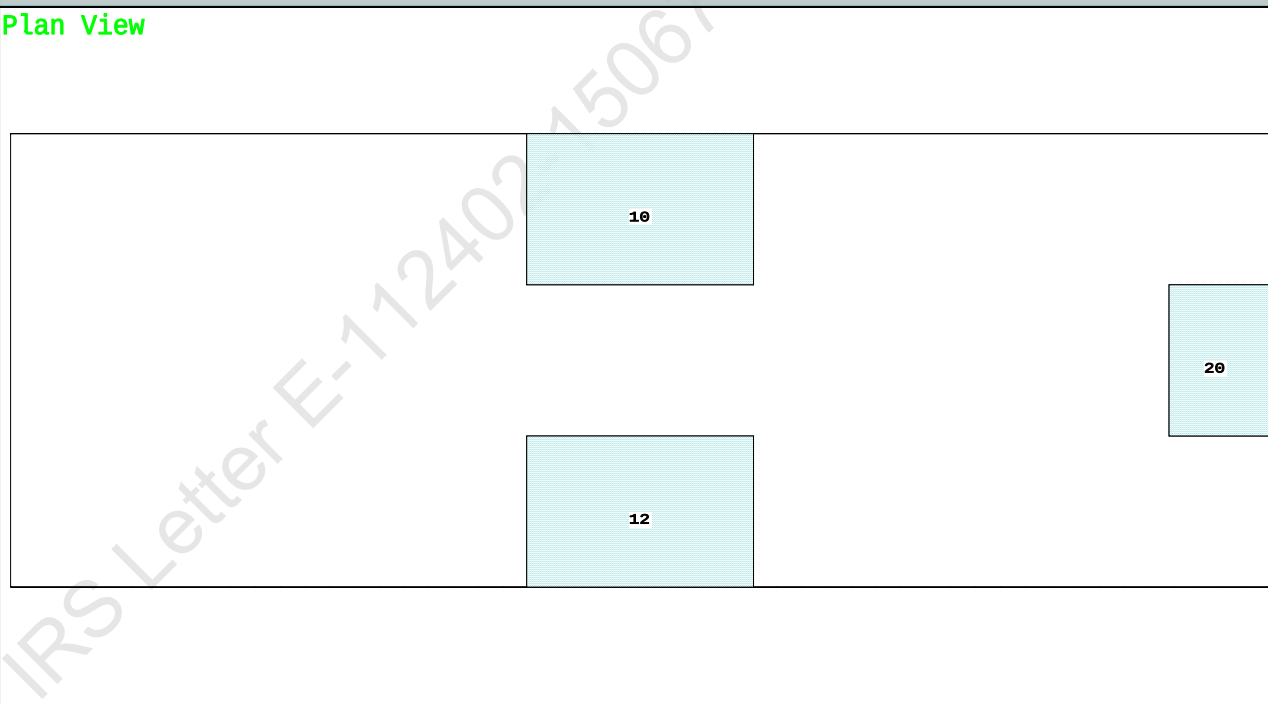
Condition 8 : Lightship + Load at Stbd + Ballast + Deck CargoCrane Lifting 51.3T @ 12m Radius Stbd

Condition Graphic - Draft: 2.198 @ 50.000f, 2.226 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

10 04_WBTK.P

12 04_WBTK.S

20 07_WBTK.C

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PONTOON

Condition 8 : Lightship + Load at Stbd + Ballast + Deck CargoCrane Lifting 51.3T @ 12m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.198 @ 50.00f, 2.212 @ 25.00f, 2.226 @ 0.00

Trim: Aft 0.028/50.000, Heel: zero

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	7.500f	0.000	33.218	
Deck Cargo			912.48	27.500f	0.000	4.000	
Total Fixed----->			1,519.75	24.183f	0.000	4.838	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p	1.509	162.73*
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s	1.509	162.73*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
Total Tanks----->			373.31	28.042f	0.000	1.558	397.78
Total Weight----->			1,893.06	24.944f	0.000	4.191	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,893.06	24.942f	0.000	1.132	

Righting Arms: 0.000 0.000

Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->	1.025	900.4	24.999f	0.000	101.45	12.967
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
		9.23	37.25	98.39	9.908	

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

364.2 Cu.M. (15%) SALT WATER

150.00 MT Crane

51.30 MT Crane Load

912.48 MT Deck Cargo

DRAFT AT FP (M) = 2.198
 DRAFT AT MS (M) = 2.212
 DRAFT AT AP (M) = 2.226
 DRAFT AT FWD MARK (M) = 2.200
 DRAFT AT AFT MARK (M) = 2.224

HYDROSTATIC PROPERTIES

Trim: Aft 0.028/50.000, No Heel, VCG = 4.191

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/				
Draft----	Weight(MT)-----	LCB-----	VCB-----	cm-----	LCF-----	cm trim-----	GML-----	GMT
2.212	1,893.06	24.942f	1.132	9.23	24.999f	37.25	98.39	9.908
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.								
Trim is per 50.00m.								

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 397.8 m.-MT

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Condition 8 : Lightship + Load at Stbd + Ballast + Deck Cargo
Crane Lifting 51.3T @ 12m Radius Stbd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 2.198 @ 50.00f, 2.212 @ 25.00f, 2.226 @ 0.00

Trim: Aft 0.028/50.000, Heel: zero

Part-----	LPA*SF-----	HCP-----	Arm----	Pressure-----	Moment
HULL	40.5	0.405	1.478	0.05500	3.29
CRANE	57.0	3.292	4.365	0.05500	13.68
DECKCARGO.C	29.4	1.801	2.875	0.05500	4.65
Total wind heeling moment to starboard----->					21.62
Distances in METERS.-----Pressure in MT/Sqm.-----					Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.944f TCG = 0.000 VCG = 4.191

Free Surface Adjustment: 0.210

Adjusted CG: LCG = 24.944f TCG = 0.000 VCG = 4.401

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm	
Depth---	Trim---	Heel---	Weight(MT)---	Area---	
			in Trim--in Heel-->	Angle	
2.214	0.03a	0.00	1,893.1	0.0000	2.530
2.214	0.03a	0.06s	1,893.1	0.0000	2.466
2.212	0.03a	2.53s	1,893.1	0.0093	0.000
2.203	0.03a	5.06s	1,893.1	0.0380	-2.534
2.306	0.05a	10.06s	1,893.1	0.1362	-7.534
2.423	0.06a	12.76s	1,893.1	0.1970	-10.229
2.539	0.06a	15.06s	1,893.1	0.2488	-12.534
2.817	0.08a	20.06s	1,893.1	0.3557	-17.534
3.088	0.10a	25.06s	1,893.1	0.4455	-22.534
3.336	0.12a	30.06s	1,893.1	0.5130	-27.534
3.559	0.15a	35.06s	1,893.1	0.5553	-32.534
3.755	0.17a	40.06s	1,893.1	0.5708	-37.534
3.766	0.17a	40.38s	1,893.1	0.5708	-37.847
3.923	0.19a	45.06s	1,893.1	0.5586	-42.534
4.060	0.21a	50.06s	1,893.1	0.5185	-47.534
4.167	0.22a	55.06s	1,893.1	0.4506	-52.534
4.242	0.24a	60.06s	1,893.1	0.3550	-57.534
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.					

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 397.8 m.-MT was applied to artificially modify the CG.

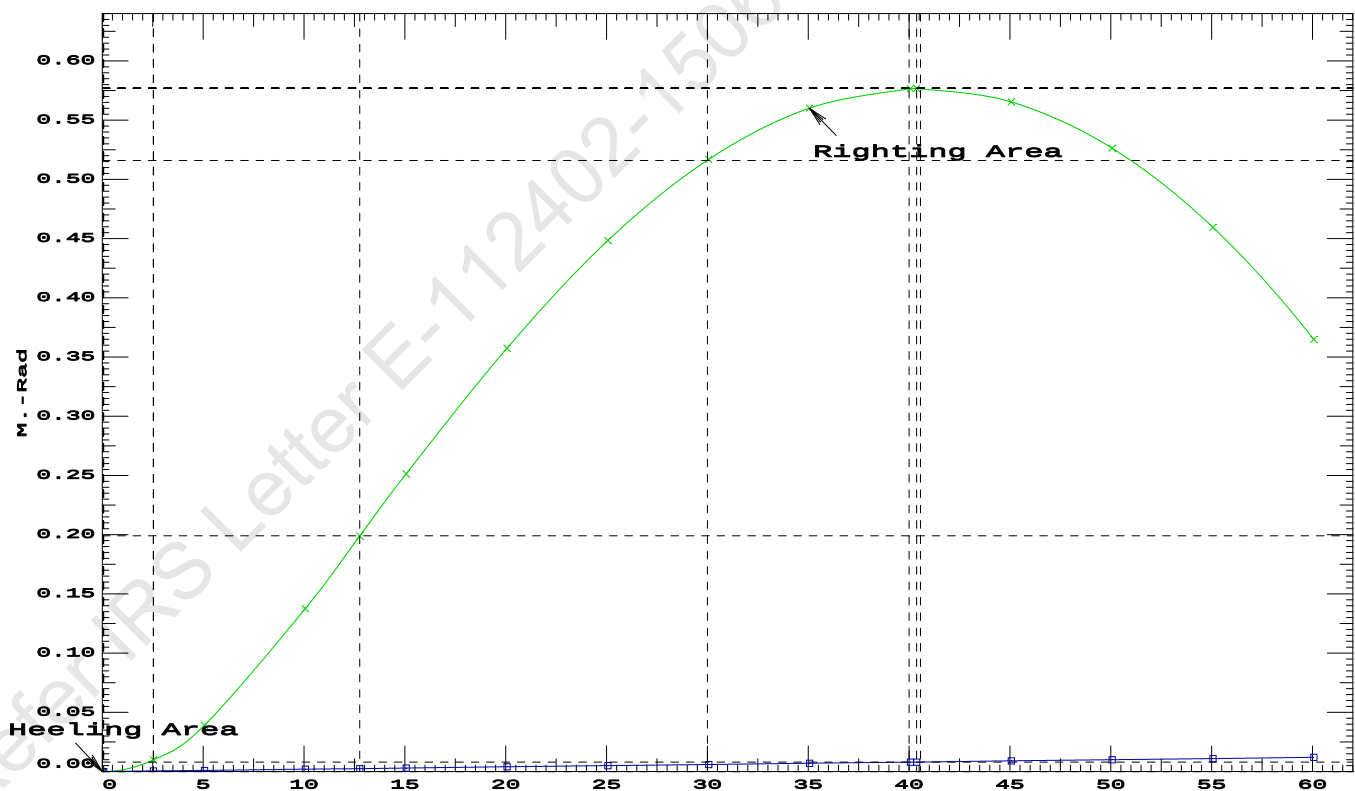
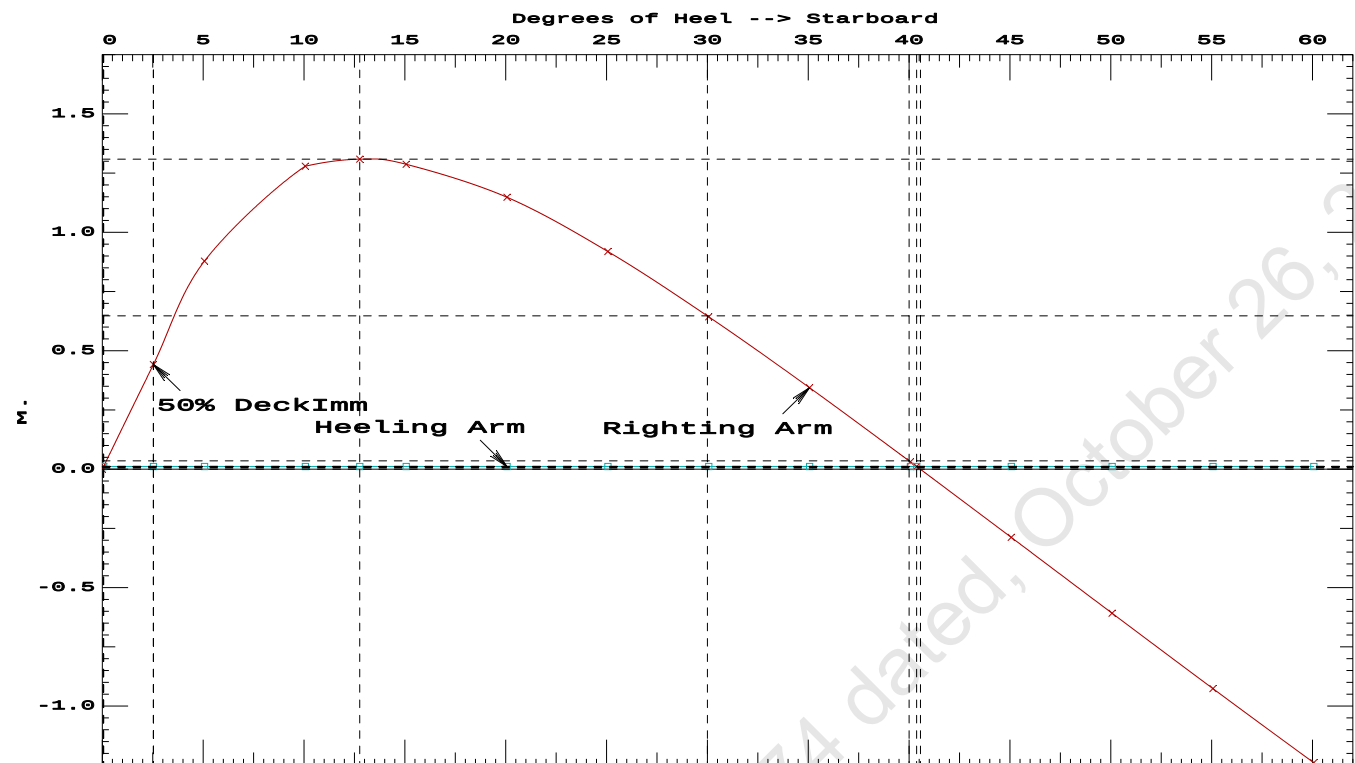
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.62 (constant)

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2-----Min/Max-----Attained

(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.1970 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	40.31 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	2.47 P

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Condition 8 : Lightship + Load at Stbd + Ballast + Deck Cargo
Crane Lifting 51.3T @ 12m Radius Stbd



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Condition 8 : Lightship + Load at Stbd + Ballast + Deck Cargo
Crane Lifting 51.3T @ 12m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.944f TCG = 0.000 VCG = 4.191

Free Surface Adjustment: 0.210

Adjusted CG: LCG = 24.944f TCG = 0.007p VCG = 4.401

Origin	Degrees of	Displacement	Residual Arms	Res.
Depth---	Trim---	Heel---	Weight(MT)---	in Trim--in Heel---> Area
2.213	0.03a	1.82s	1,893.1	0.000 0.000 0.0000
2.215	0.04a	6.82s	1,893.1	0.000 0.786 0.0343
2.379	0.05a	11.82s	1,893.1	0.000 0.987 0.1146
2.418	0.06a	12.65s	1,893.1	0.000 0.990 0.1290
2.634	0.07a	16.82s	1,893.1	0.000 0.931 0.1999
2.915	0.09a	21.82s	1,893.1	0.000 0.756 0.2747
3.178	0.11a	26.82s	1,893.1	0.000 0.507 0.3304
3.417	0.13a	31.82s	1,893.1	0.000 0.221 0.3624
3.575	0.15a	35.45s	1,893.1	0.000 0.000 0.3695
3.631	0.15a	36.82s	1,893.1	0.000 -0.085 0.3684
3.817	0.17a	41.82s	1,893.1	0.000 -0.400 0.3474
3.974	0.19a	46.82s	1,893.1	0.000 -0.721 0.2985
4.102	0.21a	51.82s	1,893.1	0.000 -1.041 0.2216
4.197	0.23a	56.82s	1,893.1	0.000 -1.358 0.1169
4.261	0.24a	61.82s	1,893.1	0.000 -1.669 -0.0153

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 397.8 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 615.60

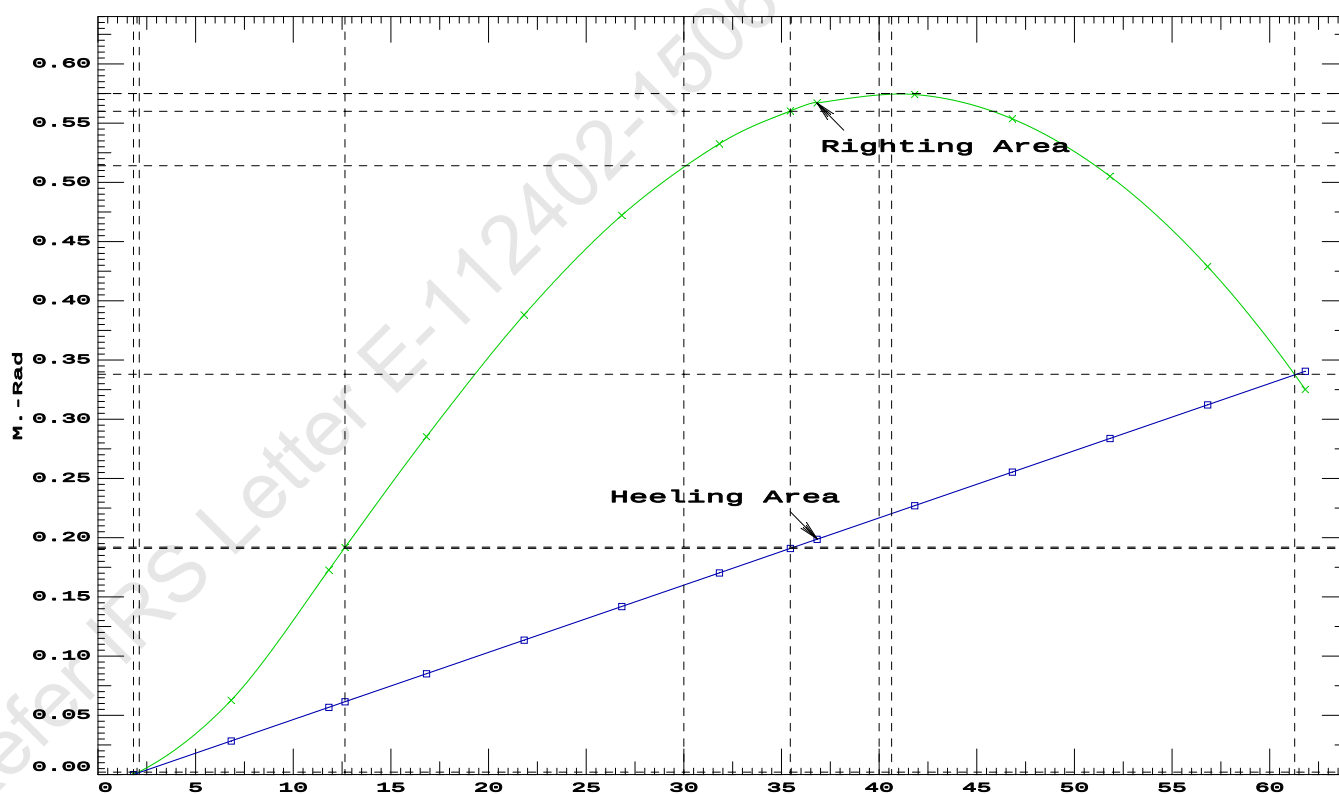
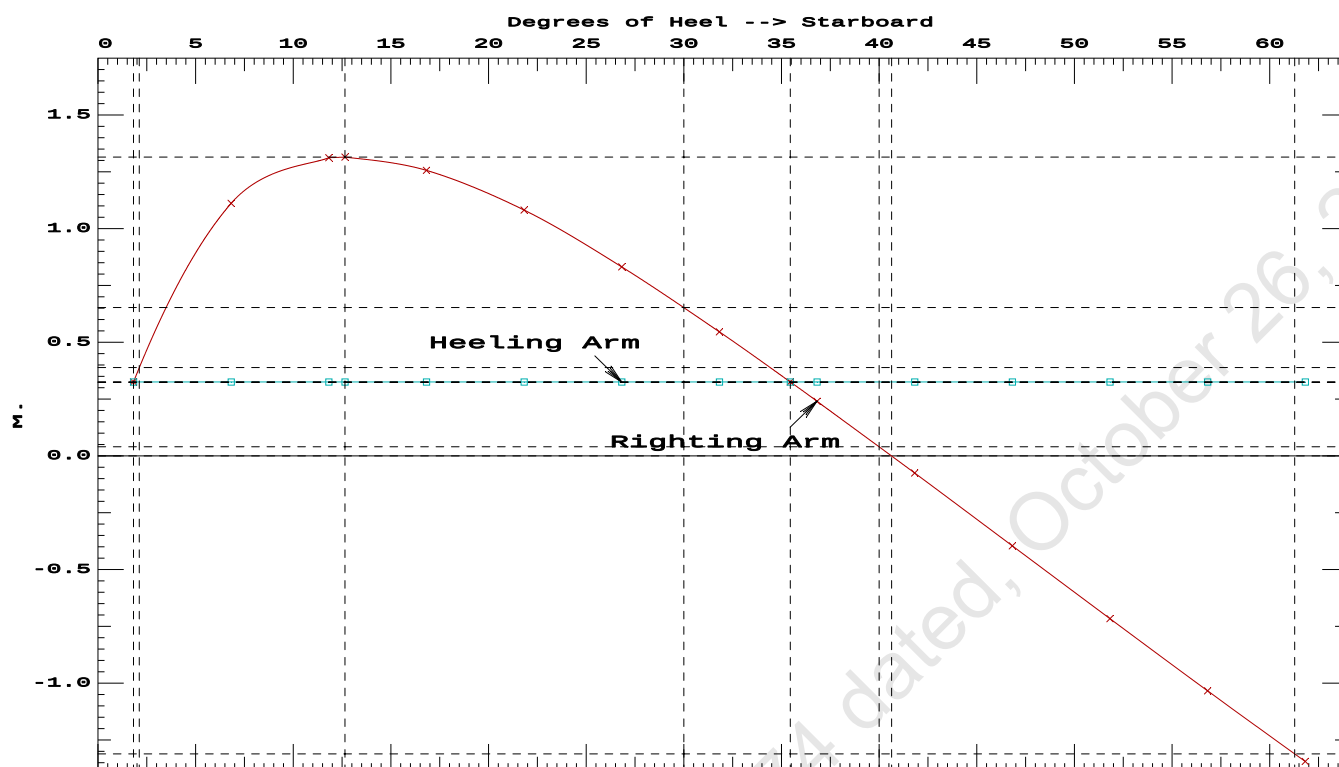
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	1.82 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	304.4 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	2.935 P

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Condition 8 : Lightship + Load at Stbd + Ballast + Deck CargoCrane Lifting 51.3T @ 12m Radius Stbd

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Condition 8 : Lightship + Load at Stbd + Ballast + Deck CargoCrane Lifting 51.3T @ 12m Radius Stbd

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.198 @ 50.00f, 2.212 @ 25.00f, 2.226 @ 0.00

Trim: Aft 0.029/50.000, Heel: Stbd 1.82 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	7.500f	0.000	33.218	
Deck Cargo			912.48	27.500f	0.000	4.000	
Total Fixed----->			1,519.75	24.183f	0.000	4.838	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p	1.509	162.73*
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s	1.509	162.73*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
Total Tanks----->			373.31	28.042f	0.000	1.558	397.78
Total Weight----->			1,893.06	24.944f	0.000	4.191	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,893.04	24.942f	0.422s	1.138	

	Righting Arms:			0.000	0.325s		
	External Arms:			0.000	0.325s		
	Residual Righting Arms:			0.000	0.000s		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		899.1	24.987f	0.022s	100.91	12.922
			MT/cm				
			9.22				

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 1.82

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After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 2.262 @ 50.00f, 2.156 @ 25.00f, 2.051 @ 0.00
Trim: Fwd 0.211/50.000, Heel: 0.00 deg.

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----
LIGHT SHIP	405.97	25.000f	0.000	3.000
Crane	150.00	7.500f	0.000	5.200
Deck Cargo	912.48	27.500f	0.000	4.000
Total Fixed----->	1,468.45	24.766f	0.000	3.846

Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p	1.509 162.73*
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s	1.509 162.73*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872 72.33*
Total Tanks----->			373.31	28.042f	0.000	1.558 397.78
Total Weight----->			1,841.76	25.430f	0.000	3.382

Displ(MT)----	LCB-----	TCB-----	VCB-----
HULL	1.025	1,841.76	25.439f 0.001 1.104

Righting Arms:	0.000	0.001s
External Arms:	0.000	0.001
Residual Righting Arms:	0.000	0.000

Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025	900.3	25.005f	0.000	104.24	13.324
		MT/cm-----	m.-MT/cm-----		GML-----	GMT-----
		9.23	37.56		101.96	11.045

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: 12404.11; t3: 30.00; t: 15.00
limmarg: 5248.14; t3: 15.00; t: 7.50
limmarg: 1560.15; t3: 7.50; t: 3.75
limmarg: 325.99; t3: 3.75; t: 1.88
limmarg: 6.64; t3: 1.87; t: 0.94
limmarg: -73.41; t3: 0.93; t: 0.47
limmarg: -40.06; t3: 1.40; t: 0.23
limmarg: -18.87; t3: 1.63; t: 0.12
limmarg: -6.55; t3: 1.75; t: 0.06
limmarg: -0.06; t3: 1.81; t: 0.03
limmarg: 3.26; t3: 1.84; t: 0.01

Theta3 is 1.84 degrees, as demonstrated below.

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 24.766f TCG = 0.000 VCG = 3.846

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
2.036	0.24f	1.82s	1,841.8	0.000	-0.360	0.0000
2.039	0.24f	0.00	1,841.8	0.000	0.000	-0.0057
2.036	0.24f	1.84p	1,841.8	0.000	0.366	0.0002
2.025	0.25f	5.00p	1,841.8	0.000	0.983	0.0374
2.053	0.35f	10.00p	1,841.8	0.000	1.561	0.1509
2.211	0.46f	15.00p	1,841.8	0.000	1.693	0.2962
2.246	0.48f	15.88p	1,841.8	0.000	1.696	0.3223
2.401	0.59f	20.00p	1,841.8	0.000	1.649	0.4432
2.568	0.75f	25.00p	1,841.8	0.000	1.500	0.5815
2.712	0.90f	30.00p	1,841.8	0.000	1.298	0.7040

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1468.45 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Res. Area Ratio from abs -1.8 deg to abs 1.8 > 1.000 1.033 P

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 24.766f TCG = 0.000 VCG = 3.846

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
2.036	0.24f	1.82s	1,841.8	0.000	-0.360	0.0000
2.039	0.24f	0.00	1,841.8	0.000	0.000	-0.0057
2.036	0.24f	1.84p	1,841.8	0.000	0.366	0.0002
2.025	0.25f	5.00p	1,841.8	0.000	0.983	0.0374
2.053	0.35f	10.00p	1,841.8	0.000	1.561	0.1509
2.211	0.46f	15.00p	1,841.8	0.000	1.693	0.2962
2.246	0.48f	15.88p	1,841.8	0.000	1.696	0.3223
2.401	0.59f	20.00p	1,841.8	0.000	1.649	0.4432
2.568	0.75f	25.00p	1,841.8	0.000	1.500	0.5815
2.712	0.90f	30.00p	1,841.8	0.000	1.298	0.7040
2.833	1.05f	35.00p	1,841.8	0.000	1.066	0.8073
2.930	1.20f	40.00p	1,841.8	0.000	0.816	0.8896
3.003	1.35f	45.00p	1,841.8	0.000	0.553	0.9494
3.052	1.49f	50.00p	1,841.8	0.000	0.282	0.9858
3.077	1.61f	55.00p	1,841.8	0.000	0.006	0.9984
3.078	1.62f	55.11p	1,841.8	0.000	0.000	0.9984
3.081	1.73f	60.00p	1,841.8	0.000	-0.271	0.9869

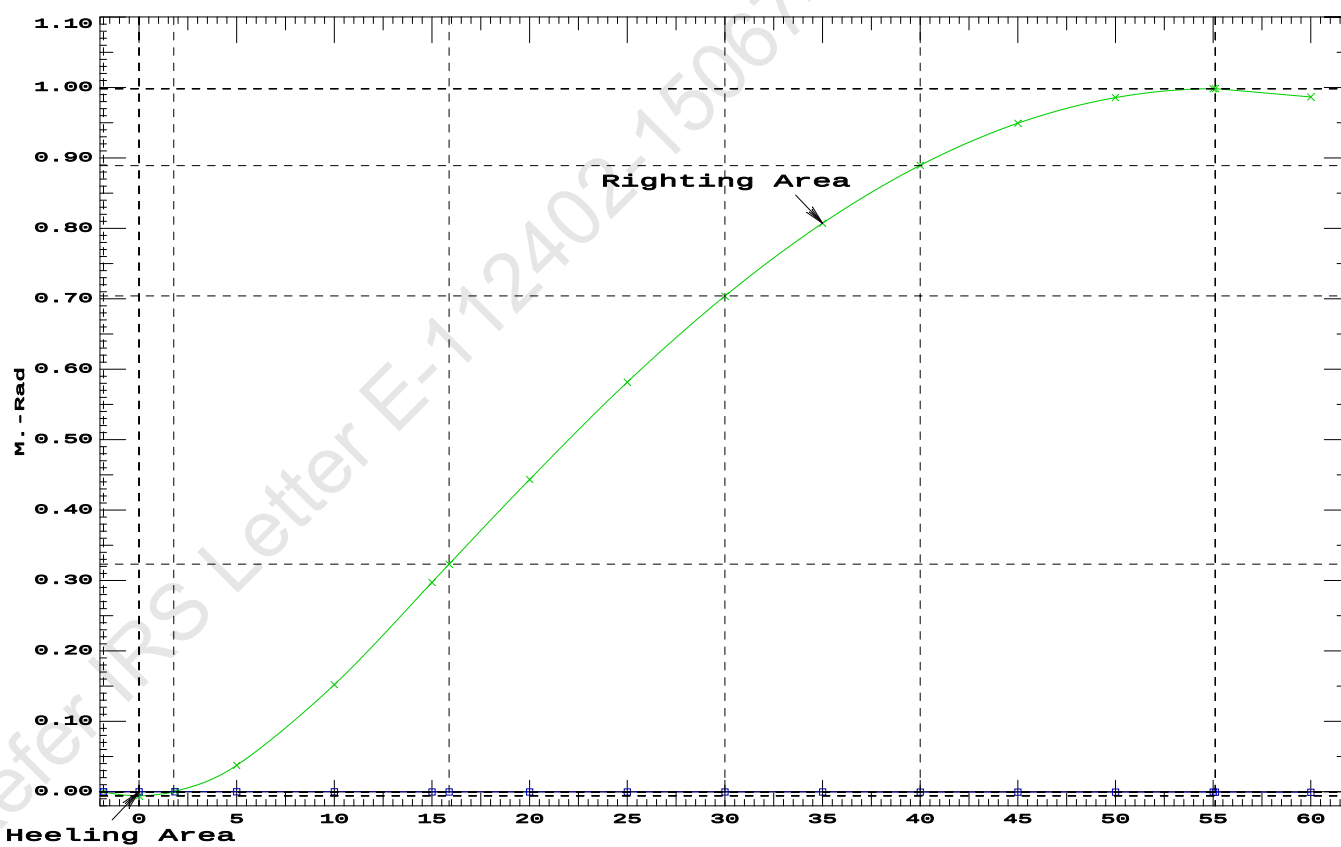
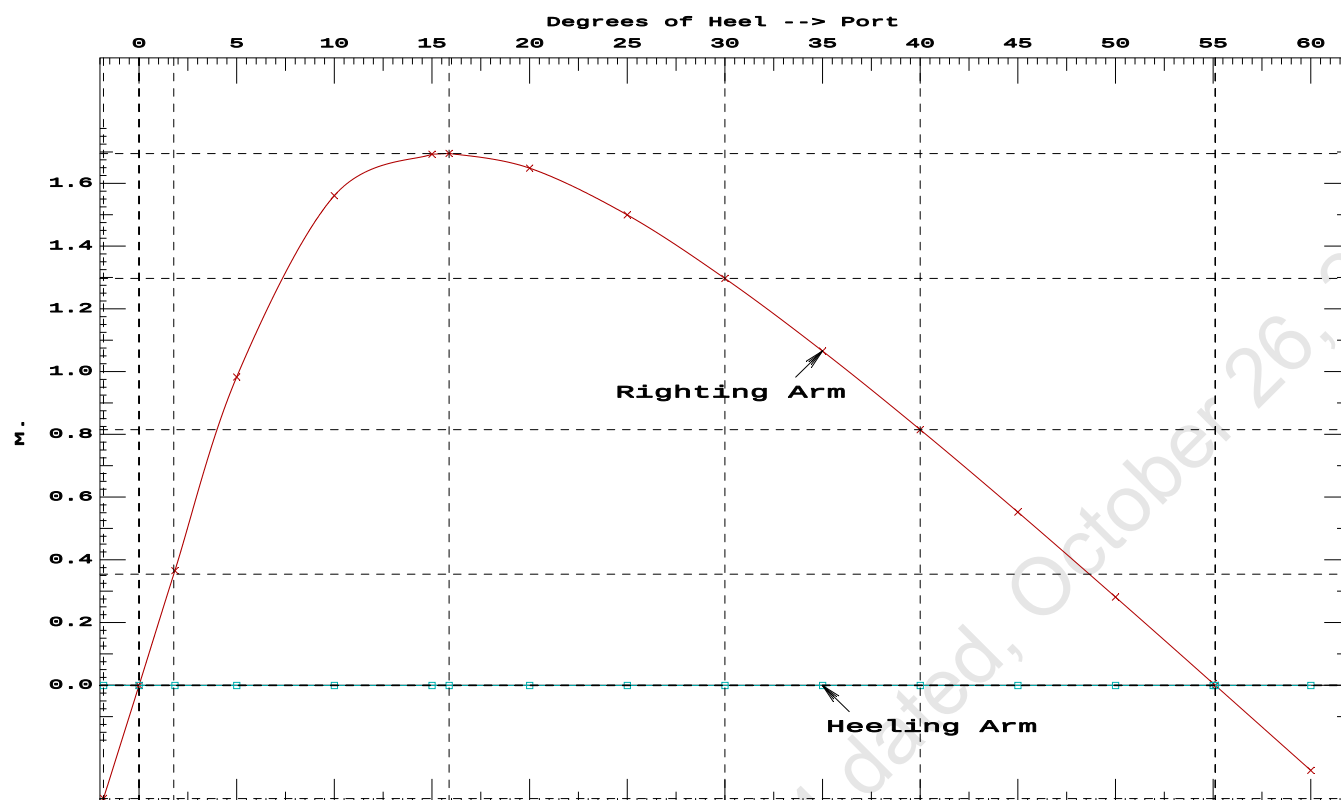
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1468.45 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Res. Area Ratio from abs -1.8 deg to RAZero > 1.000 176.213 P
(2) Angle from abs 1.8 deg to RAZero > 20.00 deg 53.27 P

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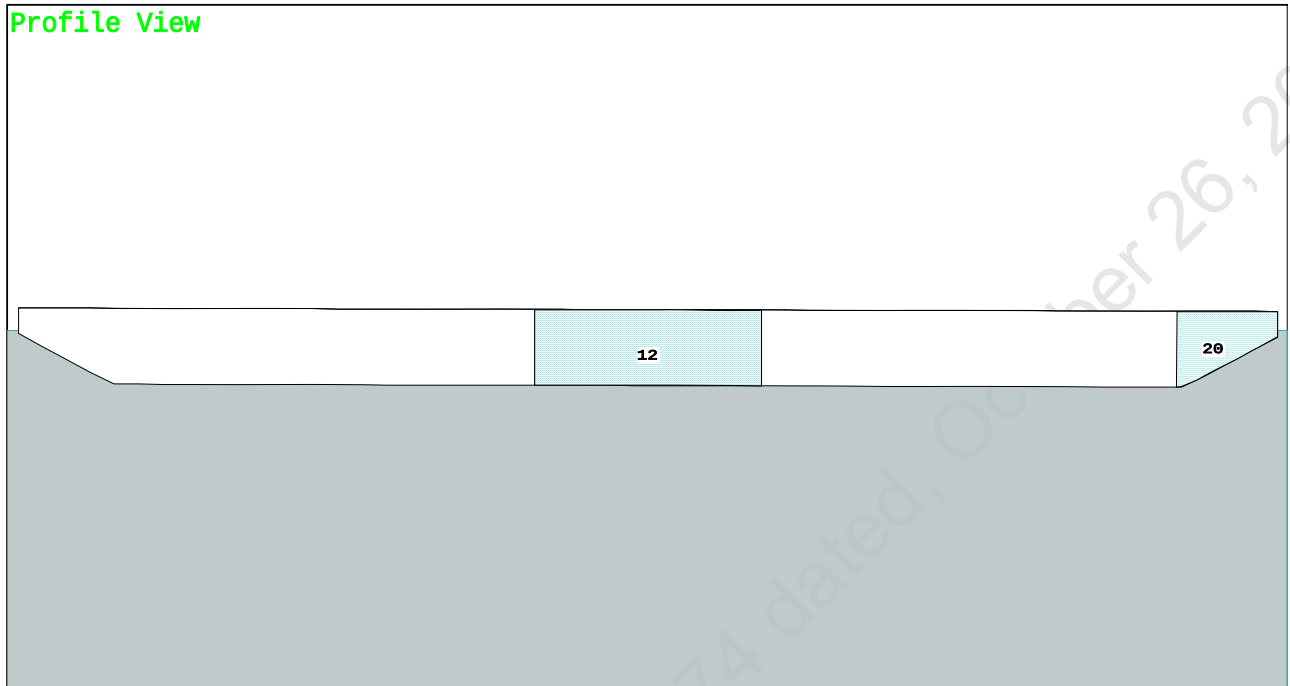


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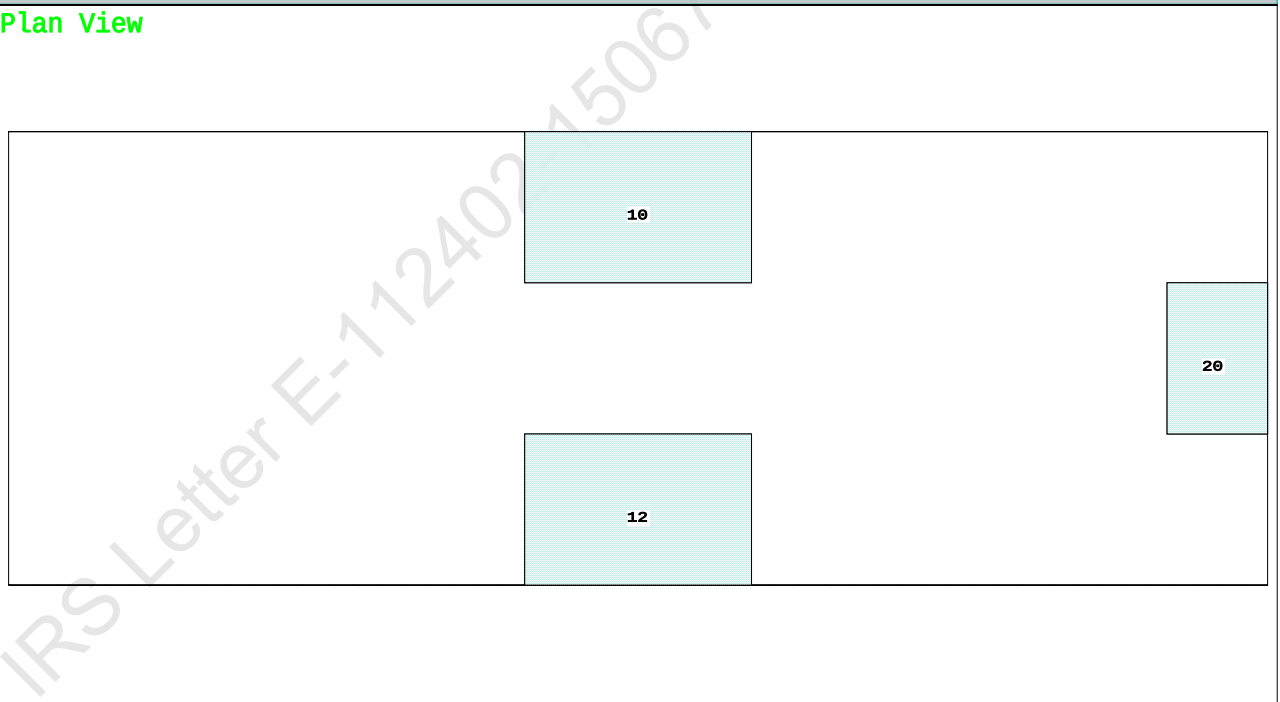
Condition 9 : Lightship + Load at Fwd + Ballast + Deck Cargo
Crane Lifting 51.3T @ 12m Radius Fwd

Condition Graphic - Draft: 2.280 @ 50.000f, 2.144 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

10 04_WBTK.P 12 04_WBTK.S 20 07_WBTK.C

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PONTOON

Condition 9 : Lightship + Load at Fwd + Ballast + Deck CargoCrane Lifting 51.3T @ 12m Radius Fwd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.280 @ 50.00f, 2.212 @ 25.00f, 2.144 @ 0.00

Trim: Fwd 0.137/50.000, Heel: zero

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	19.500f	0.000	33.218	
Deck Cargo			912.48	27.500f	0.000	4.000	
Total Fixed----->			1,519.75	24.588f	0.000	4.838	
	Load-----	SpGr-----	Weight(MT)---- <th>LCG-----</th> <th>TCG-----</th> <th>VCG-----</th> <th>FSM-----</th>	LCG-----	TCG-----	VCG-----	FSM-----
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p	1.509	162.73*
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s	1.509	162.73*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
Total Tanks----->			373.31	28.042f	0.000	1.558	397.78
Total Weight----->			1,893.06	25.269f	0.000	4.191	
			Displ(MT)---- <th>LCB-----</th> <th>TCB-----</th> <th>VCB-----</th> <td></td>	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,893.06	25.278f	0.000	1.132	

Righting Arms: 0.000 0.000

Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->	1.025	900.4	25.004f	0.000	101.45	12.967
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
		9.23	37.25	98.39	9.908	

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

364.2 Cu.M. (15%) SALT WATER

150.00 MT Crane

51.30 MT Crane Load

912.48 MT Deck Cargo

DRAFT AT FP (M) = 2.280
 DRAFT AT MS (M) = 2.212
 DRAFT AT AP (M) = 2.144
 DRAFT AT FWD MARK (M) = 2.269
 DRAFT AT AFT MARK (M) = 2.155

HYDROSTATIC PROPERTIES

Trim: Fwd 0.137/50.000, No Heel, VCG = 4.191

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight(MT)-----	LCB-----	VCB-----	cm-----
2.212	1,893.06	25.278f	1.132	9.23
				25.004f
				37.25
				98.39
				9.908

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
 Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 397.8 m.-MT

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PONTOON

Condition 9 : Lightship + Load at Fwd + Ballast + Deck Cargo
Crane Lifting 51.3T @ 12m Radius Fwd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 2.280 @ 50.00f, 2.212 @ 25.00f, 2.144 @ 0.00

Trim: Fwd 0.137/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	40.5	0.406	1.479	0.05500	3.30
CRANE	57.0	3.341	4.414	0.05500	13.84
DECKCARGO.C	29.4	1.793	2.867	0.05500	4.64
Total wind heeling moment to starboard-->					21.77
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.269f TCG = 0.000 VCG = 4.191

Free Surface Adjustment: 0.210

Adjusted CG: LCG = 25.269f TCG = 0.000 VCG = 4.401

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. 50% DeckImm Area	Angle
2.131	0.16f	0.00	1,893.1	0.000	-0.011	2.354
2.132	0.16f	0.06s	1,893.1	0.000	0.000	2.290
2.129	0.16f	2.35s	1,893.1	0.000	0.400	0.000
2.119	0.16f	5.06s	1,893.1	0.000	0.866	-2.710
2.182	0.24f	10.06s	1,893.1	0.000	1.265	-7.710
2.278	0.28f	12.76s	1,893.1	0.000	1.295	-10.407
2.377	0.31f	15.06s	1,893.1	0.000	1.273	-12.710
2.611	0.39f	20.06s	1,893.1	0.000	1.134	-17.710
2.828	0.49f	25.06s	1,893.1	0.000	0.905	-22.710
3.020	0.60f	30.06s	1,893.1	0.000	0.629	-27.710
3.188	0.70f	35.06s	1,893.1	0.000	0.330	-32.710
3.330	0.81f	40.06s	1,893.1	0.000	0.017	-37.710
3.337	0.81f	40.33s	1,893.1	0.000	0.000	-37.980
3.446	0.90f	45.06s	1,893.1	0.000	-0.302	-42.710
3.536	0.99f	50.06s	1,893.1	0.000	-0.622	-47.710
3.599	1.08f	55.06s	1,893.1	0.000	-0.940	-52.710
3.637	1.15f	60.06s	1,893.1	0.000	-1.253	-57.710
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.						

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 397.8 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.77 (constant)

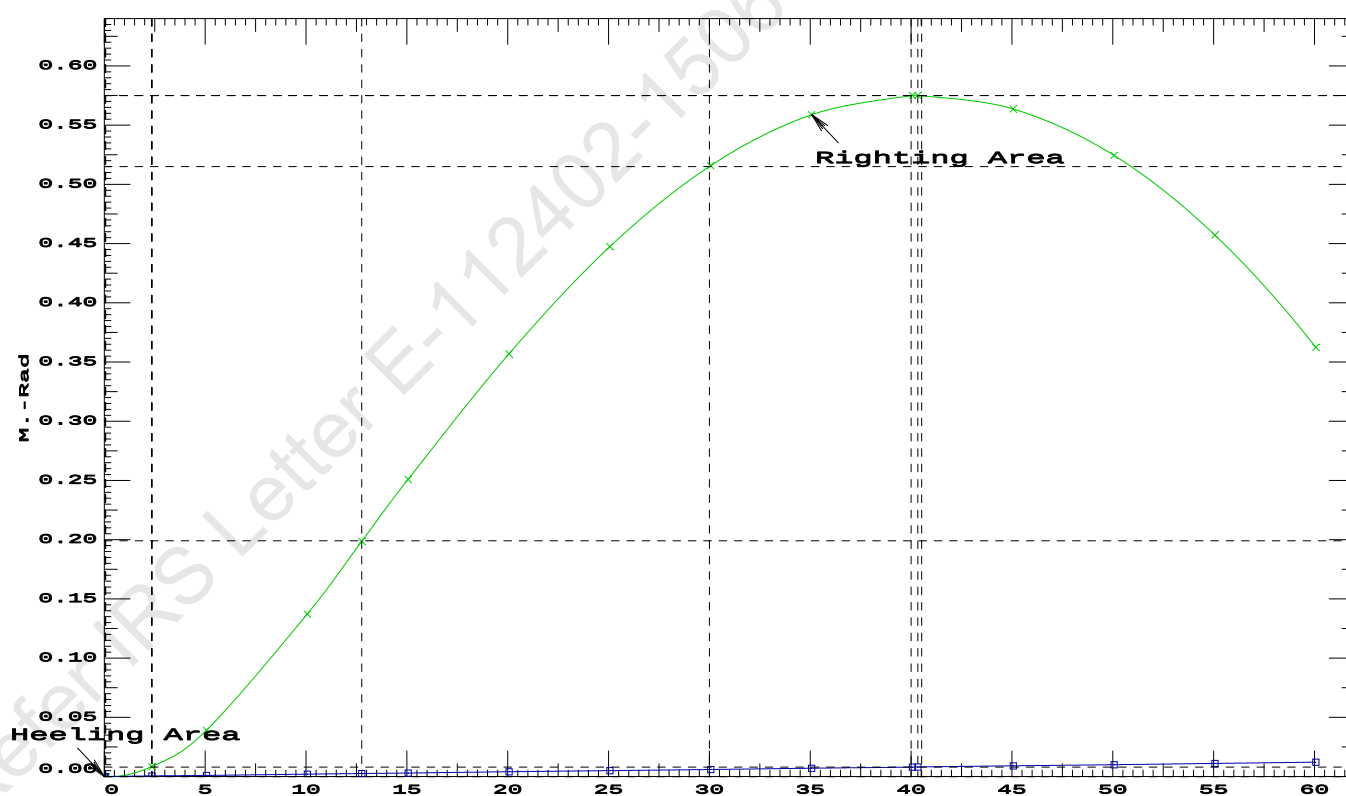
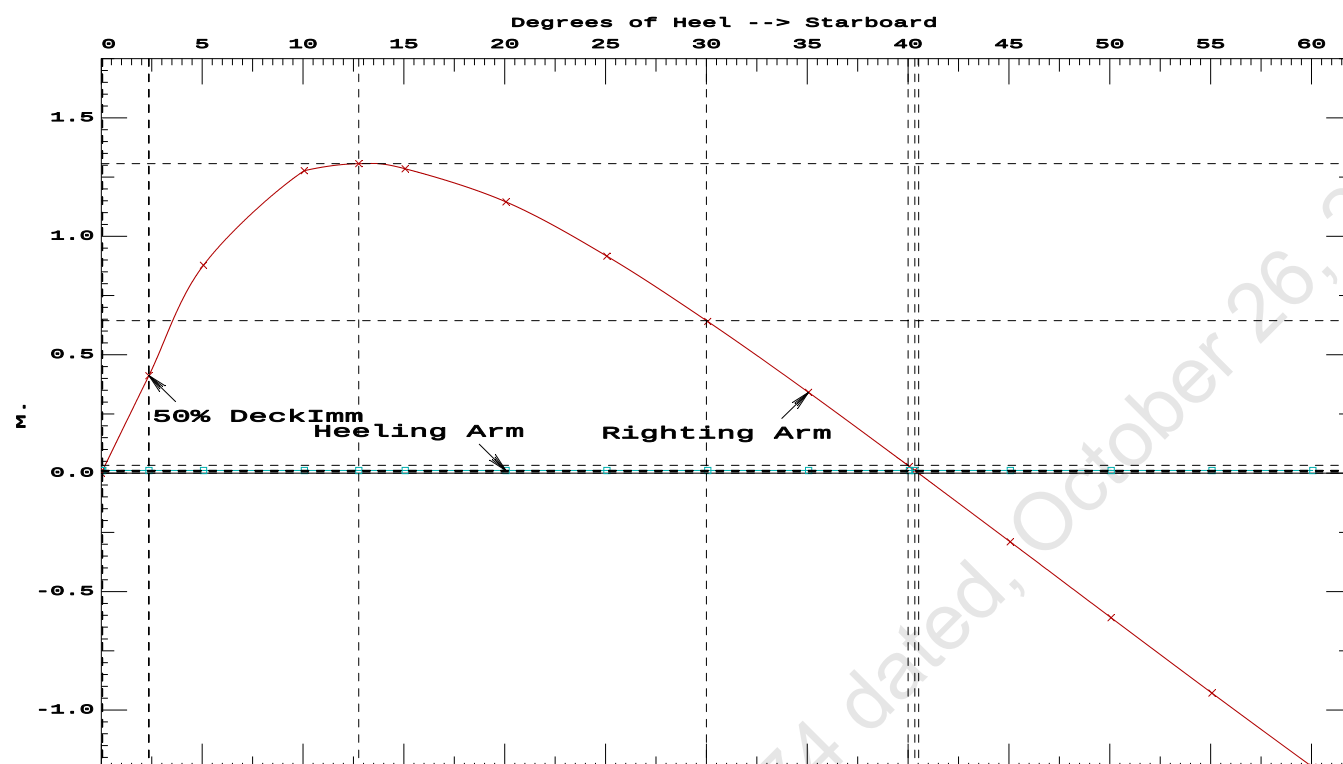
LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2-----Min/Max-----Attained

(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.1968 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	40.27 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	2.29 P

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GHS 17.22

PONT00N

Condition 9 : Lightship + Load at Fwd + Ballast + Deck CargoCrane Lifting 51.3T @ 12m Radius Fwd

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GHS 17.22 PONT00N

Condition 9 : Lightship + Load at Fwd + Ballast + Deck Cargo
Crane Lifting 51.3T @ 12m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.269f TCG = 0.000 VCG = 4.191

Free Surface Adjustment: 0.210

Adjusted CG: LCG = 25.269f TCG = 0.000 VCG = 4.401

Origin Depth---	Degrees of Trim----	Heel----	Displacement Weight(MT)---	Residual Arms in Trim--	in Heel---	Res. Area
2.131	0.16f	0.00	1,893.1	0.000	0.000	0.0000
2.119	0.16f	5.00s	1,893.1	0.000	0.867	0.0378
2.180	0.24f	10.00s	1,893.1	0.000	1.275	0.1364
2.278	0.28f	12.75s	1,893.1	0.000	1.306	0.1988
2.374	0.31f	15.00s	1,893.1	0.000	1.285	0.2497
2.609	0.39f	20.00s	1,893.1	0.000	1.147	0.3575
2.825	0.49f	25.00s	1,893.1	0.000	0.919	0.4483
3.018	0.60f	30.00s	1,893.1	0.000	0.644	0.5168
3.186	0.70f	35.00s	1,893.1	0.000	0.344	0.5601
3.329	0.80f	40.00s	1,893.1	0.000	0.032	0.5766
3.341	0.81f	40.50s	1,893.1	0.000	0.000	0.5767
3.445	0.90f	45.00s	1,893.1	0.000	-0.287	0.5655
3.535	0.99f	50.00s	1,893.1	0.000	-0.607	0.5265
3.599	1.08f	55.00s	1,893.1	0.000	-0.925	0.4596
3.636	1.15f	60.00s	1,893.1	0.000	-1.238	0.3652

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 397.8 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

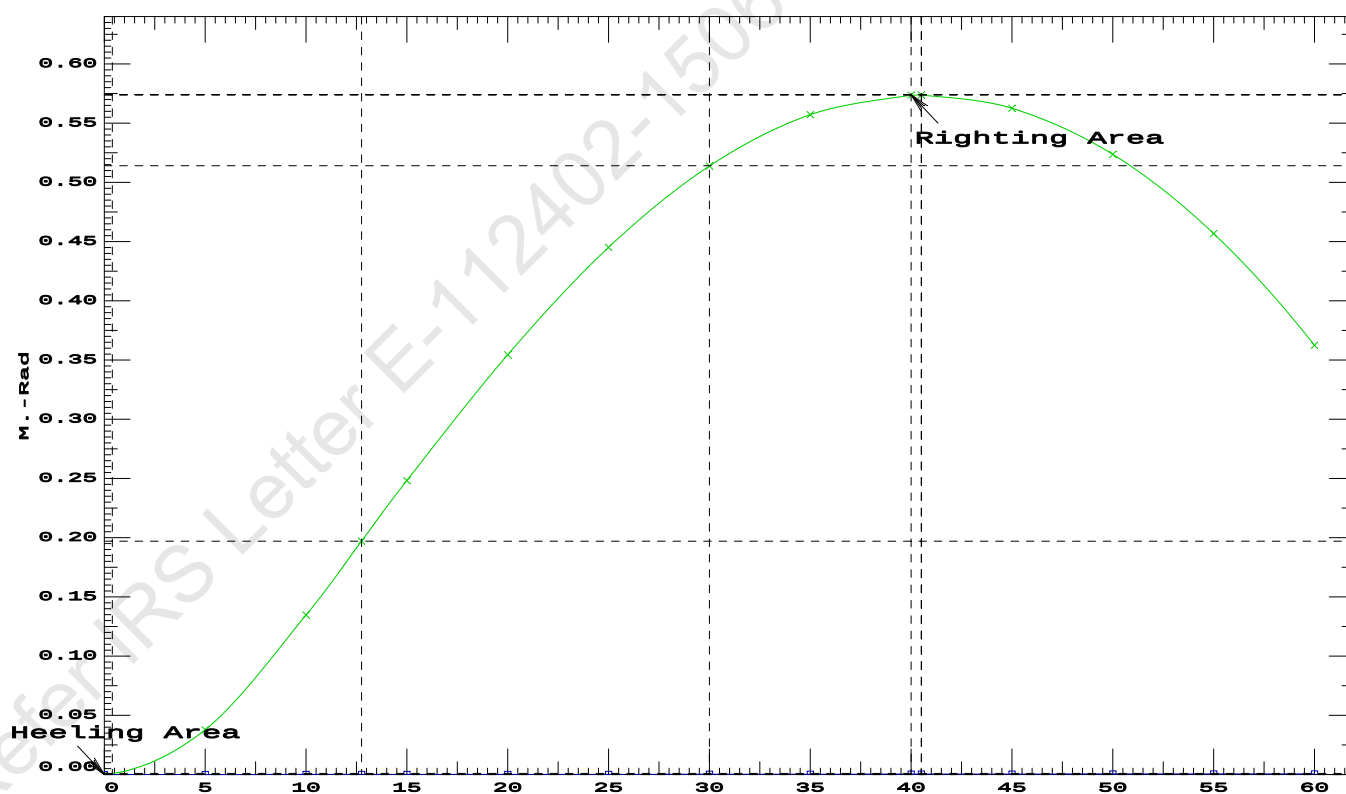
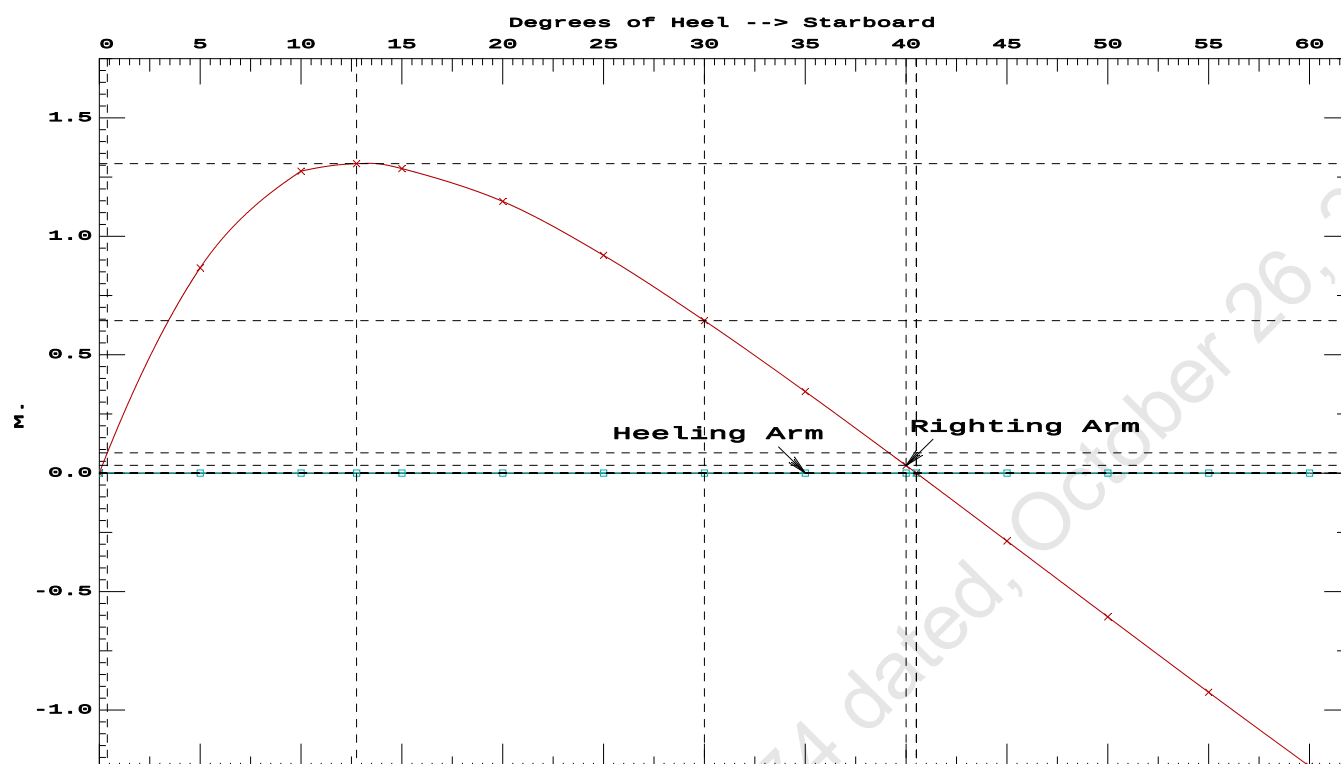
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	0.00 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	247243 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	1545.49 P

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PONTOON

Condition 9 : Lightship + Load at Fwd + Ballast + Deck CargoCrane Lifting 51.3T @ 12m Radius Fwd

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GHS 17.22 PONTON

Condition 9 : Lightship + Load at Fwd + Ballast + Deck Cargo**Crane Lifting 51.3T @ 12m Radius Fwd**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.280 @ 50.00f, 2.212 @ 25.00f, 2.144 @ 0.00

Trim: Fwd 0.137/50.000, Heel: 0.00 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	19.500f	0.000	33.218	
Deck Cargo			912.48	27.500f	0.000	4.000	
Total Fixed----->			1,519.75	24.588f	0.000	4.838	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p	1.509	162.73*
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s	1.509	162.73*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
Total Tanks----->			373.31	28.042f	0.000	1.558	397.78
Total Weight----->			1,893.06	25.269f	0.000	4.191	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,893.06	25.278f	0.001	1.132	

	Righting Arms:			0.000	0.001s		
	External Arms:			0.000	0.001		
	Residual Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		900.4	25.004f	0.000	101.45	12.967
			MT/cm-----	m.-MT/cm-----	MT/cm-----	GML-----	GMT-----
			9.23		37.25	98.39	9.908
Distances in METERS.-----				-----Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

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After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 2.262 @ 50.00f, 2.156 @ 25.00f, 2.051 @ 0.00
Trim: Fwd 0.211/50.000, Heel: 0.00 deg.

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----			
LIGHT SHIP	405.97	25.000f	0.000	3.000			
Crane	150.00	7.500f	0.000	5.200			
Deck Cargo	912.48	27.500f	0.000	4.000			
Total Fixed----->	1,468.45	24.766f	0.000	3.846			
Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----	
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p	1.509	162.73*
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s	1.509	162.73*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
Total Tanks----->			373.31	28.042f	0.000	1.558	397.78
Total Weight----->			1,841.76	25.430f	0.000	3.382	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL	1.025		1,841.76	25.440f	0.001	1.104	

Righting Arms:			0.000	0.001s			
External Arms:			0.000	0.001			
Residual Righting Arms:			0.000	0.000			
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----	
Total Waterplane---->	1.025	900.3	25.006f	0.000	104.24	13.324	
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----		
		9.23	37.56	101.96	11.045		
Distances in METERS.-----Moments in m.-MT.							

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: INFINITY; t3: 30; t: 15.00
limmarg: INFINITY; t3: 15; t: 7.50
limmarg: INFINITY; t3: 8; t: 3.75
limmarg: INFINITY; t3: 4; t: 1.88
limmarg: INFINITY; t3: 2; t: 0.94
limmarg: INFINITY; t3: 1; t: 0.47
limmarg: INFINITY; t3: 1; t: 0.23
limmarg: INFINITY; t3: 1; t: 0.12
limmarg: INFINITY; t3: 1; t: 0.06
limmarg: INFINITY; t3: 1; t: 0.03
limmarg: INFINITY; t3: 1; t: 0.01

Theta3 is 0.97 degrees, as demonstrated below.

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 24.766f TCG = 0.000 VCG = 3.846

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
2.039	0.24f	0.00	1,841.8	0.000	0.000	0.0000
2.038	0.24f	0.97s	1,841.8	0.000	0.193	0.0016
2.025	0.25f	5.00s	1,841.8	0.000	0.983	0.0430
2.053	0.35f	10.00s	1,841.8	0.000	1.560	0.1595
2.211	0.46f	15.00s	1,841.8	0.000	1.692	0.3047
2.246	0.48f	15.88s	1,841.8	0.000	1.695	0.3306
2.401	0.59f	20.00s	1,841.8	0.000	1.648	0.4515
2.568	0.75f	25.00s	1,841.8	0.000	1.499	0.5898
2.712	0.90f	30.00s	1,841.8	0.000	1.296	0.7121

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1468.45 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 1 > 1.000 LARGE

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GHS 17.22 PONTON

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 24.766f TCG = 0.000 VCG = 3.846

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
2.039	0.24f	0.00	1,841.8	0.000	0.000	0.0000
2.038	0.24f	0.97s	1,841.8	0.000	0.193	0.0016
2.025	0.25f	5.00s	1,841.8	0.000	0.983	0.0430
2.053	0.35f	10.00s	1,841.8	0.000	1.560	0.1595
2.211	0.46f	15.00s	1,841.8	0.000	1.692	0.3047
2.246	0.48f	15.88s	1,841.8	0.000	1.695	0.3306
2.401	0.59f	20.00s	1,841.8	0.000	1.648	0.4515
2.568	0.75f	25.00s	1,841.8	0.000	1.499	0.5898
2.712	0.90f	30.00s	1,841.8	0.000	1.296	0.7121
2.833	1.05f	35.00s	1,841.8	0.000	1.065	0.8154
2.930	1.20f	40.00s	1,841.8	0.000	0.814	0.8975
3.003	1.35f	45.00s	1,841.8	0.000	0.551	0.9572
3.052	1.49f	50.00s	1,841.8	0.000	0.280	0.9935
3.077	1.61f	55.00s	1,841.8	0.000	0.005	1.0060
3.078	1.62f	55.09s	1,841.8	0.000	0.000	1.0060
3.081	1.73f	60.00s	1,841.8	0.000	-0.272	0.9944

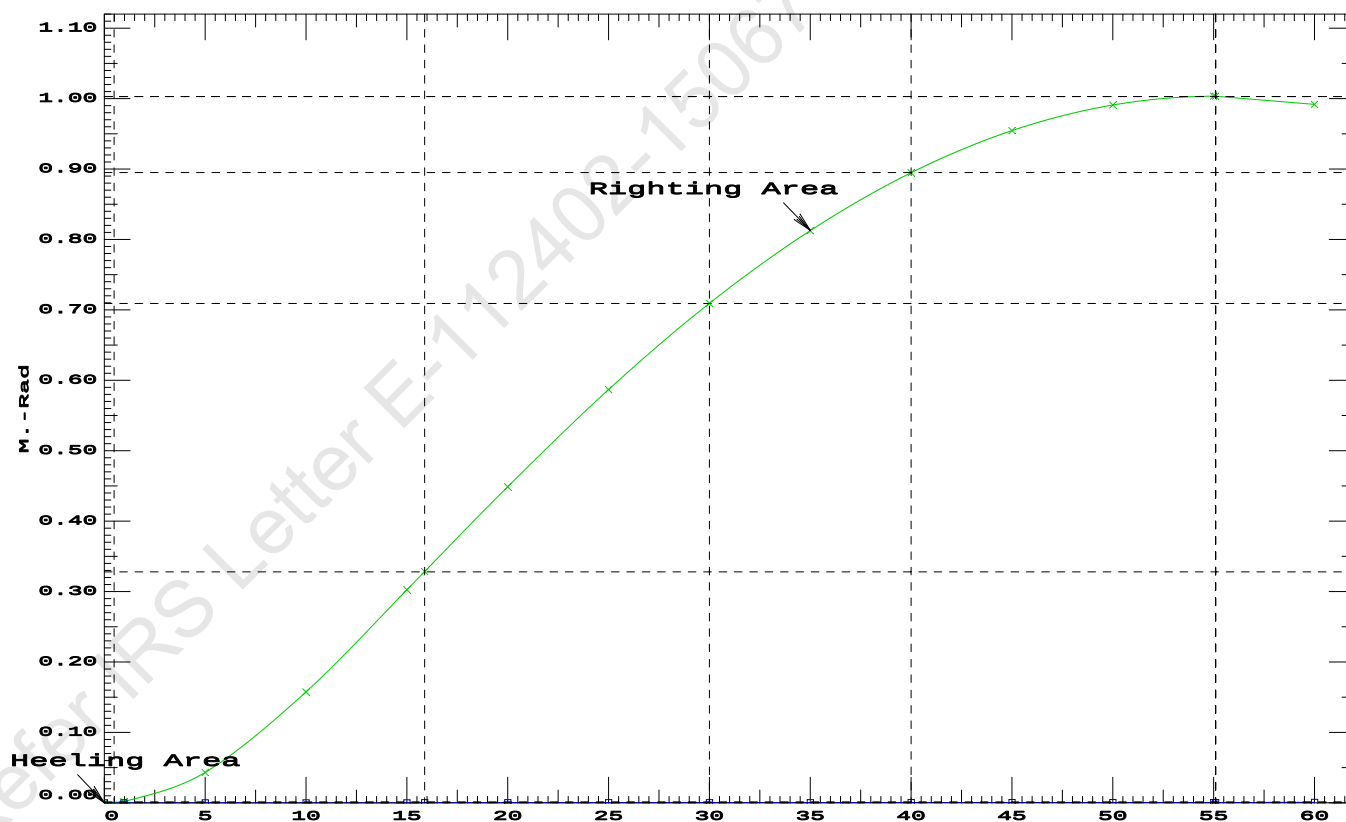
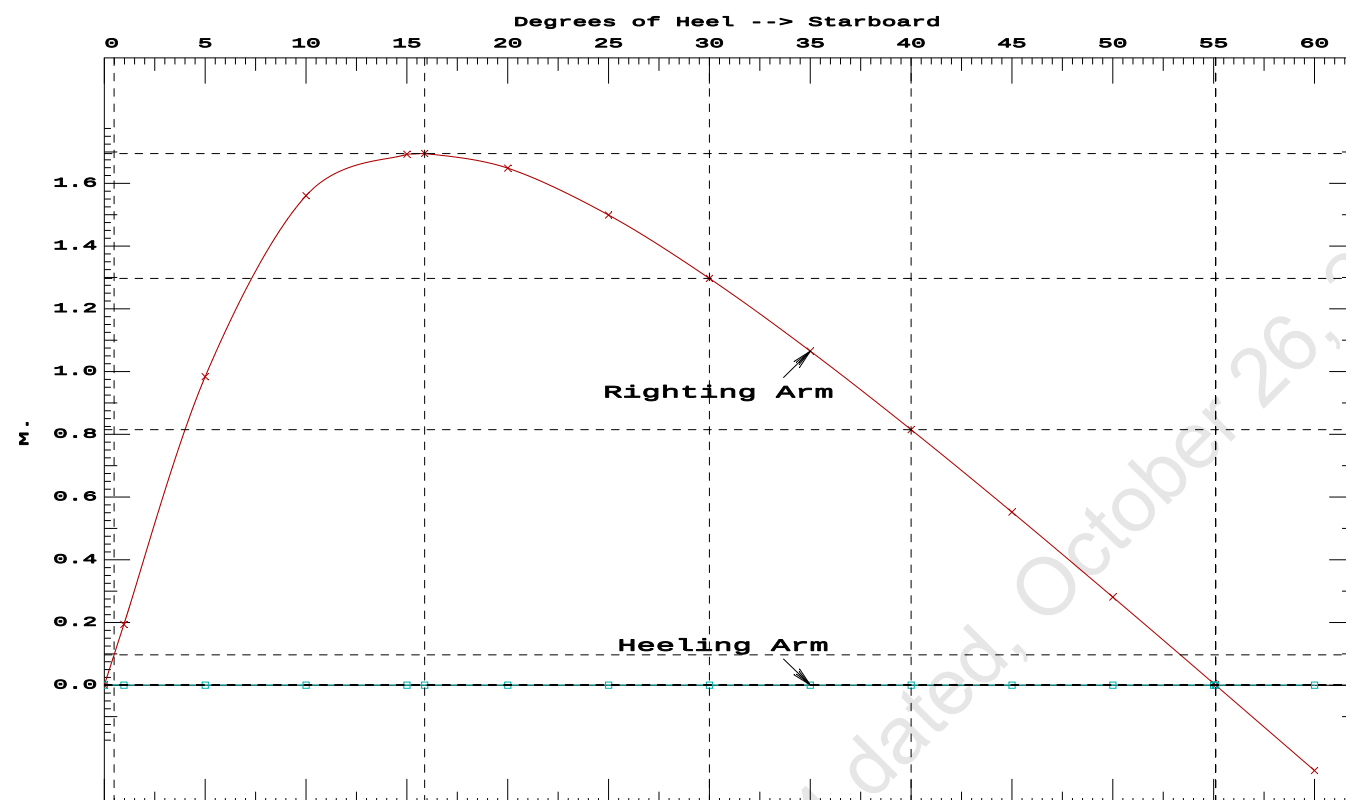
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1468.45 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero > 1.000 LARGE
(2) Angle from abs 1 deg to RAZero > 20.00 deg 54.12 P

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GHS 17.22 PONT00N



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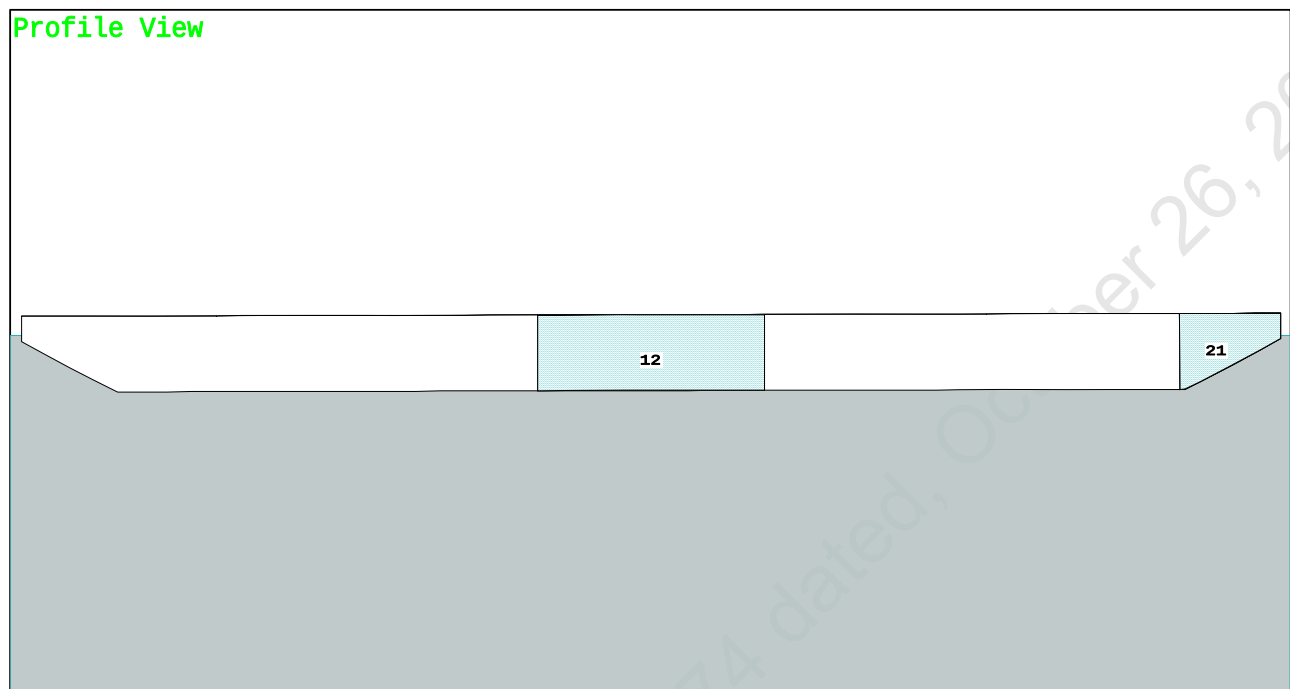
GHS 17.22

PONT00N

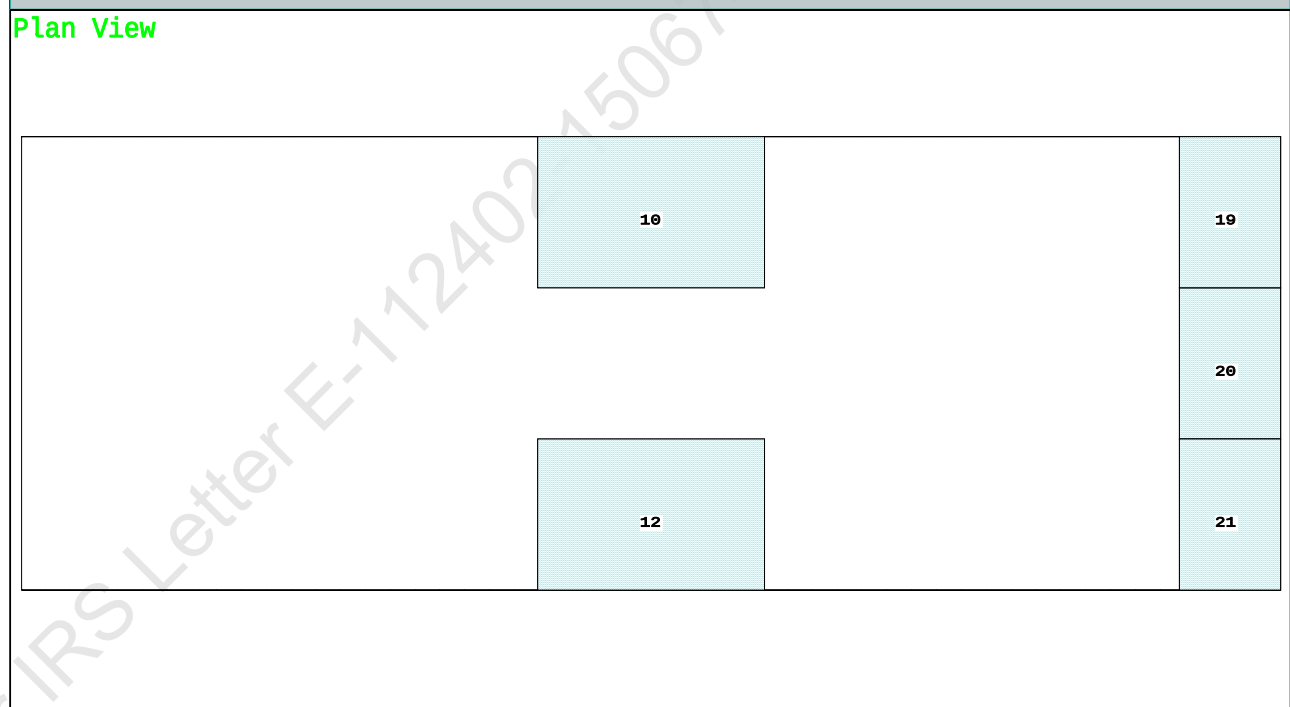
Condition 10 : Lightship + Load at Aft + Ballast + Deck CargoCrane Lifting 36.6T @ 18m Radius Aft

Condition Graphic - Draft: 2.148 @ 50.000f, 2.276 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

10 04_WBTK.P 12 04_WBTK.S 19 07_WBTK.P 20 07_WBTK.C 21 07_WBTK.S

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GHS 17.22

PONTOON

Condition 10 : Lightship + Load at Aft + Ballast + Deck CargoCrane Lifting 36.6T @ 18m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.148 @ 50.00f, 2.212 @ 25.00f, 2.276 @ 0.00

Trim: Aft 0.127/50.000, Heel: zero

Part-----	Weight(MT)-----	LCG-----	TCG-----	VCG-----	FSM-----
LIGHT SHIP	405.97	25.000f	0.000	3.000	
Crane	250.00	7.500f	0.000	5.200	
Crane Load	36.60	10.500a	0.000	76.977	
Deck Cargo	727.97	27.500f	0.000	4.000	
Total Fixed----->	1,420.54	22.287f	0.000	5.806	
Load-----	SpGr-----	Weight(MT)-----	LCG-----	TCG-----	VCG-----
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s
Total Tanks----->		472.52	32.161f	0.000	1.626
Total Weight----->		1,893.06	24.751f	0.000	4.762
		Displ(MT)-----	LCB-----	TCB-----	VCB-----
HULL	1.025	1,893.06	24.742f	0.000	1.132

Righting Arms:		0.000	0.000		
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----
Total Waterplane----->	1.025	900.4	24.997f	0.000	101.45
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----
		9.23	37.04	97.82	9.260
Distances in METERS.-----			Moments in m.-MT.		

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

461.0 Cu.M. (18%) SALT WATER

250.00 MT Crane
727.97 MT Deck Cargo

36.60 MT Crane Load

DRAFT AT FP (M) = 2.148
DRAFT AT MS (M) = 2.212
DRAFT AT AP (M) = 2.276
DRAFT AT FWD MARK (M) = 2.159
DRAFT AT AFT MARK (M) = 2.265

HYDROSTATIC PROPERTIES

Trim: Aft 0.127/50.000, No Heel, VCG = 4.762

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/				
Draft----	Weight(MT)-----	LCB-----	VCB-----	cm-----	LCF-----	cm trim-----	GML-----	GMT-----
2.212	1,893.06	24.742f	1.132	9.23	24.997f	37.04	97.82	9.260
Distances in METERS.-----					Specific Gravity = 1.025.-----			
					Moment in m.-MT.			
					Trim is per 50.00m.			

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 542.4 m.-MT

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GHS 17.22

PONTOON**Condition 10 : Lightship + Load at Aft + Ballast + Deck Cargo****Crane Lifting 36.6T @ 18m Radius Aft****HEELING MOMENT specification****Lateral Plane Method****Wind pressure toward starboard**

USK draft: 2.148 @ 50.00f, 2.212 @ 25.00f, 2.276 @ 0.00

Trim: Aft 0.127/50.000, Heel: zero

Part-----	LPA*SF-----	HCP-----	Arm----	Pressure-----	Moment
HULL	40.5	0.406	1.479	0.05500	3.30
CRANE	57.0	3.262	4.336	0.05500	13.59
DECKCARGO.C	29.4	1.806	2.880	0.05500	4.66
Total wind heeling moment to starboard----->					21.54
Distances in METERS.-----Pressure in MT/Sqm.-----					Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.751f TCG = 0.000 VCG = 4.762

Free Surface Adjustment: 0.287

Adjusted CG: LCG = 24.752f TCG = 0.000 VCG = 5.049

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm	
Depth---	Trim---	Heel---	Weight(MT)---	Area---	
			in Trim--in Heel-->	Angle	
2.264	0.15a	0.00	1,893.1	0.0000	2.369
2.264	0.15a	0.07s	1,893.1	0.0000	2.302
2.262	0.15a	2.37s	1,893.1	0.0076	0.000
2.254	0.15a	5.07s	1,893.1	0.0355	-2.698
2.382	0.22a	10.07s	1,893.1	0.1262	-7.698
2.443	0.24a	11.42s	1,893.1	0.1536	-9.055
2.638	0.29a	15.07s	1,893.1	0.2266	-12.698
2.942	0.37a	20.07s	1,893.1	0.3163	-17.698
3.246	0.46a	25.07s	1,893.1	0.3843	-22.698
3.528	0.56a	30.07s	1,893.1	0.4254	-27.698
3.756	0.65a	34.47s	1,893.1	0.4373	-32.099
3.785	0.67a	35.07s	1,893.1	0.4370	-32.698
4.015	0.76a	40.07s	1,893.1	0.4179	-37.698
4.214	0.86a	45.07s	1,893.1	0.3673	-42.698
4.382	0.94a	50.07s	1,893.1	0.2853	-47.698
4.514	1.02a	55.07s	1,893.1	0.1723	-52.698
4.611	1.08a	60.07s	1,893.1	0.0289	-57.698
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.					

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 542.4 m.-MT was applied to artificially modify the CG.

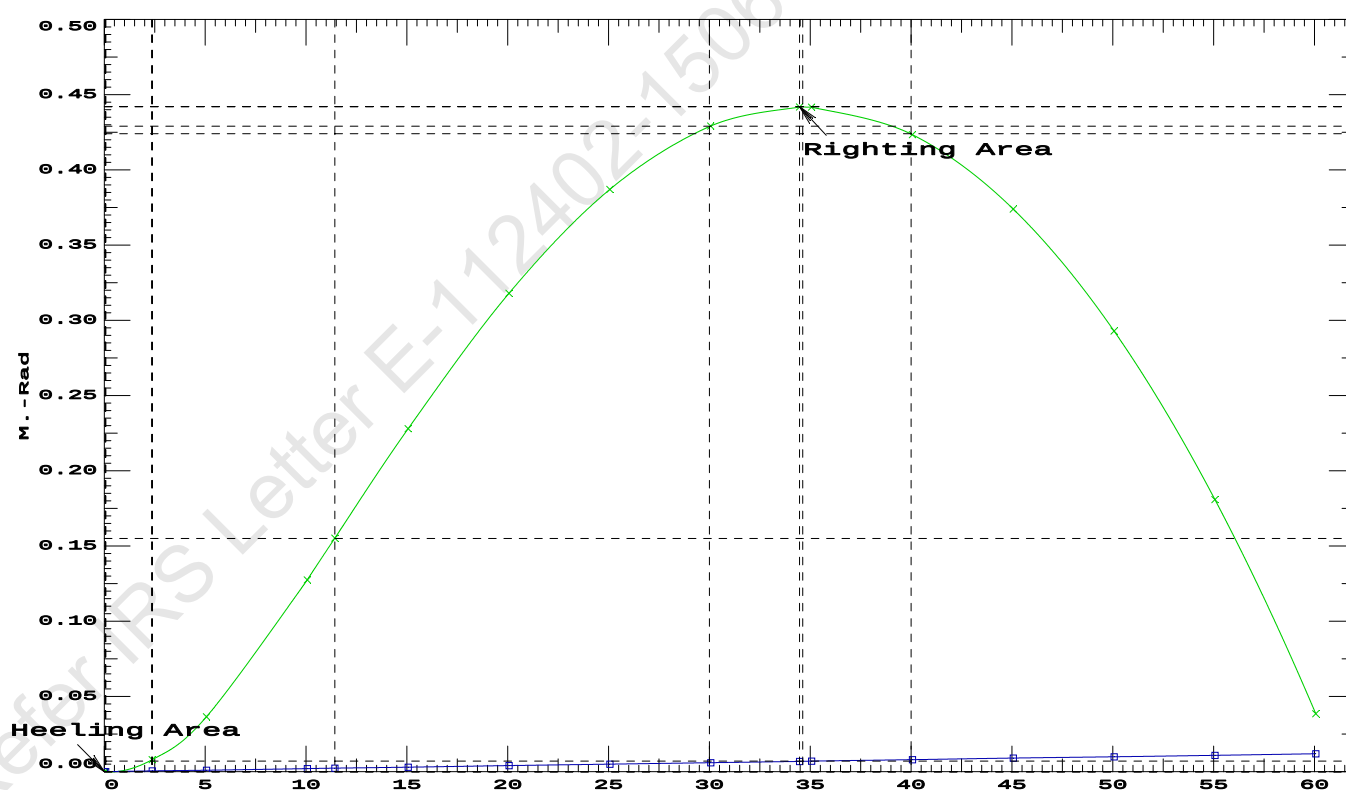
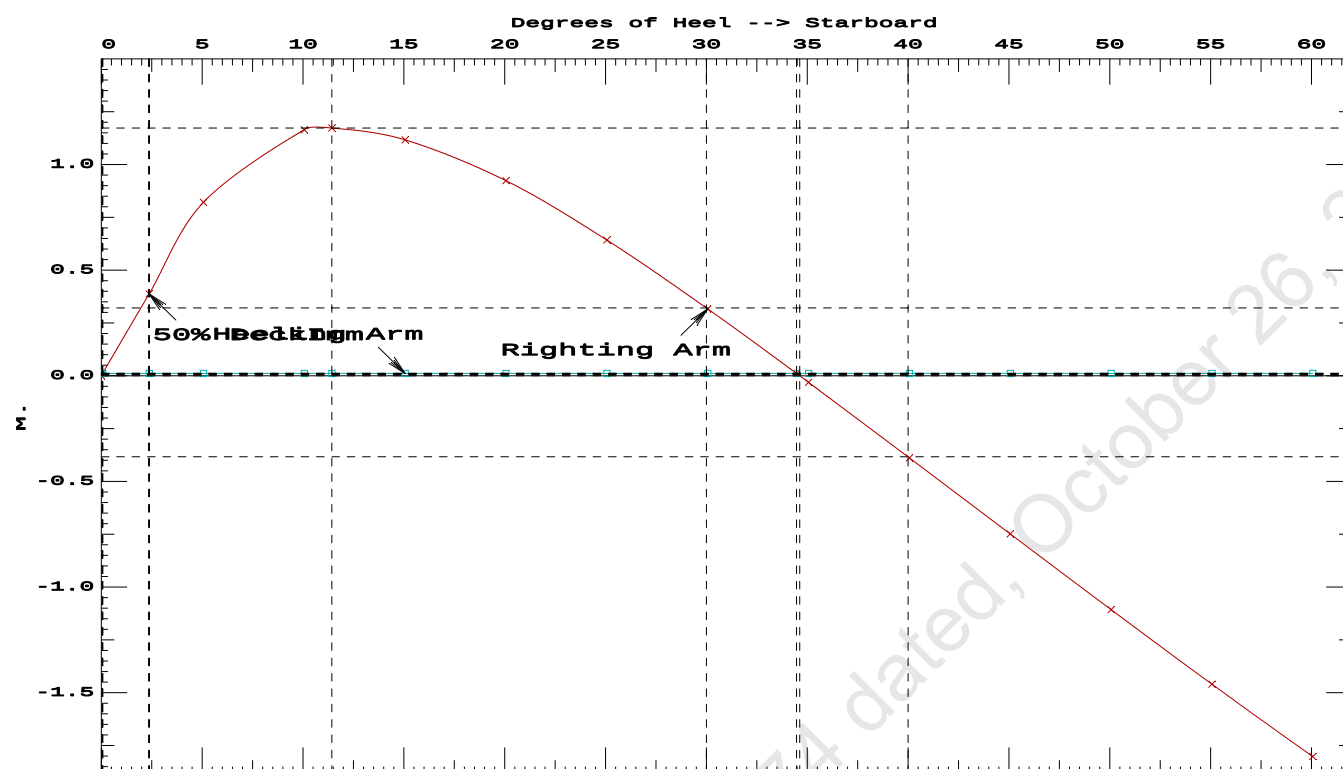
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.54 (constant)

LIM-----	IS CODE 2008 PART B CHAPTER-2 SEC.2----	Min/Max-----	Attained
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.1536 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	34.40 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	2.30 P

22-10-20 23:40:00 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONT00N

Condition 10 : Lightship + Load at Aft + Ballast + Deck CargoCrane Lifting 36.6T @ 18m Radius Aft

22-10-20 23:40:00 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON**Condition 10 : Lightship + Load at Aft + Ballast + Deck Cargo****Crane Lifting 36.6T @ 18m Radius Aft****RESIDUAL RIGHTING ARMS vs HEEL ANGLE**

Total CG: LCG = 24.751f TCG = 0.000 VCG = 4.762

Free Surface Adjustment: 0.287

Adjusted CG: LCG = 24.752f TCG = 0.000 VCG = 5.049

Origin Depth---	Degrees of Trim----	Heel----	Displacement Weight(MT)---	Residual Arms		Res. Area
				in Trim--	in Heel---	
2.264	0.15a	0.00s	1,893.1	0.000	0.000	0.0000
2.254	0.15a	5.00s	1,893.1	0.000	0.810	0.0354
2.379	0.22a	10.00s	1,893.1	0.000	1.162	0.1265
2.444	0.24a	11.43s	1,893.1	0.000	1.172	0.1556
2.634	0.29a	15.00s	1,893.1	0.000	1.118	0.2280
2.938	0.36a	20.00s	1,893.1	0.000	0.926	0.3188
3.242	0.46a	25.00s	1,893.1	0.000	0.645	0.3880
3.525	0.56a	30.00s	1,893.1	0.000	0.320	0.4304
3.763	0.66a	34.62s	1,893.1	0.000	0.000	0.4434
3.782	0.66a	35.00s	1,893.1	0.000	-0.027	0.4433
4.012	0.76a	40.00s	1,893.1	0.000	-0.384	0.4255
4.212	0.86a	45.00s	1,893.1	0.000	-0.745	0.3762
4.380	0.94a	50.00s	1,893.1	0.000	-1.103	0.2956
4.513	1.02a	55.00s	1,893.1	0.000	-1.456	0.1839
4.610	1.08a	60.00s	1,893.1	0.000	-1.799	0.0418

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 542.4 m.-MT was applied to artificially modify the CG.

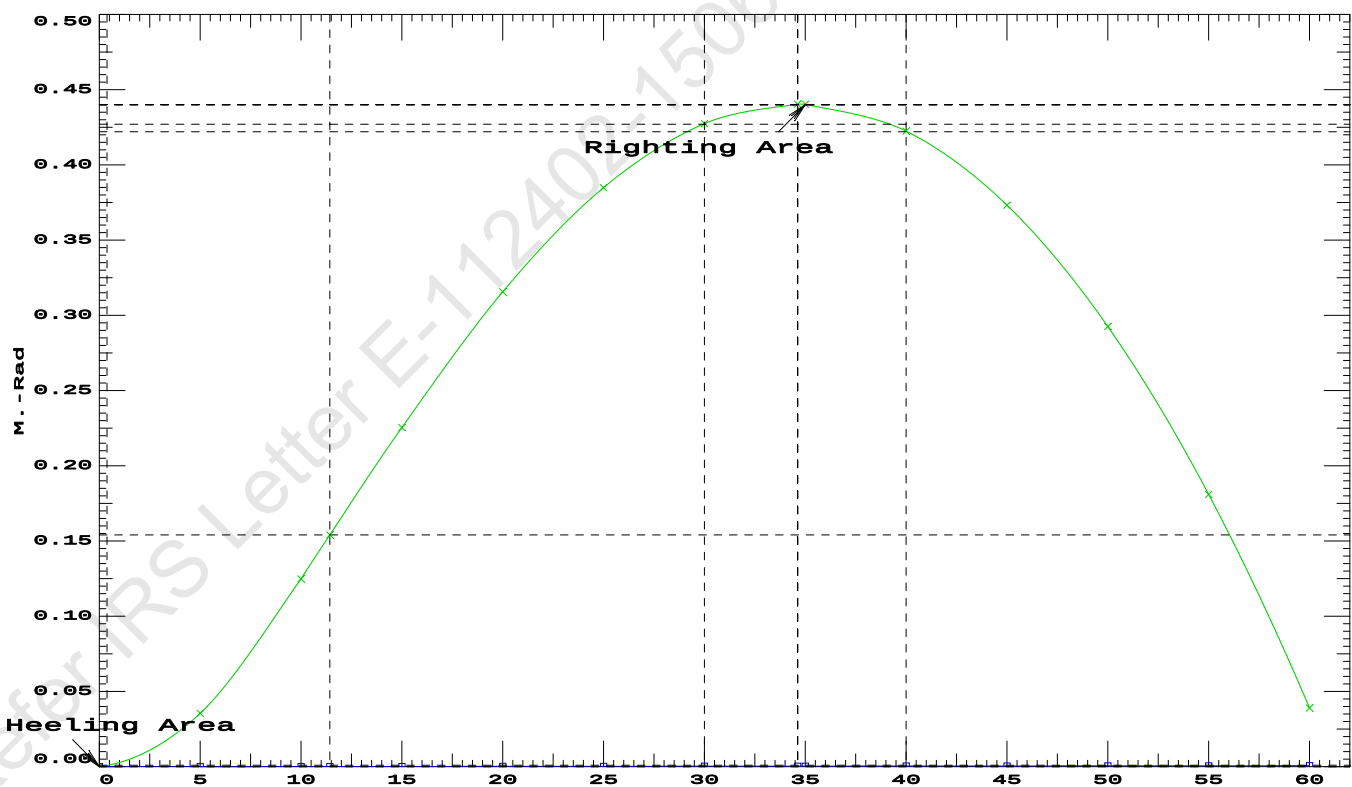
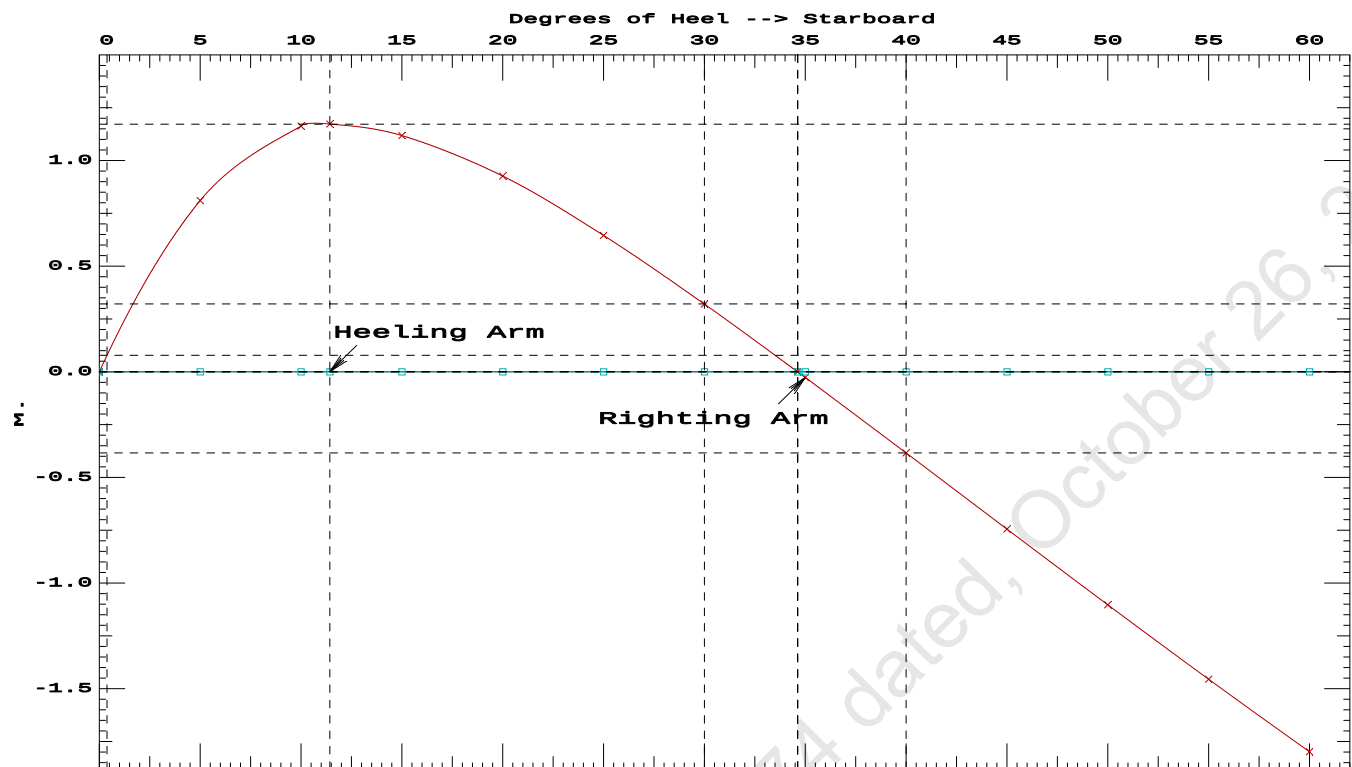
Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	0.00 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	221859 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	1390.46 P

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Condition 10 : Lightship + Load at Aft + Ballast + Deck Cargo
Crane Lifting 36.6T @ 18m Radius Aft



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Condition 10 : Lightship + Load at Aft + Ballast + Deck Cargo**Crane Lifting 36.6T @ 18m Radius Aft**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.148 @ 50.00f, 2.212 @ 25.00f, 2.276 @ 0.00

Trim: Aft 0.127/50.000, Heel: 0.00 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	10.500a	0.000	76.977	
Deck Cargo			727.97	27.500f	0.000	4.000	
Total Fixed----->			1,420.54	22.287f	0.000	5.806	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p	1.509	162.73*
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			472.52	32.161f	0.000	1.626	542.42
Total Weight----->			1,893.06	24.751f	0.000	4.762	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,893.06	24.742f	0.001	1.132	

	Righting Arms:			0.000	0.001s		
	External Arms:			0.000	0.001		
	Residual Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->		1.025	900.4	24.997f	0.000	101.45	12.891
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			9.23		37.04	97.82	9.260
Distances in METERS.-----				Moments in m.-MT			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

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After hook load drop							
WEIGHT and DISPLACEMENT and WATERPLANE STATUS							
USK draft: 2.283 @ 50.00f, 2.172 @ 25.00f, 2.062 @ 0.00							
Trim: Fwd 0.221/50.000, Heel: 0.00 deg.							
Part-----			Weight(MT)----	LCG----	TCG----	VCG	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Deck Cargo			727.97	27.500f	0.000	4.000	
Total Fixed----->			1,383.94	23.154f	0.000	3.923	
	Load----	SpGr-----	Weight(MT)----	LCG----	TCG----	VCG-----	FSM
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p	1.509	162.73*
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			472.52	32.161f	0.000	1.626	542.42
Total Weight----->			1,856.46	25.446f	0.000	3.339	
			Displ(MT)----	LCB----	TCB----	VCB	
HULL	1.025		1,856.46	25.456f	0.001	1.112	

Righting Arms:				0.000	0.001s		
External Arms:				0.000	0.001		
Residual Righting Arms:				0.000	0.000		
Part-----	SpGr-----		WPA-----	LCF-----	TCF-----	BML-----	BMT
Total Waterplane---->	1.025		900.4	25.006f	0.000	103.43	13.142
			MT/cm-----	m.-MT/cm-----	GML-----	GMT	
			9.23		37.58	101.20	10.915
Distances in METERS.-----				Moments in m.-MT			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

Theta3 is 0.07 degrees, as demonstrated below.

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 23.154f TCG = 0.000 VCG = 3.923

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
2.050	0.25f	0.00	1,856.5	0.000	0.000	0.0000
2.050	0.25f	0.07s	1,856.5	0.000	0.013	0.0000
2.036	0.26f	5.00s	1,856.5	0.000	0.979	0.0427
2.066	0.37f	10.00s	1,856.5	0.000	1.536	0.1578
2.229	0.49f	15.00s	1,856.5	0.000	1.661	0.3004
2.261	0.50f	15.82s	1,856.5	0.000	1.663	0.3241
2.424	0.62f	20.00s	1,856.5	0.000	1.618	0.4446
2.595	0.78f	25.00s	1,856.5	0.000	1.473	0.5803
2.744	0.94f	30.00s	1,856.5	0.000	1.276	0.7006

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1383.94 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 0.1 > 1.000 LARGE

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 23.154f TCG = 0.000 VCG = 3.923

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
2.050	0.25f	0.00	1,856.5	0.000	0.000	0.0000
2.050	0.25f	0.07s	1,856.5	0.000	0.013	0.0000
2.036	0.26f	5.00s	1,856.5	0.000	0.979	0.0427
2.066	0.37f	10.00s	1,856.5	0.000	1.536	0.1578
2.229	0.49f	15.00s	1,856.5	0.000	1.661	0.3004
2.261	0.50f	15.82s	1,856.5	0.000	1.663	0.3241
2.424	0.62f	20.00s	1,856.5	0.000	1.618	0.4446
2.595	0.78f	25.00s	1,856.5	0.000	1.473	0.5803
2.744	0.94f	30.00s	1,856.5	0.000	1.276	0.7006
2.869	1.10f	35.00s	1,856.5	0.000	1.049	0.8023
2.971	1.26f	40.00s	1,856.5	0.000	0.804	0.8833
3.047	1.41f	45.00s	1,856.5	0.000	0.547	0.9423
3.100	1.55f	50.00s	1,856.5	0.000	0.282	0.9785
3.128	1.68f	55.00s	1,856.5	0.000	0.012	0.9914
3.129	1.69f	55.23s	1,856.5	0.000	0.000	0.9914
3.134	1.80f	60.00s	1,856.5	0.000	-0.259	0.9806

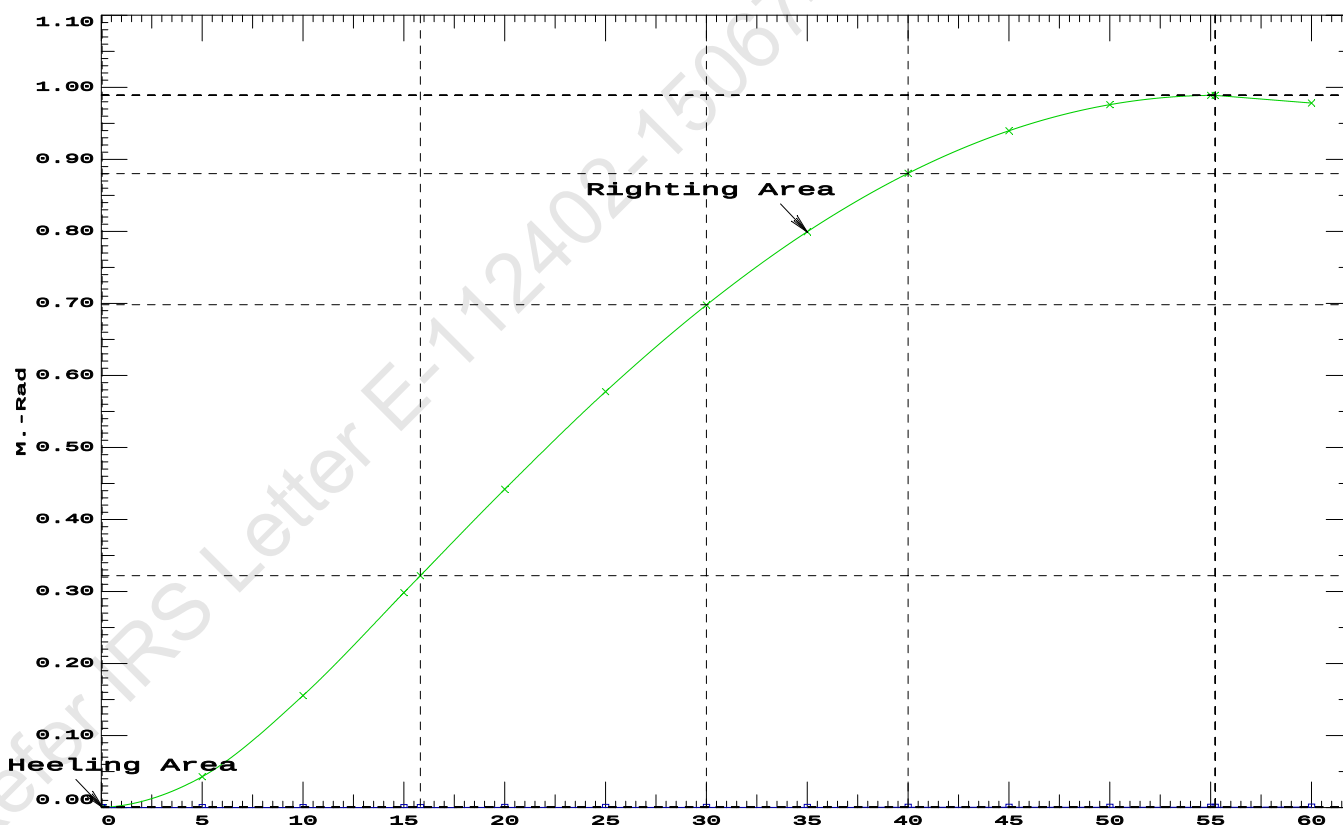
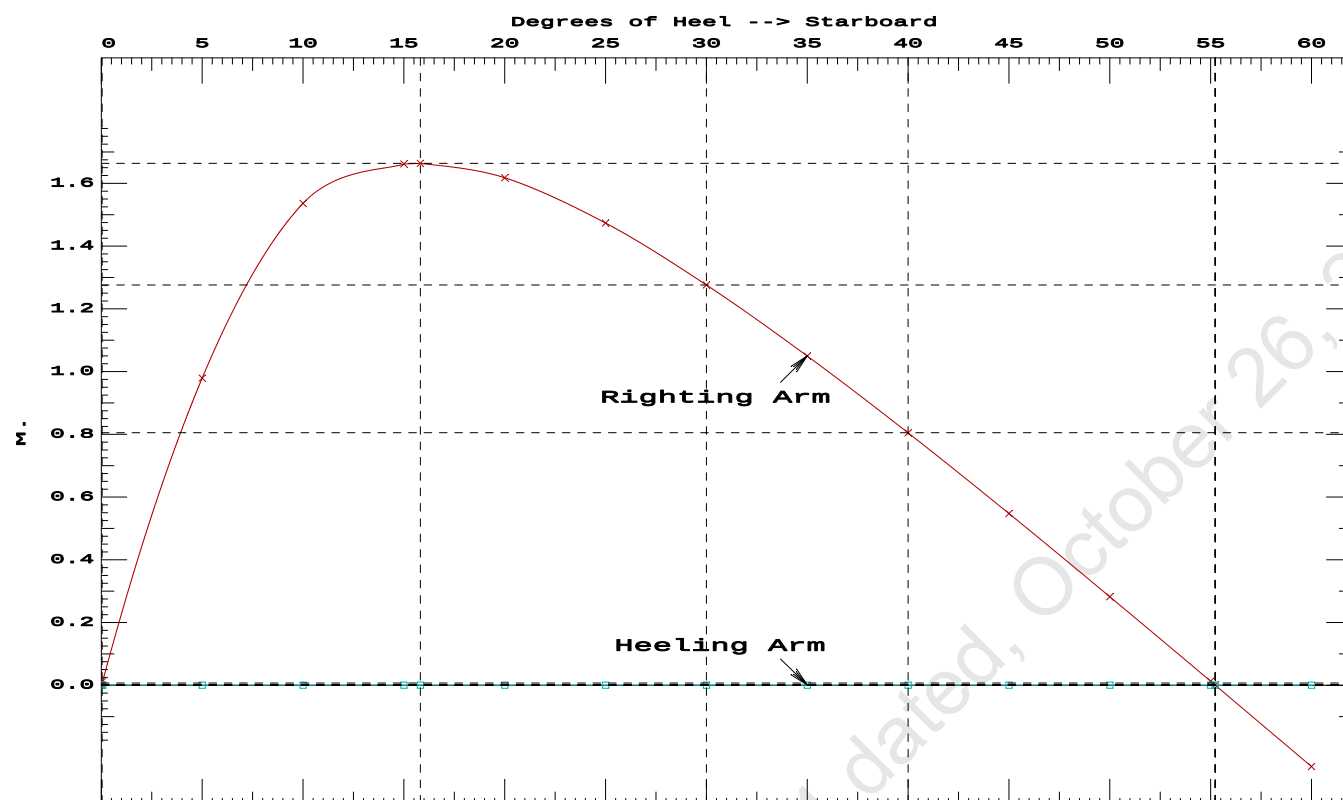
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1383.94 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero > 1.000 LARGE
(2) Angle from abs 0.1 deg to RAZero > 20.00 deg 55.16 P

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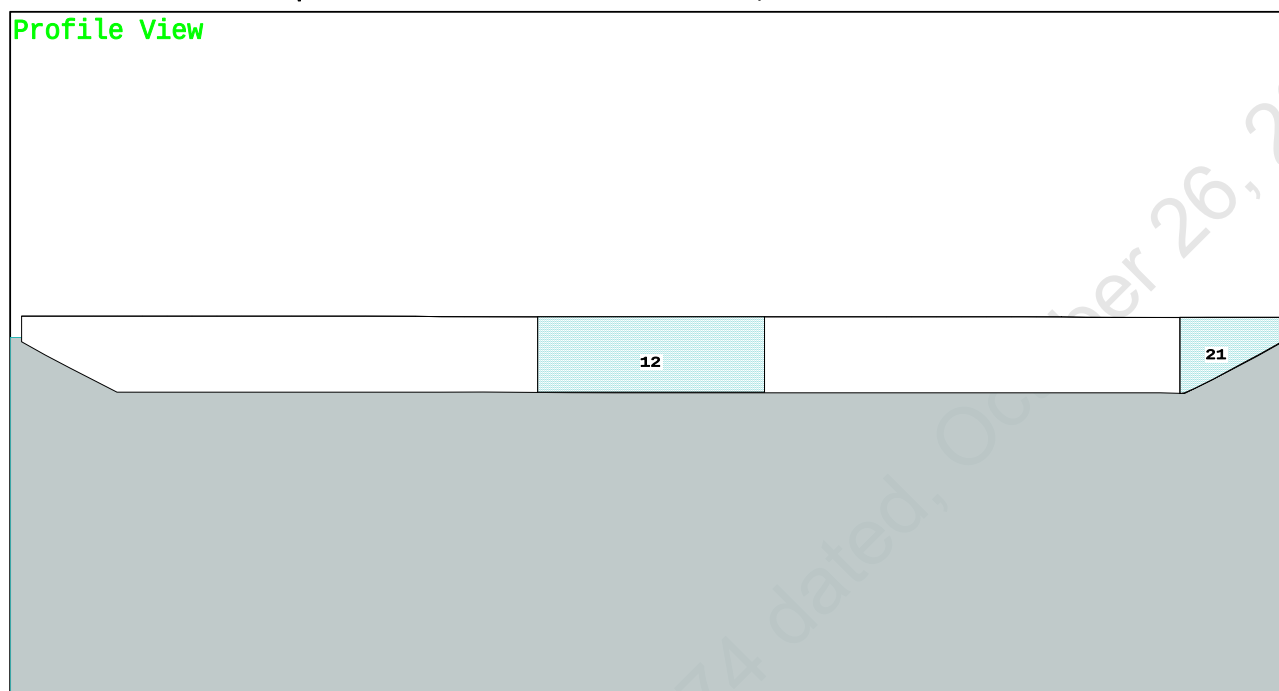
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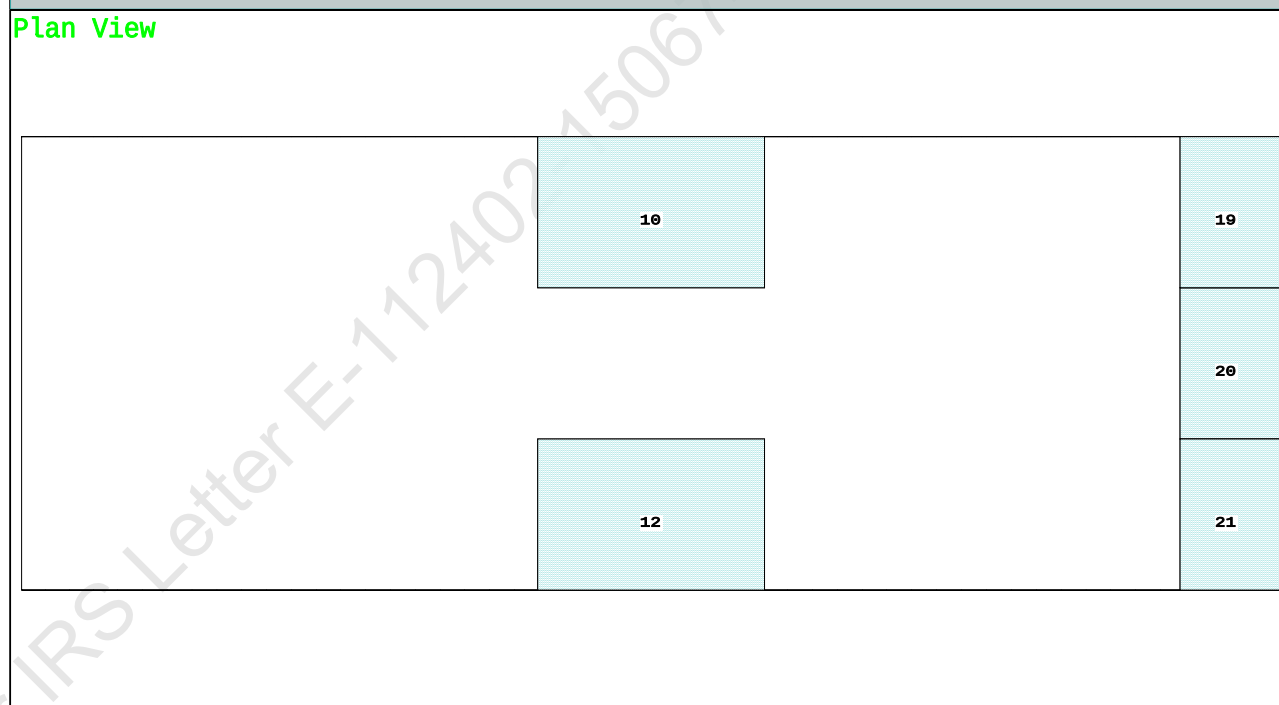
Condition 11 : Lightship + Load at Stbd + Ballast + Deck CargoCrane Lifting 36.6T @ 18m Radius Stbd

Condition Graphic - Draft: 2.237 @ 50.000f, 2.187 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

10 04_WBTK.P 12 04_WBTK.S 19 07_WBTK.P 20 07_WBTK.C 21 07_WBTK.S

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PONTOON

Condition 11 : Lightship + Load at Stbd + Ballast + Deck CargoCrane Lifting 36.6T @ 18m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.237 @ 50.00f, 2.212 @ 25.00f, 2.187 @ 0.00

Trim: Fwd 0.051/50.000, Heel: zero

Part-----	Weight(MT)-----	LCG-----	TCG-----	VCG-----	FSM-----
LIGHT SHIP	405.97	25.000f	0.000	3.000	
Crane	250.00	7.500f	0.000	5.200	
Crane Load	36.60	7.500f	0.000	76.977	
Deck Cargo	727.97	27.500f	0.000	4.000	
Total Fixed----->	1,420.54	22.750f	0.000	5.806	
Load-----	SpGr-----	Weight(MT)-----	LCG-----	TCG-----	VCG-----
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s
Total Tanks----->		472.52	32.161f	0.000	1.626
Total Weight----->		1,893.06	25.099f	0.000	4.762
		Displ(MT)-----	LCB-----	TCB-----	VCB-----
HULL	1.025	1,893.05	25.103f	0.000	1.132

Righting Arms:		0.000	0.000		
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----
Total Waterplane----->	1.025	900.4	25.001f	0.000	101.45
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----
		9.23	37.04	97.82	9.260
Distances in METERS.-----			Moments in m.-MT.		

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

461.0 Cu.M. (18%) SALT WATER

250.00 MT Crane
727.97 MT Deck Cargo

36.60 MT Crane Load

DRAFT AT FP (M) = 2.237
DRAFT AT MS (M) = 2.212
DRAFT AT AP (M) = 2.187
DRAFT AT FWD MARK (M) = 2.233
DRAFT AT AFT MARK (M) = 2.191

HYDROSTATIC PROPERTIES

Trim: Fwd 0.051/50.000, No Heel, VCG = 4.762

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight(MT)-----	LCB-----	VCB-----	cm-----
2.212	1,893.05	25.103f	1.132	9.23
				25.001f
				37.04
				97.82
				9.260

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 542.4 m.-MT

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PONTOON

Condition 11 : Lightship + Load at Stbd + Ballast + Deck Cargo
Crane Lifting 36.6T @ 18m Radius Stbd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 2.237 @ 50.00f, 2.212 @ 25.00f, 2.187 @ 0.00

Trim: Fwd 0.051/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	40.5	0.405	1.478	0.05500	3.29
CRANE	57.0	3.315	4.388	0.05500	13.76
DECKCARGO.C	29.4	1.797	2.871	0.05500	4.64
Total wind heeling moment to starboard-->					21.69
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.099f TCG = 0.000 VCG = 4.762

Free Surface Adjustment: 0.287

Adjusted CG: LCG = 25.099f TCG = 0.000 VCG = 5.049

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Res. 50% DeckImm Area	Angle
2.174	0.06f	0.00	1,893.1	0.000	-0.011	2.494
2.175	0.06f	0.07s	1,893.1	0.000	0.000	2.426
2.172	0.06f	2.49s	1,893.2	0.000	0.396	0.000
2.163	0.06f	5.07s	1,893.1	0.000	0.810	-2.574
2.246	0.09f	10.07s	1,893.0	0.000	1.154	-7.574
2.297	0.10f	11.42s	1,893.1	0.000	1.163	-8.922
2.461	0.12f	15.07s	1,893.1	0.000	1.107	-12.574
2.718	0.15f	20.07s	1,893.1	0.000	0.914	-17.574
2.963	0.19f	25.07s	1,893.1	0.000	0.633	-22.574
3.183	0.23f	30.07s	1,893.1	0.000	0.307	-27.574
3.358	0.26f	34.50s	1,893.1	0.000	0.000	-32.003
3.379	0.27f	35.07s	1,893.1	0.000	-0.040	-32.574
3.549	0.31f	40.07s	1,893.1	0.000	-0.398	-37.574
3.691	0.34f	45.07s	1,893.1	0.000	-0.758	-42.574
3.806	0.38f	50.07s	1,893.1	0.000	-1.117	-47.574
3.892	0.41f	55.07s	1,893.1	0.000	-1.469	-52.574
3.949	0.43f	60.07s	1,893.1	0.000	-1.813	-57.574
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.						

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 542.4 m.-MT was applied to artificially modify the CG.

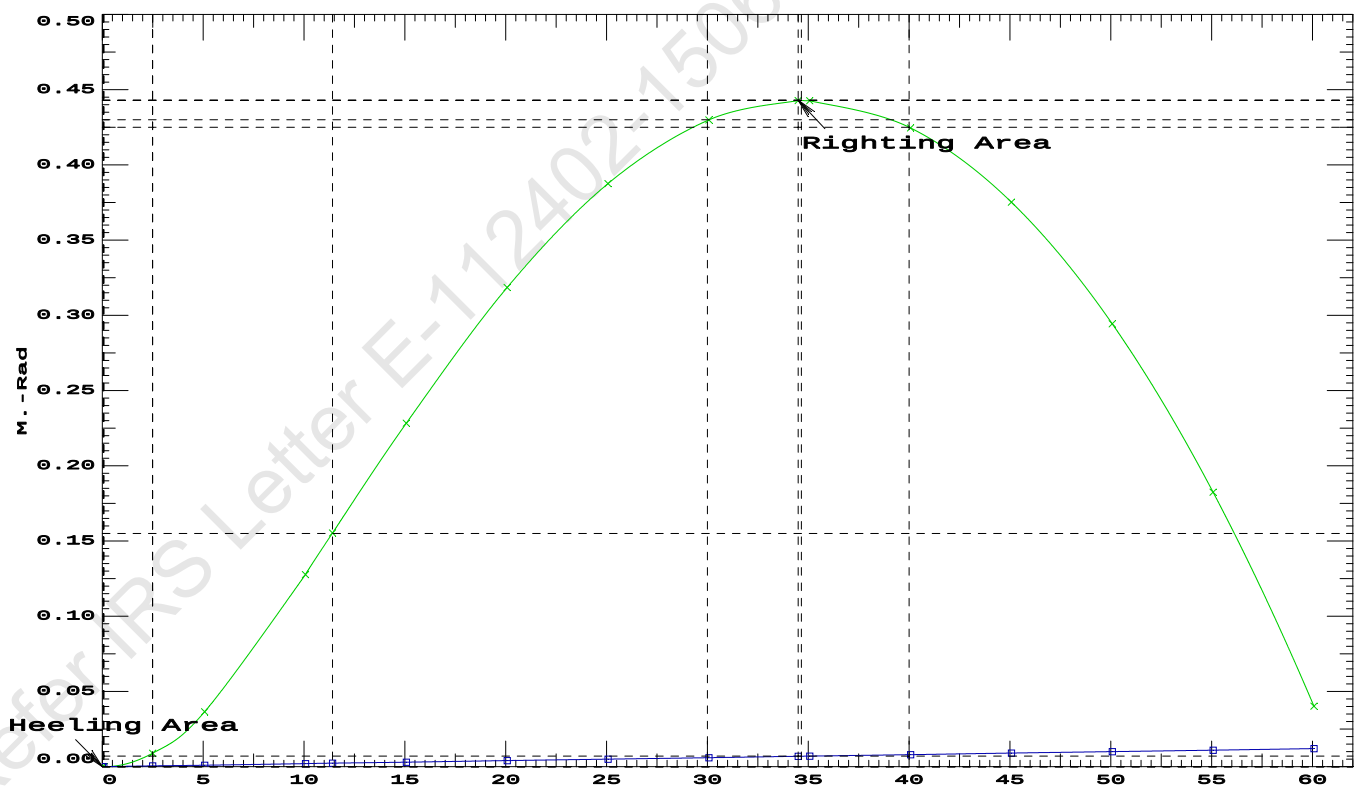
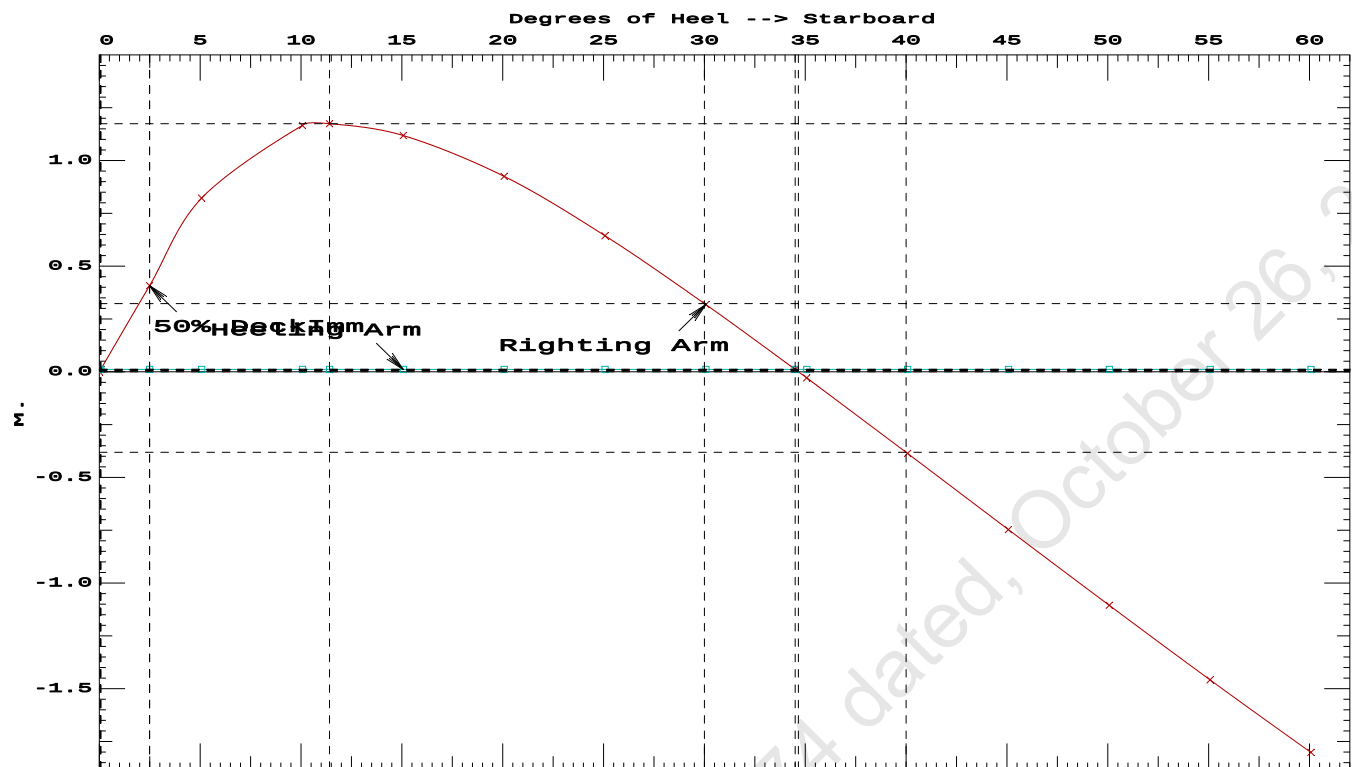
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.69 (constant)

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2-----Min/Max-----Attained

(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.1535 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	34.43 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	2.43 P

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Condition 11 : Lightship + Load at Stbd + Ballast + Deck Cargo
Crane Lifting 36.6T @ 18m Radius Stbd



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PONTOON**Condition 11 : Lightship + Load at Stbd + Ballast + Deck Cargo
Crane Lifting 36.6T @ 18m Radius Stbd****RESIDUAL RIGHTING ARMS vs HEEL ANGLE**

Total CG: LCG = 25.099f TCG = 0.000 VCG = 4.762

Free Surface Adjustment: 0.287

Adjusted CG: LCG = 25.099f TCG = 0.010p VCG = 5.049

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
2.173	0.06f	2.06s	1,893.2	0.000	0.000	0.0000
2.171	0.07f	7.06s	1,893.1	0.000	0.709	0.0309
2.295	0.10f	11.39s	1,893.1	0.000	0.836	0.0912
2.323	0.10f	12.06s	1,893.0	0.000	0.834	0.1010
2.561	0.13f	17.06s	1,893.1	0.000	0.715	0.1697
2.818	0.16f	22.06s	1,893.1	0.000	0.482	0.2228
3.053	0.20f	27.06s	1,893.1	0.000	0.179	0.2521
3.171	0.22f	29.76s	1,893.1	0.000	0.000	0.2564
3.265	0.24f	32.06s	1,893.1	0.000	-0.158	0.2532
3.450	0.28f	37.06s	1,893.1	0.000	-0.510	0.2242
3.609	0.32f	42.06s	1,893.1	0.000	-0.870	0.1640
3.740	0.36f	47.06s	1,893.1	0.000	-1.231	0.0724
3.843	0.39f	52.06s	1,893.1	0.000	-1.588	-0.0507
3.918	0.42f	57.06s	1,893.1	0.000	-1.938	-0.2046
3.963	0.44f	62.06s	1,893.1	0.000	-2.278	-0.3887

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 542.4 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 658.80

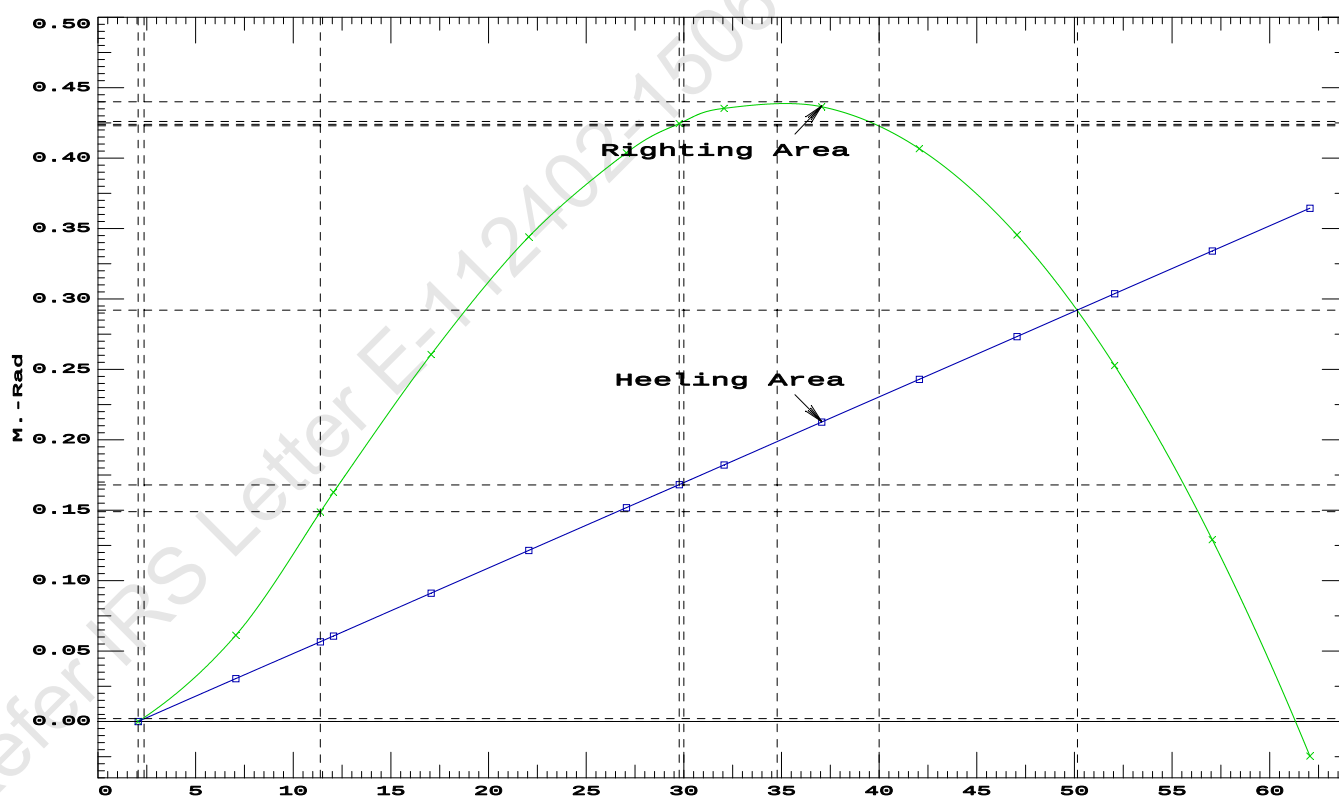
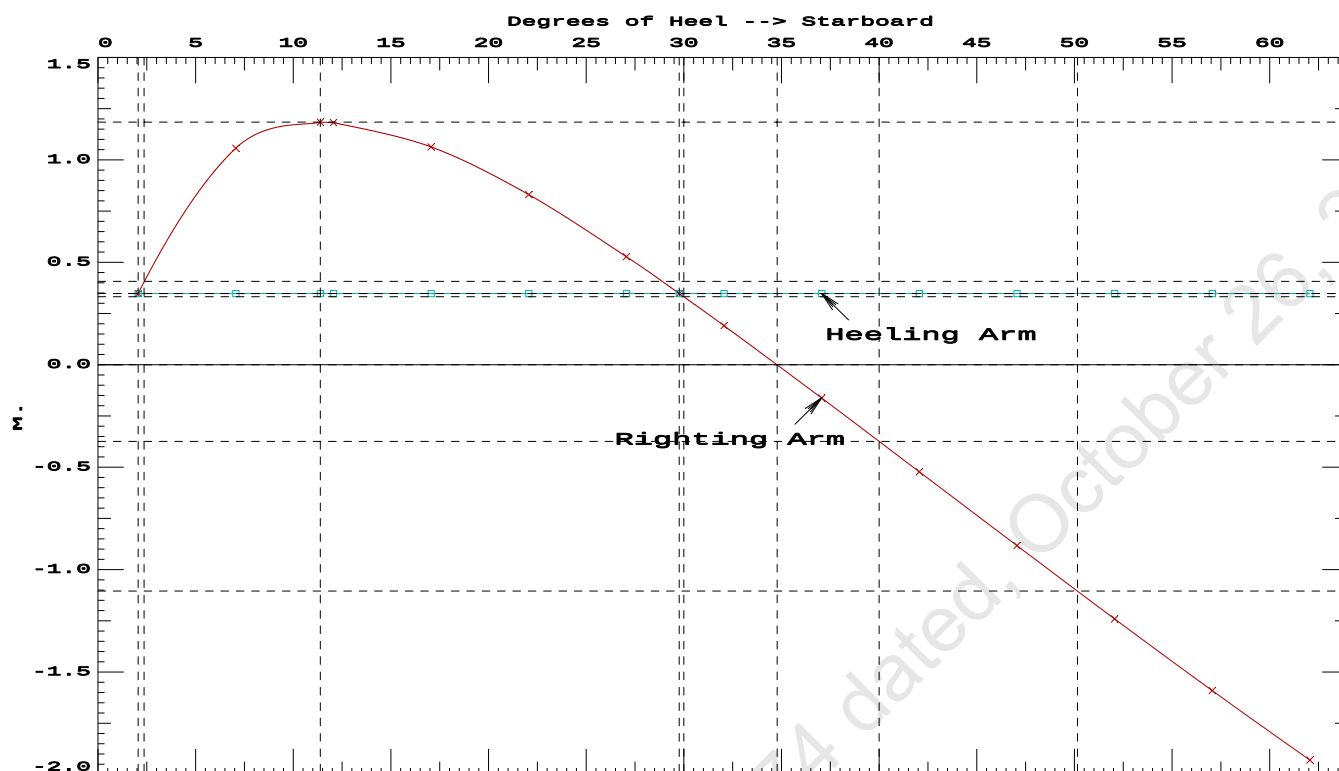
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT-----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	2.06 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	240.3 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	2.524 P

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PONTOON

Condition 11 : Lightship + Load at Stbd + Ballast + Deck CargoCrane Lifting 36.6T @ 18m Radius Stbd

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GHS 17.22 PONTON

Condition 11 : Lightship + Load at Stbd + Ballast + Deck Cargo**Crane Lifting 36.6T @ 18m Radius Stbd**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.237 @ 50.00f, 2.212 @ 25.00f, 2.186 @ 0.00

Trim: Fwd 0.051/50.000, Heel: Stbd 2.06 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	7.500f	0.000	76.977	
Deck Cargo			727.97	27.500f	0.000	4.000	
Total Fixed----->			1,420.54	22.750f	0.000	5.806	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p	1.509	162.73*
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			472.52	32.161f	0.000	1.626	542.42
Total Weight----->			1,893.06	25.099f	0.000	4.762	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,893.06	25.103f	0.479s	1.140	

	Righting Arms:			0.000	0.348s		
	External Arms:			0.000	0.348s		
	Residual Righting Arms:			0.000	0.000s		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->		1.025	898.4	25.020f	0.029s	100.65	12.828
			MT/cm				
			9.21				

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 2.06

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After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 2.283 @ 50.00f, 2.172 @ 25.00f, 2.062 @ 0.00
Trim: Fwd 0.221/50.000, Heel: 0.00 deg.

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----			
LIGHT SHIP	405.97	25.000f	0.000	3.000			
Crane	250.00	7.500f	0.000	5.200			
Deck Cargo	727.97	27.500f	0.000	4.000			
Total Fixed----->	1,383.94	23.154f	0.000	3.923			
Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----	
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p	1.509	162.73*
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			472.52	32.161f	0.000	1.626	542.42
Total Weight----->			1,856.46	25.446f	0.000	3.339	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL	1.025		1,856.46	25.456f	0.001	1.112	

Righting Arms:			0.000	0.001s			
External Arms:			0.000	0.001			
Residual Righting Arms:			0.000	0.000			
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----	
Total Waterplane---->	1.025	900.4	25.006f	0.000	103.43	13.142	
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----		
		9.23	37.58	101.20	10.915		
Distances in METERS.-----Moments in m.-MT.							

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: 9504.85; t3: 30.00; t: 15.00
limmarg: 4018.16; t3: 15.00; t: 7.50
limmarg: 1187.01; t3: 7.50; t: 3.75
limmarg: 230.93; t3: 3.75; t: 1.88
limmarg: -17.17; t3: 1.87; t: 0.94
limmarg: 86.30; t3: 2.81; t: 0.47
limmarg: 29.43; t3: 2.34; t: 0.23
limmarg: 5.34; t3: 2.11; t: 0.12
limmarg: -6.25; t3: 1.99; t: 0.06
limmarg: -0.54; t3: 2.05; t: 0.03
limmarg: 2.38; t3: 2.08; t: 0.01

Theta3 is 2.08 degrees, as demonstrated below.

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 23.154f TCG = 0.000 VCG = 3.923

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
2.047	0.26f	2.06s	1,856.5	0.000	-0.406	0.0000
2.050	0.25f	0.00	1,856.5	0.000	0.000	-0.0073
2.047	0.26f	2.08p	1,856.5	0.000	0.411	0.0002
2.036	0.26f	5.00p	1,856.5	0.000	0.979	0.0356
2.066	0.37f	10.00p	1,856.5	0.000	1.536	0.1479
2.228	0.49f	15.00p	1,856.5	0.000	1.662	0.2906
2.261	0.50f	15.82p	1,856.5	0.000	1.664	0.3144
2.423	0.62f	20.00p	1,856.5	0.000	1.619	0.4348
2.595	0.78f	25.00p	1,856.5	0.000	1.474	0.5707
2.744	0.94f	30.00p	1,856.5	0.000	1.277	0.6911

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1383.94 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Res. Area Ratio from abs -2.1 deg to abs 2.1 > 1.000 1.024 P

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 23.154f TCG = 0.000 VCG = 3.923

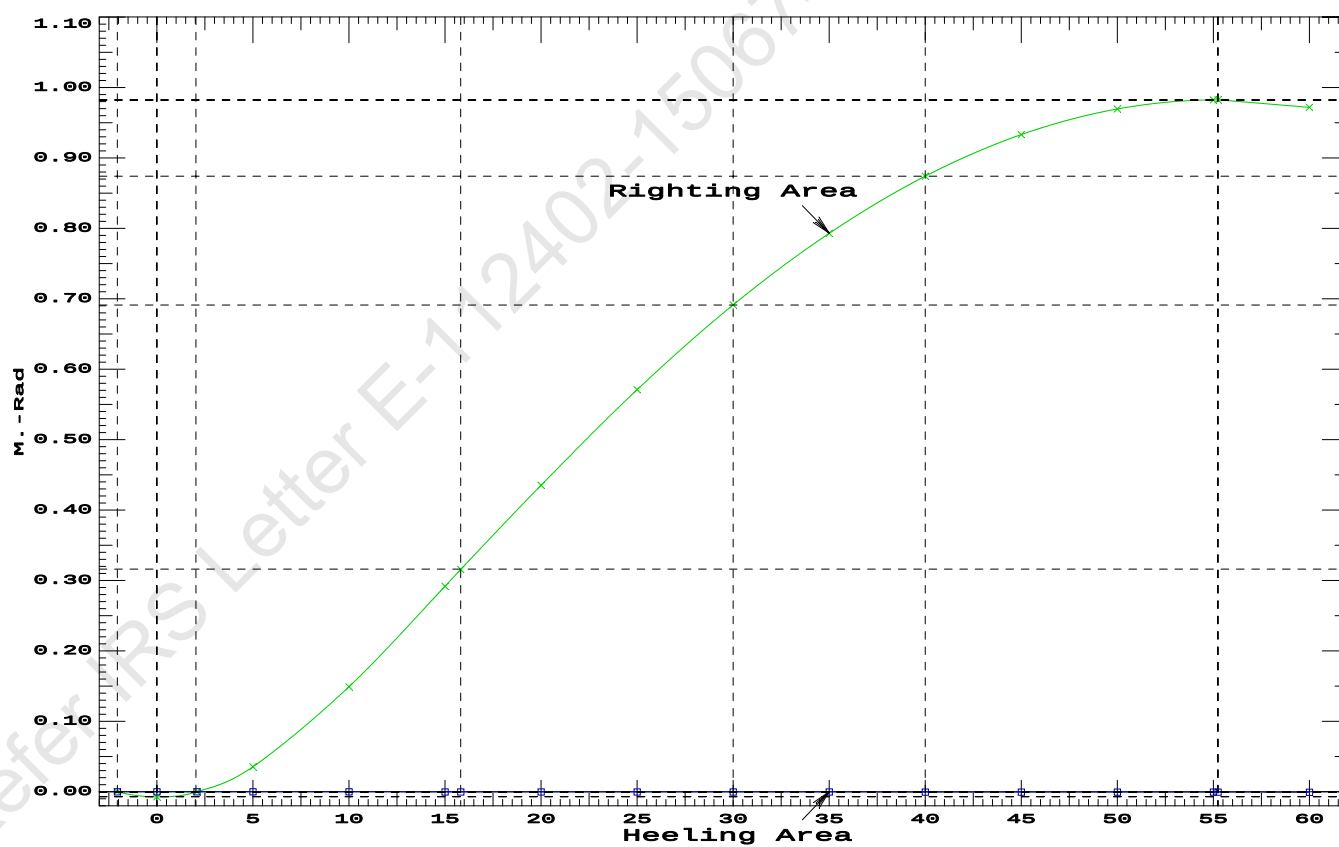
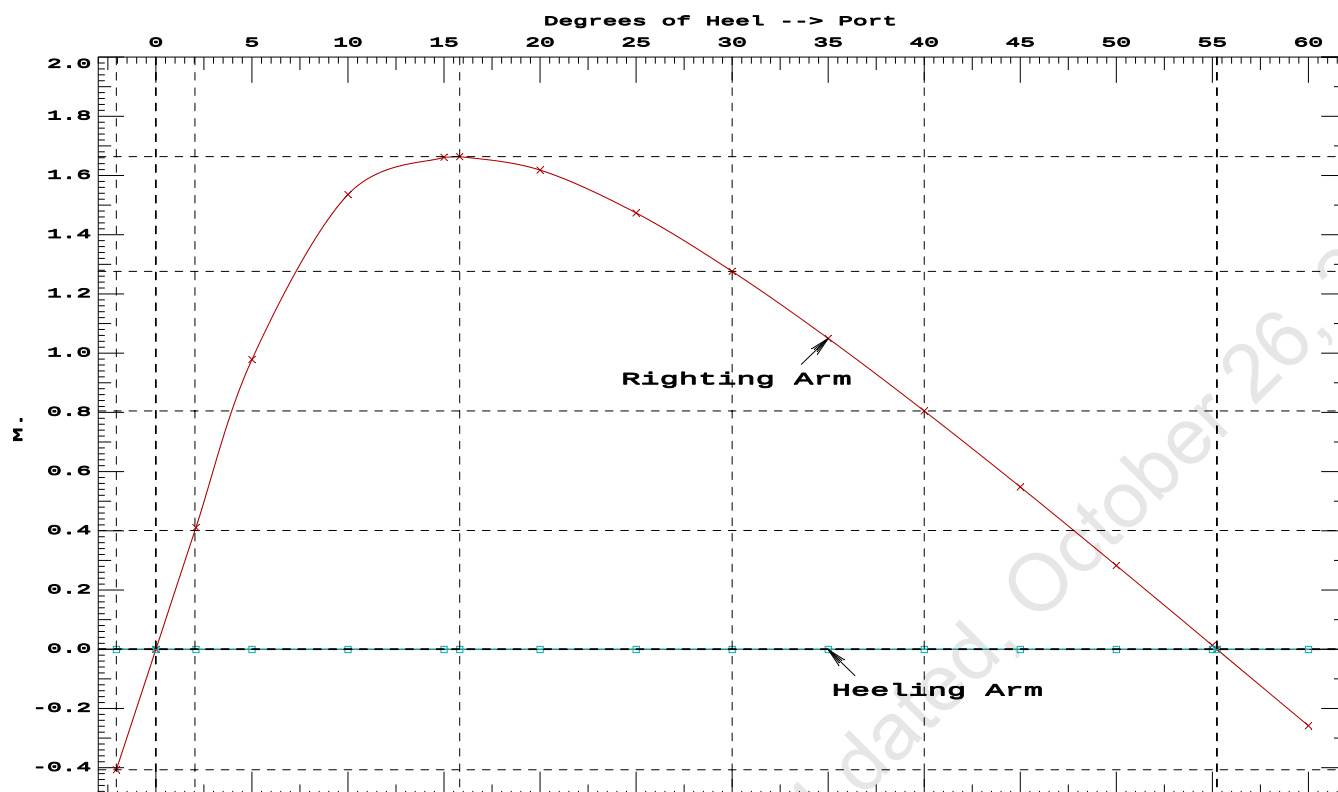
Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
2.047	0.26f	2.06s	1,856.5	0.000	-0.406	0.0000
2.050	0.25f	0.00	1,856.5	0.000	0.000	-0.0073
2.047	0.26f	2.08p	1,856.5	0.000	0.411	0.0002
2.036	0.26f	5.00p	1,856.5	0.000	0.979	0.0356
2.066	0.37f	10.00p	1,856.5	0.000	1.536	0.1479
2.228	0.49f	15.00p	1,856.5	0.000	1.662	0.2906
2.261	0.50f	15.82p	1,856.5	0.000	1.664	0.3144
2.423	0.62f	20.00p	1,856.5	0.000	1.619	0.4348
2.595	0.78f	25.00p	1,856.5	0.000	1.474	0.5707
2.744	0.94f	30.00p	1,856.5	0.000	1.277	0.6911
2.869	1.10f	35.00p	1,856.5	0.000	1.051	0.7929
2.970	1.26f	40.00p	1,856.5	0.000	0.806	0.8740
3.047	1.41f	45.00p	1,856.5	0.000	0.548	0.9332
3.100	1.55f	50.00p	1,856.5	0.000	0.283	0.9695
3.128	1.68f	55.00p	1,856.5	0.000	0.013	0.9825
3.129	1.69f	55.25p	1,856.5	0.000	0.000	0.9825
3.134	1.80f	60.00p	1,856.5	0.000	-0.258	0.9718

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1383.94 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Res. Area Ratio from abs -2.1 deg to RZero > 1.000 135.596 P
(2) Angle from abs 2.1 deg to RZero > 20.00 deg 53.17 P

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PONTON

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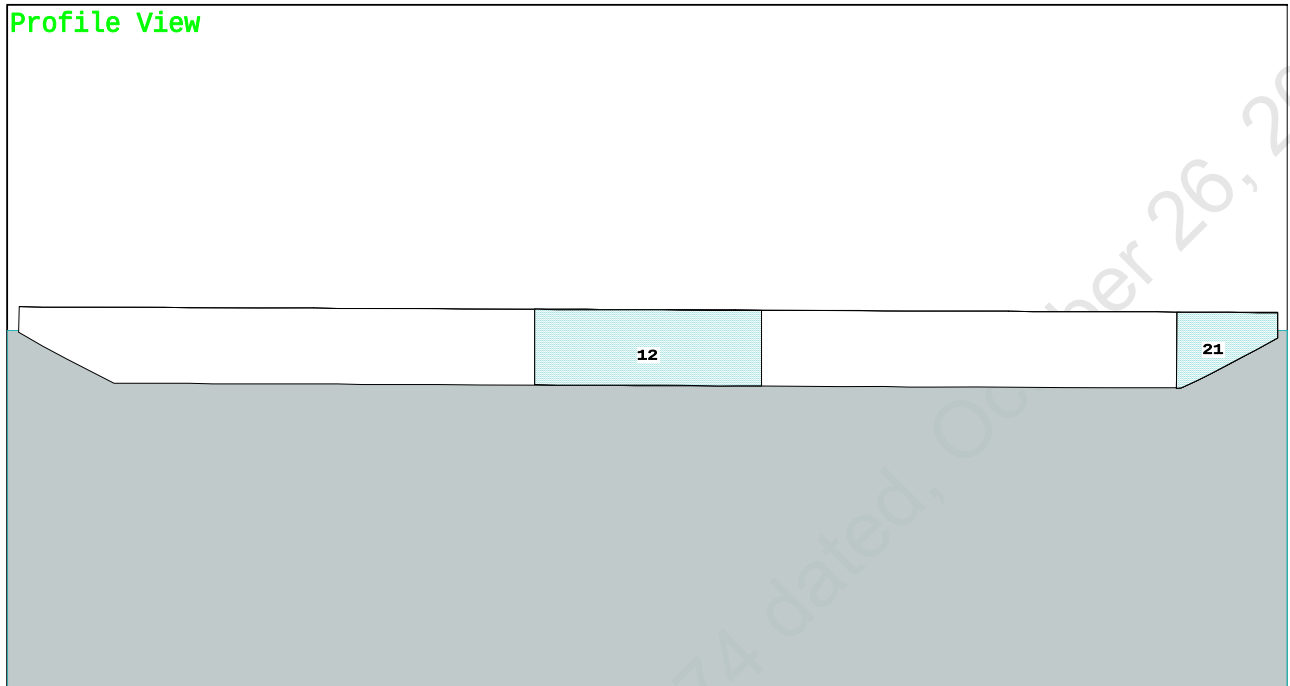
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PONT00N

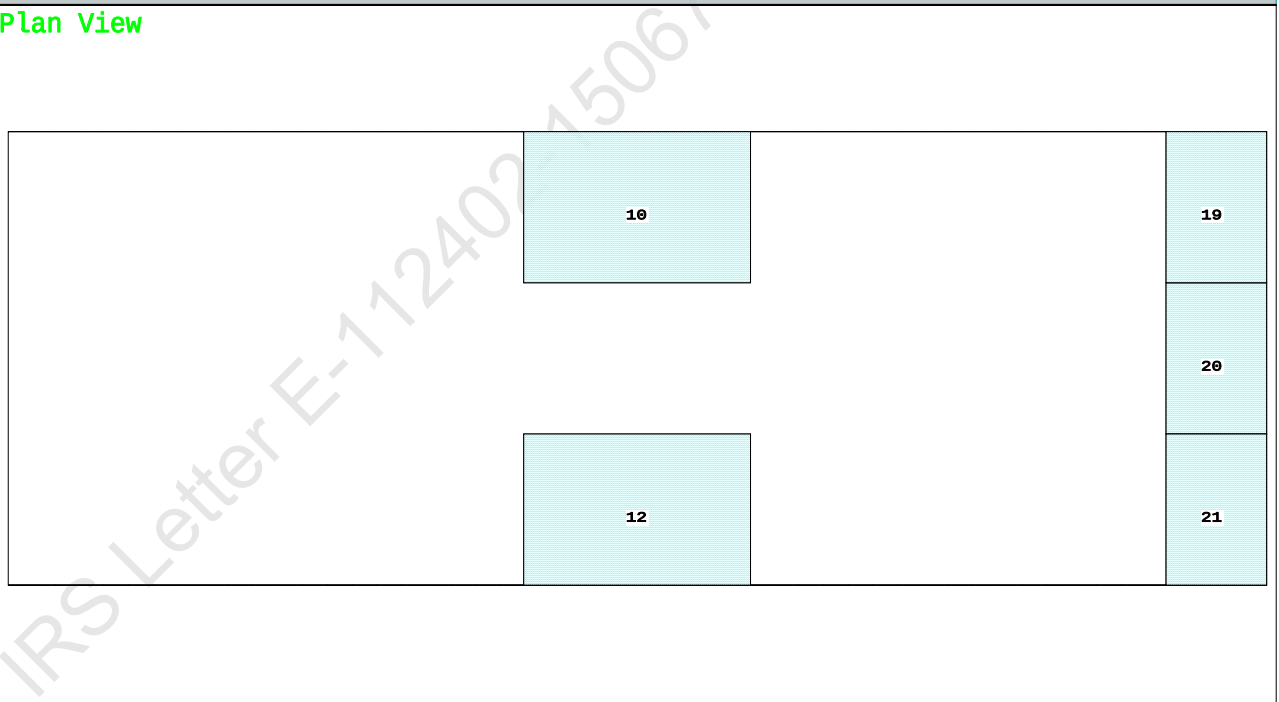
Condition 12 : Lightship + Load at Fwd + Ballast + Deck CargoCrane Lifting 36.6T @ 18m Radius Fwd

Condition Graphic - Draft: 2.326 @ 50.000f, 2.098 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

10 04_WBTK.P 12 04_WBTK.S 19 07_WBTK.P 20 07_WBTK.C 21 07_WBTK.S

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PONTOON

Condition 12 : Lightship + Load at Fwd + Ballast + Deck CargoCrane Lifting 36.6T @ 18m Radius Fwd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.326 @ 50.00f, 2.212 @ 25.00f, 2.098 @ 0.00

Trim: Fwd 0.229/50.000, Heel: zero

Part-----			Weight(MT)-----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	25.500f	0.000	76.977	
Deck Cargo			727.97	27.500f	0.000	4.000	
Total Fixed----->			1,420.54	23.214f	0.000	5.806	
	Load-----	SpGr-----	Weight(MT)-----	LCG-----	TCG-----	VCG-----	FSM-----
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p	1.509	162.73*
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			472.52	32.161f	0.000	1.626	542.42
Total Weight----->			1,893.06	25.447f	0.000	4.762	
			Displ(MT)-----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,893.06	25.464f	0.000	1.133	

Righting Arms: 0.000 0.000

Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->	1.025	900.4	25.006f	0.000	101.45	12.891
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
		9.23	37.04	97.82	9.261	

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

461.0 Cu.M. (18%) SALT WATER

250.00 MT Crane
727.97 MT Deck Cargo

36.60 MT Crane Load

DRAFT AT FP (M) = 2.326
DRAFT AT MS (M) = 2.212
DRAFT AT AP (M) = 2.098
DRAFT AT FWD MARK (M) = 2.308
DRAFT AT AFT MARK (M) = 2.116

HYDROSTATIC PROPERTIES

Trim: Fwd 0.229/50.000, No Heel, VCG = 4.762

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight(MT)-----	LCB-----	VCB-----	cm-----
2.212	1,893.06	25.464f	1.133	9.23
				25.006f
				37.04
				97.82
				9.261

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 542.4 m.-MT

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PONTOON

Condition 12 : Lightship + Load at Fwd + Ballast + Deck Cargo
Crane Lifting 36.6T @ 18m Radius Fwd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 2.326 @ 50.00f, 2.212 @ 25.00f, 2.098 @ 0.00

Trim: Fwd 0.229/50.000, Heel: zero

Part-----	LPA*SF-----	HCP-----	Arm----	Pressure-----	Moment
HULL	40.5	0.408	1.482	0.05500	3.30
CRANE	57.0	3.368	4.442	0.05500	13.93
DECKCARGO.C	29.4	1.789	2.863	0.05500	4.63
Total wind heeling moment to starboard----->					21.86
Distances in METERS.-----Pressure in MT/Sqm.-----					Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.447f TCG = 0.000 VCG = 4.762

Free Surface Adjustment: 0.287

Adjusted CG: LCG = 25.446f TCG = 0.000 VCG = 5.049

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---	Heel---	Weight(MT)---	in Trim--in Heel---	Area---Angle
2.086	0.26f	0.00	1,893.1	0.000 -0.011 0.0000 2.206
2.085	0.26f	0.07s	1,893.1	0.000 0.000 -0.0000 2.138
2.082	0.26f	2.21s	1,893.1	0.000 0.349 0.0065 0.000
2.071	0.27f	5.07s	1,893.1	0.000 0.808 0.0354 -2.862
2.113	0.40f	10.07s	1,893.1	0.000 1.149 0.1258 -7.862
2.152	0.43f	11.42s	1,893.1	0.000 1.158 0.1531 -9.219
2.286	0.52f	15.07s	1,893.2	0.000 1.101 0.2259 -12.862
2.494	0.66f	20.07s	1,893.1	0.000 0.906 0.3151 -17.862
2.679	0.84f	25.07s	1,893.1	0.000 0.625 0.3826 -22.862
2.838	1.02f	30.07s	1,893.1	0.000 0.299 0.4232 -27.862
2.956	1.17f	34.39s	1,893.1	0.000 0.000 0.4346 -32.183
2.973	1.20f	35.07s	1,893.1	0.000 -0.048 0.4343 -32.862
3.082	1.37f	40.07s	1,893.1	0.000 -0.405 0.4147 -37.862
3.167	1.54f	45.07s	1,893.1	0.000 -0.765 0.3636 -42.862
3.229	1.70f	50.07s	1,893.1	0.000 -1.123 0.2812 -47.862
3.267	1.83f	55.07s	1,893.1	0.000 -1.475 0.1678 -52.862
3.285	1.95f	60.07s	1,893.1	0.000 -1.817 0.0241 -57.862
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 542.4 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.86 (constant)

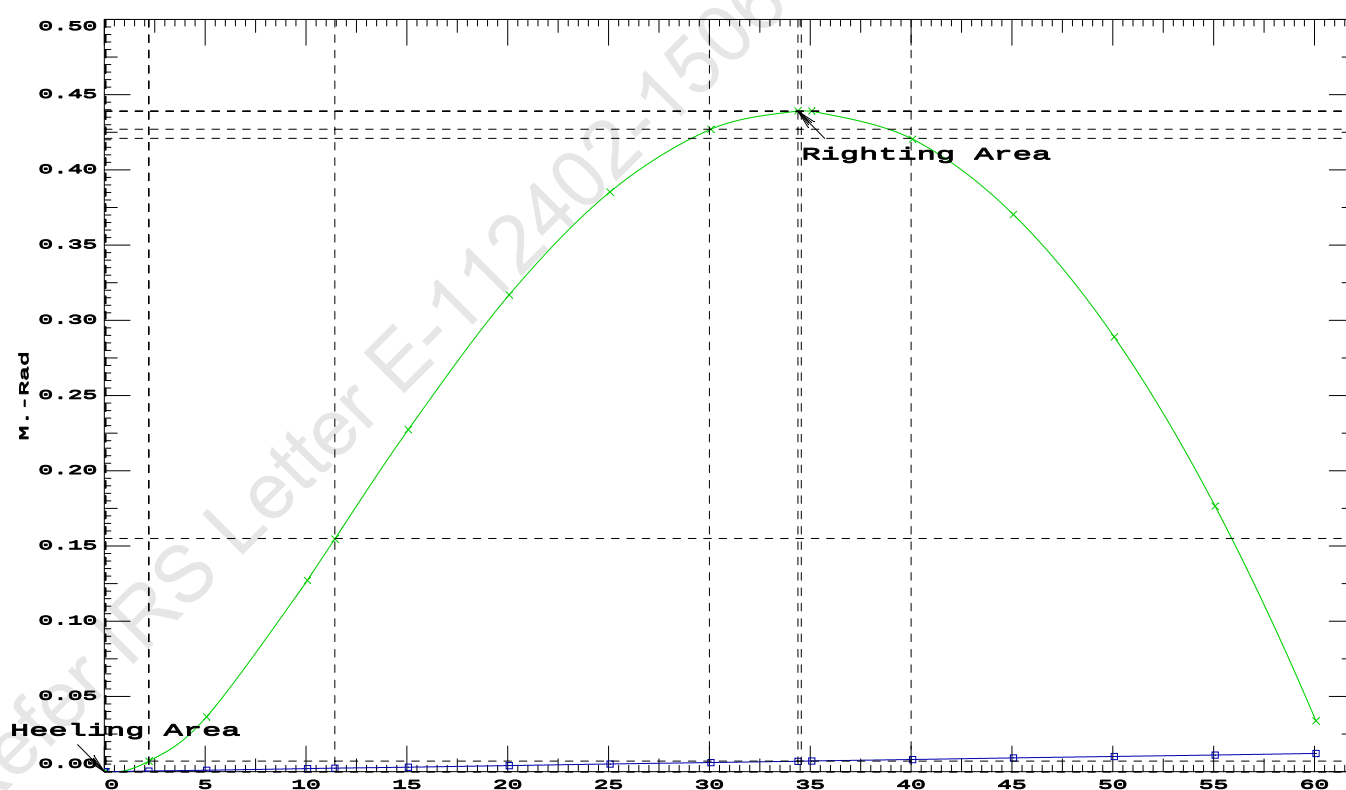
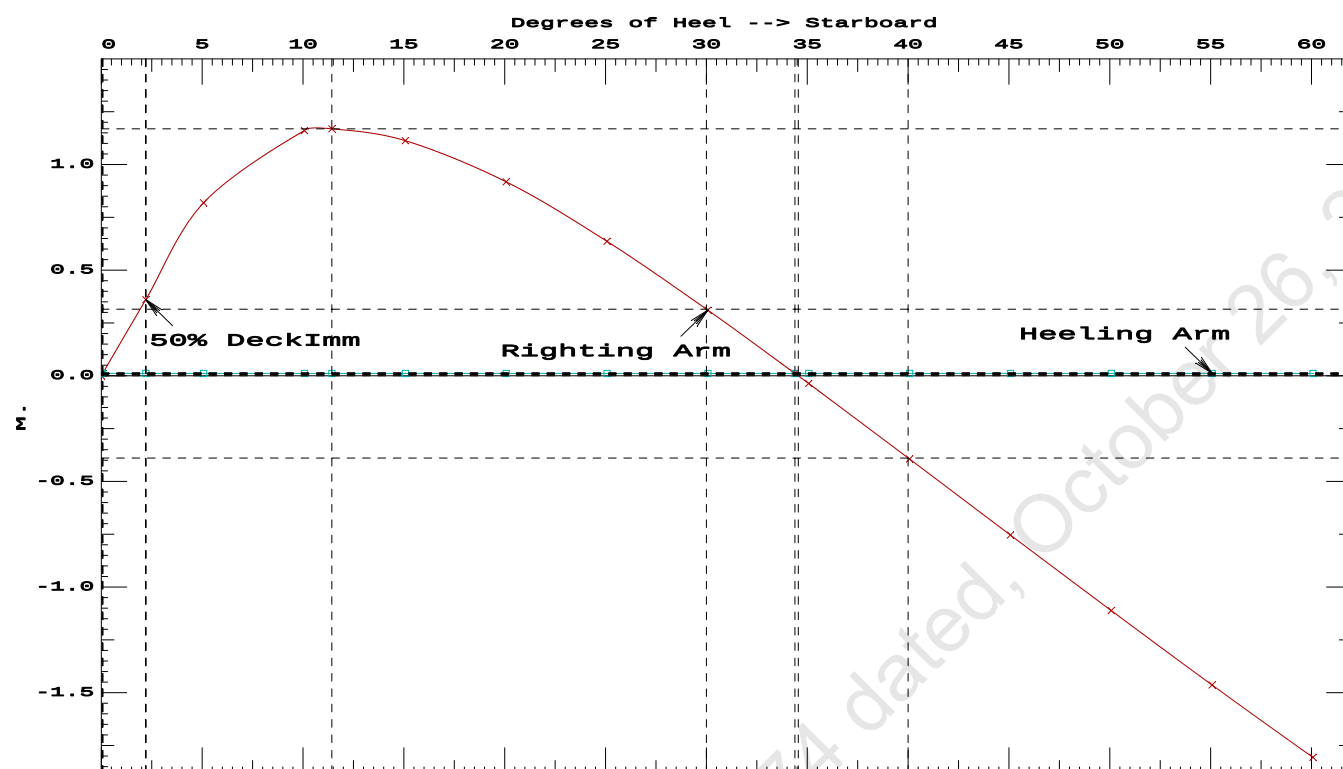
LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2-----Min/Max-----Attained

(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.1531 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	34.32 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	2.14 P

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PONTOON

Condition 12 : Lightship + Load at Fwd + Ballast + Deck CargoCrane Lifting 36.6T @ 18m Radius Fwd

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PONTOON**Condition 12 : Lightship + Load at Fwd + Ballast + Deck Cargo****Crane Lifting 36.6T @ 18m Radius Fwd****RESIDUAL RIGHTING ARMS vs HEEL ANGLE**

Total CG: LCG = 25.447f TCG = 0.000 VCG = 4.762

Free Surface Adjustment: 0.287

Adjusted CG: LCG = 25.446f TCG = 0.000 VCG = 5.049

Origin Depth---	Degrees of Trim----	Heel----	Displacement Weight(MT)---	Residual Arms		Res. Area
				in Trim--	in Heel---	
2.086	0.26f	0.00s	1,893.1	0.000	0.000	0.0000
2.072	0.27f	5.00s	1,893.1	0.000	0.808	0.0353
2.111	0.40f	10.00s	1,893.1	0.000	1.159	0.1261
2.152	0.43f	11.43s	1,893.1	0.000	1.168	0.1551
2.283	0.52f	15.00s	1,893.2	0.000	1.114	0.2272
2.492	0.66f	20.00s	1,893.1	0.000	0.920	0.3176
2.676	0.83f	25.00s	1,893.1	0.000	0.640	0.3863
2.836	1.01f	30.00s	1,893.1	0.000	0.314	0.4283
2.960	1.18f	34.54s	1,893.1	0.000	0.000	0.4409
2.971	1.19f	35.00s	1,893.1	0.000	-0.033	0.4407
3.081	1.37f	40.00s	1,893.1	0.000	-0.390	0.4224
3.166	1.54f	45.00s	1,893.1	0.000	-0.750	0.3727
3.228	1.69f	50.00s	1,893.1	0.000	-1.108	0.2917
3.267	1.83f	55.00s	1,893.1	0.000	-1.460	0.1796
3.285	1.95f	60.00s	1,893.1	0.000	-1.802	0.0372

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 542.4 m.-MT was applied to artificially modify the CG.

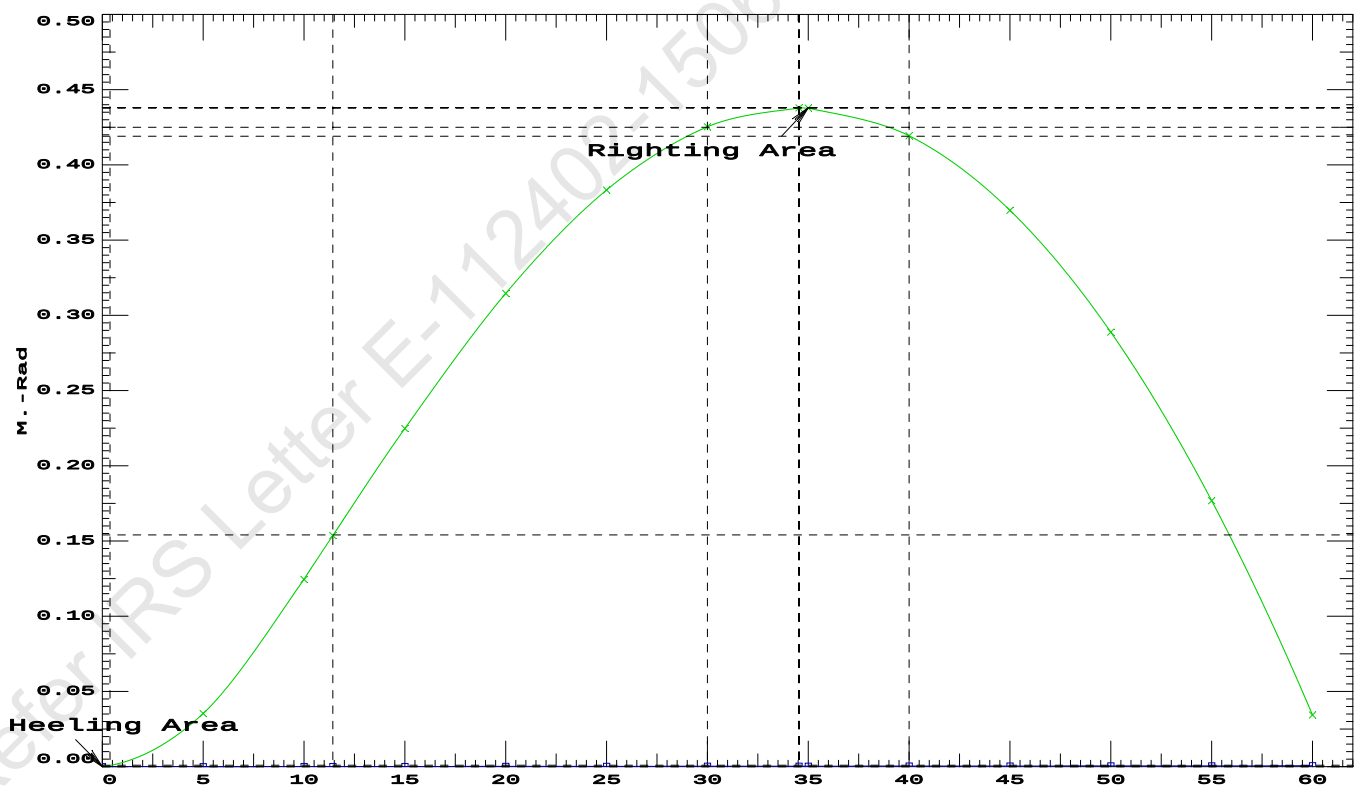
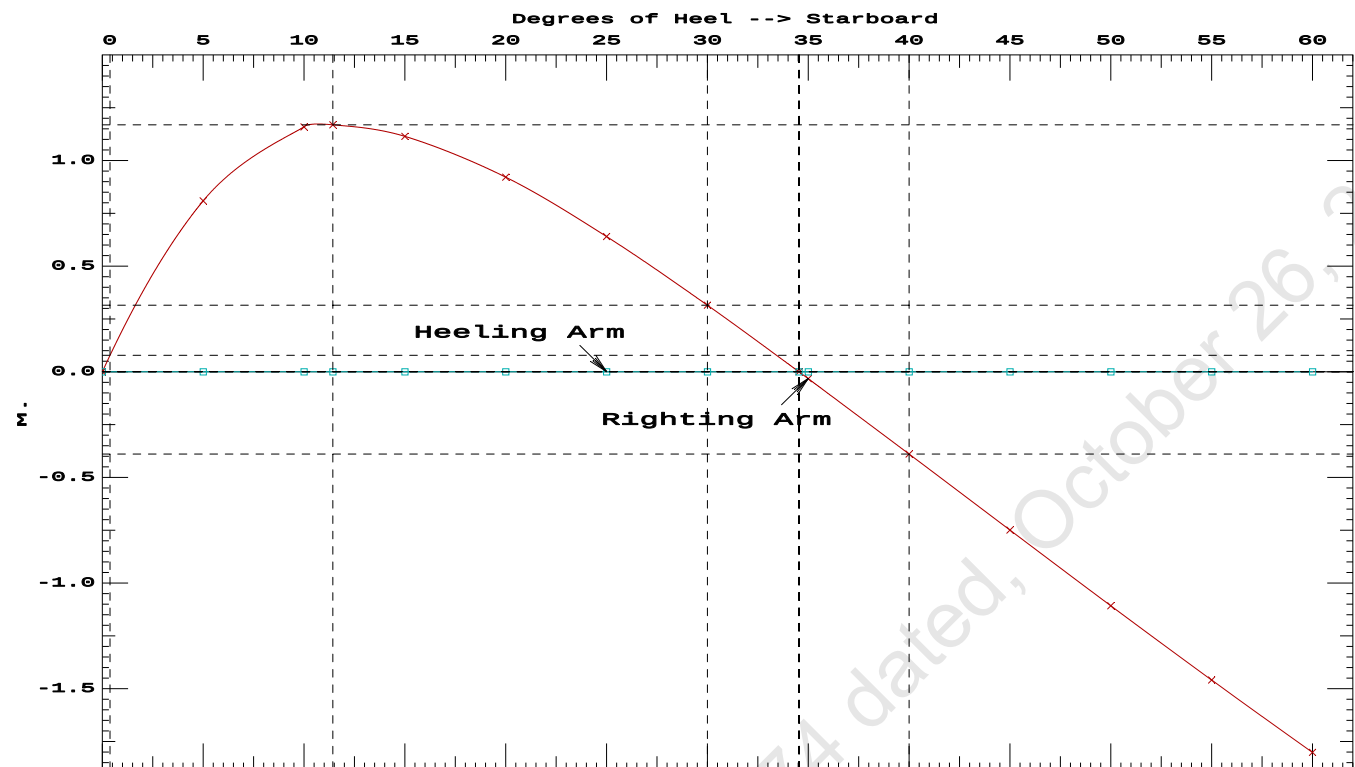
Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	0.00 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	221171 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	1385.59 P

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GHS 17.22 PONT00N

Condition 12 : Lightship + Load at Fwd + Ballast + Deck Cargo
Crane Lifting 36.6T @ 18m Radius Fwd



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GHS 17.22 PONTON

Condition 12 : Lightship + Load at Fwd + Ballast + Deck Cargo**Crane Lifting 36.6T @ 18m Radius Fwd**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.326 @ 50.00f, 2.212 @ 25.00f, 2.098 @ 0.00

Trim: Fwd 0.229/50.000, Heel: 0.00 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	25.500f	0.000	76.977	
Deck Cargo			727.97	27.500f	0.000	4.000	
Total Fixed----->			1,420.54	23.214f	0.000	5.806	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p	1.509	162.73*
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			472.52	32.161f	0.000	1.626	542.42
Total Weight----->			1,893.06	25.447f	0.000	4.762	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,893.06	25.464f	0.001	1.133	

	Righting Arms:			0.000	0.001s		
	External Arms:			0.000	0.001		
	Residual Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		900.4	25.006f	0.000	101.45	12.891
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			9.23		37.04	97.82	9.261
Distances in METERS.-----				-----Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

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After hook load drop							
WEIGHT and DISPLACEMENT and WATERPLANE STATUS							
USK draft: 2.283 @ 50.00f, 2.172 @ 25.00f, 2.062 @ 0.00							
Trim: Fwd 0.221/50.000, Heel: 0.00 deg.							
Part-----	Weight(MT)----		LCG-----	TCG-----	VCG-----		
LIGHT SHIP	405.97	25.000f	0.000	3.000			
Crane	250.00	7.500f	0.000	5.200			
Deck Cargo	727.97	27.500f	0.000	4.000			
Total Fixed----->	1,383.94	23.154f	0.000	3.923			
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
04_WBTK.P	1.000	1.025	161.60	25.000f	5.980p	1.509	162.73*
04_WBTK.S	1.000	1.025	161.60	25.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			472.52	32.161f	0.000	1.626	542.42
Total Weight----->			1,856.46	25.446f	0.000	3.339	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL	1.025		1,856.46	25.456f	0.001	1.112	

	Righting Arms:		0.000	0.001s			
	External Arms:		0.000	0.001			
	Residual Righting Arms:		0.000	0.000			
Part-----	SpGr-----		WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		900.4	25.006f	0.000	103.43	13.142
			MT/cm-----	m.-MT/cm-----		GML-----	GMT-----
			9.23		37.58	101.20	10.915
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: INFINITY; t3: 30; t: 15.00
limmarg: INFINITY; t3: 15; t: 7.50
limmarg: INFINITY; t3: 8; t: 3.75
limmarg: INFINITY; t3: 4; t: 1.88
limmarg: INFINITY; t3: 2; t: 0.94
limmarg: INFINITY; t3: 1; t: 0.47
limmarg: INFINITY; t3: 1; t: 0.23
limmarg: INFINITY; t3: 1; t: 0.12
limmarg: INFINITY; t3: 1; t: 0.06
limmarg: INFINITY; t3: 1; t: 0.03
limmarg: INFINITY; t3: 1; t: 0.01

Theta3 is 0.97 degrees, as demonstrated below.

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 23.154f TCG = 0.000 VCG = 3.923

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
2.050	0.25f	0.00	1,856.5	0.000	0.000	0.0000
2.049	0.25f	0.97s	1,856.5	0.000	0.192	0.0016
2.036	0.26f	5.00s	1,856.5	0.000	0.979	0.0428
2.066	0.37f	10.00s	1,856.5	0.000	1.536	0.1579
2.229	0.49f	15.00s	1,856.5	0.000	1.661	0.3005
2.261	0.50f	15.82s	1,856.5	0.000	1.663	0.3242
2.424	0.62f	20.00s	1,856.5	0.000	1.618	0.4447
2.595	0.78f	25.00s	1,856.5	0.000	1.473	0.5804
2.744	0.94f	30.00s	1,856.5	0.000	1.276	0.7007

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1383.94 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 1 > 1.000 LARGE

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 23.154f TCG = 0.000 VCG = 3.923

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
2.050	0.25f	0.00	1,856.5	0.000	0.000	0.0000
2.049	0.25f	0.97s	1,856.5	0.000	0.192	0.0016
2.036	0.26f	5.00s	1,856.5	0.000	0.979	0.0428
2.066	0.37f	10.00s	1,856.5	0.000	1.536	0.1579
2.229	0.49f	15.00s	1,856.5	0.000	1.661	0.3005
2.261	0.50f	15.82s	1,856.5	0.000	1.663	0.3242
2.424	0.62f	20.00s	1,856.5	0.000	1.618	0.4447
2.595	0.78f	25.00s	1,856.5	0.000	1.473	0.5804
2.744	0.94f	30.00s	1,856.5	0.000	1.276	0.7007
2.869	1.10f	35.00s	1,856.5	0.000	1.049	0.8024
2.971	1.26f	40.00s	1,856.5	0.000	0.804	0.8834
3.047	1.41f	45.00s	1,856.5	0.000	0.547	0.9424
3.100	1.55f	50.00s	1,856.5	0.000	0.282	0.9786
3.128	1.68f	55.00s	1,856.5	0.000	0.012	0.9915
3.129	1.69f	55.23s	1,856.5	0.000	0.000	0.9915
3.134	1.80f	60.00s	1,856.5	0.000	-0.259	0.9807

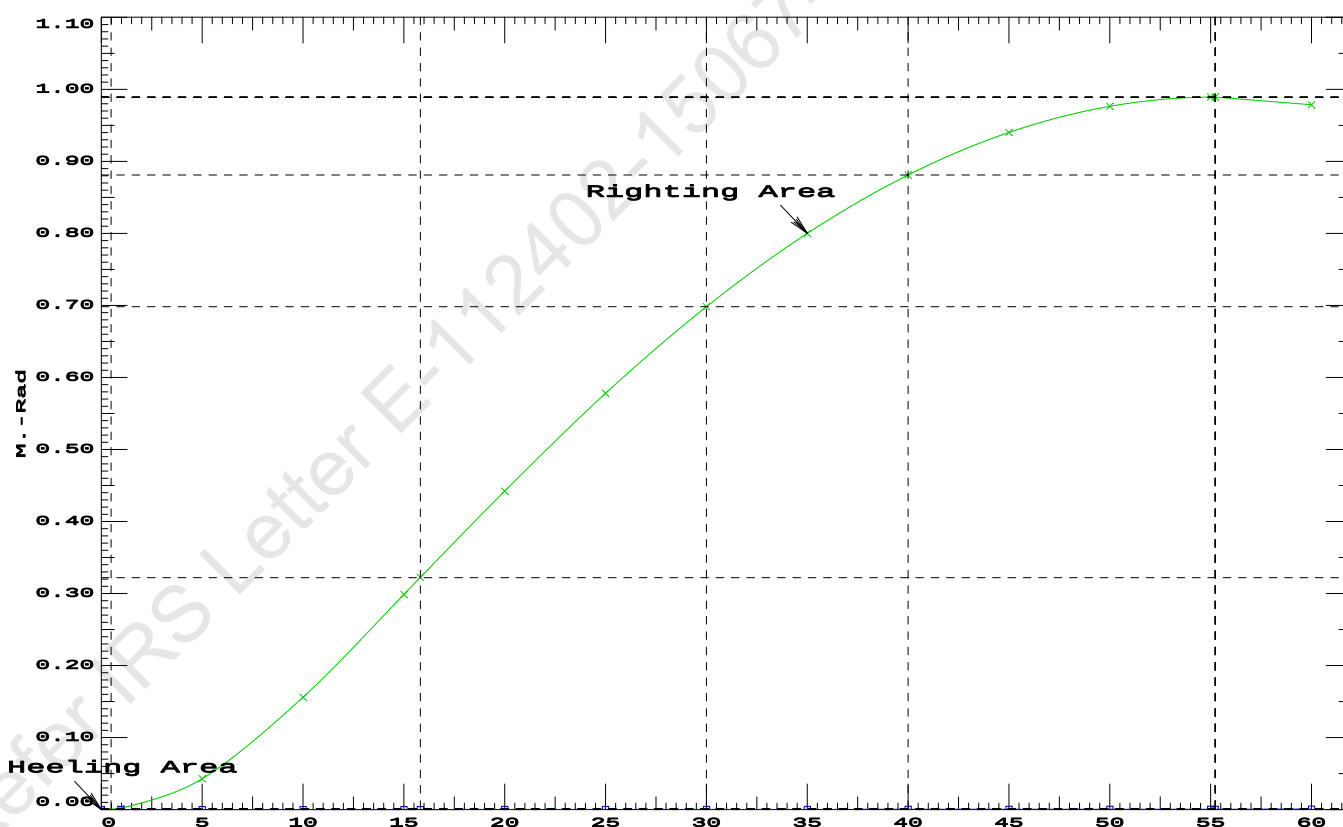
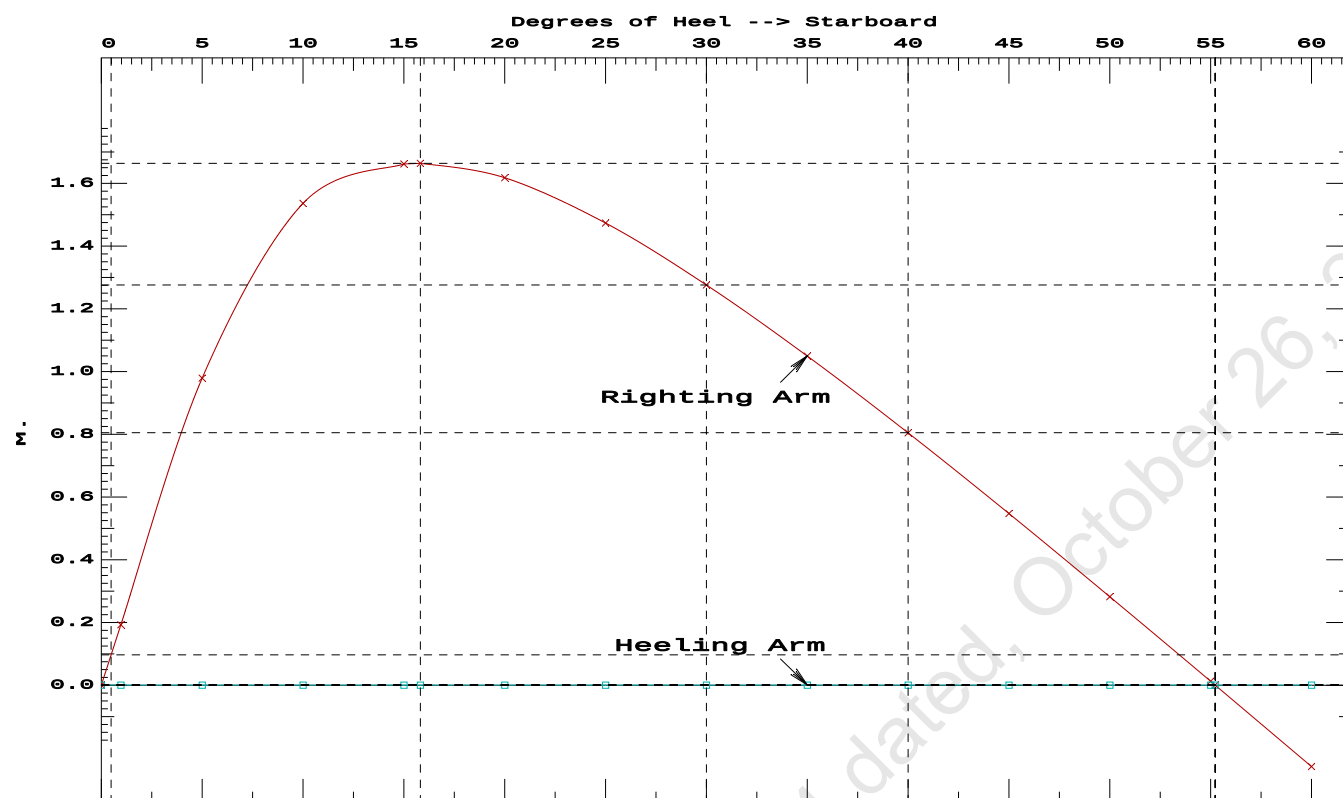
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1383.94 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero > 1.000 LARGE
(2) Angle from abs 1 deg to RAZero > 20.00 deg 54.26 P

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LIGHTSHIP PARTICULARS

Lightship Weight	=	405.97	T
VCG	=	3.000	M above Baseline
LCG	=	25.000	M fwd of AP (Fr.0)
TCG	=	0.000	M off centerline

(Reference: Lightship particulars are taken from approved draft survey report.)

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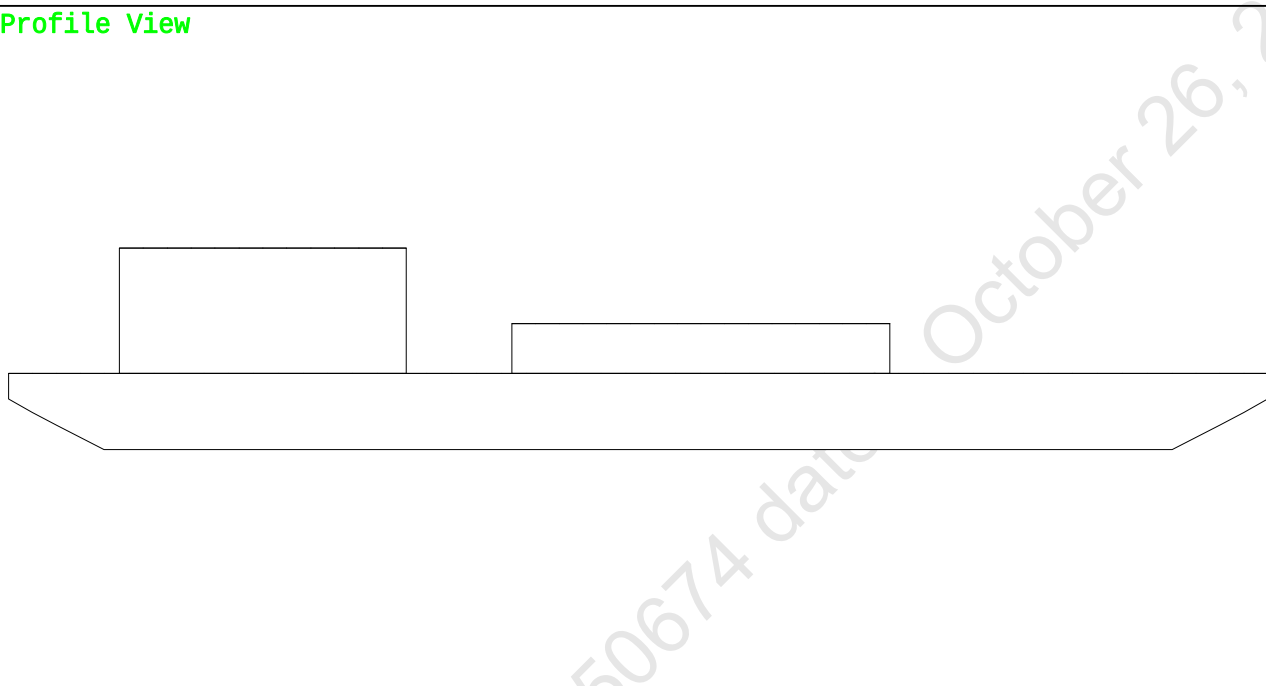
USK draft refers to the line:

0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

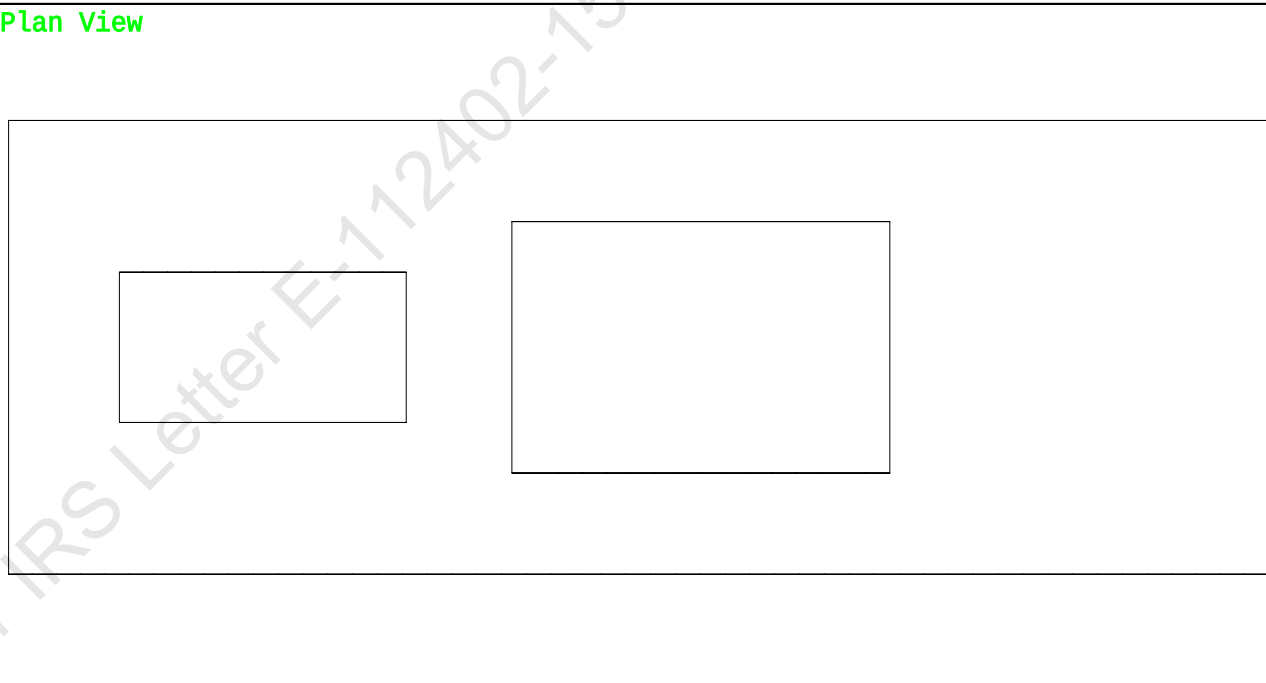
WINDAGE AREA CALCULATION AT DIFFERENT DRAFTS

Condition Graphic

Profile View



Plan View



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LATERAL PLANE STATUS

USK draft: 0.512 @ 50.00f, 0.512 @ 25.00f, 0.512 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	22.3	25.000f	-0.253	121.4	25.000f	1.281
CRANE				57.0	10.100f	5.000
DECKCARGO.C				29.4	27.500f	3.500
Total Lateral Plane->	22.3	25.000f	-0.253	207.8	21.267f	2.615
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 0.612 @ 50.00f, 0.612 @ 25.00f, 0.612 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	26.8	25.000f	-0.302	116.9	25.000f	1.229
CRANE				57.0	10.100f	4.900
DECKCARGO.C				29.4	27.500f	3.400
Total Lateral Plane->	26.8	25.000f	-0.302	203.3	21.184f	2.572
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 0.712 @ 50.00f, 0.712 @ 25.00f, 0.712 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	31.3	25.000f	-0.351	112.4	25.000f	1.176
CRANE				57.0	10.100f	4.800
DECKCARGO.C				29.4	27.500f	3.300
Total Lateral Plane->	31.3	25.000f	-0.351	198.8	21.097f	2.529
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 0.812 @ 50.00f, 0.812 @ 25.00f, 0.812 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	35.9	25.000f	-0.400	107.8	25.000f	1.123
CRANE				57.0	10.100f	4.700
DECKCARGO.C				29.4	27.500f	3.200
Total Lateral Plane->	35.9	25.000f	-0.400	194.2	21.006f	2.487
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 0.912 @ 50.00f, 0.912 @ 25.00f, 0.912 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	40.4	25.000f	-0.450	103.3	25.000f	1.070
CRANE				57.0	10.100f	4.600
DECKCARGO.C				29.4	27.500f	3.100
Total Lateral Plane->	40.4	25.000f	-0.450	189.7	20.911f	2.445
Distances in METERS.-----						

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GHS 17.22 PONTON

LATERAL PLANE STATUS

USK draft: 1.012 @ 50.00f, 1.012 @ 25.00f, 1.012 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	45.0	25.000f	-0.497	98.7	25.000f	1.018
CRANE				57.0	10.100f	4.500
DECKCARGO.C				29.4	27.500f	3.000
Total Lateral Plane->	45.0	25.000f	-0.497	185.1	20.809f	2.405
Distances in METERS.						

LATERAL PLANE STATUS

USK draft: 1.112 @ 50.00f, 1.112 @ 25.00f, 1.112 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	49.7	25.000f	-0.545	94.0	25.000f	0.966
CRANE				57.0	10.100f	4.400
DECKCARGO.C				29.4	27.500f	2.900
Total Lateral Plane->	49.7	25.000f	-0.545	180.4	20.699f	2.367
Distances in METERS.						

LATERAL PLANE STATUS

USK draft: 1.212 @ 50.00f, 1.212 @ 25.00f, 1.212 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	54.4	25.000f	-0.594	89.3	25.000f	0.914
CRANE				57.0	10.100f	4.300
DECKCARGO.C				29.4	27.500f	2.800
Total Lateral Plane->	54.4	25.000f	-0.594	175.7	20.584f	2.329
Distances in METERS.						

LATERAL PLANE STATUS

USK draft: 1.312 @ 50.00f, 1.312 @ 25.00f, 1.312 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	59.1	25.000f	-0.642	84.5	25.000f	0.863
CRANE				57.0	10.100f	4.200
DECKCARGO.C				29.4	27.500f	2.700
Total Lateral Plane->	59.1	25.000f	-0.642	170.9	20.462f	2.291
Distances in METERS.						

LATERAL PLANE STATUS

USK draft: 1.412 @ 50.00f, 1.412 @ 25.00f, 1.412 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	63.9	25.000f	-0.691	79.8	25.000f	0.810
CRANE				57.0	10.100f	4.100
DECKCARGO.C				29.4	27.500f	2.600
Total Lateral Plane->	63.9	25.000f	-0.691	166.2	20.333f	2.255
Distances in METERS.						

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LATERAL PLANE STATUS

USK draft: 1.512 @ 50.00f, 1.512 @ 25.00f, 1.512 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	68.7	25.000f	-0.739	75.0	25.000f	0.759
CRANE				57.0	10.100f	4.000
DECKCARGO.C				29.4	27.500f	2.500
Total Lateral Plane->	68.7	25.000f	-0.739	161.4	20.194f	2.220
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.612 @ 50.00f, 1.612 @ 25.00f, 1.612 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	73.6	25.000f	-0.786	70.1	25.000f	0.708
CRANE				57.0	10.100f	3.900
DECKCARGO.C				29.4	27.500f	2.400
Total Lateral Plane->	73.6	25.000f	-0.786	156.5	20.044f	2.188
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.712 @ 50.00f, 1.712 @ 25.00f, 1.712 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	78.5	25.000f	-0.834	65.2	25.000f	0.657
CRANE				57.0	10.100f	3.800
DECKCARGO.C				29.4	27.500f	2.300
Total Lateral Plane->	78.5	25.000f	-0.834	151.6	19.883f	2.157
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.812 @ 50.00f, 1.812 @ 25.00f, 1.812 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	83.4	25.000f	-0.882	60.3	25.000f	0.607
CRANE				57.0	10.100f	3.700
DECKCARGO.C				29.4	27.500f	2.200
Total Lateral Plane->	83.4	25.000f	-0.882	146.7	19.712f	2.128
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.912 @ 50.00f, 1.912 @ 25.00f, 1.912 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	88.3	25.000f	-0.930	55.4	25.000f	0.556
CRANE				57.0	10.100f	3.600
DECKCARGO.C				29.4	27.500f	2.100
Total Lateral Plane->	88.3	25.000f	-0.930	141.8	19.529f	2.100
Distances in METERS.-----						

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GHS 17.22 PONT00N

LATERAL PLANE STATUS

USK draft: 2.012 @ 50.00f, 2.012 @ 25.00f, 2.012 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	93.2	25.000f	-0.978	50.5	25.000f	0.505
CRANE				57.0	10.100f	3.500
DECKCARGO.C				29.4	27.500f	2.000
Total Lateral Plane->	93.2	25.000f	-0.978	136.9	19.333f	2.073
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 2.112 @ 50.00f, 2.112 @ 25.00f, 2.112 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	98.2	25.000f	-1.025	45.5	25.000f	0.455
CRANE				57.0	10.100f	3.400
DECKCARGO.C				29.4	27.500f	1.900
Total Lateral Plane->	98.2	25.000f	-1.025	131.9	19.118f	2.050
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 2.212 @ 50.00f, 2.212 @ 25.00f, 2.212 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	103.2	25.000f	-1.073	40.5	25.000f	0.405
CRANE				57.0	10.100f	3.300
DECKCARGO.C				29.4	27.500f	1.800
Total Lateral Plane->	103.2	25.000f	-1.073	126.9	18.887f	2.029
Distances in METERS.-----						



MEMORANDUM OF FREEBOARDS

28 September 2020

Chisti Marine Engineering Works.

Yard No. 013

Type B freeboard

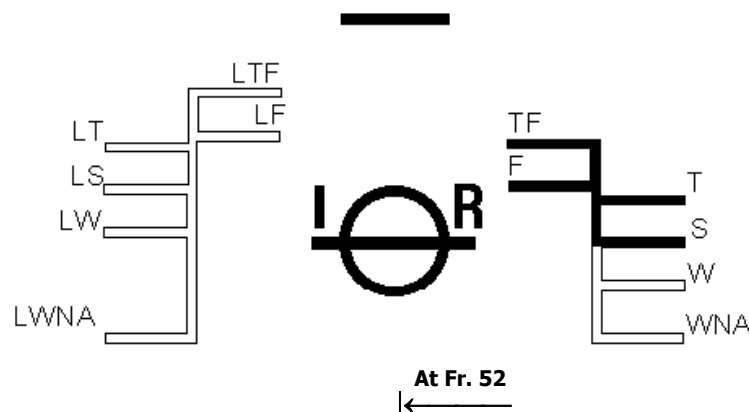
Freeboard Length : 48.000 m

Summer WL Extreme Draught : 2.212 m
- load line amidships

The upper edge of the deckline from which these freeboards are measured is 0 mm below top of
* Wood/ * steel Deck * at side / *continued to side. (*Delete words not applicable).

Freeboards from Deck Line		Seasonal, Fresh Water, or Timber allowances		
Tropical Fresh Water	713	mm.	97	mm. above summer
Fresh Water	759	mm.	51	mm. above summer
Tropical	764	mm.	46	mm. above summer
Summer	810	mm.	Upper edge of line through centre	
Winter	Not	mm.	Assigned	mm. below summer
Winter North Atlantic	Not	mm.	Assigned	mm. below summer
Timber Tropical fresh water	—	mm.	—	mm. above timber summer
Timber fresh water	—	mm.	—	mm. above timber summer
Timber Tropical	—	mm.	—	mm. above timber summer
Timber Summer	—	mm.	—	mm. above centre of ring
Timber Winter	—	mm.	—	mm. below timber summer
Timber Winter North Atlantic	—	mm.	—	mm. below timber summer

Valid only for unmanned operation while under tow along the Indian Coast, during the course of which the vessel does not go more than 20 nautical miles from the nearest land, and may cross gulfs or similar features recognised by the Administration.





IRCLASS
Indian Register of Shipping
CIN: U61100MH1975NPL018244

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Our ref: E-112248-150381

22-October-2020

Company : Cybermarine Knowledge System Pvt. Ltd

Dear Sir/Madam,

Your document (detailed below) has been reviewed.

Project Details	
Project number	NC002549
Project name	Chisti Marine Engineering Works - CHISTI -013 & CHISTI-014
Project Manager	Dhruvesh Singh
Project In-Charge	Veerendra P

Document Details	
Plan No.	CM20-V-110-0207A
Plan Title	Draft Survey Report
Rev.No.	0
Upload.No.	1
Submitted Date	20-October-2020
Review Status	Approved
Parent Plan Title	

Notes

1. The review status Approved indicates:

The following lightship particulars are approved based on the Draft survey for Yard No. 013 conducted on 16th October 2020:

Lightship Weight : 405.97 T
Lightship L.C.G : 25.00 m forward of Frame no. 0
Lightship V.C.G : 3.00 m above Baseline (VCG taken at the level of main deck)
Lightship T.C.G : 0.00 m off Centre line

Final stability documents prepared based on the approved lightship particulars should be submitted for approval.
Copy of the approved Draft survey report should form part of the Final Stability booklet.

Thanking you,
Yours Sincerely,

For INDIAN REGISTER OF SHIPPING

Ramiz Ismail
Surveyor

This is an electronically published document and does not require signature.

DRAFT SURVEY REPORT OF KB-28 (YARD NO. CHISTI / 013)

LIGHTSHIP PARTICULARS:

Lightship Displacement - 405.97 T
 Lightship L.C.G - 25.000 m from AP (Fr.0)
 Lightship V.C.G - 3.000 m above BL
 Lightship T.C.G - 0.000 m off-centerline



APPROVED

DATE: 22 Oct 2020

INDIAN REGISTER OF SHIPPING
MUMBAI

SEE LETTER E-112248-150381

Rev.	Description				Date	Prepared	Review
<u>Issued for</u>					:	<u>Approval</u>	
Client: K.B. SHIPPING & CO.,JAMNAGAR Builder: CHISTI MARINE ENGINEERING WORKS					Project: PONTOON (YARD NO. CHISTI /013)		
Doc #				Document Title: DRAFT SURVEY REPORT		Prepared	OM
CM20-V-110-0207-A						Checked	MG
Scale	Size	Sheet	Date			Review 1	
NTS	A4	1 of 14	19-10-2020			Review 2	



Cybermarine
 www.cybermarine.net info@cybermarine.net

10, Bukit Batok Crescent,
#12-01, The Spire Building,
Singapore-658079.

501, Technocity IT Premises Co-op. Soc. Ltd,
Plot No. X-5/3, Mahape,
Maharashtra- 400710, India.

22503, Katy Freeway #7,
Katy, Texas,
USA 77450.

GENERAL PARTICULARS

Name of the Vessel	KB-28
Client	K.B. SHIPPING & CO.,JAMNAGAR
Builder	CHISTI MARINE ENGINEERING WORKS
Yard No.	CHISTI/013
Type of Vessel	PONTOON
Reg. No.	JMR-0016
Port of Registry	JAMNAGAR
Class Notation	⚡ SU PONTOON,"INDIAN COASTAL SERVICE" DESCRIPTIVE NOTATION: "FOR CRANE OPERATION" "DECK STRENGTHENED FOR 10T/M²"

Main Particulars

LOA	50.000 M
LBP	50.000 M
Breadth (Mld.)	18.000 M
Depth (Mld.)	3.000 M
Draft (Mld.)	2.200 M

References

AP	at Fr.0
FP	at Fr.100
Midship	at Fr.50

Frame spacing

Throughout	500 mm
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DRAFT SURVEY REPORT : KB-28 (Yard No. CHISTI / 013)

Location of Test Sikka Port, Gujarat
 Date 16-Oct-20
 Time 2:30:00 PM
 Temperature 30 °C
 Weather Calm
 Wind Light
 Specific Gravity 1.025
 Restraining Line Loose
 Witnessed by Mr.Vaghani (Yard Rep.)
 Mr.Sanjeev Sharma (IRS Surveyor)

Drafts :**(w.r.t keel line)**

Location	Draft Port (m)	Draft Stbd (m)	Mean Draft (m)
Fwd (Fr.92)	0.520	0.520	0.520
Mid (Fr.50)	0.520	0.520	0.520
Aft (Fr.8)	0.520	0.520	0.520

Hydrostatics

Correction for Drafts

To correct for observed drafts at marks and obtain Drafts at perps.

LBP - 50.000 m

Observed Drafts (wrt keel line)

Fwd - 0.520 m
 Aft - 0.520 m

Correction to Drafts (wrt Keel line)

Location	At Fwd	At Aft	
	(Fr.92)	(Fr.8)	
Draft at Marks	0.520	0.520	m
Trim at Marks		0.000	m
Distance of Marks from PP	4.000	4.000	m
Distance between Marks		42.000	m
Draft correction factor		0.000	
Correction at PP	0.000	0.000	
Corrected Draft at PP	0.520	0.520	
Trim		0.000	m
Mean Draft		0.520	m
From Hydrostatics against above Mean Draft and Trim			
Displacement at density(T/M^3)=1.025 T/m^3		405.97	T
LCB		25.000	m
KMt		54.828	m
LCG		25.000	m
TCG		0.000	m

NOTE

1. Hydrostatic Particulars computed by Computer Program at Sp.Gr. 1.025
2. LCB, LCG are defined from AP, Fwd of AP is +ve and Aft of AP is -ve
3. Trim by Fwd is -ve and Trim by Aft is +ve.
4. KB is measured from Base Line.
5. Draft is measured from Keel Line

20-10-20 00:37:51
GHS 17.22

Cybermarine Technologies PTE. Ltd.
PONT00N

USK draft refers to the line:

0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

HYDROSTATIC PROPERTIES

No Trim, No Heel, Fixed VCG = 0.000

Draft@	Displacement	Buoyancy-Ctr.	Weight/	Moment/
25.000f	---Weight(MT)---	LCB-----VCB-----	cm-----	LCF---cm trim----
0.520	405.97	25.000f 0.252	8.21 25.000f	27.04 333.03 54.828
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.				
Trim is per 50.00m.				

Draft is from USK.

Weights to go off Board

No	Item	WEIGHT T	LCG m	L-MOM T-m	VCG m	V-MOM T-m	TCG m	T-MOM T-m
-	-	0.00	0.000	0.00	0.000	0.00	0.000	0.00
TOTAL		0.000	0.000	0.000	0.000	0.000	0.000	0.000

Weights to go on Board

No	Item	WEIGHT T	LCG m	L-MOM T-m	VCG m	V-MOM T-m	TCG m	T-MOM T-m
-	-	0.000	0.000	0.00	0.000	0.00	0.000	0.00
TOTAL		0.000	0.000	0.000	0.000	0.000	0.000	0.000

Lightship Particulars

Item	WEIGHT T	LCG m	L-MOM T-m	VCG m	V-MOM T-m	TCG m	T-MOM T-m
Particulars as Draft Survey	405.97	25.000	10149.250	3.000	1217.910	0.000	0.000
Weights to go Off board	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Weights to go on board	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Lightship Particulars	405.97	25.000	10149.25	3.000	1217.91	0.000	0.00

Lightship Particulars

Displacement	405.97	T
LCG	25.000	m From AP
TCG	0.000	m Off-Centreline
VCG	3.000	m above BL

NOTE

1. Hydrostatic Particulars computed by Computer Program at Sp.Gr. 1.025
2. LCB, LCG are defined from AP, Fwd of AP is +ve and Aft of AP is -ve
3. Trim by Fwd is -ve and Trim by Aft is +ve.
4. KB is measured from Base Line.
5. Draft is measured from Keel Line

LIGHT-SHIP SURVEY PROCEDURE AND REPORTING FORMAT

Part I



Pre requisites for Conducting Light-Ship Survey (To be provided /confirmed by Builder/Owner/Operator/Consultant prior commencement of Light-Ship Survey)

Name Of Builder/owner rep conducting trial	CHISTI MARINE ENG. WORKS
Hull Number/Vessel name	CHISTI- 013 / K.B-28
Type of Vessel (Cargo/passenger)	PONTON
Location of Vessel	SILKA
Date of Proposed Light-Ship Survey	16/10/2020
Draft of Vessel and trim	0.52m - Trim - NIL
Depth of Water	3.0 M
Anticipated tide on date of survey	5.4m. - 16/10/2020
Weather prediction for proposed date	Calm, Hot
Availability of following reports from builder/Owner/Owners rep	
Copy of Approved Inclining experiment report	☐ YES ☒ NO N/A
Details of Periodical Draft Survey report/any Draft Survey report conducted after the approval of Final TSB.	☐ YES ☒ NO N/A
Declaration of weights to go on	☐ YES ☒ NO N/A
Declaration of weights to go off - tank contents - Loose items	☐ YES ☒ NO N/A

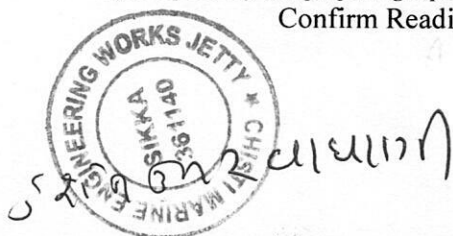


Facilities/Arrangement for conduct of Light-Ship Survey	
Availability of boat to take draft reading	☒ YES ☐ NO
Availability of draft mark plan/approved document like Prelim TSB/Final TSB for confirming location of draft marks	☒ YES ☐ NO
Safety Requirements- lighting, gas freeing, accessibility of areas requiring inspection as part of Survey	☒ YES ☐ NO
Availability of sounding tapes, hydrometer etc	☒ YES ☐ NO
Closing/isolating cross connected tanks	☒ YES ☐ NO N/A
Removal of gangway during draft reading/experiment	☒ YES ☐ NO
Permanent ballast used ? ; if any and its details	☐ YES ☒ NO N/A
If vessel has any list, reason for the list is established.	☐ YES ☒ NO N/A
A trial is conducted to see whether the vessel's draft corresponding to the loading condition is matching with the approved hydrostatics.	☒ YES ☐ NO N/A

NOTE : The responsibility of the IRS Surveyor is to verify that the survey is conducted according to accepted procedures and that all basic measurements and data are correctly taken and recorded. The responsibility for making preparation, conducting lightship survey and preparing report rest with owner/operator.

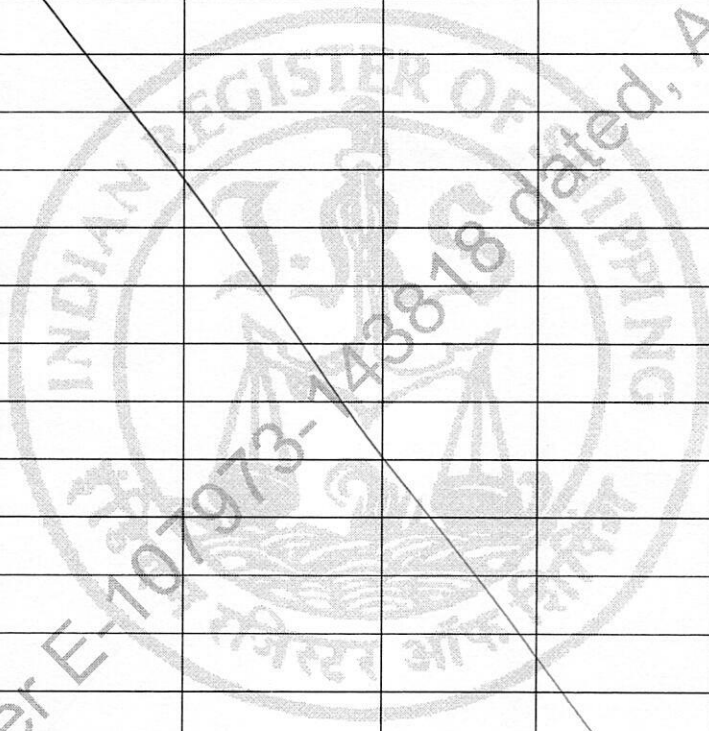
Confirm Readiness for test

Name & Signature of Shipyard /Owners rep



[illegible]

LCG: Longitudinal Center of Gravity from AP/Transom/Midship/Frame -----

[illegible]

LCG: Longitudinal Center of Gravity from AP/Transom/Midship/Frame -----



Note: Tank name should be mentioned as per the approved TSB / Capacity Plan.



Weights To Be Relocated									
Sl No.	Weight Description	Weight (t)	L.C.G (m)		V.C.G (m)		T.C.G (m)		Surveyor Verification
			Initial	Final	Initial	Final	Initial	Final	
									☐
									☐
									☐
									☐
									☐
									☐
									☐
									☐
									☐
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									☐
									☐
									☐

LCG: Longitudinal Center of Gravity from AP/Transom/Midship/Frame -----

Refer IRS Letter E-107973-143818 dated, August 08, 2020

LIGHTSHIP SURVEY PROCEDURE AND REPORTING FORMAT

Part II

REPORTING FORM FOR LIGHTSHIP SURVEY



Attending Surveyor should witness Light-Ship Survey only if the Part-I submitted by Owner/Builder/Operator/Consultant is found satisfactory

General Particulars

Name of Vessel	K. B-28
Builder/ Owner	CHISTI MARINE ENG. WORKS / K.B. SHIPPING
Yard Number/ IR Number	CHISTI-013/65347
Type of Vessel	Passenger ☒ Non Passenger ☒
Date(dd/mm/yyyy);Local Time of inspection/test	16/10/2020 - 1430 HRS
Wind / Weather condition	5 knots / CALM
Location of Test	SIRKA
Depth of water where vessel is moored, m	3.0 M
Method of reading Draft(By Boat-Y/N)	BOAT

Location of Draft Marks	Frame Number	Draft Reading (Port side),m	Draft Reading (STBD Side),m
Aft	8	0.52 M	0.52 M
Midship	50	0.52 M	0.52 M
Forward	92	0.52 M	0.52 M

Note : Recommended to attach Photographs of draft; as record

Density of Water

Results of 3 samples from Forward, Aft and Mid of the vessel		
Sample 1: (ford)	1.025	t/m3
Sample 2: (mid)	1.025	t/m3
Sample 3: (aft)	1.025	t/m3



Verification of weights to go on / go off
/re-location

SEE PART 1

Note: See Part 1 (the sheet for recording the above data) , verify and endorse.

Special Comments on Light Ship Survey, if any.

lightsup Drafft Servey Completed Satisfact
-ory. at SIKKA.

Attending surveyor to forward both forms Part 1 and Part 2 together with completed
attachments 1 and 2 to HO.

Name & designation of Attending Surveyor:

[Signature] SANDHEV SURSTMA

16/10/2020 | SIKKA



Refer IRS Letter E-107973-143818 dated, August 08, 2020